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MCMI-IIITM

MillonTM Clinical Multiaxial Inventory-III

MANUAL **Fourth Edition**

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MCMI-III Manual (Fourth Edition)

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SOUTHEAST: *Alabama*—VAH, Tuscaloosa; VAH, Tuskegee. *Florida*—University of Miami, Coral Gables; Center for Behavioral Change, Fort Myers; Grant Hospital, Miami; Harbor View Hospital, Miami; HCA Highland Park Hospital, Miami; Jackson Memorial Hospital, Miami; South Miami Hospital, South Miami; VAH, Miami. Southern Psychological Services, Tallahassee. *Georgia*—SPAR-Atlanta Police Bureau, Atlanta. *Kentucky*—Pathways, Inc., Ashland; National Institute of Mental Health Clinical Research Center, Lexington; University of Kentucky, Lexington; Central State Hospital, Louisville. *Louisiana*—VAH, New Orleans. *North Carolina*—Guilford Psychiatric Associates, Greensboro; VAH, Salisbury. *South Carolina*—Medical College of South Carolina, Charleston; Cumberland Hospital, Fayetteville; Mecklenburg CMHC, Charlotte; Marshall I. Pickens Hospital, Greenville; Self Regional Healthcare, Greenwood. *Tennessee*—Psychological Associates, McMinnville; Sullivan Associates, Greenville; Wilson County Mental Health Center, Lebanon; VAH, Memphis; VAH, Murfreesboro; Greenleaf Center, Chattanooga. *Virginia*—Medical College of Virginia, Richmond; Appalachian Counseling Center, Roanoke; Tidewater Psychiatric Institute, Virginia Beach.

MIDWEST: *Illinois*—Emerson Counseling Center, Arlington Heights; Cook County Hospital, Chicago; Edgewater-Uptown Mental Health Center, Chicago; Field Clinic, Chicago; Grant Hospital, Chicago; Illinois Masonic Hospital, Chicago; Illinois State Psychiatric Institute, Chicago; Jewish Family Service, Chicago, Highland Park, and Skokie; Loyola University Guidance Center, Chicago; Mercy Hospital and Medical Center, Chicago; Near North Family Center, Chicago; Northwestern University Memorial Hospital and Institute of Psychiatry, Chicago; Ravenswood Hospital and

Community Mental Health Center, Chicago; University of Chicago Hospitals, Chicago; University of Illinois, Medical Center and Chicago Circle, Chicago; VA Research Hospital, Chicago; VA Westside Hospital, Chicago; Weiss Memorial Hospital, Chicago; VAH, Danville; St. Mary's Hospital, Decatur; Forest Hospital, Des Plaines; Forest Institute of Professional Psychology, Des Plaines; VAH, Downey; Lincoln Center for Clinical Services, Limited, Oak Park; VAH, Hines; Human Systems, Limited, Oakbrook; Lutheran General Hospital, Park Ridge; Lutheran Pastoral Counseling Center, Park Ridge; Psychology Associates, Quincy; Northern Illinois Human Resource Development Center, Rolling Meadows; Virginia Frank Center, Skokie; Lake County Family Service, Waukegan. *Indiana*—Indiana University, Bloomington and Gary; Madison Center, South Bend; Southern Hills Mental Health Center, Jasper; GrantBlackford Mental Health, Marion; Michiana Psychological Association, Mishawaka; VAH, Marion. *Michigan*—Assessment Center, Muskegon; DeLano Clinic, Kalamazoo; VAH, Allen Park; Bethany Christian Services, Grand Rapids; Holland Community Hospital, Holland; Child and Family Center, Kalamazoo; Affiliated Psychological Services, PC, Muskegon; Pine Rest Christian Hospital, Grand Rapids; Life Transitions Family Counseling Centers, Inc., Lake Orion. *Minnesota*—Heart of Lakes Clinic, Annandale; Clinic for Attention, Learning and Memory, Minneapolis; VAH, Minneapolis; Human Services, Inc., Oakdale; VAH, St. Cloud; Consultation Services Center, St. Paul. *Nebraska*—Capstone Behavioral Health, Inc., Omaha; Premier Psychiatric Research Institute, Lincoln. *North Dakota*—Northwest Human Service Center, Williston. *Ohio*—Cleveland Psychiatric Institute, Cleveland; Comprehensive Psychiatric and Psychological Services, Columbus; VAH, Cleveland; Behavioral Consultants, Norton; Department of Corrections, Columbus. *Wisconsin*—Affiliated Psychological Resources, SC, Madison; Mendota Mental Health Institute, Madison; University of Wisconsin, Madison and Waukesha; Psychology Center, Madison; Riverhill Psychological Associates, Manitowoc; Medical Associates, Menomonee Falls; Medical College of Wisconsin, Milwaukee; Milwaukee General Hospital, Milwaukee; Psychological Evaluation and Treatment Clinic, Milwaukee; St. Croix Regional Medical Center, St. Croix Falls; Lutheran Social Services, Wausau.

NORTH AND SOUTH CENTRAL: *Arkansas*—The Bridgeway, Little Rock. *Colorado*—Colorado State University, Ft. Collins; Ft. Logan Mental Health Center, Ft. Logan. *Idaho*—Integrated Behavioral Healthcare, Coeur D'Alene; State Hospital North, Orofino. *Iowa*—French MH Center, Davenport. *Kansas*—Shawnee Mission Psychiatric Group, Shawnee Mission; Topeka State Hospital, Topeka; Central Kansas Psychological Services, Great Bend; Sumner County MH Center, Wellington; University of Kansas Medical Center, Kansas City. *Missouri*—Gray Psychological Clinic, St. Charles; Consulting Psychologists, Inc., Warrensburg. *Montana*—Northwest Counseling Center, Billings. *Nebraska*—Nova Therapeutic Center, Omaha. *Oklahoma*—Poyner Psychological Services, Choctaw; Central State Hospital, Norman; The Psychiatric Center of Oklahoma, Oklahoma City; Oklahoma State University, Stillwater; Psychiatric and Psychological Associates, Tulsa. *Texas*—Dallas County Medical Health Center, Dallas; Prima A.D.D. Corp., Dallas; University of Texas Health Sciences Center, Dallas; University of Texas Medical Branch, Galveston; Texas Tech University, Lubbock; Metro Counseling Associates, Arlington; West Houston Psychological Associates, Houston. *Utah*—VAH, Salt Lake City. *Wyoming*—Memorial Hospital, Cheyenne.

WEST COAST: *California*—Valley Psychological Services, Bakersfield; City of Hope National Medical Center, Duarte; Mid-Valley Counseling and Psychological Services, Encino; University of California, La Jolla; Garrard Center for Psychotherapy, La Mesa; Jerry L. Pettis Memorial Veterans Hospital, Loma Linda; Loma Linda Psychiatric Group, Loma Linda; Memorial Medical Center, Long Beach; VAH, Long Beach; Western Counseling Center, Long Beach; Cedars-Sinai Medical Center, Los Angeles; Psychotronics, Los Angeles; VA Outpatient Clinic, Los Angeles; VA Wadsworth Hospital, Los Angeles; VADC, Los Angeles; Guidance Center, Oceanside; Stanford University Medical Center, Palo Alto; VAH, Sepulveda; California Psychological Services, Sherman Oaks;

Institute for Human Studies, Sherman Oaks. *Hawaii*—Psychiatric and Psychological Affiliates, Honolulu. *Oregon*—Child/Adult Intervention Services, Tigard. *Washington*—VAH, Seattle; Providence Medical Center, Seattle; University of Washington Medical Center, Seattle; Center for Counseling and Psychotherapy, Yakima; Evergreen Psychological Associates, Edmonds.

CANADA: *Alberta*—Edmonton General Hospital, Edmonton; Calgary Hospital, Calgary; Drumheller Institute, Drumheller. *British Columbia*—Victoria Psychological Services, Victoria; University of British Columbia, Vancouver. *Manitoba*—Brandon Mental Health Centre, Brandon. *Ontario*—Woodstock General Hospital, Woodstock; Queen Anne's Mental Health Centre, Toronto; Clarke Institute, Toronto.

EUROPE: *Denmark*—University of Copenhagen Hospitals; Slagese Hospital, Slagese; Glostrup Psychiatric Hospital, Glostrup. *England*—Glenside Hospital Bristol, University of Liverpool; Institute of Psychiatry, London; London Hospital, London; The Royal Free Hospital, London; West Suffolk District Hospital, Suffolk. *Scotland*—Royal Edinburgh Hospital, Edinburgh.

Theodore Millon
November, 2008

Chapter 1

Introduction

The Millon™ Clinical Multiaxial Inventory (MCMI™) is intended to be an evolving assessment tool, to be refined as needed on the basis of substantive advances. Each successive version of the MCMI test has been refined and strengthened to incorporate developments in theoretical logic, research data, and professional nosology.

The current release of the MCMI-III™ test continues this tradition with updated norms and the addition of a new scale called Inconsistency to help detect random responding. These developments, described in Chapter 6, follow on the heels of the introduction of the Grossman facet scales for the Clinical Personality Patterns and Severe Personality Pathology scales in 2006.

Distinguishing Features

Several features distinguish the MCMI-III test from other inventories. These include the relative brevity of the inventory, its theoretical anchoring, its multiaxial format, its construction through three stages of validation, its use of base rate scores, and its interpretive depth.

Inventory Length

The MCMI authors have always tried to keep the

total number of items small enough to encourage the test's use in all kinds of diagnostic and treatment settings, yet large enough to permit the assessment of a wide range of clinically relevant behavior. At 175 items, the MCMI-III test is much shorter than comparable instruments. Potentially objectionable items have been screened out and vocabulary geared to an eighth-grade reading level. As a result, the great majority of patients can complete the MCMI-III test in 20 to 30 minutes, facilitating relatively simple and rapid administration while minimizing patient resistance and fatigue.

Theoretical Anchoring

Diagnostic instruments are more useful when they are systematically linked to a comprehensive clinical theory. Unfortunately, as many authors have noted (e.g., Butcher, 1972), assessment techniques and personality theory have developed almost independently. As a result, few diagnostic measures have been based on or have evolved from clinical theory. The MCMI-III test is different. Each of its Axis II scales is an operational measure of a syndrome derived from a theory of personality (Millon, 1969/1983, 1981, 1986a, 1986b, 1990; Millon & Davis, 1996).

Although the Axis I scales are not explicitly derived from the theory, they are nevertheless refined in terms of its generative framework. The scales and profiles of the MCMI-III test measure these theory-derived and theory-refined variables directly and quantifiably. Scale elevations and configurations can be used to suggest specific patient diagnoses and clinical dynamics as well as testable hypotheses about social history and current behavior.

Structural Characteristics

No less important than its link to theory is the coordination between a clinically oriented instrument and the official diagnostic system and its syndrome categories. Few currently available diagnostic instruments have been constructed to be as consonant with the official nosology as the MCMI-III test. With the advent of the *DSM-III*TM (*Diagnostic and Statistical Manual of Mental Disorders*, third edition), *DSM-III-R*TM, and *DSM-IV*[®] (American Psychiatric Association, 1994), diagnostic categories were precisely specified and operationally defined. The structure of the MCMI-III test parallels that of the *DSM* on a number of levels. First, the MCMI-III scales are grouped into the categories of personality and psychopathology to reflect the *DSM* distinction between Axis II and Axis I. Separate scales distinguish the more enduring personality characteristics of patients (Axis II) from the acute clinical disorders they display (Axis I), a distinction judged to be of considerable use by test developers and clinicians (Dahlstrom, 1972). Profiles based on all 24 clinical scales may be interpreted to illuminate the interplay between long-standing characterological patterns and the distinctive clinical symptoms currently being manifested.

Beyond the *DSM* distinction between psychiatric symptoms and enduring personality dispositions, the scales within each axis are further grouped according to the level of severity of the psychopathology. Thus, the premorbid characterological pattern of a patient is assessed independent of its degree of pathology. The Schizotypal, Borderline, and Paranoid scales represent greater levels of personality pathology and have been set off from the 11 basic personality scales, Schizoid through Masochistic (Self-Defeating). (See Table 1.1.) Similarly, the moderately

severe Clinical Syndromes are separated and assessed independently from the Severe Clinical Syndromes (Thought Disorder, Major Depression, and Delusional Disorder), which presumably have a more psychotic nature.

Table 1.1 The MCMI-III Scales

Clinical Personality Patterns

1	Schizoid
2A	Avoidant
2B	Depressive
3	Dependent
4	Histrionic
5	Narcissistic
6A	Antisocial
6B	Sadistic (Aggressive)
7	Compulsive
8A	Negativistic (Passive-Aggressive)
8B	Masochistic (Self-Defeating)

Severe Personality Pathology

S	Schizotypal
C	Borderline
P	Paranoid

Clinical Syndromes

A	Anxiety
H	Somatoform
N	Bipolar: Manic
D	Dysthymia
B	Alcohol Dependence
T	Drug Dependence
R	Post-Traumatic Stress Disorder

Severe Clinical Syndromes

SS	Thought Disorder
CC	Major Depression
PP	Delusional Disorder

Modifying Indices

X	Disclosure
Y	Desirability
Z	Debasement

Random Response Indicators

V	Invalidity
W	Inconsistency

Second, at the scale level, each axis is composed of dimensions that reflect its foremost syndromes. Thus, the Axis II scales include those personality dimensions that have been a part of the *DSM* since its third revision, and the Axis I scales reflect those syndromes that are most prominent and important in clinical work.

Three Stages of Validation

Item selection and scale development progressed through a sequence of three validation steps: (a) theoretical-substantive, (b) internal-structural, and (c) external-criterion. By using different validation strategies, the MCMI-III test upholds the standards of test developers who are committed to diverse methods of construction and validation (Hase & Goldberg, 1967).

In the theoretical-substantive stage, items for each syndrome were generated to conform to theoretical requirements and to the substance of the *DSM* criteria. In the internal-structural stage, these "rational" items were subjected to internal consistency analyses. Items that had higher correlations with scales for which they were not intended were either dropped entirely or re-examined against theoretical criteria and re-assigned or reweighted. Only items that survived each successive validation stage were included in subsequent analyses.

In the external-criterion phase, items were examined in terms of their ability to discriminate among clinical groups rather than between clinical groups and normal subjects. Normals are not an appropriate reference or comparison group (Rosen, 1962). For each generation of the MCMI test, the external phase of development has emphasized data in which target diagnostic groups were contrasted with a population of representative but undifferentiated psychiatric patients. The shift to a general psychiatric rather than a normal comparison group helps to optimize the discriminative efficiency of scales and thereby heightens differential diagnosis. A favorable ratio of valid-positive to false-positive classifications supports this strategy (see Chapter 4).

Rather than becoming a product of compromise, then, the tripartite model of test construction attempts to synthesize the strengths of each development phase by rejecting items that are found to be deficient in particular respects. This ensures that the final scales of an inventory do

not consist of items that optimize one particular parameter of test construction, but instead conjointly satisfy multiple requirements, increasing the generalizability of the end product. The final prototypal items of each MCMI, MCMI-II, and MCMI-III scale met, through sequential refinement, the basic criteria of each of these construction methods.

Diagnostic Thresholds

An important feature that distinguishes the MCMI-III test from other inventories is its use of actuarial base rate data rather than normalized standard score transformations. Because T scores are developed so that each scale of an inventory has an identical mean and standard deviation, they implicitly assume that the prevalence rates of all disorders are similar (for example, that there are roughly equal numbers of depressives and schizophrenics). In contrast, the MCMI-III test seeks to diagnose the percentages of patients who are actually found to be disordered across diagnostic settings. These data not only provide a basis for selecting optimal differential diagnostic cutting lines but also ensure that the frequency of MCMI-III-generated diagnoses and profile patterns will be comparable to representative clinical prevalence rates. Local base rates and cutting lines must still be developed for special settings. Nevertheless, validation data with a variety of populations (for example, outpatients and inpatients, clients in alcohol and drug treatment centers) suggest that the MCMI-III test can be used with a reasonable level of confidence in most clinical settings.

Interpretive Refinements

The computer-generated MCMI-III narrative report integrates a patient's personological and symptomatic features and presents them in a style that is similar to that of reports prepared by clinical psychologists. The individualized automated report synthesizes data from scale score elevations and profile configurations and is based on the results of actuarial research, the MCMI's theoretical schema (Millon, 1969/1983, 1981, 1990), and relevant *DSM* diagnoses within a multi-axial framework. Beyond providing a complex description of syndrome dynamics, the report summarizes findings along several dimen-

sions or axes: severity of disturbance, presenting clinical syndrome, basic personality pathology, psychosocial stressors, and therapeutic implications (see Chapter 7).

Advances Since the MCMI-II Test

The decision to revise the MCMI-II test in the early 1990s was motivated by theoretical, professional, and empirical concerns.

Theoretical Progress

The theory on which the MCMI and MCMI-II tests were constructed has undergone considerable development. No longer is it based primarily on the behavioral principles of reinforcement and conditioning (Millon, 1969/1983, 1981; Millon & Everly, 1985); it is instead anchored broadly and firmly to evolutionary theory (Millon, 1990; Millon & Davis, 1996). With this change, personality disorders are seen as evolutionary constructs derived from the fundamental tasks that all organisms confront, namely, the struggle to exist or survive (pleasure versus pain), the effort to adapt to the environment or adapt the environment to oneself (passive versus active), and the organism's strategy to make reproductive investments in kin or offspring versus an investment in its own personal replication (other versus self).

These three fundamental polarities form a foundation based in the larger framework of evolutionary theory that transcends any particular school or traditional perspective on personality. Accordingly, the Axis II disorders are no longer seen as being derived principally from a single clinical data level, be it behavioral, phenomenological, intrapsychic, or biophysical, that is, within one of the four traditional approaches to psychological science. Instead, personality disorders are seen as manifest across the entire matrix of the person, with expression throughout several clinical domains. Consequently, we have articulated an expanding base of diagnostic criteria and personality concepts (e.g., Millon, 1984, 1990), a framework much more extensive than the *DSM*, including the *DSM-IV*. The growing body of clinical literature provides a substantial knowledge base for the MCMI-III test. To the extent that the *DSM-IV* reflects these advances, its correspondence to the MCMI-III test has been further strengthened.

Professional Progress

In addition to theoretical progress oriented toward the understanding of Axis II, the area of personality disorders itself now enjoys worldwide scientific interest. The growth of the *Journal of Personality Disorders* and the International Society for the Study of Personality Disorders illustrates the importance attached to these syndromes as a major component of the mental disorders. These two major forums both inform and reflect the renaissance in personality theory and assessment that began in the late 1970s and 1980s (Millon, 1984, 1990) and continues today.

Moreover, the clinical field generally has seen numerous professional developments since the release of the MCMI-II test. The most significant of these was the publication of the *DSM-IV*. An increasingly solid base for making refined diagnostic decisions has been found, well beyond the literature of the late 1970s and early 1980s. To provide for additional scales, to optimize the correspondence between MCMI items and *DSM-IV* criteria, and to reflect generalization studies, 95 new MCMI-III items were introduced to replace 95 MCMI-II items. Two scales were added: one Clinical Personality Pattern (Depressive) and one Clinical Syndrome scale (Post-Traumatic Stress Disorder). Finally, a small set of items was added to strengthen the utility of the Noteworthy Responses section of the interpretive report in the areas of child abuse, anorexia, and bulimia.

Empirical Progress

Currently, more than 600 research studies have been published that employ the MCMI test as a major assessment instrument (see Appendix B of Craig, 2005, for a bibliography). This substantial empirical base, although difficult to digest in its totality, led to several major refinements in the structure of the MCMI-III test.

Numerous cross-validation and cross-generalization studies have been and continue to be executed with the goal of evaluating and improving each of the elements that make up the MCMI test: its items, scales, scoring procedures, algorithms, and interpretive text (see Choca, Shanley, & Van Denburg, 1992; Choca & Van Denburg, 1997, 2004; Craig, 1993; Hsu & Maruish, 1992; Maruish, 1994). These ongoing investigations

continue to provide an empirical grounding for further upgrading of each of these components. With the preceding information as a base, a number of changes were introduced to create the MCMI-III test.

First, the influence of the item-weighting system introduced in the MCMI-II test was moderated. Whereas before prototypal items were given a weight of 3 points, they now receive a weight of 2 points. Studies have generally shown very high correlations between scales composed of weighted and unweighted items. The authors continue to feel that the distinction between items that are more central and those that are more peripheral to the definition of a construct is an essential one and that items should be weighted according to their demonstrated substantive, structural, and external characteristics (Loevinger, 1957). However, 2 points, rather than 3, are now deemed adequate for capturing this distinction. Clinicians may still choose to inspect the prototypal items of each scale as so-called critical items when seeking support for particular criteria and when making diagnostic judgments. Abandonment of the item-weighting system, while perhaps not empirically objectionable, would have produced scales composed exclusively of singly weighted items, a characteristic that is not only incommensurate with the prototypal model that underlies the official diagnostic system but is also inconsistent with the tripartite logic that guided the development of the test itself, one that holds empirical considerations to be only one basis on which the structural features of an instrument should rest.

Second, modifications were also made to the procedures for correcting distortion effects (e.g., random responding, faking, denial, complaining). These modifications simplify the scoring procedures that were developed for the MCMI-II test.

Clinical and Research Uses

Because it is easy to administer and because computer scoring and interpretation are readily available, the MCMI-III test can be used on a routine basis in outpatient clinics, community agencies, mental health centers, college counseling programs, general and mental hospitals, independent and group practice offices, and forensic settings. The primary purpose of the MCMI-III

test is to provide information to clinicians—psychologists, psychiatrists, counselors, social workers, physicians, and nurses—who must make assessment and treatment decisions about individuals with emotional and interpersonal difficulties.

Computer-generated clinical reports for the MCMI-III test are available at two levels of detail. The MCMI-III Profile Report of scale scores is useful as a screening device to identify patients who may require more intensive evaluation or professional attention. Individual scale cutting lines on the MCMI-III test can be used to make decisions concerning primary behavior disorders or syndrome diagnoses. Similarly, elevation levels among subsets of scales can furnish grounds for judgments about impairment, severity, and chronicity of pathology. More comprehensive and dynamic interpretations of relationships among symptomatology, coping behavior, interpersonal style, and personality structure can be derived from an examination of the configural pattern of all 24 clinical scales. Alternately, the MCMI-III Interpretive Report provides a profile of the scale scores, a detailed analysis of personality and symptom dynamics, and suggestions for therapeutic management.

To maximize the clinical applicability of the Millon inventories, data are being gathered on subpopulations and minority groups. In addition, translations of the various versions of the MCMI test have been undertaken in a variety of languages, notably Chinese, Danish, Dutch, Flemish, German, Hebrew, Italian, Norwegian, Spanish, Swedish, and Thai. Several of these translations have an adequate database for research and limited clinical purposes, although none are fully established regarding their normative populations and discriminative validity. Adequate MCMI-III translations should be available within the decade.

No less significant than its clinical uses are the possibilities of employing the MCMI-III test for research purposes. As noted previously, more than 600 research studies have used the MCMI test in a significant manner. Increased conformity with *DSM-IV* criteria and the use of actuarial base rate data may be especially valuable when selecting criterion groups. Objective, quantified, and theory-grounded individual scale scores and profile patterns can be used to gener-

ate and test a variety of clinical, experimental, and demographic hypotheses. The authors continue to encourage such work, would appreciate receiving the results of studies, and would be pleased to engage in collaborative research aimed at improving the instrument's clinical and investigative utility. More detail concerning research with the MCMI-III test can be found at www.Millon.net.

Limitations and Qualifications

The following sections detail several limitations and qualifications that must be kept in mind when using the MCMI-III test.

Use in Appropriate Settings

First, the MCMI-III test is not a general personality instrument to be used with normal populations or for purposes other than diagnostic screening or clinical assessment. It differs from other more broadly applied inventories whose presumed utility for diverse populations may not be as great as is often thought.

Normative data and transformation scores for the MCMI-III test are based entirely on clinical samples and are applicable only to individuals who evidence problematic emotional and interpersonal symptoms or who are undergoing professional psychotherapy or psychodiagnostic evaluation. Although its use as an operational measure of relevant theoretical constructs is fully justified, the samples employed for such purposes are best drawn only from comparable clinical populations. To administer the MCMI-III test to a wider range of problems or class of subjects—for example, to use it in business and industry, for identifying neurologic lesions, or for assessing general personality traits among college students—is to apply the instrument to settings and samples for which it is neither intended nor appropriate.

Clinicians working with physically ill, behavioral medicine, or rehabilitation patients are directed to the Millon Behavioral Medicine Diagnostic (MBMD™; Millon, Antoni, Millon, Meagher, & Grossman, 2001). Similarly, if the individual to be assessed is a psychologically troubled teenager at the junior or senior high school or beginning college level, the clinician is advised to employ the Millon Adolescent

Clinical Inventory (MACI™; Millon, Millon, & Davis, 1993). Those who wish to appraise the psychological attributes and traits of nonclinical (i.e., normal) adults should use the Millon Index of Personality Styles *Revised* (MIPS® *Revised*; Millon, Weiss, & Millon, 2004). The MIPS test is especially suitable as a gauge of the broader theory's constructs. All of these instruments are available from Pearson.

Interpretation by Qualified Professionals

All test materials and reports offered by Pearson are assigned a qualification level according to the credentials required to purchase products at that level. The MCMI-III test is a Level C product. Level C purchasers must have a high level of expertise in test interpretation and must provide credentials indicating:

- licensure or certification to practice in their state in a field related to the purchase

OR

- a doctorate degree in psychology, education, or a closely related field with formal training in the ethical administration, scoring, and interpretation of clinical assessments related to the intended use of the assessment.

Users of the Millon computer-based reports register high levels of satisfaction with their overall quality and with their correspondence to independently derived clinical observations and judgments (Craig, 1993; Green, 1982). Nevertheless, clinicians who use the interpretive report should not be lulled over time into uncritical acceptance; they should routinely compare the statements generated against independent clinical evidence. Otherwise, test reports may take on a kind of Barnum effect, causing readers to conclude that such reports are accurate, not because they are, but because they present attributes that are so general or common that they apply to almost everyone. Moreover, the mysterious and seemingly quantitative and exact power of computers has no doubt imbued the reports, at least for some, with an undue measure of scientific merit and clinical acumen. Fortunately, the findings of investigations into the validity of MCMI interpretive reports provide strong evidence that ratings of its accuracy are

higher than can be accounted for by the Barnum effect or the computer-generated format (Craig, 1993; Moreland & Onstad, 1987; Sandberg, 1987).

In addition, MCMI-III users should note that the automated interpretive service is considered a professional-to-professional consultation. As a self-report instrument, the MCMI-III test represents only one facet of a total patient evaluation. The information contained in each interpretive report represents a series of tentative and probabilistic judgments, not a set of definitive statements. Text at the beginning of all MCMI-III interpretive reports states that the report should be evaluated in conjunction with additional clinical data (for example, current life circumstances, observed behavior, biographical history, interview responses, and information from other tests). The accuracy and richness of any self-report measure is enhanced when its findings are appraised in the context of other clinical sources. Not only does the combination of various gauges from diverse settings provide data aggregates that increase the likelihood of drawing correct inferences (Epstein, 1979, 1983), but multi-method approaches (Campbell & Fiske, 1959) provide both the experienced and the novice clinician with an optimal base for deciphering the unique features that characterize each patient. The integration of selected features of the interpretive report into management and treatment decisions is fully appropriate, but directly communicating its explicit content to patients or their relatives is strongly discouraged.

No less important than the insights provided by non-self-report sources of clinical data is the interpretive guidance furnished by a variety of nonclinical demographic indices (e.g., age, sex, marital status, vocation, ethnicity, socioeconomic factors, educational level). Although higher or lower MCMI-III scale elevations associated with these population characteristics often reflect real differences in prevalence rates (e.g., male police officers score higher than male teachers on certain scales and do so in a manner that is consistent with nontest personality data), it is important for clinicians to have a reasonable notion of what is typical for patients from particular social backgrounds. Similarly, a number of scale "modifiers" are used in the MCMI-III test to compensate for differences among patients in their distortion tendencies, notably candor and exaggeration. In

spite of these corrections, it behooves the clinician to carefully consider the impact of important demographics, not only to compensate for their effects but for the insights they may furnish and for their ability to individualize and enrich the meaning of MCMI-III clinical data.

Method Biases

As should be evident, there are distinct boundaries to the accuracy of the self-report format; by no means is it a perfect data source. Inherent psychometric limits, the tendency of similar patients to interpret questions differently, the effect of current affective states on trait measures, and the effort of patients to affect certain false appearances and impressions all lower the upper boundaries of this method's potential accuracy. However, a self-report instrument that is constructed in line with accepted techniques of validation (Loevinger, 1957) should begin to approach these upper boundaries. Given that the MCMI test has progressed through such a developmental background, we find that the results prove to be on the mark for about 55–65% of patients to whom it is administered. It is useful and generally valid, although with partial misjudgments, in an additional 25–30% of cases, and it is off target (that is, appreciably in error) approximately 10–15% of the time. These figures are in the quantitative range of five to six times greater than chance.

Although accuracy levels vary from setting to setting, these differences largely reflect difficulties in detecting the presence of a disorder at the time it is being appraised (e.g., identifying a histrionic personality disorder during a patient's depression). Also problematic is the disorder's prevalence or base rate. On purely mathematical grounds, diagnostic groups with notably low (e.g., suicide) or high (e.g., dysthymia) base rates prove statistically troublesome in that even the most optimal of selected cutoffs often produce hit rates that are only marginally better than chance (Meehl & Rosen, 1955; Rorer & Dawes, 1982).

The MCMI-III test's diagnostic scale cutoffs and profile interpretations are oriented to the majority of patients who take the inventory, that is, to those who display psychic disturbances in the mid-ranges of severity rather than those whose difficulties are either close to normal (e.g., workers' compensation litigants, patients

in marital therapy) or of manifest clinical severity (e.g., chronic schizophrenics or patients with acute psychosis). To optimize diagnostic and interpretive validity, narratives have been written to focus on moderate levels of pathology; this results in a slightly diminished degree of diagnostic and interpretive accuracy for patients in the least and the most severe ranges of psychological disturbance, a fact that users of the interpretive reports should bear in mind. Narrative analyses of patients who are experiencing ordinary life difficulties or minor adjustment disorders will tend to be construed as more troubled than they are, and conversely, those with the most serious pathologies will often be portrayed as being less severely impaired than they really are.

Diagnostic Limitations

There is no sharp division between Axis I and Axis II phenomena. The scales of every personality inventory, whether oriented primarily to personality or psychopathology, reflect in varying degrees both long-standing characteristics (i.e., traits) and more transitory characteristics (i.e., states). This fact has important implications for the construction and interpretation of psychological inventories. Noteworthy are the partial blurring effects of current depressive and anxiety states on specific personality disorder scales (Hirschfeld et al., 1983). Although these results stem in part from shared scale items (Wiggins, 1982), the level of covariation is appreciably greater than can be accounted for by item overlap alone (Lumsden, 1987). Experience with earlier versions of the MCMI test indicates that the presence of dysthymic and anxious

states accentuates scores on the personality scales while unduly diminishing those obtained on other scales, a finding that is consistent with research by Shea, Glass, Pilkonis, Watkins, and Docherty (1987).

Compensatory efforts to counter potentially distorting influences include rephrasing MCMI-III items to more clearly separate clinical and state phenomena from those of a more long-standing or trait character and employing scoring adjustments that automatically correct certain scales where marked or acute affective states are indicated. Nevertheless, clinicians should bear in mind the possibility of occasional personality disorder misrenderings when scores on the Anxiety and Dysthymia scales seem unusually or unexpectedly high or low in light of other sources of clinical evidence.

Summary

The MCMI test is an evolving assessment tool, to be revised in accordance with theoretical, professional, and empirical progress. Numerous features distinguish the MCMI-III from other inventories, including its relative brevity, its theoretical anchoring, its structural characteristics, and the use of a three-stage validation framework. Although the MCMI-III test has numerous clinical applications and has been used in many research studies, a number of limitations and qualifications must be kept in mind, including appropriate settings, interpretation by a qualified professional, and the influence of method biases and state-trait factors on test results.

Chapter 2

Theoretical-Substantive Validity

This chapter presents the rationale and strategy that guided the development of the MCMI, MCMI-II, and MCMI-III tests, focusing on the theoretical-substantive stage of test construction. Much of this material, from the theoretical model of the senior author and from *DSM-IV* criteria, served as a guide for writing the initial item pools for the various generations of the MCMI, including the MCMI-III.

The Threefold Validation Model

According to Loevinger (1957) and Jackson (1970), validation should be an ongoing process that occurs throughout all phases of test construction, not a procedure for corroborating an instrument's effectiveness after it is completed. With this principle in mind, validation of the MCMI, MCMI-II, and MCMI-III tests was an integral part of each step of development.

Once a commitment was made to this strategy, the question remained concerning which validation procedures should be used to make the final product as efficient as possible in achieving the goals of differential diagnostic and clinical interpretive utility. In her highly illuminating monograph, Loevinger (1957) proposed that development validation involves three sequential

components: substantive, structural, and external. Each of these is a necessary but not sufficient component of inventory construction. Where feasible, steps should be taken to progress from the first stage through the third. This validation sequence differs from procedures employed in the construction of past clinical inventories and should produce an instrument with optimal properties. A brief description of the rationale for and methods used with each of these three components will serve as a useful summary of the procedures followed in developing each successive version of the MCMI test.

The first validation stage, *theoretical-substantive* (more recently called the *deductive approach* by Burisch, 1984), examines the extent to which the items that make up the instrument derive their content from an explicit theoretical framework. Such a theory was developed by Millon for the MCMI test (Millon, 1969/1983, 1981, 1986a, 1990). Across all three generations of the MCMI test, this theory, first based on principles of reinforcement and now anchored to concepts fundamental to evolution, provides a series of clinically relevant constructs for personality trait and syndrome definition to be used as a guide when writing relevant scale items. Because clear boundaries and antici-

pated relationships between syndromes can be established on rational grounds, the instrument can be constructed with either distinct or inter-related scales at the initial stage of development. Theoretical-substantive validation is the subject of this chapter.

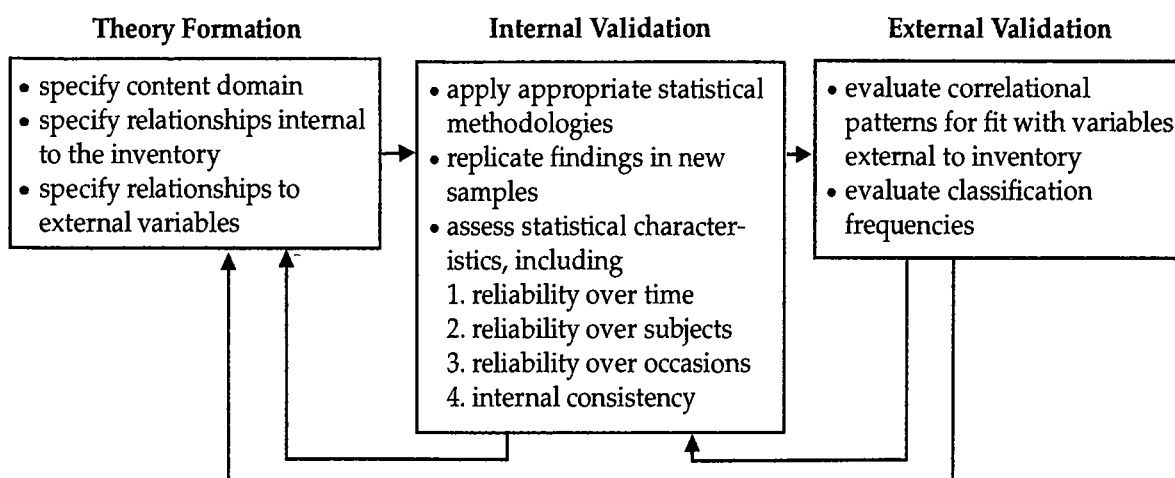
The second validation stage, *internal-structural*, refers to the model to which the instrument's items are expected to conform. In the inductive tradition, each scale is constructed as a measure of an independent trait in accordance with a factorial model. In the deductive tradition, each scale is designed to possess a high degree of internal consistency and to display considerable overlap with some of the other scales. In the structural phase, items that have already been substantively validated are administered to appropriate populations. The items that survive this second stage are those that maximize scale homogeneity, display a measure of overlap with other theoretically congruent scales, and demonstrate satisfactory levels of endorsement frequency and temporal stability. Internal-structural validation of the MCMI-III test is reviewed in Chapter 3.

The third validation stage, *external-criterion*, includes only those items and scales that have met the requirements of the substantive and structural phases of development. In this stage, the empirical correspondence between each test

scale and a variety of nonscale measures of the trait or syndrome under study is assessed. This third stage entails correlating results obtained on preliminary forms of the inventory with relevant clinical behavior. When performed in conjunction with other assessment methods and employing diverse external criteria, this procedure may also establish each scale's convergent and discriminant validity (Campbell & Fiske, 1959). External-criterion validation of the MCMI-III test is reviewed in Chapter 4.

In a set of classic papers, Hase and Goldberg (1967) and Goldberg (1972) compared alternative construction strategies and found that each displayed an equivalent level of validity across a set of diverse criteria. After reviewing several subsequent parallel studies, Burisch (1984) concluded that these findings of method equivalence continued to be supported. Nevertheless, a sequential validation strategy employing all three of these approaches is likely, at the very least, to prove equal to any single method alone and perhaps enjoy a measure of superiority. With this untested assumption in mind, developmental validation studies were begun on the MCMI test with the hope that each stage would produce increasingly refined and accurate forms of the inventory. This tripartite strategy, schematized in Figure 2.1, has served as the development framework for all subsequent Millon inventories.

Figure 2.1 Loevinger's Process Model of Validity



Note. From "Construct Validation Approach to Psychiatric Classification" by Harvey A. Skinner in *Contemporary Directions in Psychopathology* (p. 11), edited by Theodore Millon and Gerald L. Klerman, 1986, New York: Guilford Press. Copyright 1986 by Guilford Press. Adapted with permission.

The Theoretical-Substantive Framework

Theory and taxonomy are intimately linked. Philosophers of science agree that the system of kinds underlying any domain of inquiry must itself be answerable to the question that forms the very point of departure for the scientific enterprise: why does nature take this particular form rather than some other? Accordingly, one cannot merely accept any list of kinds or dimensions as given. Instead, a taxonomic scheme must be justified, and to be justified scientifically, it must be justified theoretically. Taxonomy and theory, then, are intimately linked. Quine (1977) makes a parallel case:

[O]ne's sense of similarity or one's system of kinds develops and changes . . . as one matures. . . . And at length standards of similarity set in which are geared to theoretical science. The development is away from the immediate, subjective, animal sense of similarity to the remoter objectivity of a similarity determined by scientific hypotheses . . . and constructs. Things are similar in the later or theoretical sense to the degree that they are . . . revealed by science. (p. 171)

Is such a taxonomy possible? At stake is nothing less than whether personology is to possess its own intrinsic taxonomy or remain a pseudo-science that services the larger society, establishing diagnoses according to extrinsic standards. Although the *DSM* was deliberately and appropriately formulated atheoretically, ultimately we will require some way of separating the chaff from the wheat.

The deductive approach advanced in other publications (Millon, 1969/1983, 1981, 1990; Millon & Davis, 1996) generates a true taxonomy to replace the primitive aggregation of categories that preceded it. This generative power is what Hempel (1965) meant by the "systematic import" of a scientific classification. Meehl (1978) noted that theoretical systems are made up of related assertions, shared terms, and coordinated propositions that provide fertile ground for deducing and deriving new empirical and clinical observations. What is elaborated and refined in theory, then, is understanding, an ability to see relations more clearly, to conceptualize categories more accurately, and to create greater

overall coherence in a subject and to integrate its elements in a more logical, consistent, and intelligible fashion. Pretheoretical taxonomic boundaries that were set according to clinical intuition and empirical study can now be affirmed and refined according to their constitution along underlying polarities. These polarities lend the model a holistic, cohesive structure that facilitates the comparison and contrast of groups along fundamental axes, thus sharpening the meanings of the taxonomic constructs derived. Such a framework is essential in the generation of candidate items for an inventory.

In Hempel's (1965) terms, all natural classifications are classifications, but not all classifications are natural. Not all classification systems "carve nature at its joints." In fact, an infinite number of classification systems, whether dimensional or categorical, could be proposed for the personality disorders. In this sense, the purpose of deductive theory is to extract from an infinite number of ways of grouping clinical phenomena only those that have potential to "carve nature," that is, which have systematic import with respect to further hypotheses and ultimately for methods of therapy and intervention. Hempel (1965) distinguished these two kinds of classifications:

[D]istinctions between "natural" and "artificial" classifications may well be explicated as referring to the difference between classifications that are natural and those that are not: in a classification of the former kind, those characteristics of the elements which serve as criteria of membership in a given class are associated, universally or with high probability, with more or less extensive clusters of other characteristics.

. . . a classification of this sort should be viewed as somehow having objective existence in nature, as "carving nature at the joints," in contradistinction to "artificial" classifications, in which the defining characteristics have few explanatory or predictive connections with other traits . . .

In the course of scientific development, classifications defined by reference to manifest, observable characteristics will tend to give way to systems based on theoretical concepts. (pp. 146-148)

What happens, however, when the two are confounded, that is, when more theoretically grounded taxons that carve nature exist side by side in a taxonomy with those based on superficial appearances? As noted, the difference between natural and artificial classification turns on the distinction between manifest (or empirical) and latent (or theoretical) levels of similarity. Figure 2.2 represents possible agreement and disagreement between manifest and latent levels of similarity for two individuals.

As indicated in the figure, for two patients' presentations, four possibilities exist. Going clockwise, two empirically similar presentations may, in fact, be similar. In this case, what appears to be is indeed the case. Second, two empirically similar presentations may, in fact, be different. In this case, appearances are deceiving. Two taxons are indicated, their empirical similarity making them difficult to tease apart. Third, two presentations that appear different may in fact be different. In this case, different empirical presentations legitimately indicate different category membership. Fourth, two presentations that appear different may in fact be similar. Figure 2.2 resembles other figures that are used to present the logic of diagnostic efficiency statistics: positive and negative predictive power, sensitivity, and specificity (e.g., Baldessarini, Finklestein, & Arana, 1983). Whereas these statistics capture the diagnostic dilemma associated with imperfect predictors in ignorance of the "true" state of nature for a single subject, Figure 2.2 represents the nosologic dilemma, whereby a classification system must be established on the basis of imperfect attributes in ignorance of the "true" system that carves nature at its joints.

Nosologically, the possibilities represented in quadrants II and IV of Figure 2.2 present tremendous difficulties for a classification of personality disorders. Focusing on superficial patterns of covariation (quadrant II) will lead to the institution of one taxon where two or even many more may be needed. In quadrant IV, it will lead to the institution of two (or perhaps many more) taxons where only one in fact exists. As Quine (1977) noted in his discussion of natural kinds, our attachment to a taxonomy loosens in favor of an understanding based on underlying principles because it is these very underlying principles that allowed the deduc-

tion of the taxonomy in the first place. Those who are familiar with the deductive personologic taxonomy developed by Millon know that both the narcissistic and the antisocial personality are independent types in that both are oriented toward the self. What distinguishes them is primarily their orientation toward their ecologic milieu. The narcissist is a passive-independent type, the antisocial an active-independent type. Similarly, the dependent personality is a passive-dependent type, while the narcissist is a passive-independent type. What distinguishes them is primarily their orientation toward others versus an orientation toward self. Both dependent and antisocial personalities, then, are related to the narcissist but themselves have little in common. Thus, the deductive system allows us to see the essential interrelation and differentiation of personality pathologies.

The Role of Theory in Test Construction

Reflecting on her years of experience in test construction and evaluation, Loevinger (1972) concluded:

If I were to draw a single conclusion from my own studies of personality measurement, it would be this: I consider it exceedingly unlikely that either by accident or by automation one will discover a measure of a major personality variable. There is no substitute for having a psychologist in charge who has at least a first-approximation conception of the trait he wishes to measure, always open to revision, of course, as data demand. Theory has always been the mark of a mature science. The time is overdue for psychology, in general, and personality measurement, in particular, to come of age. (p. 56)

Commenting on the development of the MMPI® (Minnesota Multiphasic Personality Inventory®), Norman (1972) notes that criterion keying was the only recourse possible because no adequate theory or body of established empirical data was available as an alternative. Meehl (1945), the most persuasive proponent of the criterion keying approach, shifted from an earlier antitheoretical position to one in which a guiding theory is seen as a valuable test development tool (Meehl, 1972). In his usual insightful fashion, Meehl states:

Figure 2.2 Manifest and Latent Similarity and the Taxonomic Situation

		LATENT SIMILARITY	
		YES	NO
MANIFEST SIMILARITY	YES	Two empirically similar presentations are in fact taxonomically similar I	Two empirically similar presentations are in fact taxonomically different PROBLEMATIC FOR STATISTICAL CLASSIFICATION SCHEMES II
	NO	Two empirically different presentations are in fact taxonomically similar PROBLEMATIC FOR STATISTICAL CLASSIFICATION SCHEMES IV	Two empirically different presentations are in fact taxonomically different III

One reason for the difficulties of psychometric personology, a reason I did not appreciate in my "dustbowl empiricist" paper of 1945, is the sad state of psychological theory.

I now think that at all stages in personality test development, from initial phase of item pool construction to a late-stage optimized clinical interpretive procedure for the fully developed and "validated" instrument, theory—and by this I mean all sorts of theory, including trait theory, developmental theory, learning theory, psychodynamics, and behavior genetics—should play an important role . . . I believe that psychology can no longer afford to adopt psychometric procedures whose methodology proceeds with almost zero reference to what bets it is reasonable to lay upon substantive personological horses. The "theory" . . . may be only weakly corroborated, but I think we have to make do with it anyway. (pp. 149–151)

[T]he preliminary item pool should be constructed in reliance upon all of the facts and theories bearing upon the test. Even one who advocates a relatively atheoretical "blind empirical criterion keying" . . . need not deprive himself of whatever theoretical insight is available at the item construction stage . . . I now believe (as I did not formerly) that an item ought to make theoretical sense. (p. 155)

Although Meehl maintained his strong commitment to the critical role played by external validation, his recognition of the value of a theoretical model served to reinforce the strategy undertaken in developing and revising the MCMI tests. Together with Loevinger, Meehl's growing appreciation of the use of theory strengthened the belief that such a course would prove both wise and fruitful.

An Evolutionary Theory of Personality Disorders

As the previous sections argue, a consistent theoretical system is extremely useful as a basis for a coherent classification system and parallel inventory scales. We will sketch the evolutionary theory below in a highly abstracted form, noting two levels at which the personality constructs may be operationalized: first in terms of the theory's fundamental polarities and second in terms of functional and structural domains of personality. We will also review the *DSM* criteria for each personality disorder and note their correspondence to MCMI-III items. This chapter is valuable in describing the point of departure for the items written for the MCMI-III; it is also a chapter to which those who interpret the inventory will want to refer often because of the depth of the descriptive material contained herein. Because the discussion of the theory itself is somewhat abstracted, interested read-

ers are referred to the following texts for further guidance: *Modern Psychopathology* (Millon, 1969/1983), *Disorders of Personality: DSM-III, Axis II* (Millon, 1981), *Contemporary Directions in Psychopathology* (Millon & Klerman, 1986), *Toward a New Personology: An Evolutionary Model* (Millon, 1990), and the revised *Disorders of Personality: DSM-IV and Beyond* (Millon & Davis, 1996). *Toward a New Personology* presents the essentials of the model as a brief treatise. The other works are more descriptive of the clinical prototypes that can be derived from the theory.

The model that follows is grounded in the principles of evolution. In essence, it seeks to explicate the structure and styles of personality with reference to deficient, unbalanced, or conflicted modes of ecological adaptation and reproductive strategy. The proposition that the development and functions of personologic traits may be usefully explored through the lens of evolutionary principles has a long if yet unfulfilled tradition. Herbert Spencer and T. H. Huxley offered suggestions of this nature around the time that Darwin's seminal *On the Origin of Species* was published (1859). In more recent times, we have seen the emergence of sociobiology, an interdisciplinary science that explores the interface between human social functioning and evolutionary biology (Wilson, 1975, 1978).

The four domains or spheres in which evolutionary principles are demonstrated are existence, adaptation, replication, and abstraction. Existence relates to the serendipitous transformation of random or less organized states into those that possess distinct structures of greater organization. Adaptation refers to the homeostatic processes employed to sustain survival in open ecosystems. Replication pertains to reproductive styles that maximize the diversification and selection of ecologically effective attributes. Abstraction concerns the emergence of competencies that foster anticipatory planning and reasoned decision making. Polarities derived from the first three phases (pleasure–pain, passive–active, other–self) are used to construct a theoretically embedded classification system of personality disorders.

These polarities have forerunners in psychological theory that may be traced as far back as the early 1900s. A number of pre–World War

I theorists, including Freud, proposed a set of three polarities that were used time and again as the raw materials for constructing psychological processes. Aspects of these polarities were “discovered” and employed by theorists in France, Germany, and Russia and in other European nations and the United States. In addition, there is a growing group of contemporary scholars whose work has begun to illuminate aspects of these polar dimensions, including Buss and Plomin (1975, 1984), Cloninger (1986, 1987), Eysenck (1957, 1967), Gray (1964, 1973), Russell (1980), and Tellegen (1985).

The first phase, existence, concerns the maintenance of integrative phenomena—whether nuclear particle, virus, or human being—against the background of entropic decompensation. Evolutionary mechanisms derived from this stage relate to the processes of life enhancement and life preservation. The former are concerned with orienting individuals toward improvement in quality of life, the latter with orienting individuals away from actions or environments that decrease the quality of life or even jeopardize existence itself. This two-part polarity may be called *existential aims*. At the highest level of abstraction, such mechanisms form, phenomenologically or metaphorically expressed, a pleasure–pain polarity. Some individuals have reversed or are conflicted regarding these existential aims (e.g., the sadistic), while others may have deficits in both of these crucial substrates (e.g., the schizoid). In terms of neuropsychological growth stages (Millon, 1969/1983, 1981), the pleasure–pain polarity is recapitulated in a developmental sensory-attachment phase, the purpose of which is the largely innate and rather mechanical discrimination of pain and pleasure signals.

Once an integrative structure exists, it must maintain its existence through exchanges of energy and information with its environment. The second evolutionary stage relates to what are called *modes of adaptation*. As with the existence domain, adaptation is framed as a two-part polarity: a passive orientation and a tendency to accommodate to one's ecological niche versus an active orientation and a tendency to modify or intervene in one's surroundings. These modes of adaptation differ from the first phase of evolution in that they are related to how that which exists can endure. In terms of neuropsychologi-

cal growth stages, these modes are recapitulated in a developmental sensorimotor-autonomy phase, during which a child either acquires an active disposition toward his or her physical and social context or perpetuates the more dependent mode of prenatal and infantile existence.

Although organisms may exist well-adapted to their environment, the existence of any life form is time-limited. To circumvent this limitation, organisms have developed *replicatory strategies* by which to leave progeny. These strategies are related to what biologists have referred to as an r- or *self-propagating* strategy at one polar extreme and a K- or *other-nurturing* strategy at the other extreme. Psychologically, the former strategy is disposed toward actions that are egotistic, insensitive, inconsiderate, and uncaring, while the latter is disposed toward actions that are affiliative, intimate, protective, and solicitous (Gilligan, 1981; Rushton, 1985; Wilson, 1978). Like pleasure-pain, the self-other polarity is not truly unidimensional. Some personality disorders are conflicted on this polarity (for example, the compulsive and negativistic or passive-aggressive). In terms of neuropsychological growth stages, an individual's orientation toward self and others is recapitulated in the pubertal-gender identity stage. A more detailed description of these polarity concepts can be found in Chapters 2 and 3 of Millon and Davis (1996), which also includes a discussion of the fourth polarity—abstraction—and its associated intracortical-initiative stage of neuropsychological development. A description of the derived personality patterns follows.

Some personalities exhibit a reasonable balance on one or other of the polarity pairs. Not all individuals fall at the center, of course. Individual differences in personality features and overall style reflect the relative positions and strengths of each polarity component. *Pleasure-deficient* personalities lack the capacity to experience or to enact certain aspects of the three polarities (e.g., the schizoid has a faulty substrate for both pleasure and pain). *Interpersonally imbalanced* personalities lean strongly toward one or another extreme of a polarity (e.g., the dependent is oriented almost exclusively toward receiving the support and nurturance of others). *Intrapsychically conflicted* personalities struggle with ambivalence toward opposing ends of a

bipolarity (e.g., the negativist vacillates between adhering to the expectations of others and doing what he or she wants to do).

Severe Personality Patterns

Personality styles reflect deeply etched and pervasive characteristics of functioning that perpetuate and aggravate everyday difficulties. They are so embedded and automatic that the individual is often unaware of their nature and self-destructive consequences. Under conditions of persistent adversity, such maladaptive styles may begin to decompensate, acquiring features of moderate or marked severity.

Three additional pathological personality patterns—schizotypal, borderline, and paranoid—have been formulated in the theory to represent more advanced stages of personality pathology. Reflecting an insidious and slow deterioration of the personality structure, these differ from the basic personality disorders by several criteria, especially deficits in social competence and frequent (but usually reversible) psychotic episodes. Less integrated in terms of personality organization and less effective in coping than their milder counterparts, they are especially vulnerable to the everyday strains of life. Nevertheless, most patients continue to display many of the major personality characteristics that were previously evident.

Explication of the Axis II Prototypes

Figure 2.3 presents the personality disorder derivatives of the evolutionary model. As first-order deductions from the theory's constructs, they are readily developed into the descriptive paragraphs that follow. These brief descriptions have the virtue of being interpretively easy to consult and understand. However, they do not reflect the more rigorous perspective of a subsequent section, where personality is seen as being anchored to and operationalized across the entire matrix of personality.

Schizoid (Scale 1)

Schizoid individuals are noted for their lack of desire and their incapacity to experience deep pleasure or pain. They tend to be apathetic, listless, distant, and asocial. Their emotions and affectionate needs are minimal, and they function

Figure 2.3 The Evolutionary Model and Its Personality Disorder Derivatives

	Existential Aim		Replication Strategy		
	Life Enhancement	Life Preservation	Reproductive Propagation	Reproductive Nurturance	
Deficiency, Imbalance, Conflict	Pleasure–Pain		Self–Other		
	Pleasure (low) Pain (low or high)	Pleasure–Pain Reversal	Self (low) Other (high)	Self (high) Other (low)	Self–Other Reversal
Adaptation Mode	Personality Disorders				
Passive: Accommodation	Schizoid (low pleasure, low pain) Depressive (high pain, low pleasure)	Masochistic	Dependent	Narcissistic	Compulsive
Active: Modification	Avoidant	Sadistic	Histrionic	Antisocial	Negativistic
Structural Pathology	Schizotypal	Borderline, Paranoid	Borderline	Paranoid	Borderline, Paranoid

as passive observers detached from the rewards and affections—as well as from the demands—of human relationships.

Avoidant (Scale 2A)

Avoidant individuals experience few positive reinforcers from themselves or others. They are vigilant and always on guard, ready to distance themselves from anxious anticipation of life's painful or negatively reinforcing experiences. Their adaptive strategy reflects fear and mistrust of others. They maintain a constant vigil to prevent their impulses and their longing for affection from resulting in a repetition of the pain and anguish they have experienced with others. Only by active withdrawal can they protect themselves. Despite their desires to relate to others, they have learned that it is best to deny these feelings and to keep a good measure of interpersonal distance.

Depressive (Scale 2B)

There are commonalities among the first three Clinical Personality Patterns, notably glumness, pessimism, lack of joy, the inability to experience pleasure, and seeming motoric retardation. For the *DSM* Axis II schizoid personality disorder, there is an incapacity for joy as well

as sadness. For the avoidant pattern, there is a hyperalertness to anticipated pain, with a consequent inattention to joy. For the depressive, there has been a significant loss, a sense of giving up, and a loss of hope that joy can be retrieved. None of these personality types experiences pleasure, but for different reasons. The depressive personality experiences pain as permanent. Pleasure is no longer even considered possible. What experiences or chemistry accounts for such persistent and characterological sadness? Clearly, there are biological dispositions to take into account. The evidence favoring a constitutional predisposition is strong, much of it favoring genetic factors. The thresholds involved in permitting pleasure or sensitizing one to sadness vary appreciably. Some individuals are inclined to pessimism and a disheartened outlook. Similarly, experience can condition a hopeless orientation to significant loss. A disconsolate family, a barren environment, and hopeless prospects can all shape the depressive character style.

Dependent (Scale 3)

Dependent individuals have learned not only to turn to others for nurturance and security but to wait passively for their leadership in providing them. They are characterized by a search for relationships in which they can lean on others

for affection, security, and guidance. Their lack of initiative and autonomy is often a consequence of parental overprotection. As a function of these experiences, they have simply learned the comforts of assuming a passive role in interpersonal relations, accepting what kindness and support they may find and willingly submitting to the wishes of others in order to maintain their affection.

Histrionic (Scale 4)

Although histrionic individuals turn to others — no less than dependents do, they appear on the surface to be quite dissimilar from their passive counterparts. This difference in overt style arises from their facile and enterprising manipulation of events, through which they maximize the attention and favors they receive and avoid the indifference and disapproval of others. These individuals often exhibit an insatiable if not indiscriminate search for stimulation and affection. Their clever and often artful social behavior gives the appearance of inner confidence and independent self-assurance; beneath this guise, however, lies a fear of genuine autonomy and a need for repeated signs of acceptance and approval. Tribute and affection must be constantly replenished and are sought from every interpersonal source and in every social context.

Narcissistic (Scale 5)

Narcissistic individuals are noted by their egotistic self-involvement, experiencing primary pleasure simply by passively being or focusing on themselves. Early experience has taught them to overvalue their self-worth. This confidence and superiority may be founded on false premises; that is, they may be unsustainable by real or mature achievements. Nevertheless, narcissists blithely assume that others will recognize their specialness. Hence, they maintain an air of arrogant self-assurance and, without much thought or even conscious intent, benignly exploit others to their own advantage. Although the tributes of others are welcome and encouraged, their air of snobbish and pretentious superiority requires little confirmation through genuine accomplishment or social approval. Their sublime confidence that things will work out well provides them with little incentive to engage in the give-and-take of social life.

Antisocial (Scale 6A)

Antisocial individuals act to counter the expectation of pain and depredation at the hands of others by engaging in duplicitous or illegal behavior designed to exploit the environment for self-gain. Their aggrandizing orientation reflects their skepticism concerning the motives of others, a desire for autonomy, and a wish for revenge and recompense for what they consider past injustices. They are irresponsible and impulsive, qualities they believe are justified because they judge others to be unreliable and disloyal. Insensitivity and ruthlessness are their only means of avoiding abuse and victimization.

Sadistic (Aggressive) (Scale 6B)

Although it was deleted from the *DSM-IV*, the sadistic construct remains part of the MCMI-III test. As a personality construct, this style or pattern extends the boundaries of the *DSM-III-R* formulation in a new and important direction, one that recognizes that individuals who are not judged to be antisocial may get personal pleasure and satisfaction in ways that humiliate others and violate their rights and feelings. Depending on social class and other moderating factors, they may parallel the clinical features of what is known in the literature as the sadistic character or they may display character styles akin to the competitively striving Type A personality. Referred to as *aggressive personalities* in the author's theory, they are generally hostile and pervasively combative, and they appear to be indifferent to or pleased by the destructive consequences of their contentious, abusive, and brutal behavior. Although many cloak their more malicious and power-oriented tendencies in publicly approved roles and vocations, they give themselves away by their dominating, antagonistic, and frequently persecutory actions.

Compulsive (Scale 7)

This ambivalent orientation coincides with the *DSM-IV* obsessive-compulsive personality disorder. Compulsive individuals have been intimidated and coerced into accepting the demands and judgments imposed on them by others. Their prudent, controlled, and perfectionistic ways derive from a conflict between hostility toward others and fear of social disapproval. They resolve this ambivalence by suppressing

their resentment and by overconforming and placing high demands on themselves and others. Their disciplined self-restraint serves to control intense, though hidden, oppositional feelings, resulting in an overt passivity and seeming public compliance. Behind this front of propriety and restraint, however, are intense anger and oppositional feelings that occasionally break through their controls.

Negativistic (Passive-Aggressive) (Scale 8A)

This ambivalent orientation approximates the *DSM-III-R* passive-aggressive personality disorder and the *DSM-IV* passive-aggressive (negativistic) construct, the latter more extensive in the number and diversity of traits it encompasses than the earlier passive-aggressive orientation. Negativistic individuals are torn between following the rewards that are offered by others and those they desire themselves. This struggle represents an inability to resolve conflicts similar to those of the obsessive-compulsive; however, the conflicts of the negativistic individual remain close to consciousness and intrude into everyday life. These individuals experience endless wrangles and disappointments as they vacillate between deference and defiance, between obedience and aggressive opposition. Their behavior is characterized by an erratic pattern of explosive anger or stubbornness intermingled with periods of guilt and shame.

Masochistic (Self-Defeating) (Scale 8B)

Although it was deleted from the *DSM-IV*, this discordant orientation corresponds to the *DSM-III-R* self-defeating (masochistic) personality disorder, a character type well described in the literature. Relating to others in an obsequious and self-sacrificing manner, masochistic individuals allow and perhaps encourage others to exploit or take advantage of them. Focusing on their very worst features, many assert that they deserve to be shamed and humbled. To compound their pain and anguish, which they may experience as comforting, they actively and repetitively recall their past misfortunes and expect problematic outcomes from otherwise fortunate circumstances. Typically acting in an unassuming and self-effacing way, they often intensify their deficits and place themselves in an inferior light or abject position.

Schizotypal (Scale S)

The *DSM-IV* schizotypal personality disorder represents a cognitively dysfunctional and interpersonally detached orientation. Schizotypal individuals prefer social isolation with minimal personal attachments and obligations. Inclined to be either autistic or cognitively confused, they think tangentially and often appear to be self-absorbed and ruminative. Behavioral eccentricities are notable, and these individuals are often perceived by others as strange or different. Depending on whether their basic pattern is active or passive, they display either an anxious wariness and hypersensitivity or an emotional flattening and deficiency of affect.

Borderline (Scale C)

The borderline personality often underlies less severe personality disorders. Each borderline variant has structural defects and experiences intense endogenous moods with recurring periods of dejection and apathy, often interspersed with spells of anger, anxiety, or euphoria. What distinguishes the borderline pattern most clearly from the two other severe patterns—schizotypal and paranoid—is the dysregulation of affect, seen most clearly in the instability and lability of mood. Additionally, many borderline individuals have recurring thoughts about self-mutilation and suicide, appear overly preoccupied with securing affection, have difficulty maintaining a clear sense of identity, and display a cognitive-affective ambivalence that is evident in conflicting feelings of rage, love, and guilt toward others.

Paranoid (Scale P)

The *DSM-IV* paranoid personality displays a vigilant mistrust of others and an edgy defensiveness against anticipated criticism and deception. There is an abrasive irritability and a tendency to precipitate exasperation and anger in others. Paranoid individuals often express fear of losing independence, leading them to vigorously resist external influence and control, whereas the other two severe patterns are noted either by the instability of their affect (borderline) or the dysregulation of their cognition (schizotypal). Paranoid individuals are distinctive in the immutability of their feelings and the inflexibility of their thoughts.

Axis II Personality Prototype Structural-Functional Domains

In what domains should personality disorders be ascribed and therefore assessed? Although personality may be considered exclusively psychodynamic or exclusively biological, the authors regard such positions as narrow and restrictive. The integrative perspective encouraged here views personality as a multidetermined and multireferential construct that may be profitably studied and assessed across a variety of content areas. The term *multireferential* is an important one for assessment purposes. One means of dealing with the conceptual and theoretical pluralism afoot in personality today is to simply assess personality in accord with a single model, effectively eliminating the eclecticism of divergent perspectives through a dismissive dogmatism. A truly comprehensive assessment, however, one that is logically consonant with the integrative nature of personality as a construct, requires that personality be assessed systematically across multiple personologic spheres if the assessment is to be content valid. Constraints on system functioning may exist in any of these spheres.

Several criteria were used to select and develop the clinical domains listed herein: (a) that they be varied in the features they embody, that is, not limited just to behavior or cognition, but instead encompass a full range of clinically relevant characteristics, (b) that they correspond to many of our profession's current therapeutic modalities (e.g., self-oriented phenomenological techniques for altering dysfunctional cognition, group treatment procedures for modifying interpersonal conduct), and (c) that they not only be coordinated with the official *DSM* schema of personality disorder prototypes, but also that each disorder be characterized by a distinctive feature within each clinical domain. As portrayed in Table 2.1, these diagnostic features are distinguished in accord with the data levels they represent—behavioral, phenomenological, intrapsychic, and biophysical. This differentiation reflects the four historic approaches that characterize the study of psychopathology. These domains can be systematically organized in a manner similar to how distinctions are drawn in the biological realm, that is, by dividing them into *structural* and *functional* attributes. The

science of anatomy investigates embedded and essentially permanent structures that serve, for example, as substrates for mood and memory. Its functional counterpart, physiology, is concerned with processes that regulate internal dynamics and external transactions. These structures and functions bind the organism together as a coherent entity.

Dividing the characteristics of the psychological world into structural and functional realms is, of course, by no means a novel notion. Psychoanalytic theory has dealt since its inception with topographic constructs such as conscious, preconscious, and unconscious, and later with structural concepts such as id, ego, and superego. Likewise, a host of quasi-stationary functional processes, such as the so-called ego apparatuses, have been posited and studied (Gill, 1963; Rapaport, 1959).

In conducting a domain-oriented assessment, clinicians should beware of regarding each domain as an independent entity, thereby falling into a naive, concretizing operationalism. Each domain is a legitimate but highly contextualized part of a single integrated whole, one that is absolutely necessary if the functional-structural integrity of the organism is to be maintained.

**Table 2.1 Functional and Structural
Domains of Personality**

Functional Domains	Structural Domains
Behavioral Level	
Expressive Acts	
Interpersonal Conduct	
Phenomenological Level	
Cognitive Style-	Self-Image
	Object Representations
Intrapsychic Level	
Regulatory Mechanisms	
	Morphologic Organization
Biophysical Level	
	Mood/Temperament

afoot - on the move, on the fast - moving
Pluralism - the state of being plural - many and varied

Nevertheless, individuals differ with respect to the domains they enact most frequently. Real patients vary not only in the degree to which they approximate each personality prototype but also in the extent to which constraints in each domain shape their overall behavior. Conceptualizing personality as a system, we might say that the constraints on the states the system may assume are present in different parts of the system for different individuals, even when those individuals share a diagnosis. Only those aspects of the system that are sufficiently crystallized possess the enduring and pervasive character we associate with personality. In this sense, the goal of an assessment is to illuminate constraints that perpetuate the system's narrow, rigid functioning and consequent adaptive inflexibility. Thus identified, the purpose of therapy should be to relax these constraints, allowing the system to assume a greater variety of states or adaptive behavior across situations.

Functional Domains

Functional characteristics represent dynamic processes that transpire within the intrapsychic world and between the individual and his or her psychosocial environment. Transactional processes take place through the medium of functional domains. For definitional purposes, we might say that functional domains represent expressive modes of regulatory action, that is, behavior, social conduct, cognitive processes, and unconscious mechanisms that manage, adjust, transform, coordinate, balance, discharge, and control the give-and-take of inner and outer life. Four functional domains relevant to the personality disorders are briefly described below.

Expressive Acts

These attributes relate to observables at the *behavioral level* of data and are usually recorded by noting how the patient acts. Through inference, observations of overt behavior enable us to deduce what the patient unknowingly reveals about him- or herself or, often conversely, what he or she wants others to think about him or her. The range and character of expressive actions are wide and diverse, and such actions convey distinctive and worthwhile clinical information,

from communicating a sense of personal incompetence to exhibiting general defensiveness to demonstrating disciplined self-control, and so on. This domain of clinical data is likely to be especially productive in differentiating patients on the passive-active polarity of Millon's (1990) theoretical model.

Interpersonal Conduct

A patient's style of relating to others is also noted essentially at the *behavioral* data level. Interpersonal conduct may be captured in a number of ways, such as how his or her actions affect others, intended or otherwise; the attitudes that underlie, prompt, and give shape to these actions; the methods by which he or she engages others to meet his or her needs; and his or her way of coping with social tensions and conflicts. Extrapolating from these observations, the clinician may construct an image of how the patient functions in relation to others: antagonistically, respectfully, aversively, secretively, and so on.

Cognitive Style

How the patient focuses and allocates attention, encodes and processes information, organizes thoughts, makes attributions, and communicates reactions and ideas to others represents data at the *phenomenological* level. These characteristics are among the most useful indices of the patient's distinctive way of functioning. By synthesizing these signs and symptoms, it may be possible to identify indications of an impoverished style, distracted thinking, cognitive flightiness, constricted thought, and so on.

Regulatory Mechanisms

Although *mechanisms* of self-protection, need gratification, and conflict resolution are consciously recognized at times, they represent data derived primarily at the *intrapsychic* level. Because defense mechanisms are internal processes, they are more difficult to discern and describe than processes that are anchored closer to the observable world. As such, they are not directly amenable to assessment by self-reflective appraisal in their pure form but only as derivatives potentially many levels removed from their core conflicts and their dynamic regulation. By definition, these dynamic regulatory mechanisms co-opt and transform both internal and external realities

before they can enter conscious awareness in a robust, unaltered form. When chronically enacted, they often perpetuate a sequence of events that intensifies the very problems they were intended to circumvent. Despite the methodological problems they present, the task of identifying which mechanisms are chosen (e.g., rationalization, displacement, reaction formation) and the extent to which they are employed is extremely useful in a comprehensive personality assessment.

Structural Domains

In contrast to functional characteristics, structural attributes represent a deeply embedded and relatively enduring template of imprinted memories, attitudes, needs, fears, conflicts, and so on that guide experience and transform the nature of ongoing life events. Psychic structures have an orienting and preemptive effect in that they alter the character of action and the impact of subsequent experiences in line with preformed inclinations and expectations. By selectively lowering thresholds for transactions that are consonant with constitutional proclivities or early learning, the individual often experiences present events as variations of the past. Of course, the residuals of the past do more than passively contribute their share to the present. By temporal precedence, if nothing else, they guide, shape, or distort the character of current events and objective realities.

For purposes of definition, structural domains may be conceived as substrates and action dispositions of a quasi-permanent nature. Possessing a network of interconnecting pathways, these structures contain the internalized residues of the past in the form of memories and affects that are associated intrapsychically with conceptions of self and others. Structures serve to close the organism to novel interpretations of the world and limit the possibilities of expression to those that have already become predominant. Their preemptive and channeling character plays an important role in perpetuating the maladaptive behavior and vicious circles of personality pathology. Four structural domains relevant to personality are overviewed next.

Self-Image

As the inner world of symbols is mastered through development, the swirl of events that

buffets the young child gives way to a growing sense of order and continuity. One major configuration emerges to impose a measure of sameness on an otherwise fluid environment: the perception of self-as-object, a distinct, ever-present, and identifiable *I* or *me*. Identity stems largely from conceptions formed at this *phenomenological* level of analysis. It is especially significant in that it serves as a guidepost and lends continuity to changing experience. Most individuals have an implicit sense of who they are, but they differ greatly in the clarity, accuracy, and complexity of their introspection (Millon, 1986b). Few can articulate the psychic elements that make up this image (e.g., that they are primarily alienated, inept, complacent, conscientious, and so on).

Object Representations

As noted previously, significant experiences from the past leave an inner imprint, a structural residue composed of memories, attitudes, and affects that serve as a substrate of dispositions for perceiving and reacting to life's events. Analogous to the various organ systems in the body, both the character and the substance of these internalized representations of significant figures and relationships from the past can be differentiated and analyzed for clinical purposes. Variations in the nature and content of this inner world can be associated with one or another personality and lead us to employ descriptive terms to represent them, such as shallow, vexatious, undifferentiated, concealed, and irreconcilable.

Morphologic Organization

The overall architecture that serves as a framework for an individual's psychic interior may display weakness in its structural cohesion, exhibit deficient coordination among its components, and possess few mechanisms to maintain balance and harmony, regulate internal conflicts, or mediate external pressures. The concept of morphologic organization refers to the structural strength, interior congruity, and functional efficacy of the personality system. *Organization* of the mind is a concept almost exclusively derived from inferences at the *intrapsychic* level of analysis. The concept is akin to and employed in conjunction with current psychoanalytic notions

such as borderline and psychotic levels, but this usage tends to be limited, relating essentially to quantitative degrees of integrative pathology, not to qualitative variations in either integrative structure or configuration. Stylistic variants of this structural attribute may be employed to characterize each of the personality prototypes: their distinctive organizational attributes are represented with descriptors such as inchoate, disjointed, and compartmentalized.

Mood/Temperament

Few observables are more clinically relevant from the *biophysical* level of data analysis than the predominant character of an individual's affect and the intensity and frequency with which he or she expresses it. The meaning of extreme emotions is easy to decode. This is not so with the more subtle moods and feelings that insidiously and repetitively pervade the patient's ongoing relationships and experiences. Not only are the expressive features of mood and drive conveyed by terms such as *distraught*, *labile*, *fickle*, or *hostile* communicated via self-report, but they are revealed as well, albeit indirectly, in the patient's activity level, speech quality, and physical appearance. Clearly, the most useful aspect of this attribute as it relates to the theory is its utility in appraising features that are relevant to the pleasure-pain and active-passive polarities.

Clinical Attributes Within Each Personality Domain

Table 2.2 presents the functional processes and structural attributes of the personality constructs derived from the theory and included in the MCMI-III test. These attributes have been constructed to be comprehensive (exhaustive of the major domains of personality) and comparable (existing at approximately equal levels of abstraction) across domains. Incorporating all the personalities of the *DSM-III-R* and *DSM-IV* yields an 8 x 14 domain-by-disorder matrix. Each cell of the matrix contains the diagnostic attribute or criterion which, in the authors' judgment, best captures the expression of that personality style within that particular domain. Vertically, the matrix facilitates ready comparison and contrast of the various disorders within content areas.

Horizontally, it suggests that criteria must be appraised conjointly across a variety of content areas in order to arrive at a comprehensive picture of each diagnostic prototype. The vertical view gets at discriminant validity, the horizontal at convergent validity. In the bottom part of Table 2.2, these attributes are described in greater detail, providing a broad descriptive picture of each of the personality disorders. For comparative purposes, the *DSM-IV* criteria for each disorder are also listed along with corresponding MCMI-III items. This should allow the reader to quickly evaluate the content validity of the MCMI-III items and of the *DSM-IV* criteria with respect to the clinical domains.

Clinical Syndromes: Axis I

The Axis I clinical syndromes are best understood as disorders embedded within the context of the Axis II styles. Accordingly, they are often seen in actual clinical cases as extensions or distortions of the patient's basic personality pattern. These syndromes tend to be relatively distinct or transient states, waxing and waning over time depending on the impact of stressful situations. In times of distress, they caricature or accentuate the basic personality style. Regardless of how distinctive they appear to be, however, they take on meaning and significance only in the context of the patient's personality and should be appraised with reference to that pattern. Despite the observation that certain disorders arise most frequently in conjunction with particular personality styles, each of these symptom states will occur in several patterns. For example, mild depression or dysthymia occurs most frequently among avoidant, dependent, and self-defeating personalities; alcohol dependence is found commonly with histrionic and antisocial patterns.

Because several covariations are possible between Axis I syndromes and Axis II personality styles, constructing a model in which these interrelationships can be specified is crucial. Most of the clinical syndromes described in this section are of the reactive kind and are thus of substantially briefer duration than the personality disorders. They usually represent states in which an active pathological process is clearly manifest, often precipitated by external events.

Typically they appear in somewhat striking or dramatic form, often accentuating or intensifying the more prosaic features of the premorbid or basic personality style. During periods of active pathology, it is not uncommon for several symptoms to covary and to change over time in their degrees of prominence. The MCMI-III Clinical Syndromes scales—A, H, N, D, B, T, and R—represent disorders of moderate severity. The Severe Clinical Syndromes scales—SS, CC, and PP—reflect disorders of marked severity. Each of these disorders is described below. Table 2.3 lists the prototypal items assigned to each Axis I scale.

Moderately Severe Clinical Syndromes (Scales A Through R)

Anxiety (Scale A)

An anxious patient often reports feeling either vaguely apprehensive or specifically phobic. He or she is typically tense, indecisive, and restless and tends to complain of various types of physical discomfort, such as tightness, excessive perspiration, ill-defined muscular aches, and nausea. A review of the patient's responses to the items on the Anxiety scale will aid in determining whether he or she is primarily phobic and, more specifically, whether the phobia is of a *simple* or a *social* variety. However, most anxious patients evidence a generalized state of tension, manifested by an inability to relax, fidgety movements, and a readiness to react and be easily startled. Somatic discomfort—for example, clammy hands or upset stomach—are also characteristic. Also notable are worrisomeness and an apprehensive sense that problems are imminent, a hyperalertness to one's environment, edginess, and generalized touchiness.

Somatoform (Scale H)

Patients with elevated scores on the Somatoform scale express their psychological difficulties through somatic channels, persistent periods of fatigue and weakness, and a preoccupation with ill health and with a variety of dramatic but largely nonspecific pains in different and unrelated regions of the body. Some patients evidence a primary somatization disorder that is characterized by recurrent, multiple somatic

complaints, often presented in a dramatic, vague, or exaggerated manner. Others have a history that may best be considered hypochondriacal; they interpret minor physical discomfort or sensations as signifying a serious ailment. If diseases are factually present, they tend to be overinterpreted despite medical reassurance. Typically, somatic complaints are employed to gain attention.

Bipolar: Manic (Scale N)

Bipolar patients evidence periods of superficial elation, inflated self-esteem, restless overactivity and distractibility, pressured speech, impulsiveness, and irritability. Also evident is unselective enthusiasm, excessive planning for unrealistic goals, an intrusive if not domineering and demanding quality to interpersonal relations, decreased need for sleep, flights of ideas, and rapid and labile shifts of mood. Very high scores may signify psychotic processes, including delusions or hallucinations.

Dysthymia (Scale D)

Patients with high scores on the Dysthymia scale remain involved in everyday life but have been preoccupied over a period of years with feelings of discouragement or guilt, a lack of initiative, behavioral apathy, low self-esteem, and frequently expressed futility and self-deprecatory comments. During periods of dejection, there may be tearfulness, suicidal ideation, a pessimistic outlook toward the future, social withdrawal, poor appetite or overeating, chronic fatigue, poor concentration, a marked loss of interest in pleasurable activities, and decreased effectiveness in performing routine life tasks. Unless Scale CC (Major Depression) is also notably elevated, there is little likelihood that psychotic depressive features will be in evidence. Close examination of the specific item responses that contribute to the patient's high score on Scale D should enable the clinician to discern the particular features of the dysthymic mood (for example, low self-esteem or hopelessness).

Alcohol Dependence (Scale B)

A patient with a high score on the Alcohol Dependence scale probably has a history of alcoholism, has tried to overcome the problem with minimal success, and as a consequence

experiences considerable discomfort in family and work settings. What is of value in this and the subsequent scale (Drug Dependence) is the opportunity to consider the problem in the context of the patient's overall style of personality functioning and coping.

Drug Dependence (Scale T)

An elevated Drug Dependence score indicates that the patient probably has a recurrent or recent history of drug abuse, tends to find it difficult to restrain impulses or keep them within conventional social limits, and is unable to manage the personal consequences of this behavior. Like the Alcohol Dependence scale, the Drug Dependence scale is composed of many subtle and indirect items and may be useful in identifying individuals who are not readily disposed to admit their drug problems.

Post-Traumatic Stress Disorder (Scale R)

Patients with elevated scores on the Post-Traumatic Stress Disorder scale have experienced an event that involved a threat to their life and reacted to it with intense fear or feelings of helplessness. Images and emotions associated with the trauma persistently result in distressing recollections and nightmares that reactivate the feelings generated by the original event. Symptoms of anxious arousal (e.g., exaggerated startle response, hypervigilance) persist, and efforts are made to avoid circumstances associated with the trauma.

Severe Clinical Syndromes (Scales SS, CC, and PP)

Thought Disorder (Scale SS)

Depending on the length and course of the problem, patients with elevated scores on this scale are usually classified as "schizophrenic" or "schizophreniform" or as having a "brief reactive psychosis." They may periodically exhibit incongruous, disorganized, or regressive behavior, often appearing confused and disoriented and occasionally displaying inappropriate affect, scattered hallucinations, and unsystematic delusions. Their thinking may be fragmented or bizarre. Their feelings may be blunted, and there may be a pervasive sense of being isolated from

and misunderstood by others. Withdrawn and seclusive or secretive behavior may be notable.

Major Depression (Scale CC)

Patients with high scores on this scale are usually incapable of functioning in a normal environment, are severely depressed, and express a dread of the future, suicidal ideation, and a sense of hopeless resignation. Some exhibit marked motor retardation, whereas others display an agitated quality, incessantly pacing and bemoaning their sorry state. Several somatic problems often occur during these periods, notably decreased appetite, fatigue, weight loss or gain, insomnia, and early rising. Problems of concentration are common as are feelings of worthlessness or guilt. Repetitive fearfulness and brooding are frequently in evidence. Depending on the underlying personality style, there may be a shy, introverted, and seclusive pattern characterized by sluggish immobility or an irritable, complaining, and whining tone.

Delusional Disorder (Scale PP)

Patients with delusional disorder are frequently considered acutely paranoid and may become periodically belligerent, voicing irrational but interconnected delusions of a jealous, persecutory, or grandiose nature. Depending on the constellation of other concurrent syndromes, there may be clear-cut signs of disturbed thinking and ideas of reference. Their moods are usually hostile, and they may feel picked on and mistreated. Suspiciousness, vigilance, and alertness to possible betrayal are typical concomitants.

Rational Validation of the MCMI Items

The derivation of the Axis II constructs and the nature of the Axis I constructs measured by the MCMI-III test have been set forth above. When the MCMI test was originally published, there was as yet no need to coordinate it with any official schema of mental disorders. Publication of the *DSM-III* was still several years in the future. We now return to the beginnings of the MCMI test in order to describe the steps taken to generate and rationally validate its preliminary items. This material is important for historical and instructional purposes and to illustrate continuity

from the MCMI test to the MCMI-III test. The theoretical-substantive validation phase in the development of the original MCMI test consisted of three sequential steps: (a) creating a theoretically based initial item pool, (b) reducing the item pool on rational grounds, and (c) selecting items to establish two equivalent provisional forms of the instrument.

Creating an Item Pool

The construction of the MCMI test went one step beyond the traditional procedure of content- or face-valid item selection; item pools were prepared for each syndrome scale on the basis of theoretically derived definitions (Gynther, Burkhart, & Hovanitz, 1979; Holden & Jackson, 1979; Wrobel & Lachar, 1982). Detailed characteristics of each syndrome, published in earlier work, provided a fund of typical response behavior for different syndromes expressed in various situational contexts. Writing and initially selecting items from this broad spectrum of characteristics ensured that the content of the initial pool was both comprehensive and representative. In addition to the major theory-based text at the time (*Modern Psychopathology*, Millon, 1969/1983), more than 80 sources—including psychiatry and psychology textbooks, rating scales, and published personality tests—were exhaustively combed to produce more than 3,500 self-descriptive, content-oriented item statements.

Next, a set of 20 theory-based clinical scales was specified for the initial classification of items. Three of the original scales were dropped and replaced with scales of potentially greater clinical use. The basic personality scales subset varied in number from 177 items for the schizoid pattern category to 432 for the antisocial group. The severe personality disorder scales ranged from 179 items for the schizotypal category to 200 for the borderline group. The clinical syndrome scales varied in number from 70 items in the somatoform category to a high of 94 in the anxiety group.

All items at this stage of development were phrased so that the True response would signify the presence of the scale syndrome. Although there was some initial concern regarding “acquiescent” bias (Jackson & Messick, 1961), subsequent research has indicated that its role was likely to be minor (Rorer, 1965). Moreover,

preliminary trials suggested that negatively worded items would lack sufficient differential specificity to be useful. That is, numerous patient syndrome types are likely to conclude that a particular behavioral item is not true of them, whereas only a few syndrome types are likely to say that the behavior is true of them. This expectation was supported in subsequent studies by others when negatively worded items proved to be of considerably less differential diagnostic use (Holden, Fekken, & Jackson, 1985).

Reducing the Item Pool on Rational Grounds

The process of rational item reduction included several stages of editing using criteria such as grammar; clarity, simplicity, content validity, and scale relevance. As a first step, readers eliminated items that were inexplicit, excessively complex, and possibly objectionable. Items were also eliminated if they contained desirability bias, were likely to be unreliable, or would probably result in extreme endorsement frequency. Most significantly, items were selected because they exemplified traits that characterized the specific scale for which they were written. Several efforts were made to select items to achieve convergent and discriminant validity, that is, to judge items not only in terms of their relevance for a specific scale but also in terms of their “lack of fit” with other selected syndrome scales (Campbell & Fiske, 1959). Another important criterion was covering the full domain of traits and behavior characterizing each syndrome. Rather than concentrating on one or two features that might be most distinctive of a personality pattern or clinical syndrome, efforts were made to provide a balanced and representative sample of all of the characteristics of the scale (Wiggins, 1973).

In addition to these preliminary item pool reduction steps, four empirical procedures for further winnowing the set were employed at this stage. First, a small sample of patients was asked to read the remaining items and judge them on clarity and ease of self-rating. Second, after the pool had been reduced using these patient judgments, eight mental health professionals who were well versed in the theory and nosology were asked to blindly sort all items into their theoretically most appropriate category.

ries. Items were retained only if six or more of these clinicians placed them correctly, that is, according to the theoretical model. Third, items that survived these initial "best-fit" clinical sortings were resorted for second- and third-best fits, first within their own subsection (except in the pathological personality subsection of three scales where only a second-best fit was sorted), and second within the other subsections. These data were stored for later decisions concerning the multiple scoring of items. Fourth, items that survived the first series of sortings were sorted once again for first and second "negative fit," that is, for those unlikely to be endorsed by individuals with certain personality styles or disorders. For example, an item signifying a strong interest in social activities would be negatively sorted for the Schizoid scale. These data were also kept as background information for the plan to key False responses at a later stage of test development.

Establishing Substantively Valid Provisional Forms

Using the criteria and steps described above, the initial pool was reduced to the 1,132 best-fit items, which were then divided into various scale categories (second- and third-best fit and negative-fit selections were not used as data for this stage of research). In addition, the pool included 36 items as a check against random, denial, and complaint distortion tendencies; protocols displaying high scores on these distortion scales were discarded. Because of the time-consuming and statistically unwieldy nature of an inventory of 1,132 items, the reduced pool was divided using item matching for representativeness into two provisional forms, each consisting of 566 items. The specific figure of 566 for each form was

determined by the availability of appropriately designed answer sheets of that length. Because the items on both forms were selected for consistency with the theoretically derived sets of syndromes, the scales were viewed as having met the criteria of a substantively valid instrument.

MCMI-III Item Generation and Refinement

Because the publication of the original MCMI test preceded that of the *DSM-III* in 1980, the tasks of item generation and rational validation were laborious and time-intensive. In contrast, Axis II is now well ensconced as a major reference point in the conception of psychopathology, with content that has been successively elaborated across three generations of the *DSM*. Hence, the desire to coordinate the MCMI-III test with the official nosology made item generation and rational selection relatively straightforward. Moreover, because the senior author was a member of the *DSM-IV* Axis II committee, the official rendering of these constructs in part reflects their conception in the evolutionary theory described earlier. Thus, the search for self-report items that would adequately capture both the official criteria and the author's more theoretical perspective was made considerably easier. A substantial pool of new items for the MCMI-III test was generated through the expert judgment of the senior author. These items were then subjected to rational scrutiny by research associates who worked on the revision project. Following their suggestions and revisions, an MCMI-II Research Form with 325 items was drafted as the foundation for further empirical study. Additional details are given in the following chapter.

Table 2.2 Expression of Personality Disorders Across the Functional and Structural Domains of Personality—Overview

Disorder	Functional Processes				Structural Attributes			
	Expressive Acts	Interpersonal Conduct	Cognitive Style	Regulatory Mechanisms	Self-Image	Object Representations	Morphologic Organization	Mood/Temperament
<u>Schizoid</u>	Impassive	Unengaged	Impoverished	Intellectualization	Complacent	Meager	Undifferentiated	Apathetic
<u>Avoidant</u>	Fretful	Aversive	Distracted	Fantasy	Alienated	Vexatious	Fragile	Anguished
<u>Depressive</u>	Disconsolate	Defenseless	Pessimistic	Aceticism	Worthless	Forsaken	Depleted	Melancholic
Dependent	Incompetent	Submissive	Naive	Introjection	Inept	Immature	Inchoate	Pacific
Histrionic	Dramatic	Attention-Seeking	Flighty	Dissociation	Gregarious	Shallow	Disjointed	Fickle
Narcissistic	Haughty	Exploitive	Expansive	Rationalization	Admirable	Contrived	Spurious	Insouciant
Antisocial	Impulsive	Irresponsible	Deviant	Acting Out	Autonomous	Debased	Unruly	Callous
Sadistic	Precipitate	Abrasive	Dogmatic	Isolation	Combative	Pernicious	Eruptive	Hostile
Compulsive	Disciplined	Respectful	Constricted	Reaction Formation	Conscientious	Concealed	Compartmentalized	Solemn
Negativistic	Resentful	Contrary	Skeptical	Displacement	Discontented	Vacillating	Divergent	Irritable
Masochistic	Abstinent	Deferential	Diffident	Exaggeration	Undeserving	Discredited	Inverted	Dysphoric
<u>Schizotypal</u>	Eccentric	Secretive	Autistic	Undoing	Estranged	Chaotic	Fragmented	Distraught or Insentient
Borderline	Spasmodic	Paradoxical	Capricious	Regression	Uncertain	Incompatible	Split	Labile
Paranoid	Defensive	Provocative	Suspicious	Projection	Inviolable	Unalterable	Inelastic	Irascible

Expression of Personality Disorders Across the Functional and Structural Domains of Personality—Prototypes

Schizoid Prototype

Functional-Structural Operationalization

Functional Processes	Structural Attributes
Expressively Impassive (e.g., appears to be in an inert emotional state, lifeless, undemonstrative, lacking in energy and vitality; is unmoved, boring, unanimated, robotic, phlegmatic; displays deficits in activation, motoric expressiveness, and spontaneity)	Complacent Self-Image (e.g., reveals minimal introspection and awareness of self; seems impervious to the emotional and personal implications of everyday social life, appearing indifferent to the praise or criticism of others)

Table 2.2 (Prototypes continued—Schizoid)

Interpersonally Unengaged (e.g., seems indifferent and remote, rarely responsive to the actions or feelings of others; chooses solitary activities, possesses minimal "human" interests; fades into the background, is aloof or unobtrusive, neither desires nor enjoys close relationships, prefers a peripheral role in social, work, and family settings)	Meager Objects (e.g., internalized representations are few in number and minimally articulated, largely devoid of the manifold percepts and memories of relationships with others, possessing little of the dynamic interplay among drives and conflicts that typify well-adjusted individuals)
Cognitively Impoverished (e.g., seems deficient across broad spheres of human knowledge and evidences vague and obscure thought processes, particularly about social matters; communication with others is often unfocused, loses its purpose or intention, or is conveyed via loose or circuitous logic)	Undifferentiated Organization (e.g., given an inner barrenness, a feeble drive to fulfill needs, and minimal pressures to defend against or resolve internal conflicts or cope with external demands, internal morphologic structures may best be characterized by their limited framework and sterile pattern)
Intellectualization Mechanism (e.g., describes interpersonal and affective experiences in a matter-of-fact, abstract, impersonal, or mechanical manner; pays primary attention to formal and objective aspects of social and emotional events)	Apathetic Mood (e.g., is emotionally unexcitable, exhibiting an intrinsic unfeeling, cold, stark quality; reports weak affectionate or erotic needs, rarely displaying warm or intense feelings; is apparently unable to experience most affects—pleasure, sadness, or anger—in any depth)

DSM/MCMI Correspondence (Schizoid)

DSM-IV Criterion	Parallel MCMI-III Item
A. A pervasive pattern of detachment from social relationships and a restricted range of expression of emotions in interpersonal settings, beginning by early adulthood and present in a variety of contexts, as indicated by four (or more) of the following:	92. I'm alone most of the time and I prefer it that way.
(1) neither desires nor enjoys close relationships, including being part of a family	105. I have little desire for close friendships.
(2) almost always chooses solitary activities	27. When I have a choice, I prefer to do things alone.
(3) has little, if any, interest in having sexual experiences with another person	46. I've always had less interest in sex than most people do.
(4) takes pleasure in few, if any, activities	148. Few things in life give me pleasure.
(5) lacks close friends or confidants other than first-degree relatives	165. Other than my family, I have no close friends.
(6) appears indifferent to the praise or criticism of others	—
(7) shows emotional coldness, detachment, or flattened affectivity	10. What few feelings I seem to have I rarely show to the outside world.

Table 2.2 (Prototypes continued—Avoidant)

Avoidant Prototype	
Functional-Structural Operationalization	
Functional Processes	Structural Attributes
Expressively Fretful (e.g., conveys personal unease and disquiet, a constant timorous, hesitant, and restive state; overreacts to innocuous events and anxiously judges them to signify ridicule, criticism, and disapproval)	Alienated Self-Image (e.g., sees self as socially inept, inadequate, and inferior, justifying thereby his or her isolation from and rejection by others; feels personally unappealing, devalues his or her achievements, and reports persistent sense of aloneness and emptiness)
Interpersonally Aversive (e.g., distances self from activities that involve intimate personal relationships and reports extensive history of social pan-anxiety and distrust; seeks acceptance but is unwilling to get involved unless certain to be liked, maintaining distance and privacy to avoid being shamed and humiliated)	Vexatious Objects (e.g., internalized representations are composed of readily reactivated, intense, and conflict-ridden memories of problematic early relations, limited avenues for experiencing or recalling gratification, and few mechanisms to channel needs, bind impulses, resolve conflicts, or deflect external stressors)
Cognitively Distracted (e.g., warily scans environment for potential threats and is preoccupied by intrusive and disruptive random thoughts and observations; an upwelling from within of irrelevant ideation upsets thought continuity and interferes with social communication and accurate appraisals)	Fragile Organization (e.g., a precarious complex of tortuous emotions depends almost exclusively on a single modality for resolution and discharge, that of avoidance, escape, and fantasy and, hence, when faced with personal risks, new opportunities, or unanticipated stress, few morphologic structures are available to deploy and few back-up positions can be reverted to, short of regressive decompensation)
Fantasy Mechanism (e.g., depends excessively on imagination to achieve need gratification, confidence building, and conflict resolution; withdraws into reveries as a means of safely discharging frustrated affectionate and angry impulses)	Anguished Mood (e.g., describes constant and confusing undercurrent of tension, sadness, and anger; vacillates between desire for affection, fear of rejection, embarrassment, and numbness of feeling)

DSM/MCMI Correspondence (Avoidant)	
DSM-IV Criterion	Parallel MCMI-III Item
A. A pervasive pattern of social inhibition, feelings of inadequacy, and hypersensitivity to negative evaluation, beginning by early adulthood and present in a variety of contexts, as indicated by four (or more) of the following:	40. I guess I'm a fearful and inhibited person.
(1) avoids occupational activities that involve significant interpersonal contact, because of fears of criticism, disapproval, or rejection	69. I avoid most social situations because I expect people to criticize or reject me.

Table 2.2 (Prototypes continued—Avoidant, Depressive)

(2) is unwilling to get involved with people unless certain of being liked	127. I won't get involved with people unless I'm sure they'll like me.
(3) shows restraint within intimate relationships because of the fear of being shamed or ridiculed	18. I'm afraid to get really close to another person because it may end up with my being ridiculed or shamed.
(4) is preoccupied with being criticized or rejected in social situations	99. In social groups I am almost always very self-conscious and tense.
(5) is inhibited in new interpersonal situations because of feelings of inadequacy	174. Although I'm afraid to make friendships, I wish I had more than I do.
(6) views self as socially inept, personally unappealing, or inferior to others	141. I feel that most people think poorly of me.
(7) is unusually reluctant to take personal risks or to engage in any new activities because they may prove embarrassing	84. I'm too unsure of myself to risk trying something new.

Depressive Prototype

Functional-Structural Operationalization	
Functional Processes	Structural Attributes
Expressively Disconsolate (e.g., appearance and posture convey an irrelievably forlorn, somber, heavy-hearted, woebegone if not grief-stricken quality; irremediably dispirited and discouraged, portraying a sense of permanent hopelessness and wretchedness)	Worthless Self-Image (e.g., judges him- or herself to be of no account, valueless to self or others, inadequate, and unsuccessful in all aspirations; barren, sterile, impotent; sees self as inconsequential and reproachable, if not contemptible, a person who should be criticized and derogated and who should feel guilty for having no praiseworthy traits or achievements)
Interpersonally Defenseless (e.g., owing to feeling vulnerable, assailable, and unshielded, will beseech others to be nurturing and protective; fearing abandonment and desertion, will not only act in an endangered manner but will seek if not demand assurances of affection, steadfastness, and devotion)	Forsaken Objects (e.g., internalized representations of the past appear forsaken as if life's early experiences have been depleted or devitalized, either drained of their richness and joyful elements or withdrawn from memory, leaving him or her to feel abandoned, bereft, discarded, cast off, and deserted)
Cognitively Pessimistic (e.g., possesses defeatist and fatalistic attitudes about almost all matters; sees things in their blackest form and invariably expects the worst; feels weighed down, discouraged, and bleak; gives the gloomiest interpretation of current events, feeling that things will never improve)	Depleted Organization (e.g., the scaffold for morphologic structures is markedly weakened, with coping methods enervated and defensive strategies impoverished, emptied, and devoid of vigor and focus, resulting in a diminished if not exhausted capacity to initiate action and regulate affect, impulse, and conflict)

Table 2.2 (Prototypes continued—Depressive, Dependent)

Asceticism Mechanism (e.g., engages in acts of self-denial, self-punishment, and self-tormenting, believing that one should exhibit penance and be deprived of life's bounties; not only is there a repudiation of pleasures but there are harsh self-judgments and self-destructive acts)	Melancholic Mood (e.g., is typically woeful, gloomy, tearful, joyless, and morose; characteristically worrisome and brooding; low spirits and dysphoric state rarely remit)
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DSM/MCMI Correspondence (Depressive)	
DSM-IV Criterion	Parallel MCMI-III Item
See <i>DSM-IV</i> , p. 733.	20. I've had sad thoughts much of my life since I was a child.
	112. I have been downhearted and sad much of my life since I was quite young.
	123. I've always had a hard time stopping myself from feeling blue and unhappy.
	151. I've never been able to shake the feeling that I'm worthless to others.
	47. I tend to always blame myself when things go wrong.
	145. I spend my life worrying over one thing or another.
	133. Even in good times, I've always been afraid that things would soon go bad.
	25. I feel guilty much of the time for no reason that I know.

Dependent Prototype

Functional-Structural Operationalization	
Functional Processes	Structural Attributes
Expressively Incompetent (e.g., withdraws from adult responsibilities by acting helpless and seeking nurturance from others; is docile and passive, lacks functional competencies, and avoids self-assertion)	Inept Self-Image (e.g., views self as weak, fragile, and inadequate; exhibits lack of self-confidence by belittling own attitudes and competencies and hence is not capable of doing things on his or her own)
Interpersonally Submissive (e.g., needs excessive advice and reassurance; subordinates self to stronger, nurturing figure, without whom he or she may feel anxiously alone and helpless; is compliant, conciliatory, and placating, afraid of being left to care for him- or herself)	Immature Objects (e.g., internalized representations are composed of infantile impressions of others, unsophisticated ideas, incomplete recollections, rudimentary drives, childlike impulses, and minimal competencies to manage and resolve stressors)

Table 2.2 (Prototypes continued—Dependent)

Cognitively Naïve (e.g., rarely disagrees with others and is easily persuaded, unsuspicious, and gullible; reveals a Pollyanna attitude toward interpersonal difficulties, watering down objective problems and smoothing over troubling events)

Inchoate Organization (e.g., owing to entrusting others with the responsibility to fulfill needs and to cope with adult tasks, there is a deficient morphologic structure and a lack of diversity in internal regulatory controls, leaving a miscellany of relatively undeveloped and undifferentiated adaptive abilities and an elementary system for functioning independently)

Introjection Mechanism (e.g., is firmly devoted to another to strengthen the belief that an inseparable bond exists between them; sacrifices independent views in favor of those of others to preclude conflicts and threats to relationship)

Pacific Mood (e.g., is characteristically warm, tender, and noncompetitive; timidly avoids social tension and interpersonal conflict)

DSM/MCMI Correspondence (Dependent)

DSM-IV Criterion	Parallel MCMI-III Item
A. A pervasive and excessive need to be taken care of that leads to submissive and clinging behavior and fears of separation, beginning by early adulthood and present in a variety of contexts, as indicated by five (or more) of the following:	16. I am a very agreeable and submissive person.
(1) has difficulty making everyday decisions without an excessive amount of advice and reassurance from others	94. People can easily change my ideas, even if I thought my mind was made up.
(2) needs others to assume responsibility for most major areas of his or her life	73. I often allow others to make important decisions for me.
(3) has difficulty expressing disagreement with others because of fear of loss of support or approval. Note: Do not include realistic fears of retribution	169. I'm always willing to give in to others in a disagreement because I fear their anger or rejection.
(4) has difficulty initiating projects or doing things on his or her own (because of a lack of self-confidence in judgment or abilities rather than a lack of motivation or energy)	35. I often give up doing things because I'm afraid I won't do them well.
(5) goes to excessive lengths to obtain nurturance and support from others, to the point of volunteering to do things that are unpleasant	45. I always try hard to please others, even when I dislike them.
(6) feels uncomfortable or helpless when alone because of exaggerated fears of being unable to care for himself or herself	135. Being alone, without the help of someone close to depend on, really frightens me.
(7) urgently seeks another relationship as a source of care and support when a close relationship ends	—

Table 2.2 (Prototypes continued—Dependent, Histrionic)

- | | |
|--|---|
| (8) is unrealistically preoccupied with fears of being left to take care of himself or herself | 108. I worry a great deal about being left alone and having to take care of myself. |
|--|---|

Histrionic Prototype

Functional-Structural Operationalization	
Functional Processes	Structural Attributes
Expressively Dramatic (e.g., is overreactive, volatile, provocative, and engaging; is intolerant of inactivity, resulting in impulsive, highly emotional, and theatrical responsiveness; describes penchant for momentary excitement, fleeting adventures, and short-sighted hedonism)	Gregarious Self-Image (e.g., views self as sociable, stimulating, and charming; enjoys the image of attracting acquaintances by physical appearance and by pursuing a busy and pleasure-oriented life)
Interpersonally Attention-Seeking (e.g., actively solicits praise and manipulates others to gain needed reassurance, attention, and approval; is demanding, flirtatious, vain, and seductively exhibitionistic, especially when wishing to be the center of attention)	Shallow Objects (e.g., internalized representations are composed largely of superficial memories of past relations, random collections of transient and segregated affects and conflicts, and insubstantial drives and mechanisms)
Cognitively Flighty (e.g., avoids introspective thought, is overly suggestible and attentive to fleeting external events, and speaks in impressionistic generalities; integrates experiences poorly, resulting in scattered learning and thoughtless judgments)	Disjointed Organization (e.g., there is a loosely knit and carelessly united morphologic structure in which processes of internal regulation and control are scattered and unintegrated, with ad hoc methods for restraining impulses, coordinating defenses, and resolving conflicts, leading to mechanisms that must, of necessity, be broad and sweeping to maintain psychic cohesion and stability, and, when successful, only further isolate and disconnect thoughts, feelings, and actions)
Dissociation Mechanism (e.g., regularly alters and recomposes self-presentations to create a succession of socially attractive but changing facades; engages in self-distracting activities to avoid reflecting on and integrating unpleasant thoughts and emotions)	Fickle Mood (e.g., displays rapidly shifting and shallow emotions; is vivacious, animated, and impetuous and exhibits tendencies to be easily excited and as easily angered or bored)

DSM/MCMI Correspondence (Histrionic)

DSM-IV Criterion	Parallel MCMI-III Item
A. A pervasive pattern of excessive emotionality and attention seeking, beginning by early adulthood and present in a variety of contexts, as indicated by five (or more) of the following:	57. I think I am a very sociable and outgoing person.

Table 2.2 (Prototypes continued—Histrionic, Narcissistic)

(1) is uncomfortable in situations in which he or she is not the center of attention	88. I never sit on the sidelines when I'm at a party.
(2) interaction with others is often characterized by inappropriate sexually seductive or provocative behavior	21. I like to flirt with members of the opposite sex.
(3) displays rapidly shifting and shallow expression of emotions	12. I show my feelings easily and quickly.
(4) consistently uses physical appearance to draw attention to self	—
(5) has a style of speech that is excessively impressionistic and lacking in detail	—
(6) shows self-dramatization, theatricality, and exaggerated expression of emotion	51. When things get boring, I like to stir up some excitement.
(7) is suggestible, i.e., easily influenced by others or circumstances	32. I am always looking to make new friends and meet new people.
(8) considers relationships to be more intimate than they actually are	80. It is very easy for me to make many friends.

Narcissistic Prototype

Functional-Structural Operationalization

Functional Processes	Structural Attributes
Expressively Haughty (e.g., acts in an arrogant, supercilious, pompous, and disdainful manner, flouting conventional rules of shared social living, viewing them as naive or inapplicable to him- or herself; reveals a careless disregard for personal integrity and a self-important indifference to the rights of others)	Admirable Self-Image (e.g., believes him- or herself to be meritorious, special, unique, and deserving of great admiration; acts in a grandiose or self-assured manner, often without parallel achievements; has a very strong sense of self-worth despite being seen by others as egotistic, inconsiderate, and arrogant)
Interpersonally Exploitive (e.g., feels entitled, is unempathic, and expects special favors without assuming reciprocal responsibilities; shamelessly takes others for granted and uses them to enhance self and indulge desires)	Contrived Objects (e.g., internalized representations are composed far more than usual of illusory and changing memories of past relationships; unacceptable drives and conflicts are readily refashioned as the need arises, while others are often simulated and pretentious)

Table 2.2 (Prototypes continued—Narcissistic)

Cognitively Expansive (e.g., has an undisciplined imagination and exhibits a preoccupation with immature and self-glorifying fantasies of success, beauty, or love; is minimally constrained by objective reality, takes liberties with facts and often lies to redeem illusions about him- or herself)	Spurious Organization (e.g., morphologic structures underlying coping and defensive strategies tend to be flimsy and transparent, appear more substantial and dynamically orchestrated than they really are, regulate impulses only marginally, channel needs with minimal restraint, and create an inner world in which conflicts are dismissed, failures are quickly redeemed, and pride is effortlessly reasserted)
Rationalization Mechanism (e.g., is self-deceptive and facile in devising plausible reasons to justify self-centered and socially inconsiderate behavior; offers alibis to place him- or herself in the best possible light, despite evident shortcomings or failures)	Insouciant Mood (e.g., manifests a general air of nonchalance, imperturbability, and feigned tranquillity; appears coolly unimpressible or buoyantly optimistic except when narcissistic confidence is shaken, at which time rage, shame, or emptiness is briefly displayed)

<i>DSM/MCMI Correspondence (Narcissistic)</i>	
<i>DSM-IV Criterion</i>	<i>Parallel MCMI-III Item</i>
A. A pervasive pattern of grandiosity (in fantasy or behavior), need for admiration, and lack of empathy, beginning by early adulthood and present in a variety of contexts, as indicated by five (or more) of the following:	—
(1) has a grandiose sense of self-importance (e.g., exaggerates achievements and talents, expects to be recognized as superior without commensurate achievements)	67. I have many ideas that are ahead of the times.
(2) is preoccupied with fantasies of unlimited success, power, brilliance, beauty, or ideal love	6. People have never given me enough recognition for the things I've done.
(3) believes that he or she is "special" and unique and can only be understood by, or should associate with, other special or high-status people (or institutions)	159. Someone would have to be pretty exceptional to understand my special abilities.
(4) requires excessive admiration	144. I guess I go out of my way to encourage people to admire the things I say or do.
(5) has a sense of entitlement, i.e., unreasonable expectations of especially favorable treatment or automatic compliance with his or her expectations	31. I think I'm a special person who deserves special attention from others.
(6) is interpersonally exploitative, i.e., takes advantage of others to achieve his or her own ends	85. I don't blame anyone who takes advantage of someone who allows it.

Table 2.2 (Prototypes continued—Narcissistic, Antisocial)

(7) lacks empathy: is unwilling to recognize or identify with the feelings and needs of others	93. There are members of my family who say I'm selfish and think only of myself.
(8) is often envious of others or believes that others are envious of him or her	26. Other people envy my abilities.
(9) shows arrogant, haughty behaviors or attitudes	5. I know I'm a superior person, so I don't care what people think.

Antisocial Prototype

Functional-Structural Operationalization	
Functional Processes	Structural Attributes
Expressively Impulsive (e.g., is impetuous and irrepressible, acting hastily and spontaneously in a restless, spur-of-the-moment manner; is short-sighted, incautious, and imprudent, failing to plan ahead, consider alternatives, or heed consequences)	Autonomous Self-Image (e.g., sees self as unfettered by the restrictions of social customs and the constraints of personal loyalties; values the image and enjoys the sense of being free, unencumbered, and unconfined by persons, places, obligations, or routines)
Interpersonally Irresponsible (e.g., is untrustworthy and unreliable, failing to meet or intentionally negating personal obligations of a marital, parental, occupational, or financial nature; actively intrudes on and violates the rights of others; transgresses established social codes through deceitful or illegal behavior)	Debased Objects (e.g., internalized representations include degraded and corrupt relationships that spur vengeful attitudes and restive impulses that are driven to subvert established cultural ideals and mores, to devalue personal sentiments, and to sully, but intensely covet, the material attainments of society denied him or her)
Cognitively Deviant (e.g., construes events and relationships in accord with socially unorthodox beliefs and morals; is disdainful of traditional ideals, fails to conform to social norms, and is contemptuous of conventional values)	Unruly Organization (e.g., inner morphologic structures to contain drive and impulse are noted by their paucity, as are efforts to curb refractory energies and attitudes, leading to easily transgressed controls, low thresholds for hostile or erotic discharge, few sublimatory channels, unfettered self-expression, and a marked intolerance of delay or frustration)
Acting-Out Mechanism (e.g., inner tensions that might accrue by postponing the expression of offensive thoughts and malevolent actions are rarely constrained; socially repugnant impulses are not refashioned in sublimated forms but are discharged directly in precipitous ways, usually without guilt or remorse)	Callous Mood (e.g., is insensitive, irritable, and aggressive as expressed in a wide-ranging deficit in social charitableness, compassion, and remorse; exhibits a coarse incivility and an offensive if not reckless disregard for the safety of self or others)

Table 2.2 (Prototypes continued—Antisocial, Sadistic)

DSM/MCMI Correspondence (Antisocial)		
DSM-IV Criterion		Parallel MCMI-III Item
A.	There is a pervasive pattern of disregard for and violation of the rights of others occurring since age 15 years, as indicated by three (or more) of the following:	53. Punishment never stopped me from doing what I wanted.
(1)	failure to conform to social norms with respect to lawful behaviors as indicated by repeatedly performing acts that are grounds for arrest	113. I've gotten into trouble with the law a couple of times.
(2)	deceitfulness, as indicated by repeated lying, use of aliases, or conning others for personal profit or pleasure	139. I'm very good at making up excuses when I get into trouble.
(3)	impulsivity or failure to plan ahead	166. I act quickly much of the time and don't think things through as I should.
(4)	irritability and aggressiveness, as indicated by repeated physical fights or assaults	—
(5)	reckless disregard for safety of self or others	—
(6)	consistent irresponsibility, as indicated by repeated failure to sustain consistent work behavior or honor financial obligations	101. I guess I don't take many of my family responsibilities as seriously as I should.
(7)	lack of remorse, as indicated by being indifferent to or rationalizing having hurt, mistreated, or stolen from another	38. I do what I want without worrying about its effect on others.
B.	The individual is at least age 18 years.	—
C.	There is evidence of conduct disorder with onset before age 15 years.	17. As a teenager, I got into lots of trouble because of bad school behavior.

Sadistic Prototype

Functional-Structural Operationalization	
Functional Processes	Structural Attributes
Expressively Precipitate (e.g., is disposed to react in sudden, abrupt outbursts of an unexpected and unwarranted nature; is recklessly reactive and daring, attracted to challenge, risk, and harm; is unflinching, undeterred by pain, and undaunted by danger and punishment)	Combative Self-Image (e.g., is proud to characterize him- or herself as assertively competitive, vigorously energetic, and militantly hardheaded; values aspects of him- or herself that present pugnacious, domineering, and power-oriented image)

Table 2.2 (Prototypes continued—Sadistic)

Interpersonally Abrasive (e.g., reveals satisfaction in intimidating, coercing, and humiliating others; regularly expresses verbally abusive and derisive social commentary and exhibits vicious if not physically brutal behavior)	Pernicious Objects (e.g., internalized representations of the past are distinguished by early relationships that have generated strongly driven aggressive energies and malicious attitudes as well as by a contrasting paucity of sentimental memories, tender affects, internal conflicts, shame, or guilt feelings)
Cognitively Dogmatic (e.g., is strongly opinionated and close-minded, unbending and obstinate in holding to his or her preconceptions; exhibits broad-ranging authoritarianism, social intolerance, and prejudice)	Eruptive Organization (e.g., despite a generally cohesive morphologic structure composed of routinely adequate modulating controls, defenses, and expressive channels, surging powerful and explosive energies of an aggressive and sexual nature threaten to produce precipitous outbursts that periodically overwhelm and overrun otherwise competent restraints)
Isolation Mechanism (e.g., can be cold-blooded and remarkably detached from an awareness of the impact of his or her destructive acts; views objects of violation impersonally, as symbols of devalued groups devoid of human sensibilities)	Hostile Mood (e.g., has an excitable and irritable temper that flares readily into contentious argument and physical belligerence; is cruel, mean-spirited, and fractious, willing to do harm or even persecute others to get his or her way)

DSM/MCMI Correspondence (Sadistic)

<i>DSM-III-R</i> ^a Criterion	Parallel MCMI-III Item
See <i>DSM-III-R</i>, p. 371.	116. I have had to be really rough with some people to keep them in line. 95. I often make people angry by bossing them. 9. I often criticize people strongly if they annoy me. 14. Sometimes I can be pretty rough and mean in my relations with my family. 64. I don't know why, but I sometimes say cruel things just to make others unhappy. 87. I often get angry with people who do things slowly. 28. I think it's necessary to place strict controls on the behavior of members of my family.

^aSadistic personality disorder was not included in the *DSM-IV*.

Table 2.2 (Prototypes continued—Compulsive)

Compulsive Prototype	
Functional-Structural Operationalization	
Functional Processes	Structural Attributes
Expressively Disciplined (e.g., maintains a regulated, highly structured, strictly organized life; perfectionism interferes with decision making and task completion)	Conscientious Self-Image (e.g., sees self as devoted to work, industrious, reliable, meticulous, and efficient, largely to the exclusion of leisure activities; is fearful of error or misjudgment and hence overvalues aspects of him- or herself that exhibit discipline, perfection, prudence, and loyalty)
Interpersonally Respectful (e.g., exhibits unusual adherence to social conventions and propriety; is scrupulous and overly conscientious about matters of morality and ethics; prefers polite, formal, and correct personal relationships, usually insisting that subordinates adhere to personally established rules and methods)	Concealed Objects (e.g., only those internalized representations, with their associated inner affects and attitudes that can be socially approved, are allowed conscious awareness or behavioral expression; as a result, actions and memories are highly regulated, forbidden impulses sequestered and tightly bound, personal and social conflicts defensively denied, kept from awareness, and maintained under stringent control)
Cognitively Constricted (e.g., constructs world in terms of rules, regulations, schedules, and hierarchies; is rigid, stubborn, and indecisive and is notably upset by unfamiliar or novel ideas and customs)	Compartmentalized Organization (e.g., morphologic structures are rigidly organized in a tightly consolidated system clearly partitioned into numerous, distinct, and segregated constellations of drive, memory, and cognition with few open channels to permit interplay among these components)
Reaction Formation Mechanism (e.g., repeatedly presents positive thoughts and socially commendable behavior that are diametrically opposed to deeper contrary and forbidden feelings; displays reasonableness and maturity when faced with circumstances that evoke anger or dismay in others)	Solemn Mood (e.g., is tense, joyless, and grim; restrains warm feelings and keeps most emotions under tight control)

DSM/MCMI Correspondence (Compulsive)	
DSM-IV ^a Criterion	Parallel MCMI-III Item
A. A pervasive pattern of preoccupation with orderliness, perfectionism, and mental and interpersonal control, at the expense of flexibility, openness, and efficiency, beginning by early adulthood and present in a variety of contexts, as indicated by four (or more) of the following:	29. People usually think of me as a reserved and serious-minded person.

^aIn the DSM-IV, this is labeled "obsessive-compulsive personality disorder."

Table 2.2 (Prototypes continued—Compulsive, Negativistic)

(1) is preoccupied with details, rules, lists, order, organization, or schedules to the extent that the major point of the activity is lost	82. I always make sure that my work is well planned and organized.
(2) shows perfectionism that interferes with task completion (e.g., is unable to complete a project because his or her own overly strict standards are not met)	114. A good way to avoid mistakes is to have a routine for doing things.
(3) is excessively devoted to work and productivity to the exclusion of leisure activities and friendships (not accounted for by obvious economic necessity)	97. I believe in the saying, "early to bed and early to rise..."
	137. I always see to it that my work is finished before taking time out for leisure activities.
(4) is overconscientious, scrupulous, and inflexible about matters of morality, ethics, or values (not accounted for by cultural or religious identification)	172. People tell me that I'm a very proper and moral person.
(5) is unable to discard worn-out or worthless objects even when they have no sentimental value	—
(6) is reluctant to delegate tasks or to work with others unless they submit to exactly his or her way of doing things	—
(7) adopts a miserly spending style toward both self and others; money is viewed as something to be hoarded for future catastrophes	59. I keep very close track of my money so I am prepared if a need comes up.
(8) shows rigidity and stubbornness	2. I think highly of rules because they are a good guide to follow.

Negativistic Prototype

Functional-Structural Operationalization	
Functional Processes	Structural Attributes
Expressively Resentful (e.g., resists fulfilling expectations of others, frequently exhibiting procrastination, inefficiency, and obstinate, contrary, and irksome behavior; reveals gratification in demoralizing and undermining the pleasures and aspirations of others)	Discontented Self-Image (e.g., sees self as misunderstood, luckless, unappreciated, jinxed, and demeaned by others; recognizes being characteristically embittered, disgruntled, and disillusioned with life)

Table 2.2 (Prototypes continued—Negativistic)

Interpersonally Contrary (e.g., assumes conflicting and changing roles in social relationships, particularly dependent and contrite acquiescence and assertive and hostile independence; conveys envy and pique toward those who are more fortunate; is obstructive and intolerant of others, expressing negative or incompatible attitudes)	Vacillating Objects (e.g., internalized representations of the past include a complex of countervailing relationships, setting in motion contradictory feelings, conflicting inclinations, and incompatible memories that are driven by the desire to degrade the achievements and pleasures of others without necessarily appearing to do so)
Cognitively Skeptical (e.g., is cynical, doubting, and untrusting, approaching positive events with disbelief and future possibilities with pessimism, anger, and trepidation; has a misanthropic view of life; whines and grumbles, exhibiting disdain for and making caustic comments about those who experience good fortune)	Divergent Organization (e.g., there is a clear division in the pattern of morphologic structures such that coping and defensive maneuvers are often directed toward incompatible goals, leaving major conflicts unresolved and full psychic cohesion often impossible by virtue of the fact that fulfillment of one drive or need inevitably nullifies or reverses another)
Displacement Mechanism (e.g., discharges anger and other troublesome emotions precipitously or by employing unconscious maneuvers to shift them from their instigator to settings or persons of lesser significance; vents disapproval by substitute or passive means such as acting inept or perplexed or behaving in a forgetful or indolent manner)	Irritable Mood (e.g., is frequently touchy, temperamental, and peevish, followed by sullen and moody withdrawal; is often petulant and impatient; unreasonably scorns those in authority and reports being easily annoyed or frustrated by others)

DSM/MCMI Correspondence (Negativistic)

DSM-IV^a Criterion	Parallel MCMI-III Item
See <i>DSM-IV</i> , pp. 733–735.	7. If my family puts pressure on me, I'm likely to feel angry and resist doing what they want.
	115. Other people often blame me for things I didn't do.
	79. I am often cross and grouchy.
	50. I strongly resent "big shots" who always think they can do things better than I can.
	126. Most successful people today have been either lucky or dishonest.
	36. I often let my angry feelings out and then feel terribly guilty about it.

^aIn the *DSM-IV*, this is labeled "passive-aggressive personality disorder (negativistic personality disorder)."

Table 2.2 (Prototypes continued—Masochistic)

Masochistic Prototype	
Functional-Structural Operationalization	
Functional Processes	Structural Attributes
Expressively Abstinent (e.g., presents him- or herself as nonindulgent, frugal, and chaste; is reluctant to seek pleasurable experiences, refraining from exhibiting signs of enjoying life; acts in an unassuming and self-effacing manner, preferring to place him- or herself in an inferior light or object position)	Undeserving Self-Image (e.g., is self-abasing, focusing on his or her very worst personal features, asserting thereby that he or she is worthy of being shamed, humbled, and debased; feels that he or she has failed to live up to the expectations of others and hence deserves to suffer painful consequences)
Interpersonally Deferential (e.g., distances him- or herself from those who are consistently supportive, relating to others where he or she can be sacrificing, servile, and obsequious, allowing if not encouraging them to exploit, mistreat, or take advantage of him or her; renders ineffectual the attempts of others to be helpful and solicits condemnation by accepting undeserved blame and courting unjust criticism)	Discredited Objects (e.g., object representations are composed of failed past relationships and disparaged personal achievements; of positive feelings and erotic drives transposed into their least attractive opposites, of internal conflicts intentionally aggravated, of mechanisms for reducing dysphoria being subverted by processes that intensify discomfort)
Cognitively Diffident (e.g., is hesitant to interpret observations positively for fear that, in doing so, they may not take problematic forms or achieve troublesome and self-denigrating outcomes; as a result, there is a habit of repeatedly expressing attitudes and anticipation contrary to favorable beliefs and feelings)	Inverted Organization (e.g., owing to a significant reversal of the pain-pleasure polarity, morphologic structures have contrasting and dual qualities—one more or less conventional, the other its obverse—resulting in repetitive undoing of affect and intention, transposing of channels of need gratification with those leading to frustration, and engaging in actions that produce antithetical if not self-sabotaging consequences)
Exaggeration Mechanism (e.g., repetitively recalls past injustices and anticipates future disappointments as a means of raising distress to homeostatic levels; undermines personal objectives and sabotages good fortune to enhance or maintain accustomed level of suffering and pain)	Dysphoric Mood (e.g., experiences a complex mix of emotions, at times anxiously apprehensive, forlorn, and mournful, or anguished and tormented; intentionally displays a plaintive and wistful appearance, frequently to induce guilt and discomfort in others)
DSM/MCMI Correspondence (Masochistic)	
DSM-III-R ^a Criterion	Parallel MCMI-III Item
See <i>DSM-III-R</i> , pp. 373–374.	19. I seem to choose friends who end up mistreating me. 70. I often think that I don't deserve the good things that happen to me.

^aMasochistic personality disorder was not included in the *DSM-IV*. In the *DSM-III-R*, it is labeled "self-defeating personality disorder."

Table 2.2 (Prototypes continued—Masochistic, Schizotypal)

43. I often feel sad or tense right after something good has happened to me.
161. I seem to create situations with others in which I get hurt or feel rejected.
104. I can't experience much pleasure because I don't feel I deserve it.
122. I seem to make a mess of good opportunities that come my way.
90. I sometimes get confused and feel upset when people are kind to me.

Schizotypal Prototype

Functional-Structural Operationalization	
Functional Processes	Structural Attributes
Expressively Eccentric (e.g., exhibits gauche and peculiar mannerisms; is perceived by others as aberrant and disposed to behave in an unobtrusively odd, aloof, curious, or bizarre manner)	Estranged Self-Image (e.g., exhibits recurrent social perplexities and illusions and experiences of depersonalization, derealization, and dissociation; sees him- or herself as forlorn, with repetitive thoughts of life's emptiness and meaninglessness)
Interpersonally Secretive (e.g., prefers privacy and isolation with a few highly tentative attachments and personal obligations; has drifted over time into increasingly peripheral vocational roles and clandestine social activities)	Chaotic Objects (e.g., internalized representations consist of a piecemeal jumble of early relationships and affects, random drives and impulses, and uncoordinated channels of regulation that are only fitfully competent for binding tensions, accommodating needs, and mediating conflicts)
Cognitively Autistic (e.g., capacity to "read" thoughts and feelings of others is markedly dysfunctional; mixes social communication with personal irrelevancies, circumstantial speech, ideas of reference, and metaphorical asides; is often ruminative, appearing self-absorbed and lost in daydreams with occasional magical thinking, bodily illusions, obscure suspicions, odd beliefs, and a blurring of reality and fantasy)	Fragmented Organization (e.g., possesses permeable ego boundaries; coping and defensive operations are haphazardly ordered in a loose assemblage of morphologic structures, leading to desultory actions in which primitive thoughts and affects are discharged directly with few reality-based sublimations and significant further disintegration into a psychotic structural level likely under even moderate stress)

Table 2.2 (Prototypes continued—Schizotypal)

Undoing Mechanism (e.g., bizarre mannerisms and idiosyncratic thoughts appear to reflect a retraction or reversal of previous acts or ideas that have stirred feelings of anxiety, conflict, or guilt; ritualistic or magical behavior serves to repent for or nullify assumed misdeeds or "evil" thoughts)

Distraught or Insentient Mood (e.g., is excessively apprehensive and ill-at-ease, particularly in social encounters; is agitated and anxiously watchful, evincing distrust of others and suspicion of their motives that persists despite growing familiarity); *or* (e.g., manifests drab, apathetic, sluggish, joyless, and spiritless appearance; reveals marked deficiencies in face-to-face rapport and emotional expression)

DSM/MCMI Correspondence (Schizotypal)	
DSM-IV Criterion	Parallel MCMI-III Item
A. A pervasive pattern of social and interpersonal deficits marked by acute discomfort with, and reduced capacity for, close relationships as well as by cognitive or perceptual distortions and eccentricities of behavior, beginning by early adulthood and present in a variety of contexts, as indicated by five (or more) of the following:	162. I often get lost in my thoughts and forget what's going on around me.
(1) ideas of reference (excluding delusions of reference)	138. I can tell that people are talking about me when I pass by them.
(2) odd beliefs or magical thinking that influences behavior and is inconsistent with sub-cultural norms (e.g., superstitiousness, belief in clairvoyance, telepathy, or "sixth sense"; in children and adolescents, bizarre fantasies or preoccupations)	76. I keep having strange thoughts that I wish I could get rid of.
(3) unusual perceptual experiences, including bodily illusions	71. When I'm alone, I often feel the strong presence of someone nearby who can't be seen.
(4) odd thinking and speech (e.g., vague, circumstantial, metaphorical, overelaborate, or stereotyped)	117. People think I sometimes talk about strange or different things than they do.
(5) suspiciousness or paranoid ideation	8. People make fun of me behind my back, talking about the way I act or look.
(6) inappropriate or constricted affect	156. I don't understand why some people smile at me.
(7) behavior or appearance that is odd, eccentric, or peculiar	—
(8) lack of close friends or confidants other than first-degree relatives	48. A long time ago, I decided it's best to have little to do with people.

Table 2.2 (Prototypes continued—Schizotypal, Borderline)

- (9) excessive social anxiety that does not diminish with familiarity and tends to be associated with paranoid fears rather than negative judgments about self
158. I get very tense with people I don't know well because they may want to harm me.

Borderline Prototype

Functional-Structural Operationalization	
Functional Processes	Structural Attributes
Expressively Spasmodic (e.g., displays a desultory energy level with sudden, unexpected, impulsive outbursts; exhibits abrupt, endogenous shifts in drive state and inhibitory controls; places activation and emotional equilibrium in constant jeopardy and engages in recurring suicidal or self-mutilating behavior)	Uncertain Self-Image (e.g., experiences the confusion of an immature, nebulous, or wavering sense of identity, often with underlying feelings of emptiness; seeks to redeem precipitate actions and changing self-presentations with expressions of contrition and self-punitive behavior)
Interpersonally Paradoxical (e.g., although needing attention and affection, is unpredictably contrary, manipulative, and volatile, frequently eliciting rejection rather than support; frantically reacts to fears of abandonment and isolation but often in angry, mercurial, and self-destructive ways)	Incompatible Objects (e.g., internalized representations include rudimentary and extemporaneously devised but repetitively aborted learnings, resulting in conflicting memories, discordant attitudes, contradictory needs, antithetical emotions, erratic impulses, and clashing strategies for conflict reduction)
Cognitively Capricious (e.g., has rapidly changing, fluctuating, and antithetical perceptions or thoughts concerning passing events; has contrasting emotions and conflicting thoughts about self and others, notably love, rage, and guilt; vacillating and contradictory reactions are evoked in others by virtue of his or her behavior, which creates conflicting and confusing social feedback)	Split Organization (e.g., inner structures exist in a sharply segmented and conflicted configuration in which a marked lack of consistency and congruency is seen among elements; levels of consciousness often shift and result in rapid movements across boundaries that usually separate contrasting percepts, memories, and affects, all of which leads to periodic schisms in what limited psychic order and cohesion may be present, often resulting in transient, stress-related psychotic episodes)
Regression Mechanism (e.g., retreats under stress to earlier developmental levels of anxiety tolerance, impulse control, and social adaptation; among adolescents, is unable to cope with adult demands and conflicts as evident in immature if not increasingly infantile behavior)	Labile Mood (e.g., fails to accord unstable mood level with external reality; has marked shifts from normality to depression to excitement or has periods of dejection and apathy, interspersed with episodes of inappropriate and intense anger and brief spells of anxiety or euphoria)

Table 2.2 (Prototypes continued—Borderline)

DSM/MCMI Correspondence (Borderline)	
DSM-IV Criterion	Parallel MCMI-III Item
A. A pervasive pattern of instability of interpersonal relationships, self-image, and affects, and marked impulsivity beginning by early adulthood and present in a variety of contexts, as indicated by five (or more) of the following:	—
(1) frantic efforts to avoid real or imagined abandonment. Note: Do not include suicidal or self-mutilating behavior covered in Criterion 5	120. I'll do something desperate to prevent a person I love from abandoning me.
(2) a pattern of unstable and intense interpersonal relationships characterized by alternating between extremes of idealization and devaluation	98. My feelings toward important people in my life often swing from loving them to hating them.
(3) identity disturbance: markedly and persistently unstable self-image or sense of self	72. I feel pretty aimless and don't know where I'm going in life.
(4) impulsivity in at least two areas that are potentially self-damaging (e.g., spending, sex, substance abuse, reckless driving, binge eating). Note: Do not include suicidal or self-mutilating behavior covered in Criterion 5	41. I've done a number of stupid things on impulse that ended up causing me great trouble.
(5) recurrent suicidal behavior, gestures, or threats, or self-mutilating behavior	154. I have tried to commit suicide.
(6) affective instability due to a marked reactivity of mood (e.g., intense episodic dysphoria, irritability, or anxiety usually lasting a few hours and only rarely more than a few days)	83. My moods seem to change a great deal from one day to the next.
(7) chronic feelings of emptiness	142. I frequently feel there's nothing inside me, like I'm empty and hollow.
(8) inappropriate, intense anger or difficulty controlling anger (e.g., frequent displays of temper, constant anger, recurrent physical fights)	30. Lately, I have begun to feel like smashing things.
(9) transient, stress-related paranoid ideation or severe dissociative symptoms	134. I sometimes feel crazy-like or unreal when things start to go badly in my life.

Table 2.2 (Prototypes continued—Paranoid)

Paranoid Prototype	
Functional-Structural Operationalization	
Functional Processes	Structural Attributes
Expressively Defensive (e.g., is vigilantly guarded, alert to anticipate and ward off expected derogation, malice, and deception; is tenacious and firmly resistant to sources of external influence and control)	Inviolable Self-Image (e.g., has persistent ideas of self-importance and self-reference, perceiving attacks on his or her character that are not apparent to others, asserting as personally derogatory and scurrilous, if not libelous, entirely innocuous actions and events; is proudly independent, reluctant to confide in others, and highly insular, experiencing intense fears, however, of losing identity, status, and powers of self-determination)
Interpersonally Provocative (e.g., bears grudges and is unforgiving; displays a quarrelsome, fractious, and abrasive attitude with recent acquaintances; precipitates exasperation and anger by testing loyalties and exhibiting an intrusive and searching preoccupation with hidden motives)	Unalterable Objects (e.g., internalized representations of significant early relationships are a fixed and implacable configuration of deeply held beliefs and attitudes that are driven by unyielding convictions which, in turn, are aligned in an idiosyncratic manner with a fixed hierarchy of tenaciously held but unwarranted assumptions, fears, and conjecture)
Cognitively Suspicious (e.g., is unwarrantedly skeptical, cynical, and mistrustful of the motives of others, including relatives, friends, and associates, construing innocuous events as signifying hidden or conspiratorial intent; reveals tendency to read hidden meaning into benign matters and to magnify tangential or minor difficulties into proof of duplicity and treachery, especially regarding the fidelity and trustworthiness of a spouse or intimate friend)	Inelastic Organization (e.g., systemic constriction and inflexibility of underlying morphologic structures, rigidly fixed channels of defensive coping, conflict mediation, and need gratification create an overstrung and taut frame that is so uncompromising in its accommodation to changing circumstances that unanticipated stressors are likely to precipitate explosive outbursts or inner shattering)
Projection Mechanism (e.g., actively disowns undesirable personal traits and motives and attributes them to others; remains blind to his or her own unattractive behavior and characteristics, yet is overalert to and hypercritical of similar features in others)	Irascible Mood (e.g., displays a cold, sullen, churlish, and humorless demeanor; attempts to appear unemotional and objective but is edgy, envious, jealous, and quick to take offense and react angrily)

Table 2.2 (Prototypes continued—Paranoid)

DSM/MCMI Correspondence (Paranoid)	
DSM-IV Criterion	Parallel MCMI-III Item
A. A pervasive distrust and suspiciousness of others such that their motives are interpreted as malevolent, beginning by early adulthood and present in a variety of contexts, as indicated by four (or more) of the following:	49. Since I was a child, I have always had to watch out for people who were trying to cheat me.
(1) suspects, without sufficient basis, that others are exploiting, harming, or deceiving him or her	103. Sneaky people often try to get the credit for things I have done or thought of.
(2) is preoccupied with unjustified doubts about the loyalty or trustworthiness of friends or associates	175. There are people who are supposed to be my friends who would like to do me harm.
(3) is reluctant to confide in others because of unwarranted fear that the information will be used maliciously against him or her	167. I take great care to keep my life a private matter so no one can take advantage of me.
(4) reads hidden demeaning or threatening meanings into benign remarks or events	146. I always wonder what the real reason is when someone is acting especially nice to me.
(5) persistently bears grudges, i.e., is unforgiving of insults, injuries, or slights	6. People have never given me enough recognition for the things I've done. 42. I never forgive an insult or forget an embarrassment that someone caused me.
(6) perceives attacks on his or her character or reputation that are not apparent to others and is quick to react angrily or to counterattack	33. If someone criticized me for making a mistake, I would quickly point out some of that person's mistakes.
(7) has recurrent suspicions, without justification, regarding fidelity of spouse or sexual partner	89. I watch my family closely so I'll know who can and who can't be trusted.

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Table 2.3 Prototypal Items for MCMI-III Axis I Scales

Anxiety (Scale A)

- 58. I've become very jumpy in the last few weeks.
- 75. Lately, I've been sweating a great deal and feel very tense.
- 124. When I'm alone and away from home, I often begin to feel tense and panicky.
- 147. There are certain thoughts that keep coming back again and again in my mind.
- 164. There are terrible events from my past that come back repeatedly to haunt my thoughts and dreams.
- 170. I repeat certain behaviors again and again, sometimes to reduce my anxiety and sometimes to stop something bad from happening.

Somatoform (Scale H)

- 4. I feel weak and tired much of the time.
- 11. I have a hard time keeping my balance when walking.
- 37. I very often lose my ability to feel any sensations in parts of my body.
- 55. In recent weeks I feel worn out for no special reason.
- 74. I can't seem to sleep, and wake up just as tired as when I went to bed.

Bipolar: Manic (Scale N)

- 3. I enjoy doing so many different things that I can't make up my mind what to do first.
- 54. There are many times, when for no reason I feel very cheerful and full of excitement.
- 96. People have said in the past that I became too interested and too excited about too many things.
- 106. I've had many periods in my life when I was so cheerful and used up so much energy that I fell into a low mood.
- 125. People sometimes get annoyed with me because they say I talk too much or too fast for them.

Dysthymia (Scale D)

- 24. I began to feel like a failure some years ago.
- 56. For some time now I've been feeling very guilty because I can't do things right anymore.
- 62. I've become quite discouraged and sad about life in the past year or two.
- 86. For some time now I've been feeling sad and blue and can't seem to snap out of it.
- 111. I seem to have lost interest in most things that I used to find pleasurable, such as sex.
- 130. I don't have the energy to concentrate on my everyday responsibilities anymore.

Alcohol Dependence (Scale B)

- 52. I have an alcohol problem that has made difficulties for me and my family.
- 77. I have a great deal of trouble trying to control an impulse to drink to excess.
- 100. I guess I'm no different from my parents in becoming somewhat of an alcoholic.
- 152. I have a drinking problem that I've tried unsuccessfully to end.
- 23. (False) Drinking alcohol has never caused me any real problems in my work.

continued on next page

Table 2.3 (continued)

Drug Dependence (Scale T)

- 13. My drug habits have often gotten me into a good deal of trouble in the past.
- 39. Taking so-called illegal drugs may be unwise, but in the past I found I needed them.
- 66. My habit of abusing drugs has caused me to miss work in the past.
- 91. My use of so-called illegal drugs has led to family arguments.
- 118. There have been times when I couldn't get through the day without some street drugs.
- 136. I know I've spent more money than I should buying illegal drugs.

Post-Traumatic Stress Disorder (Scale R)

- 109. The memory of a very upsetting experience in my past keeps coming back to haunt my thoughts.
- 129. Years later I still have nightmares about an event that was a real threat to my life.
- 149. I feel shaky and have difficulty falling asleep because painful memories of a past event keep running through my mind.
- 160. My current life is still upset by flashbacks of something terrible that happened to me.
- 173. I still feel terrified when I think of a traumatic experience I had years ago.

Thought Disorder (Scale SS)

- 34. Lately, I have gone all to pieces.
- 61. Ideas keep turning over and over in my mind and they won't go away.
- 68. Lately, I have to think things over and over again for no good reason.
- 78. Even when I'm awake, I don't seem to notice people who are near me.
- 102. Ever since I was a child, I have been losing touch with the real world.
- 168. I very often hear things so well that it bothers me.

Major Depression (Scale CC)

- 1. Lately, my strength seems to be draining out of me, even in the morning.
- 44. I feel terribly depressed and sad much of the time now.
- 107. I have completely lost my appetite and have trouble sleeping most nights.
- 128. I feel deeply depressed for no reason I can figure out.
- 150. Looking ahead as each day begins makes me feel terribly depressed.
- 171. I have given serious thought recently to doing away with myself.

Delusional Disorder (Scale PP)

- 63. Many people have been spying into my private life for years.
 - 119. People are trying to make me believe that I'm crazy.
 - 140. I believe I'm being plotted against.
 - 153. Someone has been trying to control my mind.
-

Chapter 3

Internal-Structural Validity

In any field of science, any number of theories can be put forward to account for the observations of scientists. In the more advanced physical sciences of chemistry and physics, the problem of multiple theories, each sufficient to account for the observed data, can be dealt with by constructing paradigm-breaking experiments. These experiments quickly whittle down the number of theoretical candidates so that only those that are consistent with the observed evidence remain. Experimentation plays an important role in the progression of knowledge. Empirical evidence serves as a powerful constraint on the number and kinds of theories that can be advanced as models of the subject domain.

In the social sciences, experimentation is less of a constraint on the speculative tendencies of theorists. New ideas are proposed, experience a period of enthusiasm, and then just seem to fade away (Meehl, 1978) without ever being subjected to a strong threat of falsification. Often, the fanciful proposals of theorists undergo considerable involution, evolving a unique but ultimately tautological jargon that other scientists in the field find difficult to comprehend, as with some psychoanalytic formulations.

Just as an indefinite number of theories might be proposed to account for a particular

phenomenon, any number of items might be written in reference to a particular psychological construct. Each item may be viewed as a small theory, in that it is hypothesized that those who are, for example, narcissistic or schizoid or masochistic are likely to endorse the item as characteristic of themselves. When these items are grouped to form their respective scales, some means is required to assess how the items behave. In practice, the statistic most frequently used to make this determination is Cronbach's alpha. A high alpha coefficient indicates that a scale's items are highly intercorrelated and are thus behaving consistently. This provides some assurance that the items on the scale belong together. When the scales being composed are personality scales, further statistics are necessary to ensure that their behavior is consistent with the nature of the constructs measured. For example, scale scores should be stable over time, like personality itself.

The desirability of internally consistent scales is illustrated in the history of psychological assessment inventories. When the MMPI was constructed in the late 1930s and early 1940s, an exclusively empirical approach to item selection seemed worthwhile. The philosophy of science known as *logical positivism* was in its ascendant

cy. and psychological theory was considered as yet too weak to inform inventory construction. If items of ostensibly unrelated content were found to differentiate normals and psychiatric groups, the content discrepancies could simply be considered a point of scientific interest that might be explored at some future date.

The purely external approach used for the MMPI led to the development of diagnostic scales that were not explicitly required to have face validity, that is, to contain items that were obviously related to their constructs. Instead, items were assigned primarily on the basis of their differential endorsement frequencies. Consequently, a scale could be composed of any number of independent components. This method of scale construction has contributed to interpretive difficulties. Not only are the latent elements underlying particular scale elevations and profile patterns difficult to resolve, the same elevation or profile pattern can mean different things for different patients. Over the years, attempts have been made to alleviate these interpretive difficulties by creating new MMPI scales (e.g., the MMPI®-2 content scales, Butcher, Graham, Ben-Porath, Tellegen, Dahlstrom, & Kaemmer, 2001, and scales by Harris & Lingo, 1955, and Wiggins, 1969).

Structural Validity

Internal validity is essentially a requirement of logic. A system of propositions should be free of internal contradiction, able to stand on its own terms without collapsing because of some internal conceptual flaw. Internal validity is a necessary but not sufficient condition for the viability of any theory or taxonomy. At the item level, the coherence of a scale's basic elements can be assessed statistically.

The idea of structural validity is strongly related to internal validity. The boundary between the two is rather fuzzy but can be clarified if we distinguish between inductive and deductive approaches to a subject domain, here personality and psychopathology. Stated simply, structural validity regards the system of relations between various constructs as determined by the theory, expressed operationally as a correlation matrix of the inventory's scales. If we adopt an inductive approach, we can simply collect a great

number of observations and let the methodology, typically a factor analysis with varimax rotation, extract the core dimensions or constructs that supposedly underlie our data set. The methodology drives the ultimate structure of the inventory. The final factors become scales, extracted from the very beginning to be mutually independent and internally consistent. Internal-structural considerations are co-opted by, or folded into, the methodology. The theory that emerges from the methodology does so post hoc as an assertion based on the sufficiency of the methodology; no strong a priori notions are required to get the inventory off the ground and into the hands of test users.

While an inductive framework may be described as theory-weak, a deductive framework is by definition theory-strong because the very point of departure is the theory and the system of relations that the theory posits. Where the inductive-descriptive approach folds the internal-structural stage of validation into the methodology, the deductive-explanatory approach to inventory construction has an additional burden of proof because it must show that the system of relations posited by the theory is confirmed by the inventory's internal-structural characteristics. The evolutionary theory that underlies the MCMI-III test is described in the previous chapter, where it was noted, for example, that the narcissistic personality is related to the antisocial in that both are oriented toward self rather than others. The narcissistic personality is also related to the dependent personality because both are relatively passive patterns. Any inventory that issues from a deductive model is vested in revealing the interrelation and differentiation of phenomena and must attempt to build such relationships into its structure.

In a deductive framework, structural validity is nonfactorial in that it involves much more than merely showing the internal consistency of each scale, but instead involves satisfying constraints that derive intrinsically from the deductive paradigm. If our inventory is intended to assess personality disorders, we might expect, for example, that those classified as dependent personalities will also be classified as experiencing depressive episodes more often than, say, antisocials, because dependents are likely to feel helpless and hopeless more often than antisocials, who

are more likely to take matters into their own hands and change the world around them, albeit in a self-aggrandizing fashion. Such expectations form a set of constraints that an inventory must be constructed to satisfy. The larger this set of constraints the better because if the inventory can satisfy many such constraints at the outset, a good deal of validity has been built into it. In general, the more of these theoretical constraints the test satisfies, the better the instrument and the greater confidence we can have that it will meet whatever challenges it is put to in the future. Each constraint becomes an additional point through which the instrument is triangulated with reality from the perspective of the theory.

The Nature of Theoretical Prototypes

An important area of controversy within the personality disorders area today concerns the categorical versus dimensional nature of the disorders themselves. In the *DSM-IV*, the Axis II disorders are presented as latent disease entities that can be diagnosed on the basis of a threshold number of criteria. The medicalization of Axis II has contributed to many paradigmatic misconceptions that surround the personality disorders and continues to fuel the ire of trait researchers, who feel that both the manifest and the latent structure of personality are continuously distributed in terms of their absolute quantities and the manifest pathological consequences of these quantities.

The MCMI-III test authors remains committed to the prototype construct first adopted in the *DSM-III*. A prototype is that nucleus of characteristics that captures the most relevant and essential characteristics of a particular Axis I or Axis II disorder. The prototype is neither category nor dimension but a synthesis of both. Prototypical models are currently the preferred schema for representing personality disorders and clinical syndromes for several reasons. First, most contemporary typologies neither imply nor were constructed as all-or-none categories. Most advocates of the dimensional approach to clinical practice choose to overlook the fact that the word *categories* has been used very loosely in the *DSM*. Second, the prototype construct recognizes the explicit heterogeneity of personality-

disordered patients. Pure prototypical cases are extremely rare; most patients meet criteria for multiple disorders and may have subclinical features of other personality styles as well. Indeed, the problems imputed to categorical models largely evaporate if categories are regarded as prototypes. Prototypical constructs do not assume discrete boundaries (Cantor & Genero, 1986) and have the advantage of being already implicit in the diagnostic system. Horowitz, Post, French, Wallis, and Siegelman (1981) describe the construct succinctly:

A prototype consists of the most common features or properties of members of a category and thus describes a theoretical ideal or standard against which real people can be evaluated. All of the prototype's properties are assumed to characterize at least some members of the category, but no one property is necessary or sufficient for membership in the category. Therefore, it is possible that no actual person would match the theoretical prototype perfectly. Instead different people would approximate it to different degrees. The more closely a person approximates the ideal, the more closely the person typifies the concept. (p. 575)

Explicit in this description is the use of diagnostic criteria as a heuristic method, not as a reified model. As Horowitz et al. (1981) noted, the resemblance of an individual to the prototype is necessarily a qualitative as well as a quantitative matter. Although categories and dimensions inevitably sacrifice one or the other kind of information, the prototype conserves both. That is, by its heuristic nature, the prototype asks both how and how much the individual resembles the prototype. Because patients will probably resemble more than one of these theoretical ideals, the clinician is encouraged to use prototypes constructively in order to look beyond the one-disorder, one-cause, one-therapy perspective implicit in the current diagnostic system (i.e., to get the flavor of the patient's pathology in its dynamic totality and to integrate intervention efforts with this conception). The surplus causal meaning associated with latent taxons or dimensions need not be postulated. Such an approach would seem thoroughly consonant with the atheoretical orientation of the *DSM* which, if taken to its logical conclusion, should be atheoretical not only with

regard to the various schools of psychopathology but structurally as well.

Item Development for the MCMI-II Research Form

One hundred fifty new items (beyond the 175 MCMI-II items) were selected by expert judgment from a large item pool. These items were written not only to sample developments in the theory-based structural and functional personality domains but also to parallel the substantive nature of the then forthcoming *DSM-IV* diagnostic criteria. These *DSM* criteria were rephrased with two goals in mind: first, that they be understandable and readable to the layperson as self-report items, and second, that they reflect personality attributes consonant with the theory. All items in the combined 325-item research form were judged by the authors as meeting these dual requirements. To enhance the inventory's correspondence with the *DSM-IV*, new items were written to sample the criteria of the depressive personality, a new Clinical Personality Pattern, and post-traumatic stress disorder, a new Clinical Syndrome. Items concerning child abuse, anorexia, and bulimia were also added to strengthen the utility of the Noteworthy Responses section of the interpretive report; these three areas were not added as explicit clinical scales. The resulting 325-item True-False inventory was called the MCMI-II Research Form.

Description of the Research Study

Several hundred clinicians who used the MCMI-II test for evaluating and treating adult clients were asked to participate in the revision research. Between June 1992 and February 1993, MCMI-II Research Form data were collected from 1,079 subjects in 26 states and Canada. In addition, subjects completed a number of collateral tests, and clinician's ratings were solicited for each subject for a variety of characteristics (the external-criterion validity of the MCMI-III test is reported in the next chapter).

All subjects signed an informed consent form before participating in the MCMI-II Research Form validation study. Subjects were not paid for participating, but clinicians were paid

for each person they tested. Clinicians were also able to generate MCMI-II scores and interpretive reports from the participants' responses on the MCMI-II Research Form.

From the original pool of 1,079 clinical subjects, 8 were excluded because of missing or incomplete forms. Subjects were also excluded if they met any of the following conditions:

1. Gender was not indicated ($n = 0$)
2. The score on the MCMI-II Validity index was greater than 1 ($n = 6$)
3. Age was under 18 or not recorded ($n = 12$)
4. Responses to 23 or more of the 325 items were missing ($n = 16$)
5. Base rate scores for MCMI-II Scales 1–8B were all below 60 ($n = 9$)
6. The raw score on MCMI-II Scale X (Disclosure) was below 145 or above 590 ($n = 37$)

These rules were identical to the MCMI-II invalidity conditions except that the number of omitted items was increased to account for the additional length of the MCMI-II Research Form.

A total of 73 subjects were excluded because they met one or more of these conditions (the total above is 80 because some subjects were eliminated for more than one reason). The remaining 998 subjects were divided into two groups: one ($N = 600$) was used to develop the MCMI-III scales, and the other ($N = 398$) was used for cross-validation.

After the MCMI-III scales were developed, two additional rules were applied to ensure that the exclusion rules paralleled the invalidity conditions implemented in the final MCMI-III test. These rules specified the exclusion of any subjects whose base rate scores for MCMI-III Scales 1–8B were all below 60 or whose raw score on MCMI-III Scale X (Disclosure) was below 34 or above 178. However, no additional subjects met these criteria, leaving the development and cross-validation samples at 600 and 398 subjects, respectively. Demographic and clinical information about these two samples is presented in Tables 3.1 and 3.2.

Table 3.1 Demographic and Clinical Information for the Development Sample (N = 600)

	Males		Females		Total	
	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
Race/Ethnicity						
White	246	84.0	272	88.6	518	86.3
Black	25	8.5	27	8.8	52	8.7
Hispanic	13	4.4	4	1.3	17	2.8
Asian	2	.7	0	0.0	2	.3
Native American	4	1.4	2	.7	6	1.0
Other	2	.7	1	.3	3	.5
Not Reported	1	.3	1	.3	2	.3
Marital Status						
Never Married	104	35.5	66	21.5	170	28.3
First Marriage	72	24.6	84	27.4	156	26.0
Remarried	27	9.2	47	15.3	74	12.3
Separated	21	7.2	29	9.4	50	8.3
Divorced	55	18.8	55	17.9	110	18.3
Widowed	7	2.4	10	3.3	17	2.8
Cohabiting	4	1.4	7	2.3	11	1.8
Other	0	0.0	5	1.6	5	.8
Not Reported	3	1.0	4	1.3	7	1.2
Age						
18–25	53	18.1	47	15.3	100	16.7
26–35	90	30.7	109	35.5	199	33.2
36–45	96	32.8	83	27.0	179	29.8
46–55	38	13.0	43	14.0	81	13.5
56–88	16	5.5	25	8.1	41	6.8
Not Reported	0	0.0	0	0.0	0	0.0
Setting/Status						
Outpatient (Never Hospitalized)	116	39.6	118	38.4	234	39.0
Outpatient (Previously Hospitalized)	32	10.9	29	9.4	61	10.2
Inpatient (Mental Hospital)	62	21.2	97	31.6	159	26.5
Inpatient (General Hospital)	3	1.0	4	1.3	7	1.2
Correctional Inmate	41	14.0	10	3.3	51	8.5
College Student/Counselor	2	.7	4	1.3	6	1.0
Other	34	11.6	38	12.4	72	12.0
Not Reported	3	1.0	7	2.3	10	1.7
Education						
Did Not Complete High School	46	15.7	56	18.2	102	17.0
High School Graduate	90	30.7	91	29.6	181	30.2
Some Technical or Business School	35	11.9	36	11.7	71	11.8
Some College	58	19.8	61	19.9	119	19.8
College Degree	34	11.6	35	11.4	69	11.5
Graduate School	26	8.9	24	7.8	50	8.3
Not Reported	4	1.4	4	1.3	8	1.3

Note. The sample included 293 males (48.8%) and 307 females (51.2%).

Table 3.2 Demographic and Clinical Information for the Cross-Validation Sample (N = 398)

	Males		Females		Total	
	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
Race/Ethnicity						
White	161	81.7	184	91.5	345	86.7
Black	25	12.7	8	4.0	33	8.3
Hispanic	3	1.5	5	2.5	8	2.0
Asian	0	0.0	0	0.0	0	0.0
Native American	5	2.5	1	0.5	6	1.5
Other	2	1.0	1	0.5	3	0.8
Not Reported	1	0.5	2	1.0	3	0.8
Marital Status						
Never Married	59	29.9	38	18.9	97	24.4
First Marriage	47	23.9	49	24.4	96	24.1
Remarried	30	15.2	32	15.9	62	15.6
Separated	14	7.1	23	11.4	37	9.3
Divorced	39	19.8	44	21.9	83	20.9
Widowed	3	1.5	6	3.0	9	2.3
Cohabiting	3	1.5	3	1.5	6	1.5
Other	2	1.0	5	2.5	7	1.8
Not Reported	0	0.0	1	0.5	1	0.3
Age						
18–25	29	14.7	39	19.4	68	17.1
26–35	55	27.9	60	29.9	115	28.9
36–45	77	39.1	59	29.4	136	34.2
46–55	22	11.2	30	14.9	52	13.1
56–88	14	7.1	13	6.5	27	6.8
Not Reported	0	0.0	0	0.0	0	0.0
Setting/Status						
Outpatient (Never Hospitalized)	82	41.6	103	51.2	185	46.5
Outpatient (Previously Hospitalized)	20	10.2	18	9.0	38	9.5
Inpatient (Mental Hospital)	42	21.3	49	24.4	91	22.9
Inpatient (General Hospital)	2	1.0	1	0.5	3	0.8
Correctional Inmate	29	14.7	4	2.0	33	8.3
College Student/Counselee	1	0.5	2	1.0	3	0.8
Other	16	8.1	14	7.0	30	7.5
Not Reported	5	2.5	10	5.0	15	3.8
Education						
Did Not Complete High School	36	18.3	36	17.9	72	18.1
High School Graduate	70	35.5	68	33.8	138	34.7
Some Technical or Business School	30	15.2	24	11.9	54	13.6
Some College	33	16.8	41	20.4	74	18.6
College Degree	16	8.1	21	10.4	37	9.3
Graduate School	12	6.1	11	5.5	23	5.8
Not Reported	0	0.0	0	0.0	0	0.0

Note. The sample included 197 males (49.5%) and 201 females (50.5%).

Development of the MCMI-III Scales and Indices

The MCMI-III scales were developed using the 600 subjects in the development sample. First, the endorsement frequency for each of the 325 items was examined to ensure that it fell within an acceptable range given the content of the item. Items with unexpectedly high or low endorsement rates were eliminated from consideration. The remaining items were then assigned to scales on the basis of item content. These items formed the preliminary MCMI-III scales (which consisted at this stage only of prototypal items) and were thereafter assigned a weight of 2 points when calculating scale scores.

Next, the following were calculated: internal consistency (Cronbach's alpha); corrected item-total correlations (i.e., correlation between each item and its scale score calculated without that item); correlation of each item with scores on the other preliminary MCMI-III scales, with clinician ratings, and with collateral test scores; correlations among all preliminary MCMI-III scales, between the preliminary MCMI-III scales and the MCMI-II scales, and between the MCMI-III scales and the clinician ratings and collateral test scores. Scale development was an iterative process in which all of the statistics listed above were recomputed and re-evaluated simultaneously as items were added or removed from their respective scales on the basis of statistical (internal and external) and substantive (*DSM-IV* and the evolutionary theory) criteria. At each iteration, each item was required to appear as a prototype on exactly one scale. After the first iteration, most items also appeared as non-prototypes on other scales but were given a scoring weight of 1 rather than 2. Items were then dropped through successive iterations until the target inventory length of 175 items was met. The text of the final items can be found in Appendix A. The composition of the MCMI-III scales and final item weights are presented in Appendix B.

For the MCMI-II test, the three Modifying Indices—Scale X (Disclosure), Scale Y (Desirability), and Scale Z (Debasement)—were empirically derived by asking students to take the test keeping a particular response set in mind (e.g., trying to put their best foot forward and to appear as psychologically healthy as possible

for the Desirability index). Development of the MCMI-III Modifying Indices was not based on empirical studies. Rather, all items that appeared on MCMI-II Scales X, Y, and Z and that were retained on the MCMI-III clinical scales were assigned to the appropriate MCMI-III Modifying Index. Items from the MCMI-II Modifying Indices that were not carried forward to the MCMI-III test were replaced with items with similar content.

The norming of the test, including the concept of base rate (BR) scores, is described in Chapter 6.

Internal Consistency and Test-Retest Reliability

Internal consistency is a necessary but not sufficient condition for structural validity. A complete evaluation of internal structural validity requires not only that scales be internally consistent but also that the pattern of relationships observed across scales satisfy what the theory requires. Internal consistency and scale inter-correlation data, based on the cross-validation sample ($N = 398$), are reported in the remainder of this chapter. Statistical results of higher magnitudes were generally obtained with the development sample, but because these results formed the essential basis on which items were selected and scales improved, they would be spuriously inflated in this context.

Internal Consistency

The internal consistency of a scale is a summary measure of the degree to which its items inter-correlate. Internal consistency is generally appraised though the use of a statistic called Cronbach's alpha, which is the mean of all possible split-half reliabilities. Higher alphas are generally better, although alphas that are too high may indicate that a scale is composed of items that are virtual rephrasings of one another. Table 3.3 presents the total number of scale items, alpha internal consistency, and test-retest reliability for each MCMI-III scale. Internal consistency results for the clinical scales range from .66 for the Compulsive scale to .90 for Major Depression. Alphas exceed .80 for 20 of the scales.

Table 3.3 Length, Internal Consistency, and Test-Retest Reliability of the MCMI-III Scales

		Number of Items	Internal Consistency ^a (Cronbach's Alpha)	Test-Retest ^b Reliability
Clinical Personality Patterns				
1	Schizoid	16	.81	.89
2A	Avoidant	16	.89	.89
2B	Depressive	15	.89	.93
3	Dependent	16	.85	.89
4	Histrionic	17	.81	.91
5	Narcissistic	24	.67	.89
6A	Antisocial	17	.77	— .93
6B	Sadistic (Aggressive)	20	.79	.88
7	Compulsive	17	.66	.92
8A	Negativistic (Passive-Aggressive)	16	.83	.89
8B	Masochistic (Self-Defeating)	15	.87	.91
Severe Personality Pathology				
S	Schizotypal	16	.85	.87
C	Borderline	16	.85	.93
P	Paranoid	17	.84	.85
Clinical Syndromes				
A	Anxiety	14	.86	.84
H	Somatoform	12	.86	.96
N	Bipolar: Manic	13	.71	.93
D	Dysthymia	14	.88	.91
B	Alcohol Dependence	15	.82	.92
T	Drug Dependence	14	.83	.91
R	Post-Traumatic Stress Disorder	16	.89	.94
Severe Clinical Syndromes				
SS	Thought Disorder	17	.87	.92
CC	Major Depression	17	.90	.95
PP	Delusional Disorder	13	.79	.86
Modifying Indices				
X	Disclosure	NA	NA	.94
Y	Desirability	21	.86	.92
Z	Debasement	33	.95	.82

^aCross-validation sample ($N = 398$)

^bTest-retest interval = 5–14 days ($n = 87$)

Test-Retest Reliability

Test-retest reliability is as important as internal consistency. Eighty-seven subjects were readministered the MCMI-II Research Form 5 to 14 days after the initial administration. Data from this second administration were used to estimate retest stability of the MCMI-III scales. Table 3.3 presents the test-retest reliability for each scale. Results range from .82 for Debasement (Scale Z) to .96 for Somatoform (Scale H). The median stability coefficient was .91, which strongly suggests that MCMI-III results are highly stable over short time periods.

Correlation Matrix of MCMI-III Scale BRs

As noted earlier, the interscale correlation matrix is important in appraising the structural component of test validity. Although such an appraisal is best left to expert judges who must balance theoretical constraints against the narrow frontiers of psychometric measurement, the limitations of the self-report modality, and the tendencies of particular personality styles to distort their responses, a correlation matrix of MCMI-III BR scores is presented for inspection in Table 3.4. In general, the correlation matrix exhibits a pattern of findings that is congruent with the expectations of the evolutionary model. The role of a broad "general maladjustment factor" is evident from the overall pattern of moderate to high intercorrelations.

Table 3.4 Correlations Between MCMI-III Scale Base Rate Scores

Scale	1	2A	2B	3	4	5	6A	6B	7	8A	8B	S	C	P
1	—													
2A	.71	—												
2B	.64	.72	—											
3	.48	.65	.63	—										
4	-.75	-.80	-.65	-.52	—									
5	-.43	-.65	-.48	-.39	-.25	—								
6A	.32	.28	.39	.28	.25	NS	—							
6B	.32	.27	.42	.24	-.23	NS	.65	—						
7	-.38	-.40	-.50	-.39	.46	.36	-.61	-.43	—					
8A	.57	.61	.69	.56	-.51	-.29	.56	.64	-.50	—				
8B	.59	.70	.74	.71	-.60	-.45	.42	.40	.69	.69	—			
S	.67	.68	.70	.55	-.61	-.38	.39	.40	.68	.68	.70	—		
C	.59	.62	.76	.63	-.55	-.38	.61	.57	.79	.79	.73	.70	—	
P	.57	.57	.56	.43	-.47	-.18	.36	.44	.72	.72	.54	.67	.55	—
A	.55	.55	.66	.61	-.49	-.37	.32	.44	.61	.61	.60	.64	.67	.55
H	.61	.54	.65	.55	-.48	-.36	.24	.36	.55	.55	.56	.57	.58	.46
N	.26	.28	.45	.38	-.17	NS	.45	.50	.36	.59	.47	.50	.57	.50
D	.68	.66	.79	.63	-.65	-.48	.36	.37	.51	.66	.70	.65	.74	.51
B	.35	.32	.40	.28	-.37	-.15	.78	.54	-.53	.48	.41	.40	.54	.34
T	.27	.23	.32	.20	-.22	NS	.82	.48	-.47	.43	.35	.32	.51	.34
R	.61	.57	.75	.57	-.52	-.37	.36	.39	-.43	.65	.64	.69	.74	.56
SS	.64	.60	.76	.60	-.54	-.36	.43	.46	-.48	.68	.67	.76	.79	.57
CC	.62	.58	.69	.58	-.53	-.42	.25	.34	-.41	.57	.60	.62	.68	.46
PP	.41	.35	.38	.31	-.31	NS	.26	.34	-.22	.53	.35	.52	.42	.71
X	.62	.67	.78	.68	-.60	-.38	.56	.59	-.55	.83	.79	.73	.81	.74
Y	-.60	-.75	-.69	-.61	.79	.73	-.31	-.26	.62	-.52	-.65	-.58	-.64	-.37
Z	.65	.67	.77	.66	-.60	-.45	.40	.54	-.50	.75	.70	.71	.80	.60

Table 3.4 (continued)

Scale	A	H	N	D	B	T	R	SS	CC	PP	X	Y	Z
1	.55	.61	.26	.68	.34	.27	.61	.64	.62	.41	.62	-.60	.65
2A	.55	.54	.28	.66	.32	.23	.56	.60	.58	.35	.67	-.75	.67
2B	.66	.64	.45	.79	.40	.32	.75	.76	.69	.38	.78	-.69	.77
3	.61	.55	.38	.63	.28	.20	.57	.60	.58	.31	.68	-.61	.66
4	-.49	-.48	-.17	-.65	-.37	-.22	-.52	-.54	-.53	-.31	-.60	.79	-.60
5	.37	-.36	NS	-.48	-.15	NS	-.37	-.36	-.42	NS	-.38	.73	-.45
6A	.32	.24	.45	.36	.78	.82	.36	.43	.24	.26	.56	-.31	.40
6B	.44	.36	.50	.37	.54	.48	.39	.46	.34	.34	.59	-.26	.54
7	-.39	-.35	-.36	-.51	-.53	-.47	-.42	-.48	-.41	-.22	-.55	.62	-.50
8A	.61	.55	.59	.66	.48	.43	.65	.68	.57	.53	.83	-.52	.75
8B	.60	.56	.47	.70	.41	.35	.64	.67	.60	.35	.78	-.64	.70
S	.64	.57	.50	.65	.40	.32	.69	.76	.62	.52	.73	-.58	.71
C	.66	.58	.57	.74	.54	.51	.74	.79	.68	.42	.81	-.64	.80
P	.55	.46	.50	.51	.34	.34	.56	.56	.46	.71	.74	-.37	.60
A	—	.66	.48	.68	.35	.22	.81	.77	.68	.44	.69	-.51	.77
H	.66	—	.32	.78	.26	.13*	.64	.71	.87	.39	.60	-.53	.79
N	.48	.32	—	.35	.39	.39	.48	.52	.31	.42	.62	-.22	.48
D	.68	.78	.35	—	.38	.26	.73	.79	.84	.38	.72	-.67	.85
B	.35	.26	.39	.38	—	.62	.38	.42	.27	.24	.53	-.34	.42
T	.22	.13*	.39	.26	.62	—	.29	.33	.17	.21	.50	-.26	.30
R	.81	.64	.48	.73	.38	.29	—	.76	.72	.43	.74	-.56	.77
SS	.77	.71	.52	.79	.42	.33	.76	—	.77	.46	.74	-.58	.82
CC	.68	.87	.31	.84	.27	.17	.72	.77	—	.39	.66	-.64	.85
PP	.44	.39	.42	.38	.24	.21	.43	.46	.39	—	.58	-.24	.46
X	.69	.60	.62	.72	.53	.50	.74	.74	.66	.58	—	-.62	.80
Y	-.51	-.53	-.22	-.67	-.34	-.26	-.56	-.58	-.64	-.24	.62	—	-.67
Z	.77	.79	.48	.85	.42	.30	.77	.82	.85	.46	.80	-.67	—

Note. Unless otherwise indicated, all coefficients are significant at the .01 level (two-tailed).

* $p < .05$, NS = not significant

Chapter 4

External-Criterion Validity

At the beginning of Chapter 2, we argued that the theoretical-substantive stage of validity is necessary but not sufficient for the creation of valid measurement systems. The items that operationalize a theory's constructs should also form coherent scales and should conform to the system of relationships posited by the theory, a requirement of the internal-structural stage of validation.

Like the theoretical-substantive validity stage, the internal-structural stage is necessary but not sufficient for the validation of an instrument. This fact becomes readily apparent in two examples. The first comes from mathematics, which consists explicitly of highly developed systems of propositions that stand on the strength of deductive proof. Entire subfields of mathematics concern entities that exist only in an abstract symbolism, not in the natural world. In this sense, they are highly coherent fantasies with no empirical correlates in everyday life or the physical universe, internally consistent but without empirical reality. The second example comes from patients who suffer from delusional disorders. Many of these disorders are highly systematized frames of reference that make sense in the patient's own internal world, accounting for objective events in a fashion that, while as

internally consistent as the belief system of most normal individuals, is also obviously false. The external-criterion stage of validation is intended to ensure that an instrument's scales have value in the real world, not just in the minds of the test authors.

This chapter reports on the external validity of three generations of the MCMI test. Although the MCMI-III test differs from its predecessors, information about the external validity of the original MCMI test and the MCMI-II test was deemed important to show continuity from past to present.

Relationships Between the Original MCMI Test and the MMPI Test

One of the foremost problems related to establishing an instrument in the professional community involves creating a database of empirical articles that can be consulted by users to better understand how the instrument performs with particular populations, the relationships of its scales to other test scales, and so on. An important direction that can be taken in this regard expands an instrument's nomological linkages through detailed studies with already accepted standards. Not only does this provide important

concurrent validity evidence, it also allows users to understand the new instrument in relation to something they already know and establishes that the new test adds something incremental to the field.

When the MCMI test was originally published in 1977, it was absolutely new, with only a fraction of the research that exists today. In contrast, perhaps thousands of articles had been published with the MMPI test, then already more than 35 years old. At that point in the MCMI's history, one of our goals was to show not only a measure of concordance between it and the MMPI test but also that the MCMI's strong theoretical foundation could add something to the MMPI's "dustbowl" empiricism. The idea was to illustrate the potential utility of blending clinical data from two inventories conceived and constructed through strikingly different philosophies. The MMPI test is well suited to the assessment of the Axis I syndromes. In contrast, the MCMI test is strongest in the measurement of Axis II disorders. Hence, they often prove an excellent complementary combination for tapping both clinical syndromes and personality types. Although the results presented here pertain to the MCMI test and would probably be somewhat different were this research repeated with the MCMI-III and MMPI-2 tests, they nevertheless illustrate the complementarity of the two instruments.

Figures 4.1 to 4.9 summarize the overlap between the most frequent MMPI and MCMI high-point configural codes. Data were drawn from a study of 3,283 patients from diverse clinical settings who had taken both instruments within a period of less than one month (Antoni, Levine, Tischer, Green, & Millon, 1986; Antoni, Tischer, Levine, Green, & Millon, 1985a, 1985b; Levine, Antoni, Tischer, Green, & Millon, 1987; Levine, Tischer, Antoni, Green, & Millon, 1985). The configural codes of the MCMI test are limited to the personality scales. Thus, the figures portray relationships between MCMI personality scales (Axis II) and MMPI clinical scales (Axis I). Data were gathered over an 18-month period from 46 clinicians located in a variety of settings across the United States: 27 were in independent or group practice, 12 were in mental health centers or public agencies, 3 were at Veterans Administration medical centers, 2 were at private

psychiatric hospitals, and 2 were at university medical centers. Of these 46 clinicians, 43 were psychologists and three were psychiatrists.

Complete data could not be obtained for all cases. The complete cases were used to calculate reasonably accurate estimates for the entire sample. These indicate that 86% were outpatients and 14% inpatients. Males made up 54% of the sample, females 46%. Twenty-nine percent were 20–34 years old, 30% were between 35 and 44, 22% were between 45 and 54, 15% were 55–64, and 4% were 65 or older. Diagnostic assignments were recorded in 69% of the cases. Among these cases, the most prevalent Axis I diagnoses were as follows: affective 21%, anxiety 17%, adjustment 14%, drug addiction 11%, alcoholism 10%, and schizophrenia 6% (all other Axis I syndromes were diagnosed in 4% or fewer of the cases). Axis II disorders, in descending frequency, were as follows: dependent 17%, borderline 15%, atypical/mixed 12%, histrionic 11%, passive-aggressive 10%, avoidant 8%, compulsive 6%, narcissistic 5%, antisocial 5%, paranoid 5%, schizoid 3%, and schizotypal 3%. MCMI and MMPI configural codes for cases with diagnoses were not appreciably different from those for cases without diagnoses.

Figures 4.1 to 4.9 are largely self-explanatory. Nevertheless, a brief review of a single figure may be useful. Figure 4.3 shows the results obtained from the 10.8% of the patient sample with an MMPI 28/82 code type (MMPI Scale 2 is Depression and Scale 8 is Schizophrenia). The horizontal axis shows the most frequent MCMI two-scale high-point codes for the MMPI 28/82 cases. The vertical axis shows the percentage among those in the entire sample with each of these MCMI codes who also received a 28/82 code on the MMPI. For example, 33.9% of all MCMI 1,2 (Schizoid/Avoidant) profiles in the sample obtained a 28/82 code on the MMPI, as did 30.3% of MCMI 2,1 profiles, 16.2% of MCMI 8,2 profiles, etc. The *probability ratio* across the bottom of the figure represents how much more frequently this correspondence occurred than could be explained by chance. For example, the overlap between MCMI code 1,2 and MMPI 28/82 is 9.0 times greater than chance; for MCMI profile 2,1 it is 6.1 times greater than chance, and so on. Also represented through the bar graphs is the percentage of patients in each

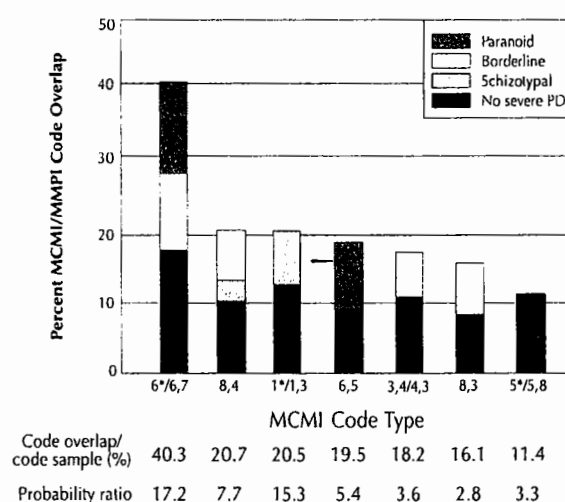
code type overlap group who scored above the cutoff for the more severe Paranoid (P), Borderline (C), and Schizotypal (S) scales. For example, the 33.9% shown in the MCMI 1,2 bar in Figure 4.3 breaks down into approximately 4% who exceeded the cutoff on the Paranoid scale, 2% on the Borderline scale, 20% on the Schizotypal scale, and 8% on none of these scales.

What may be of value to the reader are the MCMI profile codes with which, for example, the MMPI 28/82 code corresponds. As can be seen, the "standard" 28/82 MMPI interpretation can be enriched or refined by the covarying diagnoses and clinical features suggested by the overlapping MCMI codes. For example, a patient with MMPI 28/82 and MCMI 1,2 configural codes would appear somewhat similar to yet would be clinically different in significant ways from a patient with an MMPI 28/82 and an MCMI 2,3 code. A brief review of the clinical significance of various MCMI and MMPI code type combinations reflected in Figures 4.1 to 4.9 is presented next. More detailed discussions can be found in summary articles (Antoni et al., 1986; Antoni et al., 1985a, 1985b; Levine et al., 1985).

Figure 4.1 shows the most common MCMI code types obtained by the 318 patients (9.7% of the sample) who received an MMPI 24/42 profile (2 is Depression; 4 is Psychopathic Deviate). Characterized in the MMPI literature as "psychopaths in trouble," 24/42 patients are noted for recurring acting out, impulsiveness, immaturity, and narcissism. At the same time, they appear to be characterized by frustration, guilt, self-blame, and paranoid ideation. Among the MCMI profile covariations are those with 6*/6,7 and 6,5 high-point codes, patients who appear to be driven to act out interpersonally because of their mistrust of others (note the high percentage who also scored at the paranoid personality level).

Another group of MMPI 24/42 patients exhibited MCMI profile codes in which Scale 1 (Schizoid) was prominent; these Scale 1 MCMI codes suggest a pattern of social indifference and affective blandness rather than an active disdain for others. Other prominent MCMI variants of the MMPI 24/42 code are associated with the 8,4 and 8,3 profiles (passive-aggressive mixed with either histrionic or dependent elements). These patients reflect what Antoni et al. (1985a) called an "emotionally acting-out" group, best charac-

Figure 4.1 MCMI/MMPI Code Type Overlap: MMPI 24/42^a



^aDepression/Psychopathic Deviate

Note. *n* = 318, 9.7% of sample. Asterisk = a pure type.

MCMI scales: 1 = Schizoid, 3 = Dependent, 4 = Histrionic, 5 = Narcissistic, 6 = Antisocial, 7 = Compulsive, 8 = Passive-Aggressive.

terized by their intense and outwardly expressed affect, which is employed to gain attention or support. To explain the fact that the MMPI 24/42 pattern encompasses what might be seen as contradictory dispositions (depression/psychopathy), Antoni et al. (1985a) asserted that the MCMI 8,4 and 8,3 profiles represent intrinsically ambivalent behavior, a vacillation between irritable and hostile expression. It is notable that close to half of the MCMI patients in the 8,4 and 8,3 groups scored in the borderline range. Furthermore, more than 40% were assigned a borderline personality diagnosis by their clinician, and more than 20% of the remainder were diagnosed as passive-aggressive. (Where relevant and where the sample sizes are sufficiently large, mention will be made of the percentage of clinician-judged Axis II diagnoses.)

Figure 4.2 shows the most common MCMI code types for the 228 patients (6.9% of the sample) who obtained an MMPI 27/72 profile code (2 is Depression; 7 is Psychasthenia), a configuration characterized by depression, anxiety, sensitivity, feelings of helplessness, and extreme emotionality. Levine et al. (1985), examining the

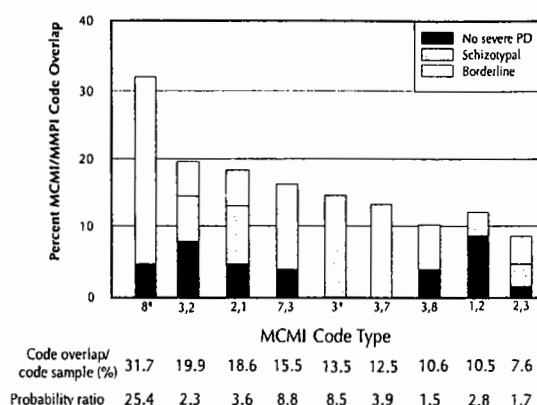
MCMI personality variants associated with the MMPI 27/72 pattern, divided them into three dependent types: fearful, ambivalent, and compulsive. The *fearful dependent* type includes those patients whose MCMI codes consist of some combination of Scales 1, 2, and 3 (Schizoid, Avoidant, and Dependent). In Figure 4.2, MCMI codes 3,2; 2,1; 1,2; and 2,3 occur roughly two to three times more often than would be expected by chance. The cardinal feature of this MCMI variant is the simultaneous existence of strong dependency needs and a fearful mistrust of others. Notably, clinicians diagnosed close to 50% of these patients as dependent personalities.

In the second MCMI variant, what Levine et al. (1985) called the *ambivalently dependent* group, we see code elevations associated with high Scale 8 (Passive-Aggressive) scores (in Figure 4.2, codes 8* and 3,8). The feature that unites these patients is their ambivalence, their struggle between independence and assertiveness on the one hand and submissive dependency on the other. Not only did most of these patients score high on the MCMI Borderline scale, but approximately 40% were diagnosed as borderline by their clinicians.

The third MCMI variant, the *compulsive dependent* group, is represented by the 7,3 and 3,7 profiles, signifying overly conscientious conformity, intropunitiveness, and behavioral rigidity. Although this is a relatively small group, it should be noted that almost 50% of these patients were diagnosed as dependent personalities.

Figure 4.3, overviewed earlier, shows the most common MCMI profile codes for the 353 patients (10.8% of the sample) with an MMPI 28/82 code type. MMPI 28/82 patients have been described in differing, if not contrasting, ways in the MMPI literature as dependent, submissive, irritable, resentful, stubborn, and docile. Most are diagnosed as manic-depressive or schizoaf-fective. Antoni et al. (1985a) divided them into three variants according to their MCMI profiles. The first, an *interpersonally acting-in* group, is seen among those with MCMI 1,2 and 2,1 codes; they “react to stress . . . through withdrawal, self-depreciation, and self-punitive responses” (p. 396). As Figure 4.3 shows, this group is six to nine times more prevalent among MMPI 28/82 code types than would be accounted for by chance, and about half of them also score in

**Figure 4.2 MCMI/MMPI Code Type Overlap:
MMPI 27/72^a**



^aDepression/Psychasthenia

Note. $n = 228$, 6.9% of sample. Asterisk = a pure type.

MCMI scales: 1 = Schizoid, 2 = Avoidant, 3 = Dependent, 7 = Compulsive, 8 = Passive-Aggressive.

the schizotypal range. External support for these findings was obtained in clinician diagnostic assignments. In a large sample, more than 25% received avoidant diagnoses, and more than 15% each were assigned a schizoid or schizotypal label. All three diagnoses were three to four times more common than expected by chance.

The second MCMI variant—*emotionally acting-out*—consists of those profiles with elevations on MCMI Scales 8 and 2 (Passive-Aggressive and Avoidant); in other words, these cases have both MCMI 2,8/8,2 and MMPI 28/82 code types. This variant is portrayed by Antoni et al. (1985a) as vacillating between irritability, self-alienation, and angry and impulsive outbursts. Found in Figure 4.3 in the 8*; 2,8; and 8,2 columns, they also score high on the MCMI Borderline scale. Interestingly, close to 35% of these patients were diagnosed as borderline by their clinicians, and approximately 20% were labeled passive-aggressive personalities.

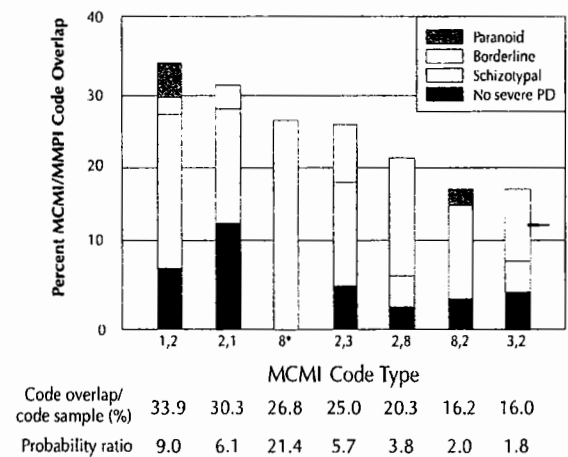
The third MCMI variant of the MMPI 28/82 profile is found by combining MCMI Scales 2 and 3 (Avoidant and Dependent). Called *emotionally acting-in*, these patients are characterized by a life of intense conflict over the

incompatibility of their dependency needs and their mistrust of others. Fearful of rejection from those on whom they depend, they typically withdraw from sources of support, leading them to feel lonely and isolated. Frequently achieving scores in the borderline and schizotypal ranges, these patients were diagnosed as avoidant, schizotypal, and borderline about three times as often as would be accounted for by chance.

Figure 4.4 shows the most common MCMI code types for the 186 patients (5.7% of the sample) who obtained an MMPI 34/43 profile code (3 is Hysteria; 4 is Psychopathic Deviate). In the MMPI literature, individuals with 34/43 codes are most frequently characterized as passive-aggressive, chronically resentful individuals who have failed to develop the skills for direct or effective expression of their angry feelings. Two primary MMPI 34/43 variants appear derivable from MCMI data: the *conforming* and *externalizing* groups. Both are associated with MCMI Scale 8 (Passive-Aggressive). The conforming group is seen most prominently in the columns for MCMI codes 4,8 and 7,8/8,7, where the percentage overlap with the MMPI 34/43 code is especially high (42.3% and 29.6%, respectively) and where the probability ratio is also very high (26.7 and 36.1 times chance, respectively). In the externalizing group, the more prominent MCMI scale is 6 (Antisocial), seen best in columns 6,8 and 4,6/6,4. Described as possessing a retaliatory mentality, these individuals appear to be unwilling to acknowledge their anger overtly. By engaging in a process called *projective identification* in the psychoanalytic literature, they create encounters that they experience as unreasonable or threatening, thereby justifying their angry rebuttals.

Figure 4.5 covers the MMPI 48/84 profile (4 is Psychopathic Deviate; 8 is Schizophrenia) and its covarying MCMI high-point codes. Encompassing 202 patients (6.2% of the sample), those with this MMPI profile have been described as individuals who do not fit in well with their environment. They are nonconforming and resentful of authority, tend to be unstable, behave in erratic and unpredictable ways, and have exaggerated needs for attention and affection. For some, the 48/84 MMPI code indicates the presence of longstanding paranoid tendencies. Antoni, Levine, Tischer, Green, and Mil-lon (1987) suggested that the MCMI code types

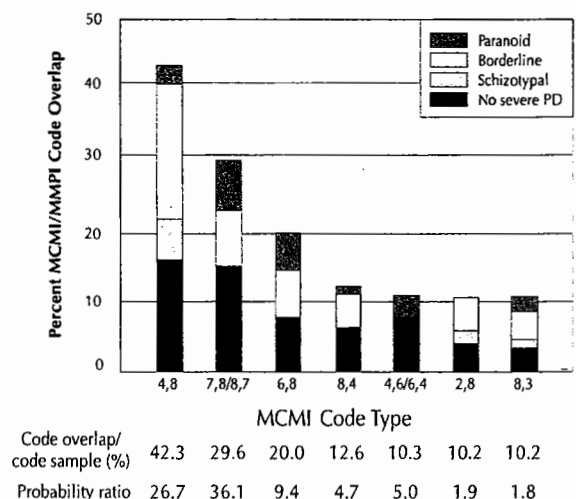
Figure 4.3 MCMI/MMPI Code Type Overlap: MMPI 28/82^a



^aDepression/Schizophrenia

Note. $n = 353$, 10.8% of sample. Asterisk = a pure type. MCMI scales: 1 = Schizoid, 2 = Avoidant, 3 = Dependent, 8 = Passive-Aggressive.

Figure 4.4 MCMI/MMPI Code Type Overlap: MMPI 34/43^a



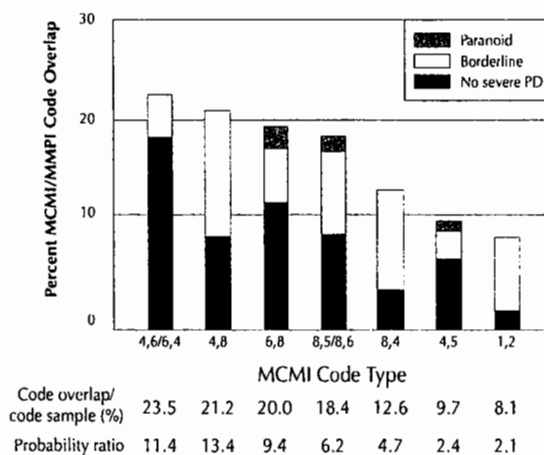
^aConversion Hysteria/Psychopathic Deviate

Note. $n = 186$, 5.7% of sample. MCMI scales: 2 = Avoidant, 3 = Dependent, 4 = Histrionic, 6 = Antisocial, 7 = Compulsive, 8 = Passive-Aggressive.

shown in Figure 4.5 fall into three groups: the first with elevations on the Passive-Aggressive (8) scale (e.g., 4,8; 6,8; 8,4), the second with high scores on the Antisocial (6) and/or Narcissistic (5) scales (e.g., 6,8; 4,5), and the third with elevations on the Schizoid (1) and Avoidant (2) scales (e.g., 1,2). The first group—labeled *emotionally ambivalent*—includes individuals who have deep conflicts over their dependency needs, often alternating between angry manipulativeness and dependent acquiescence. The second group—*emotionally contemptuous*—appears to be similar to the emotionally ambivalent group, but these individuals have much less conflict and are less concerned about issues of dependency or securing support and nurturance from others. Their manipulateness appears to stem from an uncomplicated disregard, if not contempt, for others. The distinction between these first two groups is in part supported by the fact that clinicians assigned the diagnosis of antisocial personality to more than 25% of the contemptuous group but to only 5% of the ambivalent group. Conversely, ambivalents were diagnosed as borderline about 30% of the time, whereas fewer than 10% of those in the contemptuous category received this diagnosis. The third MCMI-based category, noted in the 1,2 column, encompasses that small number of MMPI 48/84 patients who are best described as *interpersonally detached* owing to their affective deficits or their protective withdrawal behavior.

The MMPI 49/94 configuration (4 is Psychopathic Deviate; 9 is Hypomania) is a well-known code achieved by 305 patients (9.3%) in the study's population. Figure 4.6 summarizes its primary MCMI profile covariants. The literature on the 49/94 code is consistent, stressing adjectives such as impulsive, extroverted, pleasure-seeking, untrustworthy, resentful, self-centered, hostile, and manipulative. As can be seen in Figure 4.6, the majority of the MCMI covariant codes are combinations of Scales 4, 5, and 6 (Histrionic, Narcissistic, and Antisocial, respectively). Levine et al. (1987) note that three MCMI groups may be formed from these data. The first, in which Scale 4 (Histrionic) is primary (e.g., 4,5; 4,6/6,4; and 4,8), has been called the *dependency-motivated group*, notable for its insatiable need for attention, its impulsivity, and its sensation- or stimulus-seeking behavior. The

Figure 4.5 MCMI/MMPI Code Type Overlap: MMPI 48/84^a



^aPsychopathic Deviate/Schizophrenia

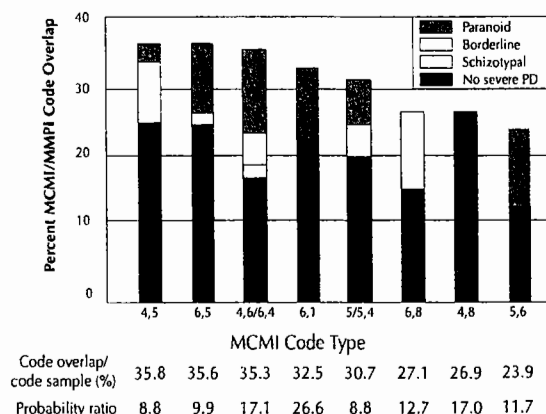
Note. $n = 202$, 6.2% of sample.

MCMI scales: 1 = Schizoid, 2 = Avoidant, 4 = Histrionic, 5 = Narcissistic, 6 = Antisocial, 8 = Passive-Aggressive.

second, called the *self-esteem-motivated group*, with a primary elevation on Scale 5 (Narcissistic; 5/5,4; 5,6), is characterized by displays of arrogance, dominance, and self-assurance. Not surprisingly, more than 25% of these cases were diagnosed as narcissistic personalities by their clinicians, more than five times what would be expected by chance. The third MCMI variant, consisting of profiles dominated by Scale 6 (Antisocial; 6,5; 6,8), is called the *aggressively motivated group*. Individuals in this group consider authority figures to be punitively threatening, which produces hostile acting-out behavior, irresponsibility, and a deficient social conscience. Consonant with this, close to 40% of the patients in this group were diagnosed as antisocial personalities, a figure that is approximately seven times higher than chance in the study population.

Figure 4.7 shows the most common MCMI code types for MMPI code 78/87 (7 is Psychasthenia; 8 is Schizophrenia), a group of 272 cases (8.3% of the sample). Individuals with the 78/87 code are described in the MMPI literature as introspective, obsessional, indecisive, worried, tense, shy, and socially inept. Diagnostically, they are thought by some to be neurotic, by oth-

**Figure 4.6 MCMI/MMPI Code Type Overlap:
MMPI 49/94^a**



^aPsychopathic Deviate/Hypomania

Note. $n = 305$, 9.3% of sample.

MCMI scales: 1 = Schizoid, 4 = Histrionic,

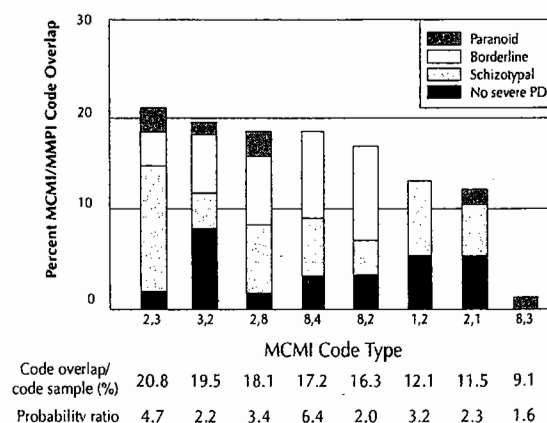
5 = Narcissistic, 6 = Antisocial, 8 = Passive-Aggressive.

ers to be psychotic. They are viewed as fitting several diagnostic categories, namely, passive-dependent, schizoid, schizotypal, and borderline. Antoni et al. (1985a), discussing covariations similar to those in Figure 4.7, subdivided the alignment into three sets, some of which were described previously: an *interpersonally acting-in* group, noted by high scores on MCMI Scales 1 and 2 (Schizoid and Avoidant); an *emotionally acting-out* group, with profiles dominated by Scale 8 (Passive-Aggressive); and an *emotionally acting-in* group, composed of high-point combinations of Scales 2 and 3 (Avoidant and Dependent). Individuals in the first covariant group—the MMPI 78/87 and MCMI 1,2/2,1 combination—exhibit withdrawal, indecision, diminished self-esteem, pervasive anxiety, and obsessional thoughts about their incompetence and unworthiness. These interpretations are consistent with the fact that patients in this group were diagnosed as avoidant somewhat more than 30% of the time by their clinician, a figure about four times that expected by chance. The second pattern (MMPI 78/87 combined with MCMI Scale 8) has as its cardinal features a mix of inner turmoil, angry defiance, and sullen moodiness. These patients

were diagnosed as borderline more than 40% of the time, close to three times the rate expected by chance. Last among the MMPI 78/87 covariants, those highest on MCMI Scales 2 and 3, were patients labeled emotionally acting-in, reflecting their frustration over the incompatibility of their strong dependency needs and their distrust of others, resulting in protective withdrawal and social isolation. Diagnostic assignments of avoidant and schizotypal for these patients were appreciably more frequent than chance.

The MCMI configurations associated with the MMPI 89/98/9* profile (8 is Schizophrenia, and 9 is Hypomania; 228 cases, 6.9% of the sample) are presented in Figure 4.8. According to Antoni et al. (1986), these patients are described in the MMPI literature as infantile in their expectations of others, demanding attention and reacting with hostile resentment if their demands are not met. They fear close relationships and tend to be socially withdrawn. They are sometimes referred to as hyperactive, emotionally labile, and even grandiose, boastful, and fickle—all in all, a rather heterogeneous mix that the MCMI codes may help tease apart. Antoni et al. (1986) identified three MCMI-based variants.

**Figure 4.7 MCMI/MMPI Code Type Overlap:
MMPI 78/87^a**



^aPsychasthenia/Schizophrenia

Note. $n = 272$, 8.3% of sample.

MCMI scales: 1 = Schizoid, 2 = Avoidant, 3 = Dependent, 4 = Histrionic, 8 = Passive-Aggressive.

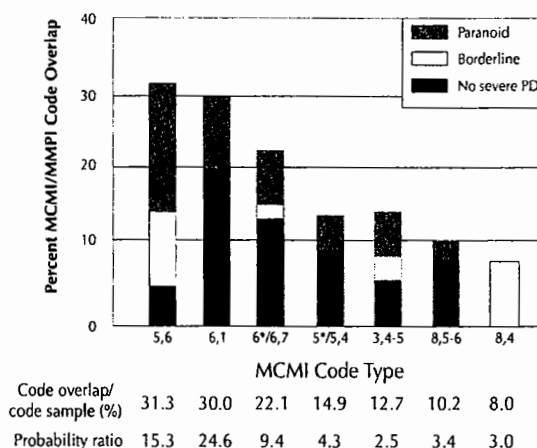
The first, the *interpersonally acting-out* group, noted by a high MCMI Scale 6 (Antisocial), is characterized by mistrust of others, hostile actions, rejection of social norms, risk-taking behavior, and drug abuse. Clinicians diagnosed about 25% of these patients as antisocial, 15% as paranoid, and 15% as narcissistic, all in the range of three to four times chance expectations. Many of these individuals have drug and/or alcohol problems.

The second group—the *interpersonally grandiose* MCMI variant—is noted by an elevated Scale 5 (Narcissistic) and has high proportions scoring above the cutoff for paranoid disorder (e.g., 57% in the 5,6 group). These patients are propelled by a strong need for high self-esteem and exhibit self-referential preoccupation, an arrogant personal style, and disregard for social constraints. More than 60% were assigned, in about equal proportions, clinician diagnoses of narcissistic, antisocial, or paranoid personality.

The third variant (*emotionally acting-out*), with primary elevations on MCMI Scale 8 (Passive-Aggressive), consists of individuals who are likely to exhibit labile, unpredictable, intense, and outwardly discharged emotions, often mixed with self-alienation, conflictual dependency, and a negative outlook on life. Patients in this group were diagnosed as borderline, passive-aggressive, and paranoid at rates that are appreciably higher than expected by chance.

The final figure in this group (Figure 4.9) portrays the most common MCMI covariants of a flat MMPI profile (170 cases, 5.2% of the sample). Individuals obtaining these profiles appear to include nonclinical respondents, those who were inclined to “fake good” or deny the presence of any problems, and those who are naive, inhibited, and defensive about psychological matters. What is most notable about these subjects’ MCMI high-point codes is the almost uniform presence of a high score on Scale 7 (Compulsive), reflecting the compulsive personality disorder. Also worthy of comment is the small proportion, across all columns, of the three severe personality disorders. Clearly the MMPI and MCMI profiles have something similar to say: these respondents were either normal, in denial, or psychologically naive. What the MCMI code types have to offer is a measure of insight into which of these conditions may be operative. An MCMI “spike 7” suggests virtuous propriety and extreme or rigid denial of problems, an inference supported in part by the fact that the

**Figure 4.8 MCMI/MMPI Code Type Overlap:
MMPI 89/98/9*^a**



^aSchizophrenia/Hypomania

Note. *n* = 228, 6.9% of sample. Asterisk = a pure type.

Hyphen = third-highest scale.

MCMI scales: 1 = Schizoid, 3 = Dependent,

4 = Histrionic, 5 = Narcissistic, 6 = Antisocial,

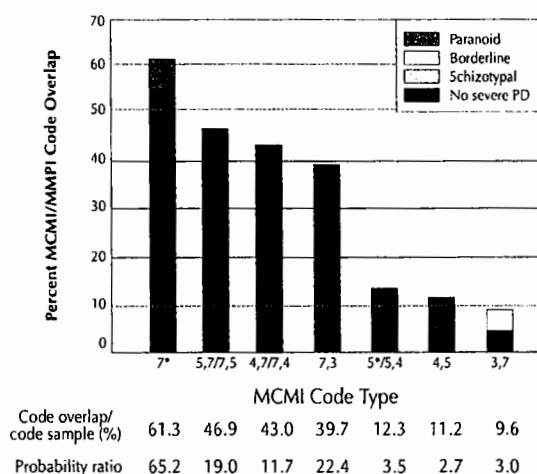
7 = Compulsive, 8 = Passive-Aggressive.

percentage of clinician diagnoses of paranoid and compulsive personalities among these patients was more than five times that expected by chance. Patients with their highest score on Scale 5 (Narcissistic) exhibit arrogance and a contemptuous attitude toward those who seek to pry into their illusions about themselves. Both narcissistic and paranoid diagnoses were present in relatively high proportions for these individuals. Among those with a flat MMPI profile whose highest MCMI scale was 4 (Histrionic), there are suggestions of a strong element of “social desirability,” a need to appear attractive and psychologically healthy. Among those with prominent elements of Scale 3 (Dependent) in the profile, we are most likely to encounter naïveté and a Pollyanna-like and guiltless form of denial, an interpretation supported in part by the high frequency of dependent personality diagnoses.

MCMI-II Profiles of Diagnostic Groups

Figures 4.11 to 4.32 show the MCMI-II profiles of patients diagnosed with various *DSM-III-R*

**Figure 4.9 MCMI/MMPI Code Type Overlap:
A Flat MMPI**



Note. $n = 170$, 5.2% of sample. Asterisk = a pure type.
MCMI scales: 3 = Dependent, 4 = Histrionic,
5 = Narcissistic, 7 = Compulsive.

disorders. The data reported in these figures were derived from three unpublished studies.

Study 1 was based on diagnostic data for the MCMI-II's normative patient population of 825 patients. Of relevance here is that clinicians made their Axis I and Axis II assignments after having seen their patients for only one or two interview sessions. They were to employ excerpts from the October 1985 draft of the *DSM-III-R* criteria, supplemented by a criterion set for the aggressive/sadistic personality disorder. Also notable is that they were given space to fill in only one diagnosis for each of the two axes. Axis II diagnoses were assigned to 479 cases, of which 52 were mixed-atypical, a group that was not included in further data analyses. The sample size for each of the 13 designated Axis II disorders is recorded (under Group A) in Figures 4.11 through 4.23. Axis I diagnoses that were directly relevant to the MCMI-II scales were assigned in 344 cases, broken down by Axis I disorder in Figures 4.24 to 4.32.

Studies 2 and 3 differed from Study 1 in that diagnostic judgments were made on the basis of extensive clinical data, that is, a substantive assessment evaluation (not including the MCMI-II

test) and/or six or more interviews or therapy sessions. Study 2 clinicians assigned diagnoses in accord with *DSM-III-R* categories and a comprehensive checklist of 240 descriptive phrases (the Millon Personality Diagnostic Checklist, MPDC; see Figure 4.10), which encompassed *DSM-III-R* criteria and criteria from the author's theoretical model of personality disorders (Millon, 1984, 1986a, 1986b). Because the clinicians used both the *DSM-III-R* and the MPDC criteria and because they had time to become acquainted with their patients' clinical status, it was assumed that their diagnostic assessments were more accurate than those made by the clinicians in Study 1. In addition, Study 2 clinicians were permitted to rank order up to three Axis I and three Axis II *DSM-III-R* diagnoses (only the first two of each, however, were included in the data analyses). One hundred seventy clinicians assigned ratings and diagnoses that served as the database for Study 2; counting the two highest-ranked diagnoses for each axis, a total of 431 Axis II and 246 Axis I diagnoses that were relevant to the MCMI-II scales were derived from this population.

Study 3 differed from Study 2 in two ways. First, all clinical participants in Study 3 were the first author's colleagues, doctoral students, or attendees at full-day workshops. Second, they were instructed to pay special attention to the selection of the primary diagnosis for Axis II and Axis I. As in Study 2, it was assumed that the clinicians' greater diagnostic "sophistication," at least with regard to Axis II, and their more extensive acquaintance with their patients' clinical status would result in more accurate *DSM-III-R* diagnoses than those assigned by the Study 1 clinicians. In addition, these participants were encouraged to have their cases independently evaluated by other clinicians for several purposes, including establishing interjudge reliability statistics (kappas for 151 cases for Axis II ranged from .59 to .86; for 132 cases for Axis I, they ranged from .43 to .93). Subsequent to recording their independent appraisals, the Study 3 clinicians discussed differences in diagnostic assignments and, where possible, arrived at a single agreed-upon primary diagnosis. Four hundred sixty-seven patients were evaluated and rated in this study, producing 458 primary Axis II and 369 primary Axis I diagnoses relevant

Figure 4.10 Millon Personality Diagnostic Checklist (MPDC)

Behavior/Activation

1. seems attracted to risk, danger, and harm
2. impulsive, restless, impetuous
3. uncooperative, contrary, and stubborn
4. avoids being taxed by ordinary activities
5. maintains regulated and highly organized lifestyle
6. refrains from pleasurable experiences
7. does self-damaging, impulsive acts
8. displays intolerance of boredom and inactivity
9. has difficulty doing things on his/her own
10. displays deficits in expressiveness
11. considers conventional lifestyles "beneath" oneself
12. has sudden, unexpected outbursts
13. highly defensive and vigilant
14. alternatively obstructive, then acquiescent
15. volunteers to do unpleasant tasks to gain approval
16. acts in a spur-of-the-moment manner
17. excessively devoted to work/productivity
18. strongly resists external influence and control
19. meticulous and perfectionistic
20. deliberately does poor job or works slowly
21. behaviorally odd, peculiar, or eccentric
22. anxiously anticipates ridicule/humiliation
23. has made suicidal threats or attempts
24. phlegmatic and lacking in spontaneity
25. disposed to be envious and jealous
26. avoids adult tasks and responsibilities
27. undermines own good fortunes
28. fails to honor financial obligations
29. seen by others as curious if not bizarre
30. places self in inferior or demeaning positions
31. lies for purpose of harming others
32. overly dramatic and theatrically reactive
33. unwilling to sustain consistent work behavior
34. ill-equipped to assume mature roles
35. acts arrogantly self-assured and confident
36. goes to great lengths to avoid being alone
37. engages in ritualistic or "magical" behaviors
38. perfectionism interferes with completing tasks
39. procrastinates and puts off things
40. scans environment for potential threats
41. conduct disorder evident before age 15

42. avoids personal obligations or routine work
43. fails to complete tasks beneficial to self
44. undeterred by pain or punishment
45. exhibits socially gauche habits and mannerisms
46. self-sabotaging and self-debasing
47. appears lethargic and lacking in vitality
48. recklessly daring and seemingly fearless
49. regresses to immature levels under stress
50. highly self-disciplined

Mood/Affect

51. emotionally impassive, and unaffectionate
52. unusually fearful of new experiences
53. exhibits shortsighted hedonism
54. ruthlessly indifferent to the welfare of others
55. quick to be annoyed or disappointed in others
56. evidences inappropriate or constricted affect
57. provokes rejection, then feels hurt or humiliated
58. tends to be churlish and irascible
59. devastated when close relationships end
60. cold and humorless, but edgy
61. describes constant tension, sadness, or anger
62. reacts to criticism with rage or shame
63. callous and insensitive
64. mood shifts from dejection to anger to apathy
65. contentious and belligerent
66. unable to experience pleasure in depth
67. fears losing control or giving into impulse
68. pursues pleasure-oriented social life
69. displays marked intolerance of frustration
70. generally apathetic and inexpressive
71. takes pleasure in the suffering of others
72. quick to take personal offense
73. restrains warmth and affection
74. shows intense affective instability
75. experiences minimal guilt or remorse for actions
76. has chronic feelings of emptiness or boredom
77. has excitable and pugnacious temper
78. unresponsive to feelings in others
79. fearful of loss or desertion

Figure 4.10 (continued)

- | | |
|--|--|
| 80. doesn't experience much sadness or anger | 115. flighty and attentive to superficialities |
| 81. lacks tender affects, shame, or guilt | 116. unimaginative and cognitively constricted |
| 82. exaggerates the risks of ordinary situations | 117. preoccupied with fantasies of success/
power |
| 83. seems mean-spirited and fractious | 118. vague, circuitous, or easily derailed
thinking |
| 84. expresses inappropriate and uncontrolled
anger | 119. disregards the truth and repeatedly lies |
| 85. highly agitated and anxiously watchful | 120. readily upset by the unfamiliar |
| 86. has low threshold for impulse discharge | 121. closed minded, dogmatic, and prejudiced |
| 87. displays exaggerated and shortlived
emotions | 122. short-sighted, fails to plan ahead |
| 88. characteristically timid and uncompetitive | 123. repetitively recalls past injustices |
| 89. show little desire for sexual experience | 124. tends to be skeptical and cynical |
| 90. generally appears nonchalant and insouciant | 125. not introspective, integrates experiences
poorly |
| 91. vacillates between being anguished and
numb | 126. suspicious and mistrustful |
| 92. enjoys being physically cruel or verbally
abusive | 127. indecisive, can't make decisions |
| 93. tense, unrelaxed, and grim | 128. drawn to violence, injury, firearms, etc. |
| 94. frequently touchy and resentful | 129. shows contempt for conventional morals |
| 95. has penchant for momentary excitements | 130. often contrite over erratic/moody
behaviors |
| 96. drawn to relationships in which he/she will
suffer | 131. has misanthropic view of life |
| 97. tries to keep emotions under tight control | 132. quick to project undesirable personal
traits |
| 98. tends to be sulky and moody | 133. mixes communications with personal
irrelevancies |
| 99. readily flares into argument and physical
attack | 134. especially critical of others in authority |
| 100. feels dejected or guilty after positive
experience | 135. preoccupied with lists, details, rules, etc. |
| | 136. searches for hidden motives in others |
| | 137. critical and negativistic |
| | 138. anticipates future difficulty and
disappointment |
| | 139. experiences ideas of reference |
| | 140. envious or disdainful of those more
fortunate |
| | 141. attitudes are capricious or antithetical |
| | 142. evidences impoverished or digressive
speech |
| | 143. overly conscientious about matters of
morality |
| | 144. reads threatening meanings into benign
events |
| | 145. has contrasting perceptions about same
person |
| Cognitive Style/Content | |
| 101. avoids reflecting on unpleasant thoughts | |
| 102. has perceptual illusions | |
| 103. Pollyanna attitude toward difficulties | |
| 104. withdraws into reveries to fulfill needs | |
| 105. has undisciplined and expansive
imagination | |
| 106. exhibits magical thinking or superstitious
beliefs | |
| 107. preoccupied with fears of being alone | |
| 108. cognitively self-absorbed, if not autistic | |
| 109. speech characterized by exaggeration and
hyperbole | |
| 110. inner preoccupations upset thought
continuity | |
| 111. fabricates events to bolster self-illusions | |
| 112. minimally introspective, with barren inner
world | |
| 113. easily persuaded and gullible | |
| 114. displays suspicious or paranoid-like
ideation | |
| Interpersonal Conduct | |
| | 146. seems socially aloof and remote |
| | 147. compliant, submissive, and placating |
| | 148. untrustworthy and unreliable |
| | 149. self-centered and socially inconsiderate |

Figure 4.10 (continued)

- | | |
|---|---|
| 150. socially pan-anxious and fearfully guarded | 187. takes advantage and exploits others |
| 151. actively seeks attention and solicits praise | 188. readily inflicts harm to achieve ends |
| 152. engages in clandestine pursuits | 189. engages in self-sacrifice and martyrdom |
| 153. elicits rejection rather than desired acceptance | 190. interpersonally erratic and paradoxical |
| 154. expects special favors without reciprocating | 191. prefers others to make major decisions |
| 155. unusually harsh toward underlings | 192. unjustly questions fidelity of spouse/friend |
| 156. obstructs others by not assuming responsibilities | 193. passively resists social expectations |
| 157. has no close friends or intimates | 194. prefers peripheral roles and solitary activities |
| 158. intensely competitive and power-oriented | 195. seems to enjoy undermining wishes of others |
| 159. shamelessly uses others to enhance self | 196. keeps distance unless certain to be liked |
| 160. competitive and power oriented | 197. irresponsible, if a parent |
| 161. seeks acceptance, but is self-protectively distant | 198. courts undeserved blame and criticism |
| 162. subordinates desires to please nurturing person | 199. tends to be provocative and fractious |
| 163. assaultive, gets into physical fights | 200. claims others make unjust demands on him/her |
| 164. doesn't sustain monogamous relationships | 201. capricious and unpredictably contrary |
| 165. adheres closely to social conventions | 202. aware of, but indifferent to, feelings of others |
| 166. abusively inflicts pain on others | 203. socially quarrelsome and abrasive |
| 167. assumes conflicting and changing social roles | 204. feels helpless or uncomfortable when alone |
| 168. enigmatic and secretive | 205. violates established social/legal codes |
| 169. shows disregard for rights of others | 206. constantly seeks recognition and admiration |
| 170. has few social desires or human interests | 207. permits others to be exploitive and mistreating |
| 171. uninterested in people who treat him/her well | |
| 172. seeks to induce guilt in others | Self-Image/Perception |
| 173. fearful of appearing publicly foolish | 208. views self as gregarious and charming |
| 174. stays in problem relationships fearing abandonment | 209. perceived as shallow, lacking in genuineness |
| 175. seductively flirtatious and exhibitionistic | 210. minimal interest in own personal life |
| 176. fails to meet social/economic responsibilities | 211. sees self as socially isolated |
| 177. intimidates, coerces, and demeans others | 212. feels entitled to be viewed as special |
| 178. interpersonally proper and respectful | 213. reports feelings of aloneness and emptiness |
| 179. rejects or undoes efforts of others to be helpful | 214. fears losing powers of self-determination |
| 180. carries grudges, does not forgive insults | 215. especially fearful of making errors |
| 181. has pattern of unstable and intense relationships | 216. viewed as vain and self-indulgent |
| 182. appears indifferent to praise or criticism | 217. seeks to be personally autonomous and unencumbered |
| 183. creates attractive but changing social facades | 218. has a grandiose sense of self-importance |
| 184. insists others do things his/her way | 219. sees self as misunderstood or unappreciated |
| 185. sulky and argumentative when others make requests | 220. sees self as inept and/or fragile |
| 186. desires affection, but fearful of rebuff | 221. cohesion of self is fragile and tenuous |
| | 222. feels rejection and unwanted by others |

Figure 4.10 (continued)

- | | |
|--|--|
| <ul style="list-style-type: none">223. values restraint, prudence and emotional control224. claims to deserve being shamed and debased225. overly concerned with physical appearance226. belittles own aptitudes and competencies227. has a confused sense of self228. presents self as domineering and/or tough229. recognizes being disillusioned and resentful230. pridefully independent and insular231. focuses on worst features of self | <ul style="list-style-type: none">232. self-perceptions are haphazard and fragmented233. has sense of high self-worth234. experiences depersonalization235. sees self as reliable and conscientious236. notably lacking in self-confidence237. has persistent identity disturbances238. exaggerates achievements and talents239. sees self as deserving to suffer240. life described in mechanical, impersonal way |
|--|--|

to the MCMI-II scales. Test data for this study were obtained using an earlier inventory form (MCMI-H Form 220R), which included all 175 MCMI items and the 45 replacement items selected for the MCMI-II test. It should be noted that data from this form were used in part to develop item weights, thereby creating a degree of capitalization on chance that slightly inflated scale-to-diagnosis correspondence.

A few comments and caveats must be noted regarding Figures 4.11 through 4.32. First, the data from Studies 2 and 3 were combined and are represented by Group B in the figures. Data from Study 1 are represented by Group A. As already discussed, clinicians in Studies 2 and 3 were significantly more familiar than the clinicians in Study 1 with the patients they rated. It is important to note that the profile shown in each figure represents the overall diagnostic group's median score for each scale. It is also important to keep in mind that a figure does not necessarily portray an MCMI-II profile exhibited by all patients in the *DSM-III-R* diagnostic group to which that profile pertains. This may be the case for some patients, but not all, because the figures are based on summary statistics. Buried in these averages are individual heterogeneity and subgroup diversity. By this we mean that each summary profile subsumes two to four distinct, polythetic subtypes. For example, many patients diagnosed as avoidant may have Avoidant and Schizoid as their highest and second-highest scales, matching the average profile shown in Figure 4.12, but other configurations will have Dependent, Self-Defeating, or Schizotypal as the second-highest scale after Avoidant. The reader

should keep in mind that each diagnostic profile collapses several configural patterns, albeit a limited set.

Figures 4.11 through 4.32 present data concerning the extent to which MCMI-II scale scores correspond to diagnoses made in accord with *DSM-III-R* criteria. Figure 4.11 presents the median MCMI-II base rate scores for the 16 patients (Group A) diagnosed in Study 1 as meeting the *DSM-III-R* criteria for schizoid personality disorder. It also presents the scale medians for the 27 patients (Group B) diagnosed in Studies 2 and 3 as schizoid. For both groups, the Schizoid scale was the highest score on the profile. Group B, whose data should be considered more reliable by virtue of its greater size and its superior clinical grounding, showed other high points on the Schizotypal and Avoidant scales, covariants expected in terms of the MCMI-II's underlying theoretical model. (However, as noted in the previous paragraph, this summary profile is likely to mask a number of profile variants.)

Figure 4.12 presents the MCMI-II data for the 60 patients in Study 1 (Group A) and the 101 patients in Studies 2 and 3 (Group B) who were diagnosed as exhibiting an avoidant personality disorder by clinicians using *DSM-III-R* criteria. The profiles are quite similar. Scale 2 (Avoidant), as expected, is highest, with secondary high scores on Scales 1 (Schizoid), 8A (Passive-Aggressive), 8B (Self-Defeating), and S (Schizotypal). The overlap with Scale 8B is not surprising given that avoidant and self-defeating personalities share a sense of being humiliated and abused, although for different reasons. The elevations of Axis I "state" Scales A (Anxiety)

Figure 4.11 Schizoid Personality Disorder: Profile of Median Scores

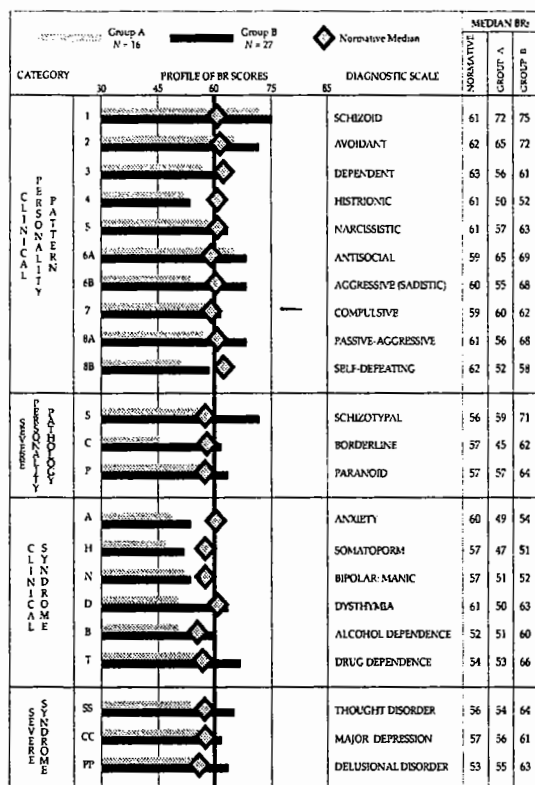
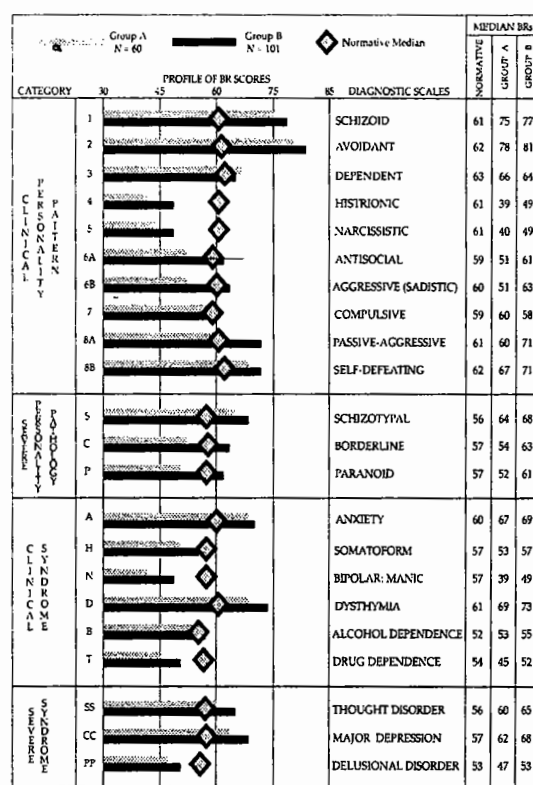


Figure 4.12 Avoidant Personality Disorder: Profile of Median Scores

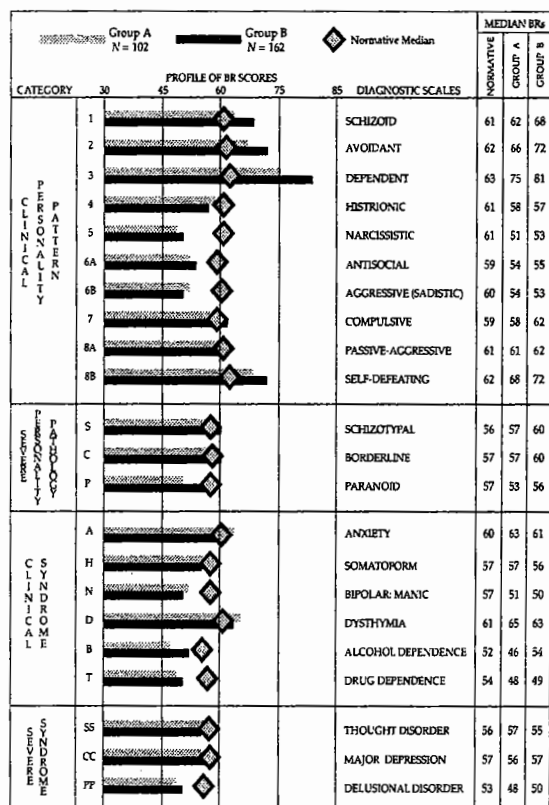


and D (Dysthymia) are likewise understandable given the chronic fear and dejection that characterize much of the avoidant person's life.

As can be seen in Figure 4.13, the Dependent scale is the predominant scale among those with a diagnosis of dependent personality disorder in both Groups A and B. The concurrent high scores on the Avoidant and Self-Defeating scales probably reflect a shared sense of low self-esteem. Subtypes pairing the Dependent scale with these two scales and with the Schizoid scale are likely.

A pattern worthy of note emerges in Figure 4.14. The Histrionic scale (Scale 4) fails to show up as a high MCMI-II score among patients diagnosed as histrionic in Group A but does so rather prominently among those so diagnosed in Group B. It is possible that the MCMI-II test portrays acutely upset histrionics accurately but cannot elicit their premorbid personality picture when they present for treatment. Histrionics enter treatment looking rather forlorn and dejected, exhibiting clinical features at variance with their more characteristic behavior (Millon, 1981). Once the more acute phase of their despondency abates, their more typical style emerges or their clinicians have had an opportunity to make a

Figure 4.13 Dependent Personality Disorder: Profile of Median Scores



more thorough assessment of their premorbid character structure. Consonant with these speculations is the MCMI-II profile obtained by Group A, in which we see that patients diagnosed as histrionic did not on average have high scores on the Histrionic scale but did give evidence of acute depression and turmoil, reflected in high scores on the Dysthymia, Thought Disorder, and Major Depression scales. By contrast, these latter scales have much lower scores among patients diagnosed as histrionic in Group B, where Scale 4 is highest on average.

There is consistency between the profiles for Groups A and B in Figure 4.15, with the Narcissistic scale achieving the highest standing among patients diagnosed with narcissistic personality disorder according to *DSM-III-R* criteria. Among the other most prominent scales are Histrionic and Aggressive (Sadistic), each of which is likely to play a major role in forming a narcissist subtype.

Although the sample sizes are relatively small for Groups A and B in Figure 4.16, the results show strong support for the validity of the MCMI-II's reconstructed Antisocial scale. This scale was changed based on theory and to

bring it more in line with the *DSM-III* (Widiger & Frances, 1987). The Antisocial scale's elevation above the others in the profile is somewhat more distinct for Group A than for Group B, seemingly reflecting more accurate judgments in Group A than in Group B, despite the fact that the Group B clinicians had more time and data to make their diagnoses. Perhaps "purer" antisocials are quickly discerned, whereas those

Figure 4.15 Narcissistic Personality Disorder: Profile of Median Scores

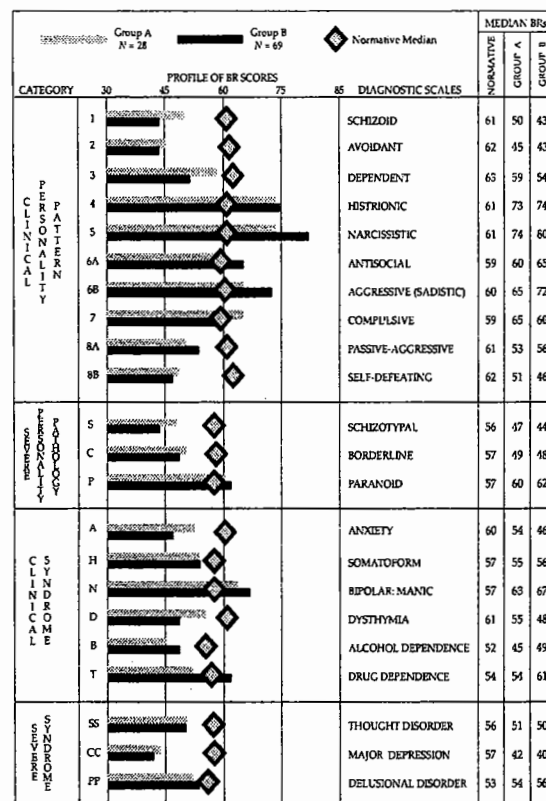
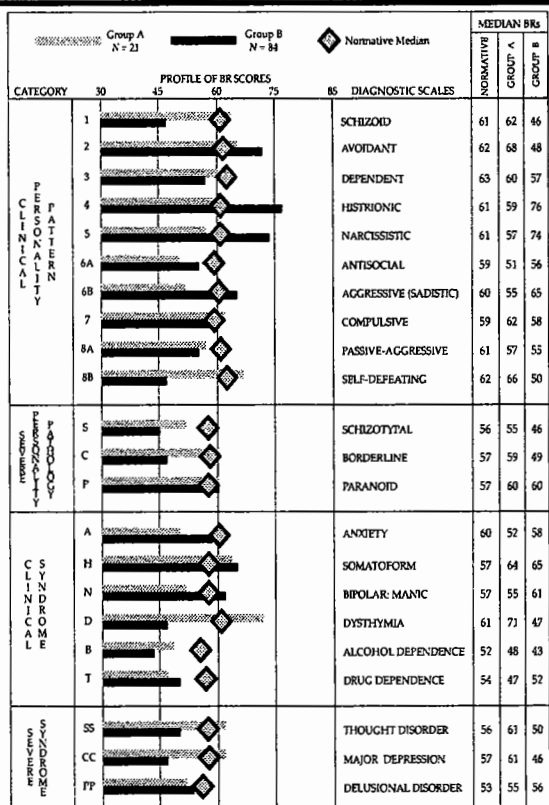


Figure 4.14 Histrionic Personality Disorder: Profile of Median Scores



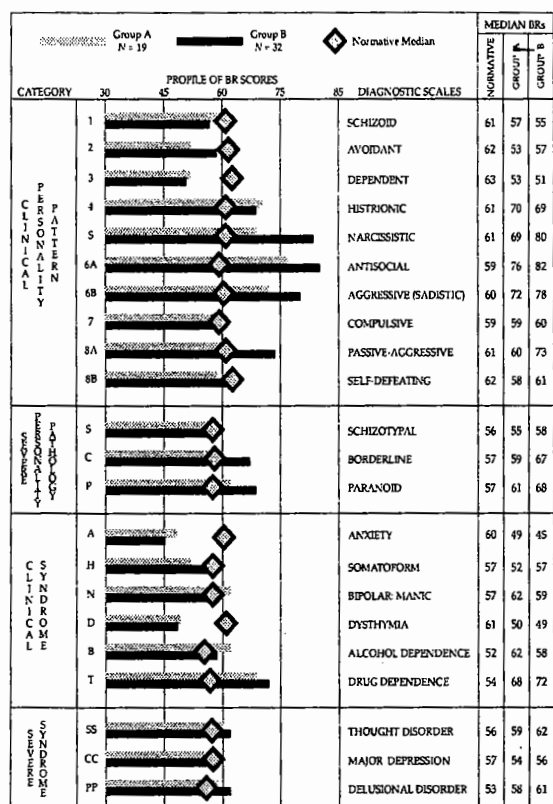
with more complicated or "devious" styles become apparent only after extensive evaluations. It would be this latter group that would show a blend of high scores on the Narcissistic and Antisocial scales, accounting for the Group B results seen in Figure 4.16.

The small sample (only four cases) for Group A in Figure 4.17 makes its data too unreliable for serious consideration, although they are in general accord with the Group B data. As expected, the Aggressive (Sadistic) personality scale ranks highest among the Group B patients diagnosed using *DSM-III-R* criteria. Other notable high points are on the Passive-Aggressive and Antisocial scales. The relative peaks on the

Borderline, Paranoid, and Drug Dependence scales are also worth noting for future examination.

Figure 4.18 presents the findings for individuals diagnosed in line with *DSM-III-R* obsessive-

Figure 4.16 Antisocial Personality Disorder: Profile of Median Scores



compulsive personality disorder criteria. The contrast between Groups A and B in Figure 4.18 is similar to that found in Figure 4.14 for the histrionic personality, although it is somewhat less pronounced. Some compulsives when initially troubled look quite different from their usual selves. Such patients may understandably mislead clinicians. In these cases, rigid emotional controls may have given way, resulting in the outcropping of previously repressed resentment and hostility and producing short-term but uncharacteristic abusive behavior. Hence, for Group A, there are high points on both the Aggressive (Sadistic) and Compulsive scales, an atypical mix but one that is understandable dynamically during times of turmoil. The upsetting phase may pass after a period of time. Thus, in Group B, clinicians may be appraising more compensated cases that not only had high scores on the

Figure 4.17 Aggressive (Sadistic) Personality Disorder: Profile of Median Scores

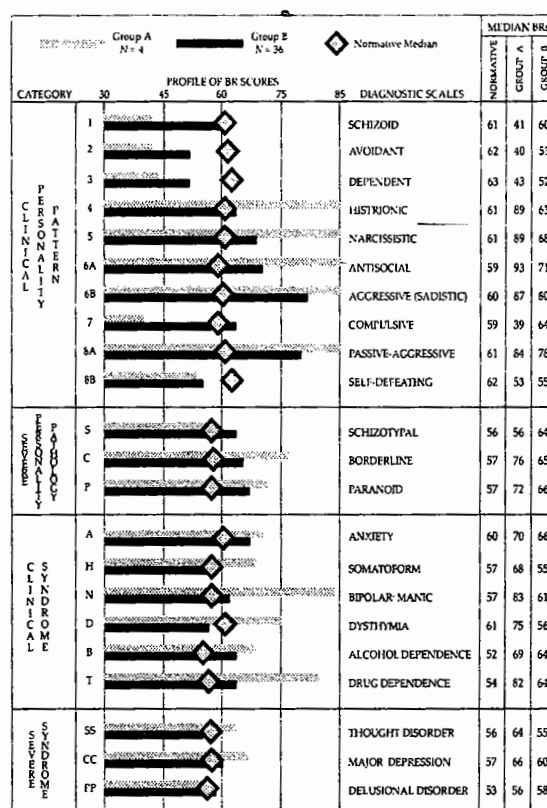
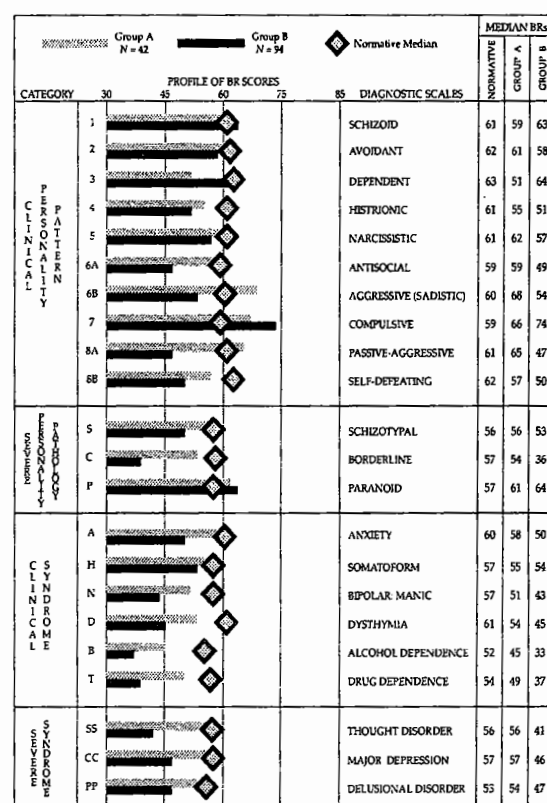


Figure 4.18 Obsessive-Compulsive Personality Disorder: Profile of Median Scores



Compulsive scale but are once again able to deny their less attractive traits, thereby producing the low scores found across all other scales.

Figure 4.19 shows an interesting but not unexpected pattern for those diagnosed according to *DSM-III-R* as passive-aggressive, which is one of the least reliable categories in that version of the *DSM*. Those patients who were appraised after only one or two sessions (Group A) scored highest on the Aggressive (Sadistic) scale. Perhaps some clinicians were disinclined to assign the new and more maligning label "sadistic" to individuals who fit these criteria. On the other hand, with more and better evidence about a patient's characteristic style (Group B), we see that the Passive-Aggressive scale achieves the expected level of prominence, although the Aggressive (Sadistic) scale remains high.

Patients diagnosed with the *DSM-III-R* self-defeating personality disorder are profiled in Figure 4.20. For both Groups A and B, Self-Defeating (8B) is the highest personality scale. The data for Group B are outstanding. Concurrent

elevations with 8B are likely for the Dependent and Avoidant scales, a pattern noted in the earlier discussion of Figure 4.13. Those diagnosed as self-defeating also have high scores on the Axis I Anxiety, Dysthymia, Thought Disorder, and Major Depression scales.

The MCMI-II group profiles shown in Figure 4.21 for patients diagnosed as schizotypal are also consistent with expectations. Although Scale S (Schizotypal) is high, in line with the theory, Scales 1 (Schizoid), 2 (Avoidant), and 3 (Dependent) are also elevated, generally exceeding Scale S in magnitude. Note the overall high scores for these patients—except for Scales 4 (Histrionic), 5 (Narcissistic), 7 (Compulsive), and N (Bipolar: Manic)—suggesting that patients who are diagnosed as schizotypal report a wide range of disturbing symptomatology.

Individuals with a borderline personality disorder diagnosis (whose MCMI-II scores are summarized in Figure 4.22) appear to be no less accurately appraised after one or two sessions (Group A) than after several sessions (Group B).

Figure 4.19 Passive-Aggressive Personality Disorder: Profile of Median Scores

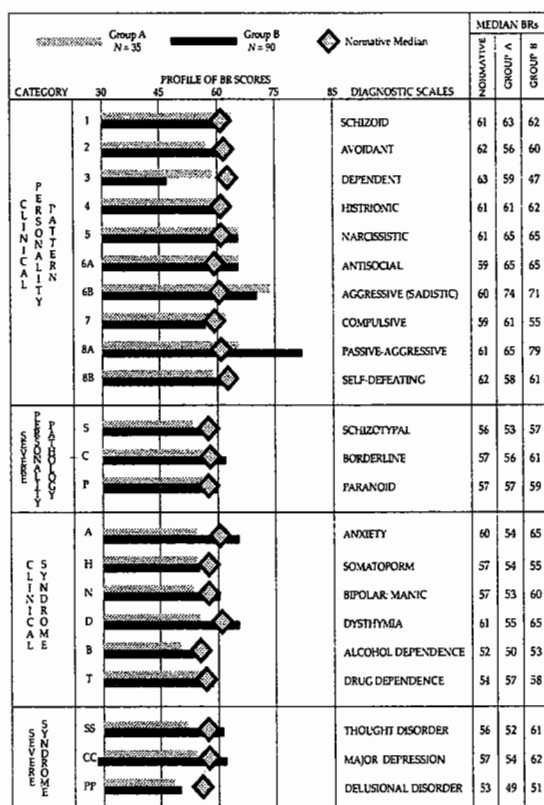
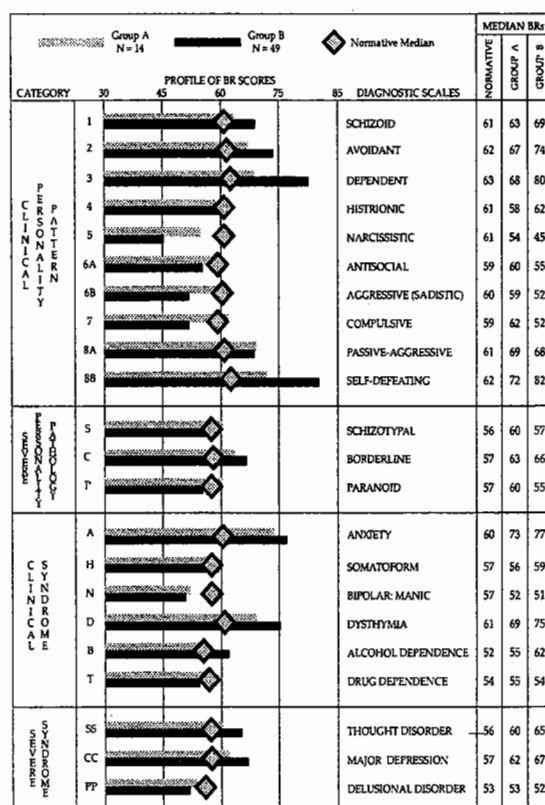


Figure 4.20 Self-Defeating Personality Disorder: Profile of Median Scores



The highest personality disorder score for Group B is on the Self-Defeating scale, followed closely by the Borderline and Passive-Aggressive scales; each of these scales captures significant elements of the borderline pattern. Equally consistent with the mixed if not discordant features of this disorder are the high scores on the Axis I Dysthymia, Anxiety, Alcohol Dependence, and Drug Dependence scales.

Figure 4.23 shows the results for the relatively small sample of individuals diagnosed as paranoid personalities. Although the Paranoid scale has one of the highest points on the profile, it is slightly exceeded for Group B by the Aggressive (Sadistic) scale, which is also in line with the theoretical model.

Turning to the syndromes of Axis I, Figure 4.24 shows the median MCMI-II scale scores for individuals diagnosed with a primary generalized anxiety disorder. Worthy of note are some differences between Group A and Group B. Given the relatively flat profile for Group A, we might infer that many patients evidence high levels of anxiety during the early phases of their turmoil and that anxiety of a more enduring and

Figure 4.22 Borderline Personality Disorder: Profile of Median Scores

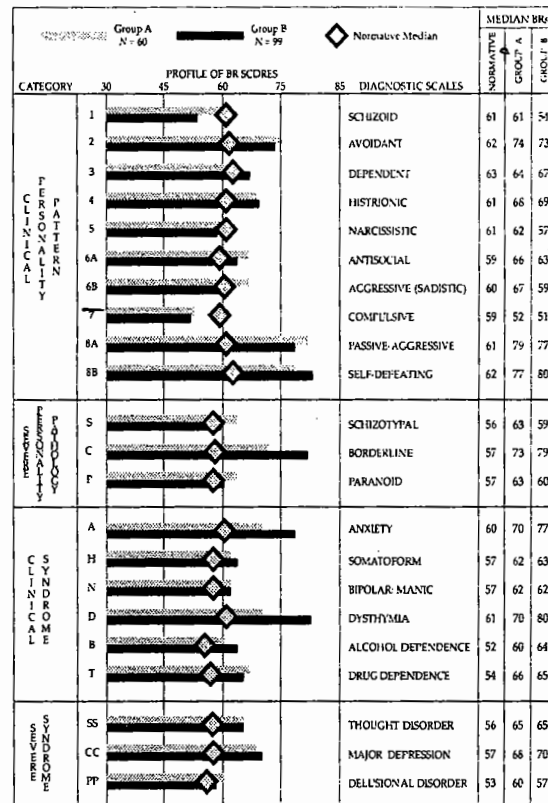


Figure 4.21 Schizotypal Personality Disorder: Profile of Median Scores

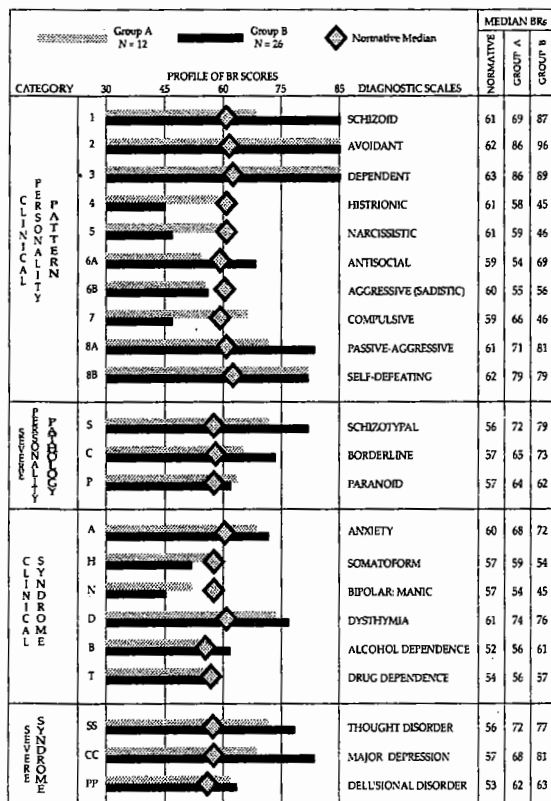
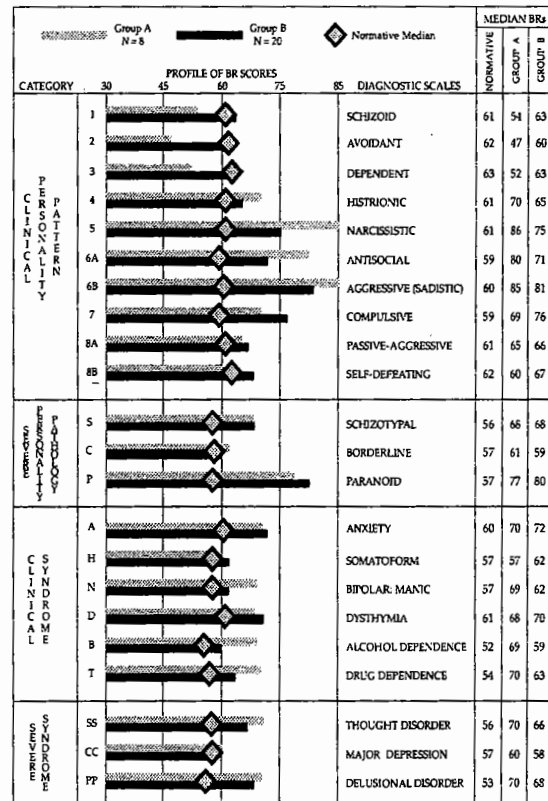


Figure 4.23 Paranoid Personality Disorder: Profile of Median Scores



pervasive nature may be difficult to differentiate. After clinicians become better acquainted with these patients (and after patients whose anxiety has had time to subside can be identified), clinicians may be better able to make accurate diagnostic assignments. Thus, among the individuals in Group B, we see that the Anxiety scale is most prominent, with concurrent elevations on the Avoidant, Passive-Aggressive, and Self-Defeating scales (Scales 2, 8A, and 8B), of which anxious feelings are often intrinsic elements.

A somatoform diagnosis (Figure 4.25) appears to correspond, as expected, with elevations on the Somatoform scale (Scale H) and the Histrionic and Compulsive scales (Scales 4 and 7), where somatic discomfort and tension displacement are characteristic.

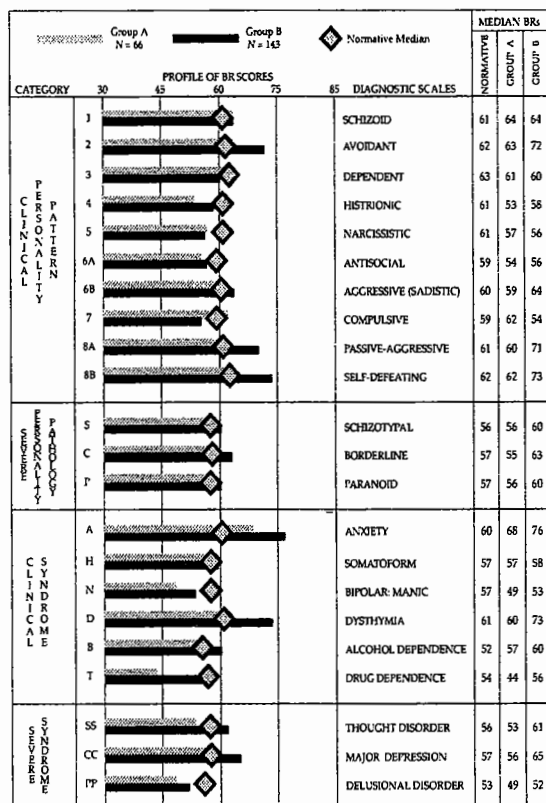
Figure 4.26 shows the results for the two groups (A and B) of *DSM-III-R*-diagnosed bipolar patients. The consistency across the two groups is modest at best, with divergences most notable among the personality scales. In the

larger and more "diagnostically credible" sample (Group B), we see the classic manic depression pattern reflected in elevations on the Bipolar: Manic and Dysthymia scales. The low score in this group on the Schizoid personality scale and the high scores on the Histrionic and Passive-Aggressive scales, which are both gauges of an active expression of affect, support the underlying theory.

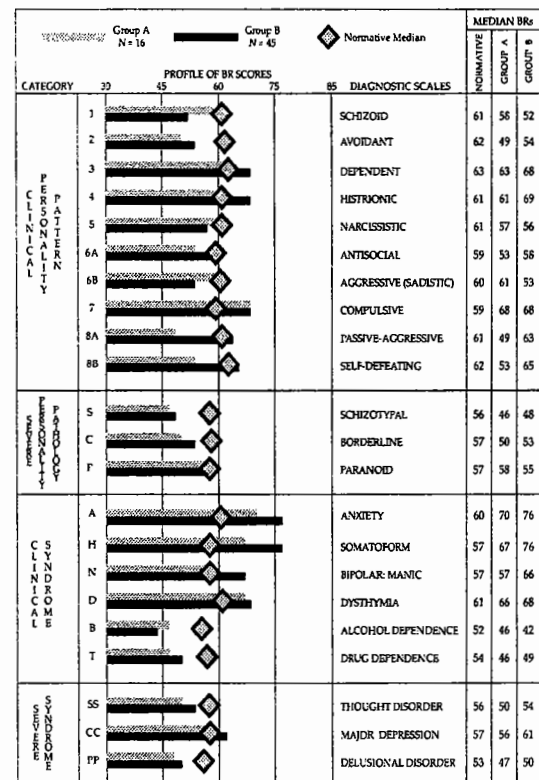
The large samples reflected in Figure 4.27 (patients diagnosed as dysthymic) add to the reliability of the results. Also raising our level of confidence is the relatively high consistency between the profiles of Groups A and B. Most relevant to the MCMI-II's validation against *DSM-III-R* criteria is that the Dysthymia scale is the highest point on both profiles.

It is clear from Figure 4.28 that the MCMI-II Alcohol Dependence scale corresponds extremely well with a *DSM-III-R* diagnosis of alcohol dependence. Elevations are also seen on the Drug Dependence and Schizoid scales in Group

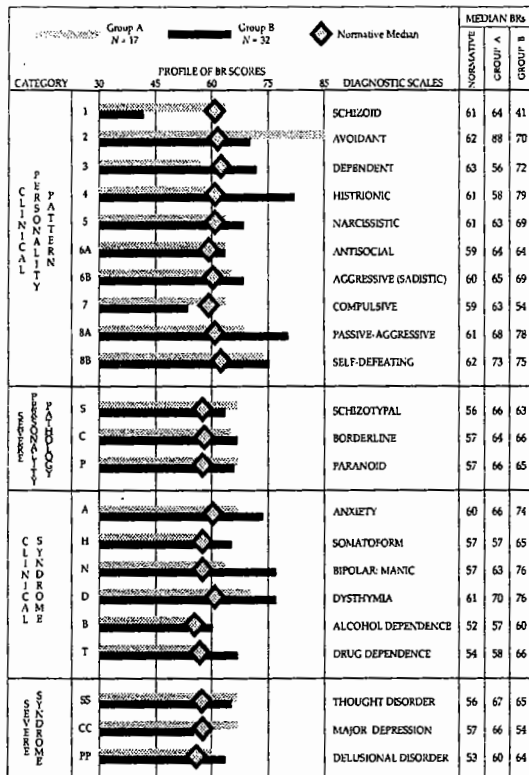
**Figure 4.24 Generalized Anxiety Disorder:
Profile of Median Scores**



**Figure 4.25 Somatoform Disorder:
Profile of Median Scores**



**Figure 4.26 Bipolar Disorder:
Profile of Median Scores**

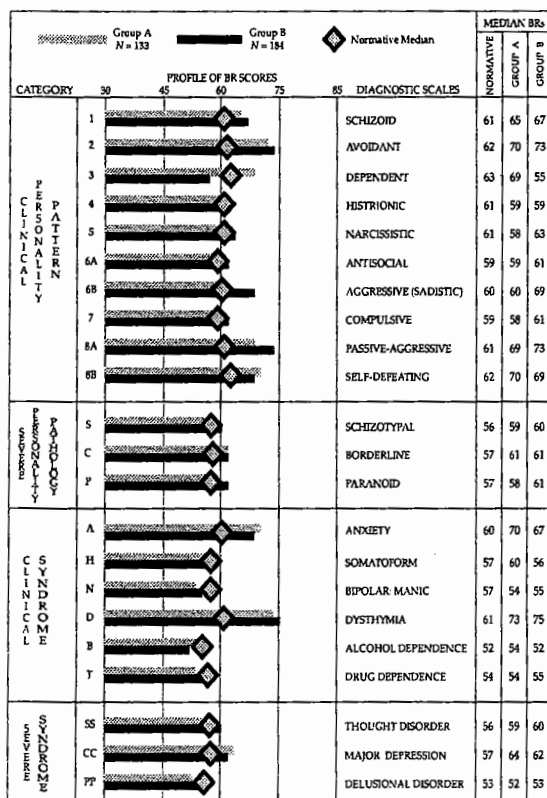


B and on the Antisocial and Aggressive (Sadistic) scales in Group A.

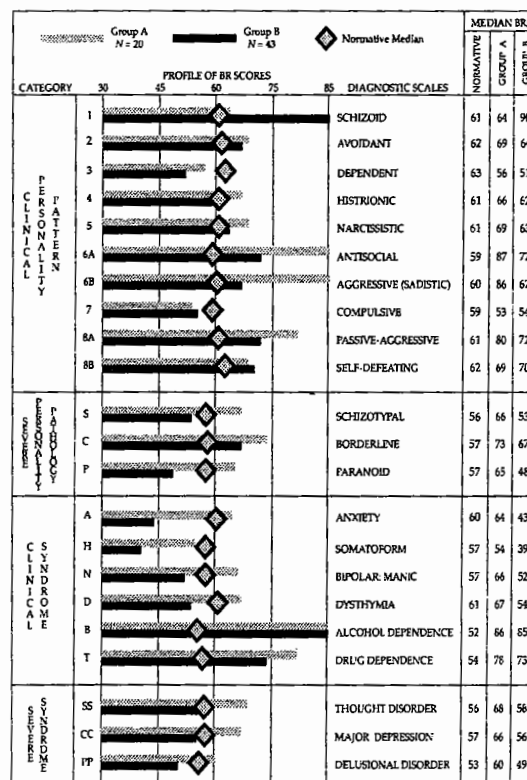
Figure 4.29 shows the high correspondence between the MCMI-II Drug Dependence scale and a *DSM-III-R* diagnosis of drug dependence. It should be noted that these patients also had high scores on the Alcohol Dependence scale, a finding that probably reflects a broad substance dependency. Also noteworthy (and consistent with the profiles in Figure 4.16) is the covariation between those diagnosed as drug dependent and those diagnosed as antisocial.

Although Figure 4.30 shows that the Thought Disorder (SS) scale is elevated in the profiles of Groups A and B, patients diagnosed as schizophrenic according to the *DSM-III-R* scored higher on average on the Avoidant (2), Self-Defeating (8B), and Schizotypal (S) scales than on SS. The relative lack of salience of Scale SS may be a product of the fact that the thinking of these patients, and hence their responding, is disturbed. Also plausible is that clinicians may be unable to differentiate the social withdrawal and odd communication style of avoidants and schizotypals from the disordered thinking and asocial behavior of schizophrenics. Neverthe-

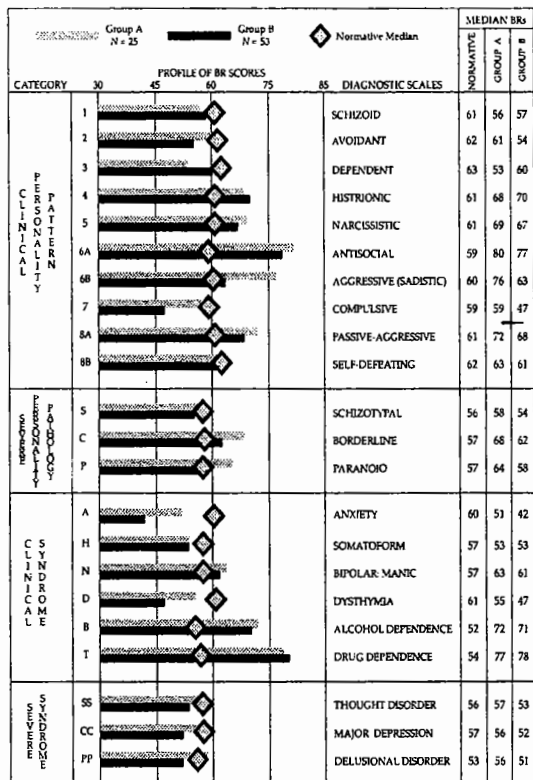
Figure 4.27 Dysthymia: Profile of Median Scores



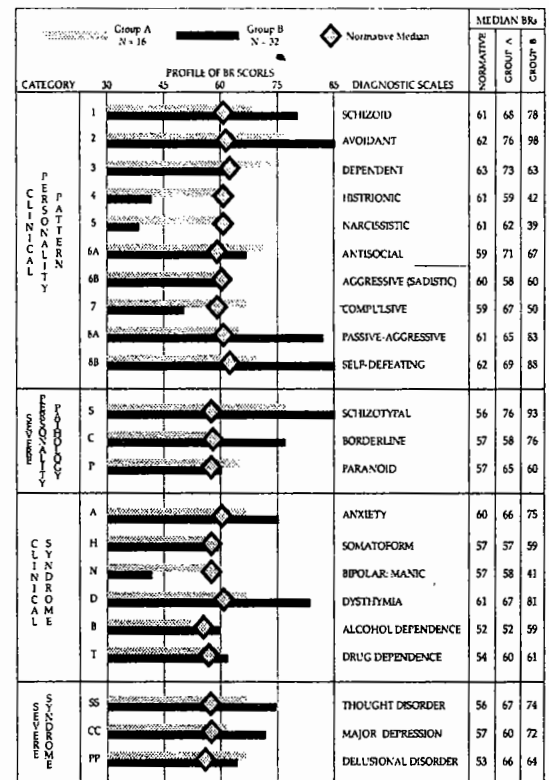
**Figure 4.28 Alcohol Dependence:
Profile of Median Scores**



**Figure 4.29 Drug Dependence:
Profile of Median Scores**



**Figure 4.30 Schizophrenia:
Profile of Median Scores**



less, the fact remains that high scores on Scales 2, 8B, and S may be as good a diagnostic gauge of schizophrenia as a high score on SS. Future research may clarify these results.

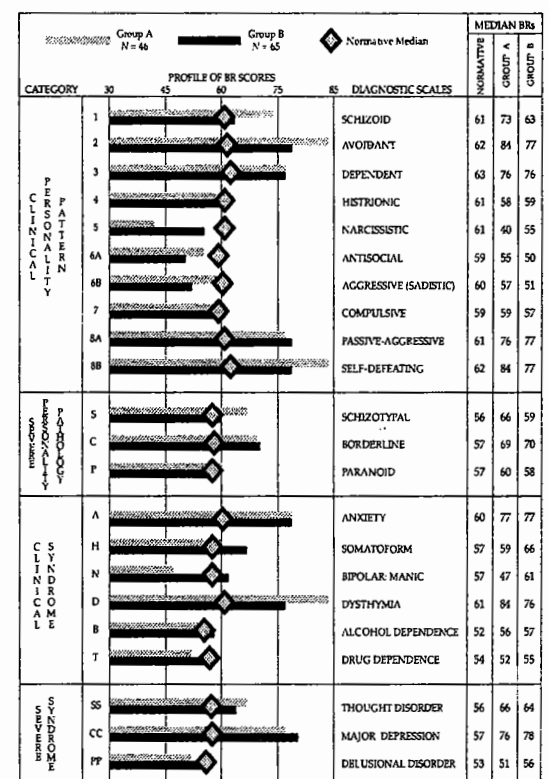
Figure 4.31 shows the average profiles of the Group A and B patients who were diagnosed as having major depression according to *DSM-III-R* criteria. For both groups, the Major Depression scale is among the highest profile points, with additional high scores on the Dysthymia, Anxiety, Dependent, Passive-Aggressive, Self-Defeating, and Avoidant scales.

Finally, Figure 4.32 shows that the Delusional Disorder scale is a high point among patients diagnosed with delusional disorder according to *DSM-III-R*. Also notable are high scores on the Thought Disorder, Paranoid, Schizotypal, Avoidant, and Passive-Aggressive scales.

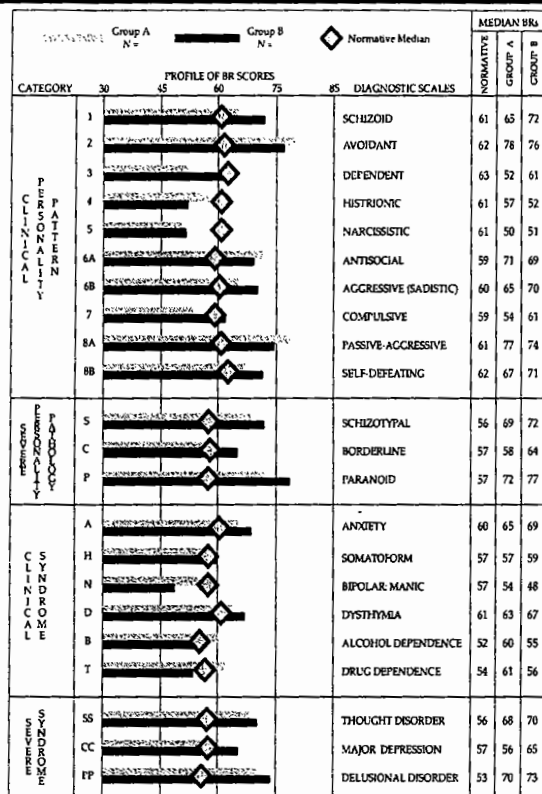
MCMI-III Validity Versus Diagnostic Judgments

In the MCMI-III clinician rating study that was reported in the first edition of this manual, the clinical judges typically saw their subjects only

**Figure 4.31 Major Depression:
Profile of Median Scores**



**Figure 4.32 Delusional Disorder:
Profile of Median Scores**



once. Ratings were most often made when the MCMI-II Research Form was administered, usually at intake. Although this design constraint created a natural upper bound on the diagnostic efficiency statistics derived from the study (Retzlaff, 1996), the data analyses nevertheless served an important screening role, a necessary but not sufficient criterion against which to evaluate the external validity of the MCMI-III test in a preliminary way.

In the clinician rating study reported below, clinicians were explicitly required to rate only subjects they knew well. Thus, these validity results for the MCMI-III test are derived from a design that more closely approximates the external validity study reported in the MCMI-II manual, allowing better comparisons to be made across the various generations of the instrument. The purpose of this section is to report these newer results and to compare the diagnostic efficiency of the MCMI-III test to that of the MCMI-II test, and where possible, to that of the MCMI test.

Method

Approximately 75 clinicians who were well acquainted with the MCMI test and with Millon's (1990) evolutionary theory and who had participated in earlier MCMI research projects were invited to participate in an MCMI-III diagnostic validity study. Of these, about 30 expressed a willingness to participate. Most indicated that they had several associates (group practice or clinic colleagues) who would join in the clinical judgment task. In all, 67 primary clinicians and associates participated in rating patients according to the directions and guidelines in the Clinician's Rating Reference Booklet (described below). In contrast to the study reported in the first edition of this manual, all of these clinicians reported having substantial direct contact with the patients they were to evaluate. The minimum contact time required was "at least three therapeutic or counseling sessions." Seven sessions was modal, and the number of hours of contact ranged from 3 to more than 60.

The first part (Part A) of the Clinician's Rating Reference Booklet is reproduced in its entirety in Figure 4.33. Also included in the figure are representative examples from its Axis I and Axis II sections. The Clinician's Rating Reference Booklet was divided into two parts. Part A consisted of an overview of the study, inclusion and exclusion criteria, and detailed steps describing what was required of the participating clinicians. For example, Step 1 asked clinicians to "Review the descriptions of the Axis I syndromes and Axis II disorders given in Part B of this booklet." It also noted that all clinicians have pet notions about some of the disorders and explains why it was necessary for the participants to use a single set of criteria or descriptions to rate patients. Also notable in Part A were the detailed anchor points that were developed for the rating tasks. The severity of each Axis I clinical syndrome, for example, was rated on a five-point scale. Rather than let each clinician determine the meaning of the labels *Slight*, *Mild*, *Syndrome*, *Marked Syndrome*, and *Extreme Syndrome*, descriptive paragraphs were provided to increase the standardization of the meaning of these points across raters. This ensured that the participating clinicians were rating the same dimensions in the same way, that is, that the meaning of any particular point along a given dimen-

Figure 4.33 Clinician's Rating Reference Booklet

PART A

I. GENERAL PROJECT INFORMATION

Thank you for participating in the current project concerning the MCMI-III. You will be playing a key role in the refinement of one of the most widely used objective diagnostic instruments in the mental health field. To ensure the validity and clinical utility of the MCMI-III, we have included sites such as yours from settings across the country to assist in gathering a large and representative clinical sample. Such careful test validation and refinement is a major undertaking that could not succeed without the participation of psychologists such as yourself and your associates—we thank you for the time, the interest, and the effort that you and your associates will be investing.

The purpose of this packet is to provide details about participation in this project. If you have any questions or would like to discuss any aspect of the research process, please call either Ted Millon, Roger Davis, or Andrew Wenger.

II. INCLUSION AND EXCLUSION CRITERIA

As an integral part of this research project, it is assumed 1) that the patients chosen for this study have already taken (or will soon take) and produced a valid MCMI-III (not the MCMI-II), 2) that the rater has personally seen the patient for at *least* 3 therapeutic or counseling sessions, and 3) that the rater has a good understanding of the patient's clinical features and personality characteristics, and 4) that the rater has reviewed the *DSM-IV* clinical syndrome summaries and *DSM-IV* personality disorder summaries, as well as the Millon Clinical Domain Descriptions given in Part B of this booklet. These are the inclusion criteria for the study.

Ideally, your Axis I syndrome and Axis II personality ratings should be made blind to MCMI-III results. To ensure a sample of adequate size, we are encouraging the inclusion of patients whose MCMI-III results have already been obtained. Do not, however, include patients for whom you have a clear recollection of their MCMI-III scores before completing the Clinical Rating Form. This is the exclusion criterion for the study.

III. CLINICIAN'S RATINGS

We ask that you and your associates complete a separate Clinician's Rating Form for each patient evaluated. Detailed information regarding the criteria of each of the Clinical Syndromes and Personality Styles or Disorders to be evaluated is included in Part B of this packet. This information is to serve as a standardizing guide for your ratings.

Completion of study consists of five steps: First, you and your associates should review descriptions of the clinical syndromes and personality features in *Part B* of this booklet. You should feel free to refer to these descriptions at any point while making your ratings. Second, you are asked to evaluate each patient with respect to a number of the Axis I disorders included in the *DSM* and assessed by the MCMI, and then to rate the severity of these disorders. Third, you are asked to select up to 3 personality patterns that characterize the patient, and to rate the severity of these patterns. Fourth, you are asked to record your confidence in each of the preceding ratings. Fifth, after completing the above, you are asked to locate the MCMI-III BR scores achieved by this patient, and to record ALL of them in the boxes at the bottom of the Clinical Rating Form. To assure uniformity in procedure, we will detail these steps in the following sections.

Figure 4.33 (continued)

STEP 1: REVIEW THE DESCRIPTIONS OF THE AXIS I SYNDROMES AND AXIS II DISORDERS GIVEN IN PART "B" OF THIS BOOKLET.

Clinicians differ in their interpretation of the various Axis I clinical syndromes and Axis II personality disorders. What one clinician considers a Depressive Personality may be interpreted by another as the characteristics of a Masochistic Personality. Accordingly, it becomes necessary to define or operationalize each of the disorders more precisely. Consequently, while each clinician may continue to hold his or her own pet notions about the various disorders, his or her conceptions are given a measure of standardization for research purposes. For the Axis I clinical syndromes, we suggest that you closely follow the listed criteria from the *DSM-IV*, summarized in the lists of Part B of this booklet. For the Axis II personality patterns, however, we have provided both the list of DSM criteria for each disorder, as well as descriptions for each of 8 Clinical Domains of personality, derived from Millon's theory. Our view here is that personality is not expressed only in the cognitive, or behavioral, or psychodynamic, or interpersonal realms, but is instead manifested across all these clinical domains, and that the DSM is markedly incomplete with regard to its sampling of many domains of clinical expression. You should attempt to use both DSM and Millon criteria when making your personality rating decisions.

STEP 2: RATE UP TO THREE OF THIS PATIENT'S AXIS I CLINICAL SYNDROMES AND THEIR SEVERITY.

All clinical syndromes included in the MCMI-III are listed in the upper left column of the Clinician's Rating Form. Fill in the encircled number under the *first* column for the Clinical Syndrome that calls for greatest professional concern or treatment. Note that two syndromes cannot both be rated as 1; the first column will contain only one darkened circle. Next, fill in the encircled number that is aligned with the *second* most clinically significant syndrome. Note that two syndromes cannot both be rated as 2; the second column should contain only one darkened circle. If a *third* syndrome is present, however slight, fill in its appropriate number under the third column as well.

Next, rate the *severity* of those clinical syndromes that you judged above as 1st, 2nd, and 3rd; these severity ratings are to be placed in the middle section of the grid. Note that severity judgments are not exclusive; that is, two clinical syndromes may both be judged Marked, for example. Use the following definitions as a guide when making your judgments:

SLIGHT	Symptoms are <i>present</i> , but modest and only on occasion. They usually are of no clinical concern. Clinical intervention is <i>not</i> called for on the basis of these symptoms alone.
MILD (Sub-Syndrome)	These symptoms are not sufficiently problematic to require active treatment in-and-of themselves. However, they are <i>troublesome</i> to the patient and/or others, usually in a more-or-less transient way. These symptoms may occasionally be associated with minor social, occupational, or family difficulties. <i>Treatment</i> should be viewed as <i>optional</i> .
SYNDROME	Symptoms are sufficiently problematic to <i>justify</i> a clinical syndrome <i>diagnosis</i> . Difficulties in social, occupational, and/or family functioning are <i>clearly present</i> and <i>treatment</i> is <i>advisable</i> .
MARKED SYNDROME	Signifies the presence of <i>serious</i> and/or <i>prolonged</i> clinical symptoms that markedly <i>impair</i> all <i>psychosocial functions</i> . They call for careful professional assessment and determination as to whether <i>treatment</i> should be in an <i>inpatient</i> or <i>outpatient</i> setting.
EXTREME SYNDROME	Symptoms are <i>intense</i> and <i>dysfunctional</i> , as well as socially and occupationally <i>debilitating</i> . They generally necessitate treatment in an <i>inpatient/residential</i> setting.

Figure 4.33 (continued)

STEP 3: RATE UP TO THREE OF THIS PATIENT'S MOST PROMINENT PERSONALITY FEATURES AND THEIR SEVERITY. Next, turn to the Personality grid in the upper right column. Identify the *one* personality pattern that most closely "fits" or captures the patient's characteristic way of functioning according to the descriptions given in Part B. Typically, few patients fit only one personality type; they often possess features or domain characteristics of 2 or 3 patterns. As you have seen in the Ratings section for the Clinical Syndromes, there are columns for a second and third choice. The *second* best fit can be recorded under the second column. If a *third* "close fit" also seems applicable, fill in the appropriate circle under the third column.

Next, rate the severity of these personality patterns in the bottom part of the grid. Note that severity judgments are not exclusive; that is, two patterns may both be judged as being disorders or traits, for example.

Use the following definitions as a guide when making severity judgments:

TRAIT	The presence of <i>minor</i> , but <i>well-defined</i> , personality features. Currently present are a mix of manifest, but <i>clinically insignificant</i> traits that fall clearly in the range of <i>normality</i> .
STYLE	A distinctive and characteristic <i>configuration</i> of personality traits that is essentially <i>subclinical</i> in nature, i.e., although there are <i>occasional adaptive difficulties</i> , treatment on these grounds alone is <i>not indicated</i> .
DISORDER	The personality pattern is sufficiently problematic to justify a <i>clinical diagnosis</i> . Characteristics <i>definitely impair life functioning</i> , resulting in periodic, but <i>significant, adaptive difficulties</i> . (Note: schizotypal, paranoid, and borderline personality ratings should be judged at least at the level of "disorders" rather than "styles" or "traits.") Outpatient <i>treatment is indicated</i> .
MARKED DISORDER	Personality characteristics are of a <i>severe</i> and <i>persistent</i> nature. They <i>markedly and repeatedly impair</i> psychosocial functioning. <i>Treatment is definitely called for</i> , mostly on an outpatient basis, but frequently in an inpatient setting.
EXTREME DISORDER	Personality features are <i>intense, chronic, and pervasive</i> , often of an idiosyncratic and graphic character, and invariably <i>debilitating</i> . <i>Inpatient/residential treatment is required</i> .

STEP 4: RECORD YOUR CONFIDENCE IN THE PRECEDING RATINGS. The degree to which you feel sure about your rating depends in large measure on the complexity of the case, and the extent to which you have had an opportunity to study its symptoms and characteristics. Using the guide that follows, indicate in the appropriate grids the degree of your confidence regarding each of your rated syndromes and disorders.

HIGH	Judgment based on considerable information and a secure understanding of the symptoms/features rated above; these judgments may be assumed to be accurate.
MEDIUM	Judgment based on reasonably good information and an adequate clinical understanding of the symptoms/features rated above. More precise appraisals, however, may result with additional information.
LOW	Judgments are modestly informed; an understanding of the symptoms/features rated above is only of a surface nature at this time. Additional information is likely to increase the validity of these judgments.

Figure 4.33 (continued)

STEP 5: RECORD THE ACTUAL MCMI-III BR SCORES OF THIS PATIENT. *Recall:* Do not review the MCMI-III BR scores of your patients before you rate them in the sections below the dark line. Write in their BR score numbers in both columns *after* completing your ratings.

IV. ILLUSTRATIVE CLINICAL RATING FORM

On the next page we have printed a sample of a completed Clinician's Rating Form. As you can see, the patient's gender and age were completed by the clinician. Also completed were 3 judgments in the Clinical Syndromes Axis I grid. "Dysthymia" was placed in the first column to signify that it was judged the most prominent or clinically significant syndrome. Alcohol Abuse was placed in the second column; and Somatoform was rated in the third column. As you can note, Dysthymia was also judged to be a "marked syndrome," whereas the other 2 syndromes were rated "mild." In the Personality grid, Schizotypal was judged the best fit and hence was placed in the first column. Avoidant, judged to be second "best fit," was recorded in the second column, and Negativistic was assigned to the third column; all other personalities were left blank. Both Schizotypal and Avoidant personalities were rated as "Disorders," whereas Negativistic was judged a "Style."

All of the clinician's confidence ratings were marked as "medium," except for Schizotypal, which was judged to have been "high." At the bottom of the form are the MCMI-III BR scores for this patient as completed by the clinician.

Figure 4.33 (continued)

Clinician's Rating Form MCMI VALIDATION STUDY																																																																																																																																																																																																																																
Instructions: Use the <i>Clinician Rating Reference Booklet</i> to guide your thinking about the clinical syndrome and personality ratings, to be made on the two upper columns. Below the dark line, fill in the MCMI-III BR scores obtained by the patient you have just rated. Also fill in the patient's age and gender. We will record a letter to represent your setting and patient's number.																																																																																																																																																																																																																																
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<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="text-align: center;">MCMI SCALE</th> <th style="text-align: center;">Actual BR</th> </tr> </thead> <tbody> <tr><td>Anxiety</td><td style="text-align: center;">78</td></tr> <tr><td>Somatoform</td><td style="text-align: center;">84</td></tr> <tr><td>Bipolar</td><td style="text-align: center;">23</td></tr> <tr><td>Dysthymia</td><td style="text-align: center;">88</td></tr> <tr><td>Alcohol Abuse</td><td style="text-align: center;">95</td></tr> <tr><td>Drug Abuse</td><td style="text-align: center;">67</td></tr> <tr><td>PTSD</td><td style="text-align: center;">70</td></tr> <tr><td>Thought Disorder</td><td style="text-align: center;">80</td></tr> <tr><td>Major Depression</td><td style="text-align: center;">70</td></tr> <tr><td>Paranoid Delusions</td><td style="text-align: center;">47</td></tr> </tbody> </table>							MCMI SCALE	Actual BR	Anxiety	78	Somatoform	84	Bipolar	23	Dysthymia	88	Alcohol Abuse	95	Drug Abuse	67	PTSD	70	Thought Disorder	80	Major Depression	70	Paranoid Delusions	47																																																																																																																																																																																																				
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Figure 4.33 (continued)

PART B AXIS I CRITERIA

Anxiety Disorder

The parallel MCMI-III scale is intended to assess the general symptoms of anxiety and to operate as a screener for the more specific Anxiety Disorders. No one *DSM-IV* criteria set exactly parallels the item content of this scale. High-scoring subjects report such cognitive symptoms as excessive worry (Generalized Anxiety Disorder) and obsessional thoughts (Obsessive-Compulsive Disorder), such cognitive-somatic symptoms as feeling tense and panicky when away from home (Agoraphobia and Panic Disorder), somatic symptoms such as sweating, and behavioral symptoms such as compulsive checking to reduce anxiety (Obsessive-Compulsive Disorder).

Somatoform Disorder

The MCMI-III Somatoform Disorder scale represents a constellation of physical symptoms, including difficulty keeping balance, feeling weak and tired much of the time, loss of sensation of various parts of the body, difficulty sleeping, and loss of appetite.

DSM-IV criteria for Undifferentiated Somatoform Disorder are as follows:

1. One or more physical complaints (e.g., fatigue, loss of appetite, gastrointestinal or urinary complaints).
2. Either (1) or (2):
 - (1) after appropriate investigation, the symptoms cannot be fully explained by a known general medical condition or the direct effects of a substance (e.g., a drug of abuse, a medication)
 - (2) when there is a general medical condition, the physical complaints or resulting social or occupational impairment is in excess of what would be expected from the history, physical examination, or laboratory findings.
3. The symptoms have endured at least 6 months and have caused clinically significant distress or impairment in social, occupational, or other important areas of functioning.

Figure 4.33 (continued)

PART B AXIS II CRITERIA SCHIZOID PERSONALITY DISORDER	
Millon Clinical Domains	DSM-IV Criteria
Behavioral Level: (F)^a Expressively Impassive (e.g., appears to be in an inert emotional state, lifeless, undemonstrative, lacking in energy and vitality; is unmoved, boring, unanimated, robotic, phlegmatic, displaying deficits in activation, motoric expressiveness, and spontaneity). (F) Interpersonally Unengaged (e.g., seems indifferent and remote, rarely responsive to the actions or feelings of others, chooses solitary activities, possesses minimal "human" interests; fades into the background, is aloof or unobtrusive, neither desires nor enjoys close relationships, prefers a peripheral role in social, work, and family settings). Phenomenological Level: (F) Cognitively Impoverished (e.g., seems deficient across broad spheres of human knowledge and evidences vague and obscure thought processes, particularly about social matters; communication with others is often unfocused, loses its purpose or intention, or is conveyed via a loose or circuitous logic). (S)^b Complacent Self-Image (e.g., reveals minimal introspection and awareness of self; seems impervious to the emotional and personal implications of everyday social life, appearing indifferent to the praise or criticism of others). (S) Meager Objects (e.g., internalized representations are few in number and minimally articulated, largely devoid of the manifold percepts and memories of relationships with others, possessing little of the dynamic interplay among drives and conflicts that typify well-adjusted persons). Intrapsychic Level: (F) Intellectualization Mechanism (e.g., describes interpersonal and affective experiences in a matter-of-fact, abstract, impersonal or mechanical manner; pays primary attention to formal and objective aspects of social and emotional events). (S) Undifferentiated Organization (e.g., given an inner barrenness, a feeble drive to fulfill needs, and minimal pressures either to defend against or resolve internal conflicts or cope with external demands, internal morphologic structures may best be characterized by their limited framework and sterile pattern). Biophysical Level: (S) Apathetic Mood (e.g., is emotionally unexcitable, exhibiting an intrinsic unfeeling, cold, and stark quality; reports weak affectionate or erotic needs, rarely displaying warm or intense feelings, and apparently unable to experience most affects—pleasure, sadness, or anger—in any depth).	A. A pervasive pattern of detachment from social relationships and a restricted range of expression of emotions in interpersonal settings, beginning by early adulthood and present in a variety of contexts, as indicated by four (or more) of the following: (1) neither desires nor enjoys close relationships, including being part of a family (2) almost always chooses solitary activities (3) has little, if any, interest in having sexual experiences with another person (4) takes pleasure in few, if any, activities (5) lacks close friends or confidants other than first-degree relatives (6) appears indifferent to the praise or criticism of others (7) shows emotional coldness, detachment, or flattened affectivity B. Does not occur exclusively during the course of Schizophrenia, a Mood Disorder With Psychotic Features, or another Psychotic Disorder and is not due to the direct physiological effects of a general medical condition. Note: If criteria are met prior to the onset of Schizophrenia, add "Premorbid," e.g., "Schizoid Personality Disorder (Premorbid)."
^aF = Functional ^bS = Structural	

sion was similar for all raters. In order to obtain an adequate sample size, clinicians were allowed to include subjects who had already taken the MCMI-III test, but they were specifically instructed not to include subjects for whom they had a clear recollection of the results. This was done to minimize criterion contamination.

Part B of the Clinician's Rating Reference Booklet standardized the prototypes against which patients were rated. This part was divided into two sections, Axis I and Axis II. For the Axis I section, it was suggested that clinicians closely follow the *DSM-IV* criteria. Where no exact *DSM-IV* parallel to a particular MCMI-III scale existed (as, for example, with Anxiety; see the booklet excerpt in Figure 4.33), descriptive paragraphs were written and provided to clinicians.

In contrast to the constructs in the Axis I section of the booklet, the Axis II constructs were operationalized through both the *DSM-IV* diagnostic criteria and the theoretically generated Millon domain descriptions presented in Chapter 2 of this manual. This approach was chosen because, from the perspective of the evolutionary theory, the *DSM-IV* is only partially representative of domains in which personality is expressed. It was important for clinicians to be aware of the full range of content relevant to the Axis II constructs when completing their ratings. Otherwise, their ratings would standardize around the incomplete criteria of the *DSM-IV* or, worse, around content to which each clinician was sensitized by his or her academic and clinical training, whether cognitive, behavioral, psychodynamic, or other.

Results: Axis II

The results of the diagnostic validity study for the MCMI-III Axis II scales are presented in Tables 4.1 and 4.2. Results for the Axis II scales are discussed first, followed by results for the Axis I scales, and then by statistics summarizing diagnostic performance across three generations of the MCMI test. Although the methods of data gathering for each MCMI version have differed somewhat in terms of the precise instructions given to clinical raters, the diagnostic standards in use at the time, sampling procedures, and so on, the *essential* requirement of using "informed" clinical raters was standard and uni-

form. That is, studies across all three generations were based on judgments made by clinicians with substantial knowledge of the patients who were rated (in contrast to the MCMI-III diagnostic validity study that was reported in the first edition of this manual).

Frequency Statistics

Table 4.1 reports the raw diagnostic frequencies and prevalence rates for the patients included in this external validity study. To illustrate, the first column of Table 4.1 indicates that 18 patients were rated as *primarily* schizoid by their clinician, 34 were rated as primarily avoidant, and so on. The most frequently rated primary personality disorder was dependent ($n = 41$), followed by borderline ($n = 40$), followed by the depressive personality diagnosis ($n = 37$) that was new to *DSM-IV*. The second column of Table 4.1 translates these raw frequencies into prevalence or base rates (for example, dependent at 13%, borderline and depressive at 12% each). The third and fourth columns report the frequency and prevalence rates for personalities judged by clinicians as second most prominent. Interestingly, the *DSM-IV*'s newest personality category (depressive) and its deleted category (masochistic) were among the top three disorders judged as playing a frequent but secondary role for these patients. The fifth and sixth columns of Table 4.1 summarize the combined raw frequencies and prevalence rates for the primary and secondary diagnoses as judged by participating clinicians. The avoidant, depressive, dependent, and borderline personalities rank highest. Those judged least prevalent were the paranoid, schizotypal, and sadistic personalities.

The final three columns of Table 4.1 report the frequency with which each MCMI-III personality scale was the highest or second highest in the patient's personality scale profile. Depressive ($n = 43$) and Borderline ($n = 34$) were most frequently the highest MCMI-III Axis II scale. The Depressive ($n = 85$) and Avoidant ($n = 70$) scales were most often the highest or second highest, as shown in the far right column. Those scales that were least often the highest or second highest were the Sadistic ($n = 21$), Schizotypal ($n = 24$), and Paranoid ($n = 27$) scales. By comparing the fifth and ninth columns of Table 4.1, we can see that there is general agreement

Table 4.1 Primary and Secondary Diagnoses Based on Clinician Judgment and the MCMI-III Test (Axis II Disorders)

	Primary Diagnosis		Secondary Diagnosis		Primary or Secondary Diagnosis		MCMI Base Rates		
	Clinician Judgment (N = 321)	Prevalence (%)	Clinician Judgment (N = 317)	Prevalence (%)	Clinician Judgment (N = 638)	Prevalence (%)	Primary (N = 322)	Secondary (N = 322)	Primary or Secondary (N = 644)
Schizoid	18	6	20	6	38	6	15	21	36
Avoidant	34	11	37	12	71	11	30	40	70
Depressive	37	12	32	10	69	11	43	42	85
Dependent	41	13	28	9	69	11	27	23	50
Histrionic	23	7	19	6	42	7	27	13	40
Narcissistic	22	7	25	8	47	7	18	27	45
Antisocial	18	6	25	8	43	7	22	24	46
Sadistic	7	2	16	5	23	4	7	14	21
Compulsive	30	9	9	3	39	6	28	12	40
Negativistic	16	5	28	9	44	7	18	21	39
Masochistic	12	4	42	13	54	8	23	41	64
Schizotypal	11	3	11	3	22	3	15	9	24
Borderline	40	12	18	6	58	9	34	23	57
Paranoid	12	4	7	2	19	3	15	12	27

between the clinical judges and the MCMI-III results concerning which personality disorders were most and least prevalent in this sample.

Diagnostic Efficiency

To establish concurrent validity, a newly constructed continuous scale is correlated with already established and accepted measures of the same construct. If the new scale correlates highly with these older measures, it is said to have concurrent validity.

Although dimensional scales lend themselves readily to the calculation of concurrent validity correlations, the *DSM-IV* casts psychological disorders as disease entities to be operationalized as present or absent through the use of diagnostic criteria and their associated diagnostic efficiency statistics. The MCMI diagnostic validity results presented here focus on sensitivity and positive predictive power. Because these tend to be inversely related, an intelligent evaluation of an instrument requires that they be considered con-

jointly. Either can be almost perfect while the other is atrocious.

For example, assume that 100 patients in a sample of 1,000 are labeled by clinical judges as thought disordered. If all 100 are identified using a thought disorder scale, the sensitivity of the scale is perfect. If, however, the other 900 patients are also identified by the scale as thought disordered, then the appraisal of its utility is changed dramatically. Here, perfect sensitivity is gained at the expense of a large number of false positives.

Conversely, if all the patients identified as thought disordered by the scale are also labeled as such by the clinical judges, the positive predictive power of the instrument is perfect. If, however, the scale succeeds in identifying only 1 out of a total of 100 thought-disordered patients in the sample, the appraisal is again dramatically changed. These two examples are merely illustrative. In general, when the sensitivity of an indicator increases, false positives also increase, and positive predictive power declines. Thus,

sensitivity and positive predictive power must be considered together to get a full picture of an instrument's diagnostic efficiency.

Diagnostic efficiency statistics for the MCMI-III Axis II scales are reported in Table 4.2. Statistics were calculated in terms of clinician-judged *primary* personality disorder diagnoses and their correspondence to the MCMI-III results. The positive predictive power (PPP) statistic represents the percentage of patients having a particular MCMI-III personality scale as highest in their profile who were in fact judged by clinicians as having that personality disorder as primary. Consequently, PPP is equal to the number of true positives (positive according to both clinical judgment and the test) divided by the total number of test positives. For example, 67% of the patients with Schizoid as their most elevated MCMI-III personality scale were judged as primarily schizoid by their clinician (see the row labeled Schizoid and the column labeled PPP in Table 4.2). Likewise, 79% of those having Compulsive as their highest MCMI-III Axis II scale were judged primarily compulsive by their clinician. The best PPP results were obtained for the Dependent (81%), Paranoid (79%), and Compulsive (79%) scales. The least impressive PPP values were obtained for the

Masochistic (30%), Negativistic (39%), and Depressive (49%) scales. The lower PPP results for these scales may reflect the tenuous manner in which the *DSM-IV* has treated these constructs. The self-defeating (masochistic) disorder was completely deleted from the *DSM-IV*, and the negativistic (passive-aggressive) and depressive personality disorders were relegated to the *DSM-IV* appendix.

Sensitivity refers to how well an indicator performs in detecting a particular personality pattern or disorder. The sensitivity (SENS) statistic reported in Table 4.2 is the percentage of those patients who were identified by clinicians as having a particular personality as their primary pattern who also had their highest MCMI-III personality score on the scale corresponding to that pattern. Thus, sensitivity is the number of patients found to be positive by both the test and the clinician divided by the total number judged positive by clinicians. For example, 57% of those judged as having a primary depressive personality by clinicians also had Depressive as the highest scale in their MCMI-III personality profile.

The positive predictive ratio (PPR) is intended as a rough measure of increment over chance. For example, if the prevalence rate of a disorder is 12% and an indicator yields no im-

Table 4.2 Diagnostic Efficiency of the Axis II MCMI-III Scales

		PPP ^a (%)	SENS ^b (%)	PPR ^c
1	Schizoid	67	56	11.89
2A	Avoidant	73	65	6.92
2B	Depressive	49	57	4.24
3	Dependent	81	54	6.38
4	Histrionic	63	74	8.79
5	Narcissistic	72	59	10.54
6A	Antisocial	50	61	8.92
6B	Sadistic	71	71	32.76
7	Compulsive	79	73	8.41
8A	Negativistic	39	44	7.80
8B	Masochistic	30	58	8.14
S	Schizotypal	60	82	17.41
C	Borderline	71	60	5.66
P	Paranoid	79	92	21.02

^aPPP = positive predictive power

^bSENS = sensitivity

^cPPR = positive predictive ratio

provement over chance at all, it will correctly identify 12% of patients. An indicator or test that is performing well should obviously perform at better than chance levels. The PPR statistic is calculated by dividing the PPP by the prevalence rate. For example, the PPR shown in Table 4.2 for the Avoidant scale is 6.92, or close to 7 times greater than what would be produced by chance alone. Obviously, where the prevalence rate is high, PPR has a low upper bound. For example, if the prevalence rate for a particular disorder was 50%, a test could be 100% accurate in identifying the presence of the disorder and yet have a PPR of only 2 (100/50). In contrast, where the prevalence rate of a disorder is 2% and the test obtains a PPP of 10%, the PPR is 5. Accordingly, the PPR incremental accuracy values shown in Table 4.2 tend to be highest for the disorders with low prevalence rates, for example, sadistic (≈ 33), paranoid (≈ 21), schizotypal (≈ 17), and schizoid (≈ 12). Lesser yet quite impressive PPR values are obtained for disorders with high prevalence rates, for example, depressive (≈ 4), borderline (≈ 6), dependent (≈ 6), and avoidant (≈ 7). As an index of how much more accurate the MCMI-III test (or any other instrument) is compared to uninformed diagnostic assignments, the PPR is a most useful diagnostic efficiency statistic. We will expand our discussion of the diagnostic efficiency of the Axis II scales later in this chapter.

Results: Axis I

Tables 4.3 and 4.4 present Axis I diagnostic efficiency statistics for MCMI scales at $BR \geq 85$ and $BR \geq 75$, respectively, across three generations of the MCMI test. We begin by continuing our discussion of results of the MCMI-III diagnostic validity study, focusing specifically on the positive predictive power of the MCMI-III Axis I scales. Two methods of calculation are presented. In the rightmost column of Table 4.3 (headed M-III^b), the BR 85 cutting score is evaluated against what clinicians judged to be the patient's most prominent Axis I disorder. The question is this: to what extent does the BR 85 cutting score correctly predict the patient's most prominent Axis I disorder? Positive predictive power statistics range from 33% for Delusional Disorder to 93% for Drug Dependence. The scales for the two Axis I disorders that were most frequently diagnosed in this sample (as shown in the Preva-

lence Rates section of Table 4.3), Anxiety and Dysthymia, achieved PPP values of 75% and 81%, respectively.

Sensitivity statistics (in the M-III^b column) for this method of calculation ranged from 24% for Somatoform Disorder to 100% for Thought Disorder. The Sensitivity column labeled M-III^a evaluates the following question: to what extent does the BR 85 cutting score detect a disorder judged by clinicians as present, regardless of whether or not that disorder is the patient's most prominent disorder? Sensitivity values here are generally somewhat lower.

Table 4.4 evaluates the same diagnostic questions with reference to the BR 75 cutting score. As expected, sensitivity values are higher and positive predictive power values are somewhat lower than for BR 85.

Three Generations of the MCMI Test: Axis I

Tables 4.3 and 4.4 also provide comparisons of sensitivity and positive predictive power statistics at $BR \geq 85$ and $BR \geq 75$ across three generations of MCMI Axis I scales. These statistics must be viewed as generally favorable for all versions of the MCMI test; however, their comparability across the three versions requires assumptions that may not be sustainable. The changes in diagnostic criteria that have occurred across three generations of the *DSM*, together with changes that sometimes occur in the fashionableness of certain diagnoses, make absolute comparisons difficult.

In Table 4.3, the columns labeled M-III^a present prevalence, sensitivity, and positive predictive power at $BR \geq 85$ for all disorders judged to be present by clinicians (prevalence was derived from clinical judgments) in the MCMI-III validity study. The prevalence statistics here are higher than for the MCMI and MCMI-II validity studies. This helps explain why the MCMI-III sensitivity values shown in the M-III^a column are lower for several scales than those shown for the MCMI and MCMI-II tests. In particular, the sensitivity of the Somatoform scale is relatively low. However, the sensitivity and positive predictive power of the severe Axis I syndrome scales—Thought Disorder, Major Depression, and Delusional Disorder—are higher than they were for the MCMI and MCMI-II tests.

The columns labeled M-III^b in Table 4.3 present prevalence, sensitivity, and positive

predictive power at $BR \geq 85$ for those Axis I disorders judged by clinicians as being most prominent. When calculated in this manner, the prevalence rates for the Axis I disorders in the MCMI-III validity study are decreased to more closely approximate those found in the MCMI and MCMI-II studies, thus allowing a better comparison among the three instruments. Although the MCMI-III sensitivity statistics are mixed relative to those for the MCMI-II test, positive predictive power statistics are generally higher, and in some cases much higher, than for the MCMI-II test.

Table 4.4 presents the Axis I comparison across three MCMI generations at $BR \geq 75$. Statistics were calculated using all disorders judged by clinicians to be present. Accordingly, the prevalence rates reported for the MCMI-III study in Table 4.4 are the same as those shown in column M-III^a of Table 4.3.

In summary, it would appear from this set of analyses that the diagnostic criterion validity of the MCMI-III test is comparable and, for some Axis I scales, marginally superior to that of the MCMI-II test.

Three Generations of the MCMI Test: Axis II

We begin this section by again noting that comparing across generations of MCMI validity studies is problematic. The structure and content of the MCMI Axis II scales have often been informed by changes in Millon's theory and in the official *DSM* nosology. Additions have included the Aggressive (Sadistic) and Self-Defeating scales from the MCMI test to the MCMI-II test, and the addition of the Depressive scale to the MCMI-III test. Moreover, the addition of these scales has been accompanied by a degree of repositioning of the content of others. The Antisocial and Passive-Aggressive scales from the MCMI test were repositioned to make room for the Aggressive (Sadistic) and Self-Defeating scales in the MCMI-II test. Also, the item content of the MCMI-III test reflects the parallel *DSM* criteria much more than was the case with the MCMI-II test. Accordingly, the comparability of diagnostic efficiency statistics across time is not something that can be directly quantified but instead becomes a matter of judgment. Obviously, comparisons between the MCMI and

Table 4.3 Prevalence, Sensitivity, and Positive Predictive Power of MCMI Axis I Scales at $BR \geq 85$

		Prevalence Rates (%)				Sensitivity (%)				Positive Predictive Power (%)			
		M	M-II	M-III ^a	M-III ^b	M	M-II	M-III ^a	M-III ^b	M	M-II	M-III ^a	M-III ^b
A	Anxiety	24	23	46	28	67	69	57	64	73	74	80	75
H	Somatoform	13	5	24	12	37	43	25	24	44	33	53	39
N	Bipolar: Manic	4	3	6	4	49	40	44	64	50	53	57	58
D	Dysthymia	26	29	38	23	73	76	55	65	76	72	88	81
B	Alcohol												
	Dependence	12	8	30	17	56	79	65	80	66	88	91	88
T	Drug												
	Dependence	6	9	19	11	52	62	78	82	63	78	92	93
R	Post-Traumatic												
	Stress	—	—	18	6	—	—	53	88	—	—	73	67
	Disorder ^c												
SS	Thought												
	Disorder	3	2	10	4	15	20	52	100 ^d	19	30	59	52
CC	Major												
	Depression	3	6	20	11	24	47	65	84	27	50	72	66
PP	Delusional												
	Disorder	2	1	9	2	15	25	40	50	23	40	63	33

Note. M = MCMI, M-II = MCMI-II, M-III = MCMI-III.

^aStatistic calculated using all disorders judged by clinicians as present.

^bStatistic calculated using disorder judged by clinicians as most prominent.

^cPost-traumatic stress disorder was not assessed by the MCMI and MCMI-II.

^dOnly 12 thought-disordered patients participated in the study.

Table 4.4 Prevalence, Sensitivity, and Positive Predictive Power of MCMI Axis I Scales at BR ≥ 75

		Prevalence Rates (%)			Sensitivity (%)			Positive Predictive Power (%)		
		M	M-II	M-III	M	M-II	M-III	M	M-II	M-III
A	Anxiety	34	39	46	88	80	88	79	84	70
H	Somatoform	18	13	24	42	57	59	51	53	49
N	Bipolar: Manic	8	4	6	56	42	67	55	55	55
D	Dysthymia	41	46	38	91	81	85	84	80	61
B	Alcohol Dependence	17	15	30	74	87	86	63	92	83
T	Drug Dependence	11	13	19	78	72	92	71	82	89
R	Post-Traumatic Stress Disorder ^a	—	—	18	—	—	82	—	—	52
SS	Thought Disorder	6	4	10	39	37	74	45	50	51
CC	Major Depression	6	14	20	41	68	85	46	73	51
PP	Delusional Disorder	5	5	9	32	34	77	36	61	61

Note. M = MCMI, M-II = MCMI-II, and M-III = MCMI-III. Statistics calculated using all disorders judged by clinicians as present.

^aPost-traumatic stress disorder was not assessed by the MCMI and MCMI-II.

MCMI-II tests and between the MCMI-II and MCMI-III tests are likely to be less problematic than comparisons between the MCMI and MCMI-III tests.

Table 4.5 presents sensitivity and PPP values for the Axis II scales using clinical judgment of primary diagnosis. Table 4.6 presents the same information for clinical judgment of primary and secondary diagnosis combined. A comparison across the columns of Tables 4.5 and 4.6 shows, with some exceptions, generally good conformity between the MCMI results and the expert judges across all three generations of the test. In terms of individual scales, the lowest MCMI-III PPP values using the primary-diagnosis-only criterion (Table 4.5) were obtained for the Masochistic scale (30%; this disorder was deleted from the *DSM-IV*) and for the scales corresponding to the two disorders that were moved to the *DSM-IV* appendix: Negativistic (39%) and Depressive (49%). The lowest MCMI-III sensitivity statistics using this criterion were for the Negativistic (44%) and Dependent (54%) scales. Using the combined primary or secondary diagnosis criterion (Table 4.6), the sensitivity statistics are higher.

Because it is difficult to form conclusions about MCMI-III validity compared to MCMI-II validity by digesting the results presented in Tables 4.5 and 4.6 in toto, these statistics have been

expressed as frequency distributions in Table 4.7. These distributions show that the sensitivity and PPP values for the MCMI-III test are somewhat better overall.

In summary, insofar as Axis II is concerned, a direct comparison of diagnostic efficiency analyses of the MCMI, MCMI-II, and MCMI-III tests shows that each form of the instrument achieved a satisfactory to high level of predictive and clinical accuracy. When clinicians are well informed about the theoretical and clinical grounding of personality constructs and, most important, when they know the patients quite well, the results of the MCMI-III test are very impressive. The shift in Axis II diagnostic efficiency statistics from the MCMI test to the MCMI-II and MCMI-III tests shows a modest but general upward trend in the instrument's sensitivity and positive predictive power.

Correlations Between MCMI-III Scores and Collateral Scale Scores

Diagnostic efficiency is only one means of assessing external validity, one that suffers from a number of weaknesses, not the least of which is the dichotomous nature of the comparisons to be made. In contrast, the use of self-report collateral instruments not only allows correlations to be

Table 4.5 Diagnostic Efficiency Statistics for Axis II Primary Diagnosis by Clinical Judgment Over Three Generations of the MCMI Test

		SENS ^a (%)			PPP ^b (%)		
		MCMI	MCMI-II	MCMI-III	MCMI	MCMI-II	MCMI-III
1	Schizoid	38	33	56	50	38	67
2A	Avoidant	67	67	65	70	72	73
2B	Depressive	— ^c	— ^c	57	— ^c	— ^c	49
3	Dependent	63	72	54	69	70	81
4	Histrionic	58	53	74	65	48	63
5	Narcissistic	50	59	59	65	55	72
6A	Antisocial	42	60	61	55	68	50
6B	Sadistic	— ^c	68	71	— ^c	58	71
7	Compulsive	53	66	73	53	61	79
8A	Negativistic	47	59	44	62	50	39
8B	Masochistic	— ^c	48	58	— ^c	46	30
S	Schizotypal	58	38	82	74	47	60
C	Borderline	60	67	60	64	60	71
P	Paranoid	54	31	92	79	40	79

^aSENS = Sensitivity

^bPPP = Positive predictive power

^cThis construct was not assessed in this version of the MCMI test.

Table 4.6 Diagnostic Efficiency Statistics for Axis II Primary and Secondary Diagnoses by Clinical Judgment Over Three Generations of the MCMI Test

		SENS ^a (%)			PPP ^b (%)		
		MCMI	MCMI-II	MCMI-III	MCMI	MCMI-II	MCMI-III
1	Schizoid	88	62	68	68	71	72
2A	Avoidant	88	76	62	80	79	63
2B	Depressive	— ^c	— ^c	75	— ^c	— ^c	61
3	Dependent	80	79	56	80	76	78
4	Histrionic	79	68	75	79	65	79
5	Narcissistic	74	74	72	59	69	77
6A	Antisocial	62	71	81	61	80	76
6B	Sadistic ^d	— ^c	78	74	— ^c	71	81
7	Compulsive	58	73	74	57	67	76
8A	Negativistic ^d	78	72	59	73	64	67
8B	Masochistic ^d	— ^c	66	85	— ^c	58	73
S	Schizotypal	74	57	73	68	69	67
C	Borderline	77	72	79	71	66	81
P	Paranoid	71	50	89	68	65	65

^aSENS = Sensitivity

^bPPP = Positive predictive power

^cThis construct was not assessed in this version of the MCMI test.

^dSadistic, Negativistic, and Masochist were referred to as Aggressive (Sadistic), Passive-Aggressive, and Self-Defeating, respectively, in the MCMI-II test.

**Table 4.7 Frequency of SENS and PPP Levels for the MCMI-II and MCMI-III Tests
(Axis II Diagnoses)**

Value of SENS and PPP Statistics (%)	Combined Primary and Secondary Diagnoses				Primary Diagnosis			
	SENS MCMI-II	SENS MCMI-III	PPP MCMI-II	PPP MCMI-III	SENS MCMI-II	SENS MCMI-III	PPP MCMI-II	PPP MCMI-III
30 to 39					3		1	2
40 to 49					1	1	4	
50 to 59	2	2	1		3	4	3	1
60 to 69	2	2	7	4	5	3	3	3
70 to 79	9	6	4	7	1	3	2	6
80 to 89		3	1	2		1		1
≥90						1		

Note. The Depressive personality is not included in these results because it was not part of the MCMI-II test.

calculated from continuous measures but also allows the comparison of the MCMI-III test to traditional gauges that purport to measure similar constructs. Clinicians participating in the MCMI-III development project required their subjects to complete any of several collateral tests, listed with their subscales below:

Beck Depression Inventory (BDI; Beck & Steer, 1987)

General Behavior Inventory (GBI; Depue et al., 1981)

Depression
Hypomania
Total

Michigan Alcoholism Screening Test (MAST; Selzer, 1971)

Impact of Event Scale (IOES; Horowitz, Wilner, & Alvarez, 1979)

State-Trait Anxiety Inventory (STAI; Spielberger, 1983)

State Anxiety
Trait Anxiety

Symptom Checklist-90-Revised (SCL-90-R; Derogatis, 1994)

Somatization (SOM)
Obsessive-Compulsive (O-C)
Interpersonal Sensitivity (INT)
Depression (DEP)
Anxiety (ANX)
Hostility (HOS)

Phobic Anxiety (PHOB)

Paranoid Ideation (PAR)

Psychoticism (PSY)

Global Severity Index (GSI)

Positive Symptom Distress Index (PSDI)

Positive Symptom Total (PST)

MMPI-2

Hypochondriasis (Hs)

Depression (D)

Hysteria (Hy)

Psychopathic Deviate (Pd)

Masculinity-Femininity (Mf)

Paranoia (Pa)

Psychasthenia (Pt)

Schizophrenia (Sc)

Hypomania (Ma)

Social Introversion (Si)

Table 4.8 shows the correlations between MCMI-III base rate scores and scores from these collateral tests. The results are discussed in the following paragraphs.

A general maladjustment factor produced fairly high correlations between the Beck Depression Inventory (BDI) and a large number of MCMI-III scales. Notable exceptions were the Histrionic (−.49), Narcissistic (−.40), and Compulsive (−.30) scales. The highest correlations between the BDI and the MCMI-III scales were, as expected, with Major Depression and Dysthymia at .74 and .71, respectively. Among the personality scales, the highest correlations were obtained with the Depressive (.56), Borderline

(.56), Masochistic (.53), and Schizoid (.53) scales. Weak relationships were found between the BDI and the Antisocial (.20) and Sadistic (.22) scales.

Results with the General Behavior Inventory (GBI) showed a similar pattern. The GBI Depression scale correlated above .60 with several relevant MCMI-III measures, including the Major Depression, Borderline, Dysthymia, and Depressive scales.

The Michigan Alcoholism Screening Test (MAST) corroborates the validity of the MCMI-III Alcohol Dependence scale with a .67 correlation. The MAST and the MCMI-III Antisocial scale correlated .39, reflecting the frequent covariation of antisocial traits with alcoholism.

The Impact of Event Scale (IOES) typically serves as a measure of the presence of persistent stressful memories. The IOES correlated highest with the MCMI-III Post-Traumatic Stress Disorder (.61), Dependent (.60), and Masochistic (.61) scales. Negative correlations were obtained with the Histrionic (−.37), Narcissistic (−.36), Drug Dependence (−.26), and Antisocial (−.16) scales.

Somewhat surprisingly, the state and trait anxiety gauges of the State-Trait Anxiety Inventory (STAI) correlated more highly with the MCMI-III depression scales (Dysthymia and Major Depression) than with the Anxiety scale. As anticipated, the relevant MCMI-III personality disorder scales correlated more highly with the STAI Trait Anxiety scale than with STAI State Anxiety.

The Symptom Checklist-90-Revised (SCL-90-R) instrument is a widely used screening measure that encompasses a range of symptom clusters. The MCMI-III Somatoform scale correlated .55 with the SCL Somatization scale. A .56 correlation was obtained between the SCL Somatization scale and the MCMI-III Major Depression scale, not a surprising result given the vegetative characteristics of severe depressive disorders. Although the SCL Obsessive-Compulsive scale does not appraise personality traits, it does reflect the presence of thought obsessions and behavioral compulsions, symptoms that are increasingly recognized as being biologically driven. This scale correlated most highly with the MCMI-III Schizotypal and Thought Disorder scales, at .66 and .67, respectively. The SCL Interpersonal Sensitivity measure was most highly

related to the MCMI-III Schizotypal (.62) and Avoidant (.60) scales. Among the highest correlations for the SCL Depression scale were those with the MCMI-III Major Depression (.74), Dysthymia (.70), and Depressive (.65) scales. The SCL Anxiety scale was highly correlated with the MCMI-III Major Depression (.62), Dysthymia (.58), Thought Disorder (.64), and Anxiety (.54) scales. The SCL Phobic Anxiety scale correlated .51 and .43 with the MCMI-III Dysthymia and Anxiety scales, respectively. The SCL Paranoid Ideation scale and the MCMI-III Paranoid, Schizotypal, and Delusional Disorder scales were moderately related, with correlations of .64, .58, and .56, respectively. The SCL Global Severity Index was strongly correlated with the MCMI-III Thought Disorder (.69), Major Depression (.69), and Schizotypal (.66) scales.

The historical importance of the MMPI-2 test and the philosophical differences between it and the MCMI-III test (the MMPI-2 test was constructed according to the criterion groups method, the MCMI-III test using the integrated model of validity described in this manual) make any study of their relationship especially interesting. The heterogeneity of the MMPI-2 clinical scales attenuates the extent to which high correlations can be expected between them and the MCMI-III's more internally consistent scales. Nevertheless, there are some obvious areas of content similarity. MMPI-2 Hypochondriasis (Hs), for example, correlated highest with MCMI-III Somatoform at .63, followed by MCMI-III Major Depression at .60. These results mirror those obtained with the SCL-90-R test and can be attributed to the interrelationship of somatization and depression because of their shared vegetative features. The MMPI-2 Depression (D) scale was most strongly related to the MCMI-III Major Depression and Dysthymia scales at .71 and .68, respectively.

As for the MMPI-2 test and the MCMI-III personality disorders scales, correlations were fairly low across the board, reflecting the appropriate use of the MMPI-2 test as a measure of Axis I rather than Axis II phenomena. MMPI-2 Depression (D) correlated most highly with the MCMI-III Depressive (.59) and Avoidant (.56) personality scales. The pattern of correlations with the MMPI-2 Psychopathic Deviate (Pd) scale was mixed and is difficult to explicate.

Table 4.8 Correlations Between MCMI-III Base Rate Scores and Collateral Test Scores

MCMI-III Scale	BDI ^a	GBI ^b			MAST ^c	IOES ^d	STAI ^e	
	Total	Depression Scale	Hypo- mania Scale	Total	Total	Total	State Anxiety	Trait Anxiety
1 Schizoid	.53**	.54**	.43**	.53**	.22	.25	.45**	.48**
2A Avoidant	.45**	.62**	.43**	.58**	.25	.49*	.52**	.59**
2B Depressive	.56**	.66**	.52**	.64**	.32*	.41*	.53**	.62**
3 Dependent	.50**	.53**	.38**	.51**	.20	.60**	.42**	.48**
4 Histrionic	-.49**	-.46**	-.32*	-.44**	.36**	-.37	-.40**	-.54**
5 Narcissistic	-.40**	-.43**	-.21	-.38**	-.10	-.36	-.27*	-.44**
6A Antisocial	.20*	.29*	.28*	.29*	.39**	-.16	.23	.26*
6B Sadistic	.22**	.20	.25	.22	.20	.02	.20	.24
7 Compulsive	-.30**	-.29*	-.23	-.28*	-.43**	.14	-.27*	-.42**
8A Negativistic	.43**	.63**	.48**	.61**	.23	.35	.44**	.55**
8B Masochistic	.53**	.58**	.43**	.55**	.35*	.61**	.44**	.59**
S Schizotypal	.51**	.55**	.51**	.56**	.26	.51**	.48**	.55**
C Borderline	.56**	.64**	.53**	.64**	.35*	.47*	.49**	.59**
P Paranoid	.42**	.60**	.50**	.60**	.10	.55**	.42**	.40**
A Anxiety	.61**	.57**	.54**	.59**	.22	.54**	.55**	.58**
H Somatoform	.63**	.55**	.47**	.55**	.09	.59**	.54**	.52**
N Bipolar: Manic	.24**	.38**	.48**	.43**	.12	.19	.25*	.27*
D Dysthymia	.71**	.61**	.48**	.59**	.30*	.56**	.60**	.68**
B Alcohol Dependence	.36**	.24	.28*	.26	.67**	.12	.25*	.22
T Drug Dependence	.18*	.26	.27	.27	.33*	-.26	.11	.17
R Post-Traumatic Stress Disorder	.63**	.66**	.52**	.64**	.30*	.61**	.49**	.55**
SS Thought Disorder	.62**	.56**	.54**	.58**	.33*	.48*	.56**	.59**
CC Major Depression	.74**	.70**	.52**	.67**	.20	.59**	.65**	.67**
PP Delusional Disorder	.23**	.39**	.51**	.45**	.17	.19	.19	.17

^aBDI = Beck Depression Inventory (N = 149)

^bGBI = General Behavior Inventory (N = 51)

^cMAST = Michigan Alcoholism Screening Test (N = 53)

^dIOES = Impact of Event Scale (N = 25)

^eSTAI = State-Trait Anxiety Inventory (N = 59)

* $p \leq .05$, ** $p \leq .01$ (two-tailed)

Table 4.8 (continued)

SCL-90-R							
		Obsessive- Compulsive		Interpersonal Sensitivity			
MCMI-III Scale		Somatization		Depression	Anxiety	Hostility	
1	Schizoid	.30**	.53**	.48**	.54**	.38**	.36**
2A	Avoidant	.27**	.49**	.60**	.51**	.35**	.40**
2B	Depressive	.34**	.56**	.53**	.65**	.41**	.40**
3	Dependent	.33**	.54**	.58**	.54**	.42**	.31**
4	Histrionic	-.27*	-.52**	-.57**	-.54**	-.39**	-.35**
5	Narcissistic	-.27**	-.40**	.55**	-.45**	-.36**	-.32**
6A	Antisocial	.10	.29**	.24*	.23*	.14	.36**
6B	Sadistic	.22*	.28**	.23*	.22*	.10	.38**
7	Compulsive	-.08	-.37**	-.30**	-.38**	-.27**	-.39**
8A	Negativistic	.33**	.47**	.49**	.47**	.33**	.50**
8B	Masochistic	.29**	.58**	.58**	.58**	.42**	.41**
S	Schizotypal	.41**	.66**	.62**	.64**	.53**	.47**
C	Borderline	.27**	.52**	.51**	.58**	.43**	.45**
P	Paranoid	.41**	.42**	.55**	.48**	.41**	.48**
A	Anxiety	.52**	.50**	.47**	.56**	.54**	.41**
H	Somatoform	.55**	.55**	.41**	.57**	.43**	.32**
N	Bipolar: Manic	.28**	.40**	.28**	.33**	.26*	.30**
D	Dysthymia	.48**	.63**	.59**	.70**	.58**	.43**
B	Alcohol Dependence	.19	.32**	.36**	.32**	.23*	-.37**
T	Drug Dependence	.06	.20	.24*	.24*	.16	.30**
R	Post-Traumatic Stress Disorder	.46**	.53**	.49**	.62**	.48**	.38**
SS	Thought Disorder	.48**	.67**	.58**	.71**	.64**	.50**
CC	Major Depression	.56**	.60**	.56**	.74**	.62**	.41**
PP	Delusional Disorder	.38**	.37**	.41**	.39**	.35**	.38**

Note. $N = 92$

* $p \leq .05$, ** $p \leq .01$ (two-tailed)

Table 4.8 (continued)

SCL-90-R						
MCMI-III Scale	Phobic Anxiety	Paranoid Ideation	Psychoticism	Global Severity Index	Positive Symptom Distress Index	Positive Symptom Total
1 Schizoid	.35**	.48**	.47**	.51**	.43**	.50**
2A Avoidant	.38**	.48**	.45**	.51**	.36**	.54**
2B Depressive	.36**	.47**	.51**	.57**	.37**	.59**
3 Dependent	.45**	.44**	.49**	.54**	.38**	.57**
4 Histrionic	-.33**	-.44**	-.50**	-.52**	-.34**	-.55**
5 Narcissistic	-.36**	-.32**	-.46**	-.46**	-.27**	-.44**
6A Antisocial	.08	.24*	.25*	.24*	.23*	.25*
6B Sadistic	.11	.27*	.18	.25*	.26*	.27**
7 Compulsive	-.18	-.30**	-.33**	-.34**	-.25*	-.36**
8A Negativistic	.31**	.50**	.42**	.50**	.40**	.52**
8B Masochistic	.43**	.52**	.52**	.57**	.42**	.57**
S Schizotypal	.50**	.58**	.58**	.66**	.49**	.64**
C Borderline	.36**	.51**	.48**	.53**	.36**	.55**
P Paranoid	.46**	.64**	.48**	.56**	.46**	.52**
A Anxiety	.43**	.42**	.45**	.58**	.39**	.60**
H Somatoform	.38**	.34**	.39**	.55**	.42**	.54**
N Bipolar: Manic	.30**	.39**	.26*	.35**	.33**	.35**
D Dysthymia	.51**	.47**	.56**	.67**	.49**	.67**
B Alcohol Dependence	.22*	.29**	.35**	.34**	.19	.37**
T Drug Dependence	.14	.29**	.26*	.23*	.21*	.19
R Post-Traumatic Stress Disorder	.43**	.46**	.48**	.58**	.42**	.57**
SS Thought Disorder	.52**	.52**	.58**	.69**	.54**	.66**
CC Major Depression	.52**	.47**	.57**	.69**	.55**	.62**
PP Delusional Disorder	.38**	.56**	.42**	.47**	.39**	.42**

Note. $N = 92$

* $p \leq .05$, ** $p \leq .01$ (two-tailed)

Table 4.8 (continued)

MMPI-2					
MCMI-III Scale	Hs	D	Hy	Pd	Mf
1 Schizoid	.37**	.46**	.17	.29**	.06
2A Avoidant	.28**	.56**	.16	.34**	.16
2B Depressive	.37**	.59**	.25**	.41**	.08
3 Dependent	.30**	.53**	.22*	.23**	.06
4 Histrionic	-.27**	-.52**	-.11	-.36**	—
5 Narcissistic	-.25**	-.52**	-.20*	-.27**	-.05
6A Antisocial	-.02	.03	-.13	.41**	.08
6B Sadistic	.04	.01	-.05	.20*	.02
7 Compulsive	-.14	-.34**	-.13	-.51**	-.06
8A Negativistic	.22**	.25**	-.01	.30**	.00
8B Masochistic	.25**	.49**	.14	.45**	.20*
S Schizotypal	.32**	.45**	.15	.43**	.19*
C Borderline	.28**	.47**	.19*	.44**	.07
P Paranoid	.10	.14	-.15	.22*	.03
A Anxiety	.42**	.52**	.28**	.38**	.09
H Somatoform	.63**	.65**	.52**	.30**	.11
N Bipolar: Manic	.09	.06	-.03	.11	-.03
D Dysthymia	.45**	.68**	.37**	.44**	.04
B Alcohol Dependence	.01	.07	-.07	.35**	.01
T Drug Dependence	-.08	-.01	-.16	.34**	.07
R Post-Traumatic Stress Disorder	.39**	.50**	.25**	.41**	.07
SS Thought Disorder	.44**	.58**	.33**	.35**	.08
CC Major Depression	.60**	.71**	.52**	.37**	.11
PP Delusional Disorder	.16	.13	-.01	.24**	.07

Note. $N = 132$. Hs = Hypochondriasis, D = Depression, Hy = Hysteria, Pd = Psychopathic Deviate, Mf = Masculinity-Femininity.

* $p \leq .05$, ** $p \leq .01$ (two-tailed)

Table 4.8 (continued)

MMPI-2					
MCMI-III Scale	Pa	Pt	Sc	Ma	Si
1 Schizoid	.43**	.47**	.50**	.16	.57**
2A Avoidant	.47**	.53**	.55**	.00	.74**
2B Depressive	.47**	.60**	.60**	.17*	.60**
3 Dependent	.42**	.66**	.55**	.11	.60**
4 Histrionic	-.45**	-.52**	-.55**	-.06	-.77**
5 Narcissistic	-.35**	-.51**	-.46**	.17	-.70**
6A Antisocial	.31**	.24**	.25**	.40**	.09
6B Sadistic	.27**	.15	.16	.33**	.08
7 Compulsive	-.44**	-.47**	-.51**	-.23**	-.33**
8A Negativistic	.38**	.40**	.46**	.36**	.41**
8B Masochistic	.48**	.60**	.60**	.25**	.54**
S Schizotypal	.54**	.52**	.60**	.37**	.47**
C Borderline	.54**	.57**	.55**	.28**	.46**
P Paranoid	.29**	.20*	.35**	.38**	.31**
A Anxiety	.54**	.61**	.63**	.36**	.45**
H Somatoform	.55**	.68**	.63**	.12	.45**
N Bipolar: Manic	.26**	.21*	.25**	.43**	.07
D Dysthymia	.55**	.70**	.67**	.16	.57**
B Alcohol Dependence	.31**	.23**	.22*	.30**	.16
T Drug Dependence	.19*	.19*	.16	.38**	-.05
R Post-Traumatic Stress Disorder	.54**	.57**	.61**	.38**	.42**
SS Thought Disorder	.59**	.62**	.61**	.25**	.51**
CC Major Depression	.60**	.73**	.67**	.08	.50**
PP Delusional Disorder	.38**	.22*	.42**	.44**	.16

Note. $N = 132$. Pa = Paranoia, Pt = Psychasthenia, Sc = Schizophrenia, Ma = Hypomania, Si = Social Introversion.

* $p \leq .05$, ** $p \leq .01$ (two-tailed)

Moderate correlations were found between the Pd scale and the MCMI-III Antisocial (.41) and Borderline (.44) scales, as expected. A more interesting relationship exists between the Pd scale and the MCMI-III Masochistic (.45), Schizotypal (.43), and Depressive (.41) scales. An examination of the MMPI-2 Harris-Lingoes subscales revealed that these relationships may be mediated by different facets of the Pd scale, in particular the Social Alienation and Self-Alienation subscales in relation to the MCMI-III Schizotypal and Masochistic scales.

Interestingly, the MMPI-2 Paranoia (Pa) scale correlated only moderately with the MCMI-III Paranoid and Delusional Disorder scales (.29 and .38, respectively) and more highly with the Thought Disorder (.59), Major Depression (.60), and Post-Traumatic Stress Disorder (.54) scales. Correlations were also noted between the Pa scale and the Schizotypal and Borderline scales (both at .54). MMPI-2 Psychasthenia (Pt) had its closest correspondence with MCMI-III Somatoform (.68), Anxiety (.61), Thought Disorder (.62), Major Depression (.73), and Dysthymia (.70).

Although the relationships between MMPI-2 Schizophrenia (Sc) and the MCMI-III Schizotypal (.60) and Thought Disorder (.61) scales were high, equally strong correlations were obtained with the Masochistic and Depressive scales (both .60). Also notable were high MMPI-2 Sc correlations with such MCMI-III Axis I measures as

the Dysthymia (.67) and Major Depression (.67) scales.

The correlations between MMPI-2 Hypomania (Ma) and its MCMI-III parallels, the Bipolar: Manic and Delusional Disorder scales, were .43 and .44, respectively. Finally, the MMPI-2 Social Introversion (Si) scale was closely related to the MCMI-III Avoidant and Schizoid scales, at .74 and .57, respectively, and to the Depressive and Dependent scales, both at .60. Also notable are the negative correlations between MMPI Si and the MCMI-III Histrionic (–.77) and Narcissistic (–.70) scales.

Summary

In this chapter we reviewed the external validity of the MCMI test across three generations of research. We began with research on the original MCMI and MMPI tests, putting this information into historical context and showing the complementarity of the two instruments. Next we looked at the criterion validity of the MCMI-III test using experienced diagnostic judges and compared this to results obtained with the MCMI and MCMI-II tests. Finally, we examined concurrent validity correlations of the MCMI-III test with other major instruments. Our overall conclusion is that the external validity of the MCMI-III test is equal to or slightly better than that of the MCMI-II test.

Chapter 5

Grossman Facet Scales

The MCMI-III Grossman facet scales represent an effort to elucidate finer distinctions than are expressed through the primary personality scales. For each of the Clinical Personality Patterns scales (1–8B) and Severe Personality Pathology scales (S, C, and P), three scales represent “facets” of the primary personality pattern, drawn from the structural-functional domains described in Chapter 2. Thus, there are 42 total facet scales tied to the 14 primary personality scales. The facet scales were developed with the aim of maximizing the assessment specificity and therapeutic utility of the MCMI-III test and its overarching evolutionary theory (Millon, 1969/1983, 1983, 1990, 1999; Millon & Davis, 1996, 1997; Millon, Grossman, Meagher, Millon, & Everly, 2000; Millon, Millon, Meagher, Grossman, & Ramnath, 2004).

Millon and Davis (1997) encouraged researchers and clinicians who use the MCMI-III test to view it as a constantly evolving instrument intended to mirror the field’s growing knowledge base. They were concerned that while the configural structure offered by the MCMI-III test facilitated personality mixtures or subtype variants, the *DSM-IV* maintained a semi-rigid categorical system that contravened the more prototypal or dimensional format seen in MCMI-

III profiles. It was their hope that future diagnostic standards would yield a more prototypal approach that is congruent with the instrument’s evolutionary theory. They also speculated that future development of the MCMI series might facilitate this advance by adding clinical domain distinctions and dimensions such as may be found in a set of new facet scales. The Grossman facet scales were developed with these goals in mind.

Both normal and abnormal personalities are multifaceted in their composition. As singular or reified entities, they can be captured using a categorical diagnostic label, but to be maximally useful clinically they require an in-depth understanding of their diverse compositional domains and the interactions among those domains. The facet scales uncover these clinically salient domains as a means of identifying and measuring specific problematic personality qualities.

Scale Development

Two key decisions to be made when constructing “post hoc” scales (i.e., new measures based on an established instrument) are summarized well by Davis (1994). One involves whether to construct the new scales with rational or statistical criteria as the initial consideration. An

investigator choosing a rational route will most likely have in mind a functional need that the scales will serve. This methodology invariably fits existing items to some set of concepts. On the other hand, researchers who choose a statistically guided route will develop new scales using an established multivariate technique such as factor or cluster analysis to identify domains of communality or clusters of items. This latter approach rests on the assumption that statistical methodology is sufficient to uncover the latent dimensions underlying an existing item pool, thus leaving the definition of new scales to methodological formalities rather than to the researcher's theoretical orientation or utilitarian desire. As described in Chapters 2 and 3, the MCMI-III clinical scales were rationally initiated and statistically refined. In keeping with this philosophy, a rational process based on Millon's theory was used as the initial basis for the development of the Grossman facet scales.

The second key decision in Davis's (1994) schema involves whether the established inventory's entire item pool will serve as the raw material for the new scales or whether items will be drawn from a rationally selected subset of the scales. In constructing the MCMI-III facet scales, a further question pertained to the level of inquiry that would be most meaningful and desirable in terms of clinical efficacy and consonance with the parent instrument. Millon's theory specifies several possible delineations, including macro-groupings of personality tendencies (i.e., pleasure-deficient, interpersonally imbalanced, intrapsychically conflicted, and structurally defective) and explication of the three evolutionary motivating aims (existence, adaptation, replication). The most molecular choice involved using items from each principal personality scale as its own discrete item pool. At this level of analysis, the logical meaningfulness of the resultant scales is maximized by virtue of the structure of the overarching theory. Such "facet scales" would seek to represent elements of each of the 14 prototypal personality patterns.

A further utilitarian purpose that the facet scales were designed to serve was the ability to examine discrepancies between elevations on the primary personality scales and their associated facet scales for purposes of treatment planning. Thus, the clinician would be in a position to

determine not only what problematic personologic elements were most evident but also what *subtype* or admixture of personality expressions the individual may present. To maintain congruence with the theory and to satisfy the goal of maximal clinical value, a logical within-scale delineation was to base the facet scales on the eight functional and structural personologic domains identified by the theory: expressive acts, interpersonal conduct, cognitive style, regulatory mechanisms, self-image, object representations, morphologic organization, and mood/temperament. These domains are described in detail in Chapter 2.

After it was decided to use a rationally initiated approach to develop facet scales that reflect the functional and structural personologic domains of each of the 14 MCMI-III personality patterns, scale development followed a three-stage process. The first stage involved deciding, for each of the 14 primary personality scales, which of the eight structural-functional domains would be represented with facet scales. For each prototypic personality disorder, Millon (1990; Millon & Davis, 1996) posits that only two or three of its personologic domains will be most salient, one to three others will probably be of moderate (supportive) importance, and the remaining domains will probably be of only occasional importance. Which domain presents as the most salient according to prototypal structure differs among the prototypic personality disorders. For example, the prototypal histrionic is primarily identified by his or her interpersonal conduct and mood/temperament, with expressive behavior and cognitive style features playing a secondary role. In contrast, the depressive prototype, whose cognitive style and mood/temperament are likely to be most salient, displays three other domains in more moderate form.

The MCMI-III item pool accords with the theory and reflects judgments regarding the salience of each structural and functional domain for each personality pattern. Facet scale construction began with predictions about which item domains in each primary personality scale were likely to match the domains of the theory. For some scales, secondary domains were better represented than expected in the scale's item composition. Nevertheless, preliminary choices at this first stage were made based on which

three domains for each personality pattern fit the theory best and on a rational examination of each scale's item composition.

The second stage involved testing the predictions generated in the first stage through statistical methodology. This work is described in detail in Grossman (2004). As with the content or facet scales that have been developed for other inventories, factor analytic procedures were used to empirically substantiate the Grossman facet scales. Because of item overlap across the primary MCMI-III scales, the relatively small number of items on each primary scale, and the high covariance expected owing to the polythetic nature of the theory, most statistical methods were not expected to yield highly parsimonious results (Choca, 1998). To optimize the analysis, an alpha factoring method was chosen. This procedure maximizes the internal consistency of the extracted factors. It differs from principal components analysis, a more widely used method, in that it extracts factors with the specific goal of maximizing internal consistency. An oblique rotation method was employed to allow for correlated factors, consonant with the polythetic model that undergirds the *DSM* personality disorders. Among the available choices for oblique rotations, the Promax method was selected because of the large size of the data set (the original MCMI-III standardization sample) and the endorsement of this method by previous investigators of polythetic personality attributes (e.g., Goldberg & Digman, 1994).

The factor analysis procedure was not strictly exploratory but rather was designed to test the theoretically generated predictions from the first stage of scale development. Thus, the most widely used criteria for determining the number of factors to extract in exploratory analyses, such as Kaiser's (1960) stopping rule of eigenvalues over 1.0 or Cattell's (1966) graphical scree test procedure, were deemed inappropriate. Hair, Anderson, Tatham, and Black (1992) suggested that an *a priori criterion* would be useful in cases where researchers were guided by a predetermined theory that specified a given number of factors. This is the procedure that was followed, using the first-stage predictions as a guide for specifying the number of factors to extract for each primary scale.

The third scale development stage involved

subjecting the tentative scales that emerged from the factor analyses to final rational and statistical refinements. The factor analytic results were scrutinized for their concordance with the predicted personologic domains, and scale adjustments were made based on item domain content and levels of factor loadings. A major decision made at this point was to augment the facet scales for each personality pattern with items from related scales within the instrument (e.g., items from Scale D: Dysthymia were added to the cognitive style facet for Scale 2B: Depressive). This decision allowed the range of scores for some very short facet scales to increase, thereby stabilizing those scales, and increased the breadth of coverage of some of the theoretical constructs. It was also determined during this stage that, unlike for the primary MCMI-III scales, items would not be weighted in the scoring of the facet scales because the facets are more unidimensional constructs and do not retain a prototypal structure.

The 42 final facet scales (3 facets for each of the 14 primary personality scales) that were created using this combined rational/statistical procedure are listed along with their coefficient alpha internal consistency reliabilities in Table 5.1. Appendix J provides descriptions of the 42 facet scales. The scoring of the facet scales can be found in Appendix D. The internal consistency values, which were based on the MCMI-III standardization sample, are impressive overall, especially given the brevity of many of the scales. Alpha exceeded .80 for several scales, and the majority of the remaining alpha values were between .60 and .80. Those few facet scales with alphas falling below this level were considered core elements of their "parent scale" construct and were therefore retained. The mean alpha across all scales was .71, which was considered satisfactory in light of the brevity of several scales and the polythetic nature of clinical personality constructs.

Norms for the facet scales were updated as part of the overall normative update, using the sample described in Chapter 6. BR scores for the facet scales range between 0 and 100 rather than 0 and 115. The transformations for raw to base rate scores for the facet scales can be found in Appendix E.

Unique Contributions of the Facet Scales

Clinicians who are familiar with the MCMI-III's undergirding theory and who use the instrument regularly are often able to use scale elevation patterns to deduce important dynamics about the individual's "motivating aims." That is, the clinician may be able to use a patient's score profile to conclude that he or she demonstrates specific qualities of the three polarities (pleasure-pain, active-passive, self-other) that give rise to personality expression. For example, a clinician examining a protocol with elevations on Scales 2A (Avoidant), 2B (Depressive), and 8A (Negativistic) might deduce that the individual will prob-

ably demonstrate high risk aversion and pain sensitivity and perpetuate negative cognitions through the creation of "impossible" interpersonal transactions (self-other conflict) but be unsure about what or how much can be done to modify his or her circumstances (active-passive ambivalence). Using the theory behind the test, a more contextual clinical profile emerges via the interaction of these personality domains, even in the absence of an Axis II diagnosis. These interactions also point to directions for intervention.

The Grossman facet scales can further refine these efforts. Each primary personality pattern is derived from a configuration of the three basic polarities. The theory specifies eight specific

Table 5.1 Grossman Facet Scales With Alpha Reliabilities

		Items	Alpha			Items	Alpha
1	Schizoid			6B	Sadistic		
1.1	Temperamentally Apathetic	8	.73	6B.1	Temperamentally Hostile	8	.65
1.2	Interpersonally Unengaged	10	.79	6B.2	Eruptive Organization	8	.59
1.3	Expressively Impassive	8	.60	6B.3	Pernicious Representations	11	.71
2A	Avoidant			7	Compulsive		
2A.1	Interpersonally Aversive	9	.79	7.1	Cognitively Constricted	8	.51
2A.2	Alienated Self-Image	10	.85	7.2	Interpersonally Respectful	6	.48
2A.3	Vexatious Representations	8	.78	7.3	Reliable Self-Image	7	.53
2B	Depressive			8A	Negativistic		
2B.1	Temperamentally Woeful	7	.83	8A.1	Temperamentally Irritable	10	.77
2B.2	Worthless Self-Image	9	.83	8A.2	Expressively Resentful	7	.68
2B.3	Cognitively Fatalistic	8	.82	8A.3	Discontented Self-Image	7	.78
3	Dependent			8B	Masochistic		
3.1	Inept Self-Image	8	.84	8B.1	Discredited Representations	8	.75
3.2	Interpersonally Submissive	8	.64	8B.2	Cognitively Diffident	8	.78
3.3	Immature Representations	6	.71	8B.3	Undeserving Self-Image	10	.85
4	Histrionic			S	Schizotypal		
4.1	Gregarious Self-Image	7	.69	S.1	Estranged Self-Image	11	.85
4.2	Interpersonally Attention-Seeking	8	.61	S.2	Cognitively Autistic	10	.79
4.3	Expressively Dramatic	7	.55	S.3	Chaotic Representations	8	.78
5	Narcissistic			C	Borderline		
5.1	Admirable Self-Image	10	.79	C.1	Temperamentally Labile	10	.81
5.2	Cognitively Expansive	7	.50	C.2	Interpersonally Paradoxical	9	.77
5.3	Interpersonally Exploitive	10	.65	C.3	Uncertain Self-Image	9	.83
6A	Antisocial			P	Paranoid		
6A.1	Expressively Impulsive	9	.56	P.1	Cognitively Mistrustful	8	.74
6A.2	Acting-Out Mechanism	10	.71	P.2	Expressively Defensive	7	.64
6A.3	Interpersonally Irresponsible	8	.61	P.3	Projection Mechanism	9	.64

and comparable functional and structural domains through which each prototypal personality pattern is expressed. A fully contextualized personality description will include the person's biophysical (mood/temperament), conscious phenomenologic (self-image, cognitive style, object representations), subconscious intrapsychic (regulatory mechanisms, morphologic organization), and behavioral (expressive acts, interpersonal conduct) expressions. These four broad categories are congruent with the main branches of therapeutic philosophy (pharmacologic/neurologic, humanistic/cognitive, psychodynamic, and behavioral, respectively), while the eight domains on which the facet scales are based are aligned with more specific therapeutic approaches within each philosophy. From these foundations, clinicians may find a sound basis for a synergistic psychotherapy (Millon et al., 2000), that is, one that draws on all of the major attributes of the whole personality.

By examining the facet scale scores for the primary personality scales that are elevated, it is possible to identify the patient's most troublesome or clinically significant domains. For example, an individual with a high score on the Avoidant scale may be most troubled by social interaction (interpersonally aversive), self-esteem issues (alienated self-image), or his or her perception of others (vexatious object representations). According to the theory, these are three of the most salient domains of this prototypal personality pattern. The facet scales directly reflect these constructs and help identify the relative salience of alternate domains. Domain salience distinctions begin the process of assessing what subtype of avoidant pattern an individual presents.

Further, two individuals diagnosed with the same Axis II syndrome are highly unlikely to present identically (Millon & Davis, 1996). There is considerable heterogeneity among individuals with the same *DSM* diagnosis. Patients with the same Axis II diagnosis may be expected to demonstrate substantially different expressions of the clinical domains within that diagnostic category. The addition of the facet scales allows for a more molecular and differentiated view of the primary diagnosis, highlighting salient features at a more discriminating level than can be achieved with broad categorical labels. The

scales can help clinicians identify the clinically most important functional and structural domains composing a person's Axis II diagnosis.

A further use of the facet scales goes beyond within-scale or within-syndrome differentiation, often providing information that may not be obvious upon initial examination. Regardless of the number of clinically significant elevations across the primary personality scales, the facet scales will probably identify specific domain elevations across several primary scales. Millon's evolutionary theory predicts that most personality presentations are not prototypal in nature. Rather, they are admixtures of two or more personality patterns that form what Millon terms subtypes (e.g., narcissistic with antisocial features, dependent with avoidant features, etc.). More than 60 subtype patterns have been identified in Millon's writings over the past decade (Millon, 1999; Millon & Davis, 1996; Millon et al., 2004), and descriptions of additional subtypes are forthcoming. Given that the theory specifies eight functional and structural domains, it is possible that a subtype pattern may be detected by examining the configuration of facet scale scores. For example, it would be possible to characterize a patient as "antisocial with histrionic features" (identified by the theory as a risk-taking antisocial) by viewing elevations on the primary scales and examining facet scales such as Expressively Impulsive (one of the Antisocial facets) and Interpersonally Attention-Seeking (one of the Histrionic facets). As a result of integrating facet scale findings across several primary scales, a more detailed clinical picture with great interpretive specificity and therapeutic utility emerges.

Unfortunately, the *DSM-IV* provides little basis for integrating the multiple Axis II features with which patients commonly present. *DSM-IV* criteria are not consistent across diagnoses in that one diagnosis may be based mostly on behavioral criteria while another may be based mostly on cognitive criteria. Many patients meet the criteria for more than one diagnosis, and the *DSM-IV* does not specify which features are most important to consider. Because Millon's prototypal domain criteria are explicitly comparable (i.e., all prototypal patterns evidence the same eight domains, with different qualitative descriptions), a meaningful clinical picture may be deduced when there are two or three eleva-

tions across the primary MCMI-III personality scales, as is often the case. For example, an individual with elevations on Scales 4 (Histrionic) and 5 (Narcissistic) may have cognitions that are more consonant with the narcissistic prototype and may have interpersonal dynamics and expressive acts that are more in line with the histrionic prototype. Interpretation of joint personality scale elevations is augmented because the facet scales reflect these eight explicitly comparable domains of the theory.

Significantly, this contextual picture is relevant even in the absence of a diagnosable Axis II disorder. As long as a primary personality scale is at a BR score of at least 65 (rather than the diagnostic benchmarks of 75 and 85), its facet scales are considered interpretable. This is highly beneficial to clinicians who are interested in conceptualizing their cases in terms of personality dynamics (i.e., personality trends that do not meet criteria for diagnosable disorders). Such an approach permits a much more holistic and useful approach to psychotherapy. It can be helpful in treating Axis I disorders or in dealing with general counseling cases that do not evidence an Axis II disorder. In these situations, the clinician may be able to explicate problematic personologic disturbances even without any diagnostic-level elevation on the primary personality scales.

Implications for Treatment Planning and Intervention

Possibly the greatest benefit that the facet scales provide clinicians is the ability to deduce theoretically substantiated, clinically sound case conceptualizations, treatment plans, and therapeutic directions. Because the facet scales are consonant with Millon's eight personologic domains, which in turn are congruent with contemporary therapeutic philosophies and treatment techniques, the clinician may use the facet scale scores to assist in generating effective, fully contextual, and integrative personologic therapeutic interventions.

In a mature science, a theory serves as the generative source for a system of classifying the phenomena within the theory's domain and specifies a method for measuring and substantiating the content of the classification system. What makes such a science clinical in nature,

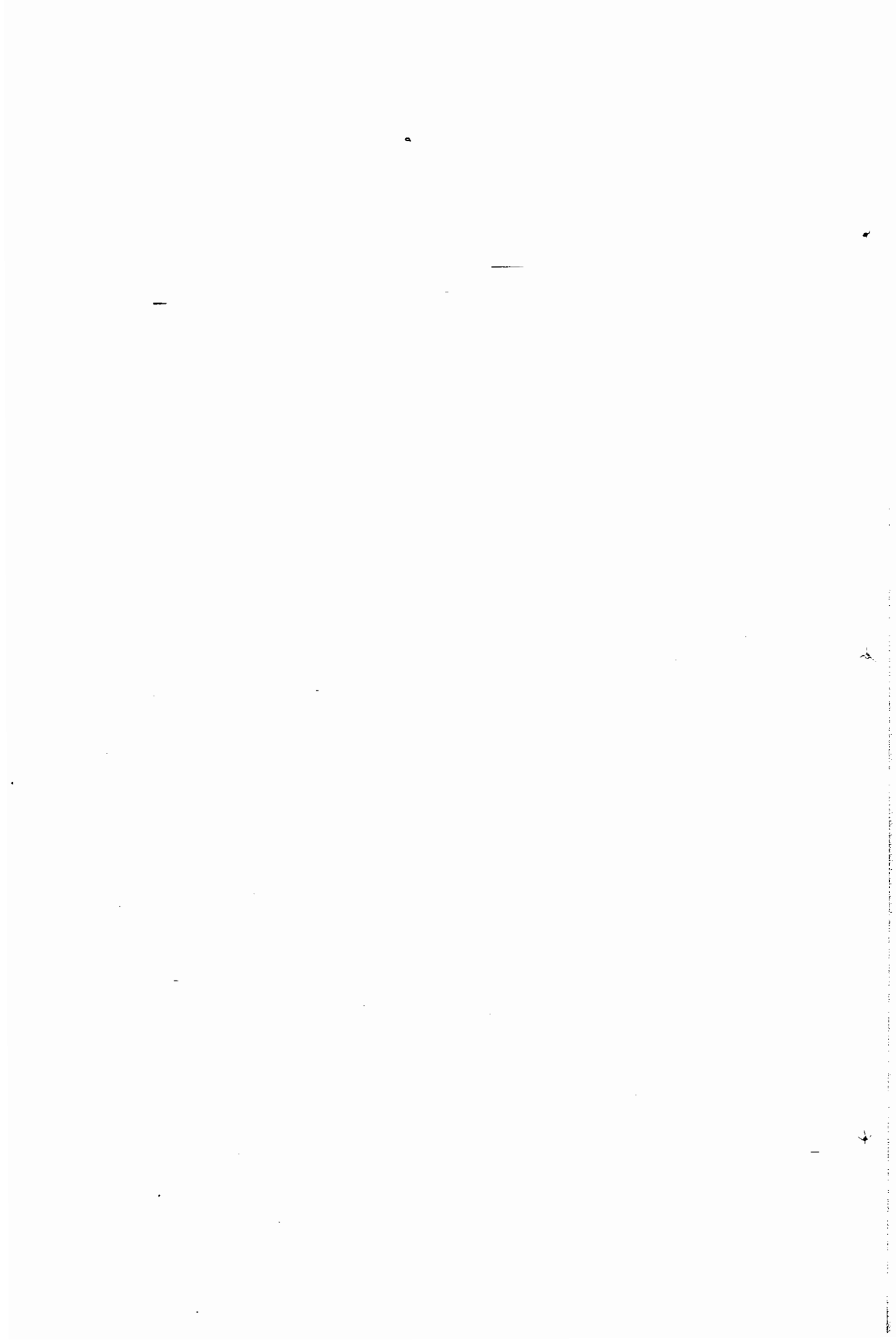
however, is the additional ability to manipulate and modify those phenomena (Millon, 1990). Clinical psychology is no exception; the function that defines it as a clinical science is the ability to intervene with patients to produce beneficial change. The Grossman facet scales can help facilitate this goal.

Perhaps the clearest example of the clinical utility of the facet scales can be found in the personologic intervention method (e.g., Millon et al., 2000; Millon & Grossman, in press) that represents a direct outgrowth of the same guiding theory from which the facet scales were generated. This paradigm is integrative in nature, recognizing the eight personologic domains measured by the facet scales as key guides to appropriate therapeutic strategies. Each domain is aligned with one of the principal schools of thought in modern psychotherapy (e.g., cognitive, psychodynamic, pharmacologic, etc.). Taken together, however, these eight domains comprise the complex, interwoven spectrum of the individual's personality, which may be viewed as the "psychic immune system." Any given personality pattern will have certain vulnerabilities that may, given constitutional dispositions and environmental conditions, allow for problematic psychological functioning. By intervening at the personality level (i.e., choosing strategies from the various available therapeutic modalities to match the problematic personologic domains) while tactically orienting treatment toward balancing the "motivating aims" polarities (i.e., pain-pleasure, passive-active, self-other), this system of psychotherapy works to bolster the individual's personologic functioning rather than merely treating the symptoms separately and out of context. The facet scales can help identify a patient's problematic personologic domains, thereby serving as a catalyst to efficient, synergistic psychotherapeutic treatment planning and intervention.

Although the Grossman facet scales are directly allied with one of the many theories found in clinical psychology, their use is in no way limited to this paradigm. Strict adherents to the various unilateral schools of psychotherapy do not accept the myriad choices offered by competing therapeutic paradigms; however, there is a growing number of integrative and eclectic approaches to treatment in both traditional and

short-term modalities that could probably benefit from the assessment information provided by the facet scales. For example, cognitive-pharmacologic and cognitive-behavioral modalities (e.g., Beck, Freeman, & Davis, 2004; Young, 1990) are now widely used for treating personality and mood disorders. Generally speaking, these models seek not only to ascertain an individual's cognitive schema (which are measured by the

cognitive style and self-image facet scales) but also to understand and modify mood difficulties (the mood/temperament facet scales) and behavioral tendencies (the expressive acts and interpersonal conduct facet scales). In these and other contemporary examples, the greater specificity afforded by the addition of personality facet scales can enhance the goal of efficient and pragmatic intervention.



Chapter 6

2008 Update: New Norms and Inconsistency Scale

In 2008, the MCMI-III norms were updated and a new scale designed to help detect random responding, the Inconsistency scale, was added to the test. This chapter describes these updates.

Updated Norms

The original MCMI-III norms were constructed as part of the development of the test in 1992 and 1993 (described in Chapter 3). The norms were updated in 2008 to reflect changes that may have occurred in the test scores obtained by clinical patients and/or in their clinical characteristics over the 15 years since the test was developed. The latter is important because the MCMI-III base rate scores (standardized scores) are scaled to reflect the varying prevalences of the characteristics measured by the test, as discussed in detail below. Also, following the trend established with the two most recently developed Millon inventories, the Millon Pre-Adolescent Clinical Inventory (M-PACI; Millon, Tringone, Millon, & Grossman, 2005) and the Millon College Counseling Inventory (MCCI; Millon, Strack, Millon, & Grossman, 2006), as well as with the latest version of the MMPI, the Minnesota Multiphasic Personality Inventory-2-Restructured Form™ (MMPI-2-RF™; Ben-Porath & Tellegen, 2008), a single, combined-gender norm group was used in the 2008 update, replacing the original separate-gender norms.

The 2008 Norm Sample

The 2008 norm sample comprises 752 individuals who completed the MCMI-III in an outpatient or inpatient treatment setting. The collection of this sample began with the mailing of a solicitation letter to a large number of MCMI-III users who were identified using the Pearson customer database. The letter asked sites to volunteer to contribute patients' MCMI-III test records along with demographic information and ratings of those patients' Axis I and Axis II characteristics.

Sites that agreed to participate were mailed data collection instructions and scannable forms for recording their cases' demographic and clinical characteristics. Sites were instructed to submit cases that had recently completed the MCMI-III, but not newly tested individuals; in other words, the test data that were used for norms development were archival. The sites coded all cases with ID numbers to preserve subject anonymity. A complete case consisted of the MCMI-III test record (submitted either electronically or by mailing in the paper-and-pencil answer sheet) and a scannable information form. As compensation for their participation, sites received free MCMI-III computerized reports for each complete case they submitted.

A total of 840 complete cases were submitted from 40 sites across the United States. Cases were excluded if one or more of the following

were true: the individual was neither an out-patient nor an inpatient; the test protocol had more than three missing item responses; the individual's raw score on the Disclosure scale (X) was less than 34 or greater than 178 (one of the test invalidity criteria described in Chapter 7); the individual's raw score on the Invalidity scale (V) was 1 or higher, or his or her raw score on the new Inconsistency scale (W) was 8 or higher, suggesting questionable protocol validity due to random responding (these criteria are described in detail later in this chapter). A few additional cases were excluded so that no site contributed more than 40 cases (5.3%) to the final sample. This left a final sample of 752 cases for norms development.

Demographic characteristics and patient status for the sample members are shown in Table 6.1. As the table shows, the sample was about equally composed of males and females. Most patients were in their 20s, 30s, or 40s, although all adult age groups up to age 79 were represented. The sample was roughly three-fourths white, with African Americans and Hispanics/Latinos each making up about 10%. Educational level and marital status were diverse. Finally, the sample was composed of approximately four-fifths out-patients and one-fifth inpatients.

The Rationale for Base Rate Scores

Scale scores can be transformed through norm referencing or criterion referencing. Norm referencing is widely used and easy to understand. In that system, norms are established using a sample that is thought to represent a larger population. When a person from that population is tested, his or her performance is compared to the mean and distribution of the standardizing sample. A *standard score* of some sort is then typically used to describe the individual being tested relative to the norm group.

The T score is the most popular standard score used with personality inventories. However, the T score and the norm-referencing system that it exemplifies pose two substantial problems when used with a personality inventory. The first of these problems is that the shape of the distribution of scores is very likely to vary from one personality scale to the next. Differences in the shapes of the distributions rob the standard score and the established cutoff points of some of their meaning.

Imagine, for instance, that the distributions of three different scales are represented by the three curves in Figure 6.1. If the cutoffs for these scales are the same and are represented by a line crossing vertically at a particular point (line D in Figure 6.1), the percentage of the population to the left of that cutoff will vary from scale to scale as a function of the shape of the distributions. The result is that the same T-score elevation defines different percentiles of the population for different scales. In other words, a T score of 80 on one scale is associated with a different percentile rank than the same T score on another scale. This effect is undesirable because the meaning of the relative elevations of the different scales is lost. Consider, for example, the scores represented by the asterisks in Figure 6.1. The T score on Scale B is the highest ranking T score, and the T score on Scale C is the second highest ranking T score. However, the percentile rank associated with the score on Scale C is higher than the percentile rank associated with the score on Scale B.

Figure 6.1 Hypothetical Distributions of Three Scales

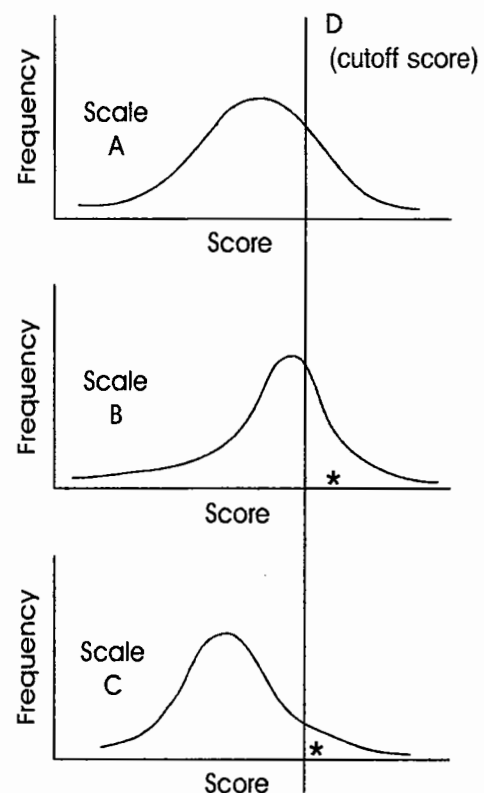


Table 6.1 Demographic Characteristics and Patient Status for the 2008 Norm Sample

	<i>n</i>	Percent
Gender		
Female	397	52.8
Male	355	47.2
Age		
18–20	62	8.2
21–29	173	23.0
30–39	180	23.9
40–49	163	21.7
50–59	113	15.0
60–69	46	6.1
70–79	15	2.0
Race/Ethnicity		
African American	83	11.1
American Indian	4	0.5
Asian American	11	1.5
Hispanic/Latino	70	9.3
White	571	76.0
Other	12	1.6
Not reported ^a	1	
Highest Education Achieved		
No high school diploma or GED	109	15.0
High school diploma or GED	203	27.8
Technical school graduate	21	2.9
Some college	202	27.7
Bachelor's degree	115	15.8
Graduate or professional degree	79	10.8
Not reported ^a	23	
Marital Status		
Never married	252	33.6
First marriage	216	28.8
Remarried	88	11.7
Separated	48	6.4
Divorced	113	15.1
Widowed	13	1.7
Cohabiting	13	1.7
Other	7	0.9
Not reported ^a	2	
Patient Status		
Outpatient, never hospitalized	521	69.3
Outpatient, previously hospitalized	86	11.4
Inpatient	145	19.3

Note. Total *N* = 752.

^a Excluded from the calculation of percentages.

The other problem with T scores and norm referencing is that they do not take into account the prevalence of the attribute being measured. T scores basically reflect how subjects compare to the members of the standardizing population. Typically the mean is set at 50 and the standard deviation at 10. Therefore, the usual cutoff of 70 is two standard deviations above the mean and defines approximately the top 2% of the population. (This 2% figure is only approximate; large departures from a normally distributed population will mean greater departures from 2%). However, this 2% figure almost never corresponds to the actual prevalence rate of a particular disorder in a particular population and therefore introduces an automatic error. For instance, if 15% of the population has a dependent personality, using a T score of 70 as a cutoff on a dependent personality scale can be expected to identify only 2% and to miss at least 13% of the true-positive cases.

The problems with T scores and norm referencing can be avoided by using criterion referencing (Retzlaff, 1996). The goal of criterion referencing is to create a standard score that is anchored to the prevalence rate of the characteristic being measured rather than to an invariable percentile of the standardizing sample that will seldom, if ever, coincide with the actual prevalence rate.

Like the MCMI and MCMI-II tests, the MCMI-III test uses the base rate (BR) score as the standard score into which raw scores are translated. The BR score was designed to anchor the cutoff points at 75 and 85 to the prevalence of a particular attribute in the psychiatric population. BR scores define a continuum of the pervasiveness and severity of a psychological attribute against which any individual can be evaluated. Using a continuum is an acknowledgment that the difference between a clinical disorder and normal functioning, especially with personality scales, is one of degree rather than kind, represented as shades of increasingly problematic adaptive inflexibility and tenuous stability under stress (Millon, 1981). Ideally, a profile of personality scales, transformed to reflect different absolute frequencies and ratios of disorders in the population, will be equated not merely in terms of the quantity of a characteristic but also in terms of the implications of that particular

quantity for psychological functioning. The base rate score represents the most direct way of getting at what is explicitly of clinical interest in the first place.

Realistically, of course, a number of factors constrain the possibilities in this regard. Perhaps foremost among these is that the components that make up personality disorders overlap and are intrinsically intercorrelated and that many patients may be diagnosed with two, three, or even more personality disorders with several covarying clinical (Axis I) symptoms. Accordingly, in addition to the absolute magnitude of any one personality scale elevation, the rank order of any personality score in the overall profile must be considered when making a diagnostic judgment. The presence of personality pathology in some patients may be so pervasive that it cannot be readily categorized in any one pattern. As a result, a significant subset of individuals may receive numerous elevations above the MCMI-III cutoff scores of 75 and 85 for disorders or traits. Obviously, not all of these elevations have equal diagnostic validity. The approach of diagnosticians should be to distinguish the most prominent and clinically significant elevations from those of lesser relevance. Clinical judgment simply cannot be removed from the assessment process for all individuals, despite the gains achieved by transforming raw scores into base rate scores.

BR Score Transformations for the Clinical Scales

The BR score transformations for the updated norms were developed following essentially the same procedure that was used to develop the original norms. This procedure involved linking the raw score frequency distributions for the various MCMI-III clinical scales to estimates of the base rates of the characteristics measured by those scales. Details are provided below.

Base rate estimates were obtained from the clinician ratings of the 752 individuals in the new normative sample. Clinicians were asked to record ratings of the individuals' Axis I and Axis II characteristics on a scannable information form. Specifically, clinicians were instructed to choose each individual's most prominent and (if applicable) second and third most prominent syndromes from a list of all of the Axis I syndromes measured by

the MCMI-III Clinical Syndromes and Severe Clinical Syndromes scales. Similarly, clinicians were instructed to choose each individual's most prominent and (if applicable) second and third most prominent personality patterns from a list of the MCMI-III Axis II personality patterns (the Clinical Personality Patterns and Severe Personality Pathology scales).

For the Axis I syndromes, the percentage of the sample for which each syndrome was rated most prominent was used as the initial estimate of the base rate with which that syndrome is *prominent* in the population. The percentage of the sample for which each syndrome was rated second or third most prominent was used as the initial estimate of the base rate with which that syndrome is *present*, though not prominent, in the population. For the Axis II personality patterns, the percentage of the sample for which each pattern was rated most prominent was used as the initial estimate of the base rate with which that pattern constitutes a personality *disorder* in the population. The percentage of the sample for which each pattern was rated second or third most prominent was used as the initial estimate of the base rate with which that pattern constitutes a personality *trait* in the population. These empirically derived estimates were then adjusted to account for epidemiological data on the prevalence of these characteristics to arrive at the final base rate targets.

The final base rate targets were linked to the raw scale score distributions of the MCMI-III clinical scales in the new norm sample to determine the transformations from raw scores to BR scores. First, initial raw score equivalents were chosen for the key BR anchor points of 85 and 75. For each Axis I scale, BR 85 was set at the raw score that came closest to cutting off the percentage at the top of the raw score distribution that matched the *prominent* base rate for that syndrome. For example, the prominent base rate for Major Depression (i.e., the percentage of the norm sample for whom Major Depression was rated the most prominent Axis I syndrome) was 27%. Thus, BR 85 was initially set at the Major Depression scale raw score that came closest to cutting off the top 27% of scorers in the norm sample. BR 75 was then set at the raw score that came closest to cutting off the percentage at the top of the raw score distribution that matched the

sum of the *prominent* and *present* base rates. In the Major Depression example, the sum of the *prominent* and *present* base rates (27% and 13%, respectively) was 40%, so BR 75 was initially set at the Major Depression scale raw score that came closest to cutting off the top 40% of scorers in the norm sample. The same process was followed to establish the initial raw score equivalents of BR 85 and 75 for the Axis II personality scales, except that BR 85 was tied to the *disorder* base rate and BR 75 was tied to the sum of the *disorder* and *trait* base rates.

For each of the clinical scales, the raw scores were linked to three additional anchor points: BR 0 was assigned to a raw score of 0, BR 60 was assigned to the median raw score, and BR 115 was assigned to the highest raw score attained in the sample. Linear interpolation was then used to assign BR scores to the raw scores between the five anchor points of 0, 60, 75, 85, and 115. This initial set of raw-to-BR score transformations was then used to determine the clinical scale BR scores for all members of the new norm sample. The score adjustments described in Appendix G were then applied to the scores, and frequency distributions for the adjusted BR scores were calculated. Next, for each scale, the percentage of sample members scoring at or above BR 85 and BR 75 after the score adjustments was compared to the base rate targets. BR score transformations were then revised as necessary to improve the match between the base rate targets and the percentage of the norm group scoring BR 85 and higher and BR 75 and higher, resulting in the final set of transformations provided in Appendix C.

Changes in Clinical Scale Elevation Patterns Between Original and Updated Norms

At the beginning of this chapter it was noted that updating the norms would account for any changes that may have occurred in the test scores and/or the clinical characteristics of clinical patients over the past 15 years. There is evidence that both of these factors were at play. Consequently, users should expect some changes in the scores that are obtained using the updated norms.

The most prominent change that many users will see is an increase in the number of individuals who have elevated scores on the Major Depression (CC) scale. This mirrors a very large increase in the estimated base rate of this syndrome

(based on clinician ratings) between the original norming of the MCMI-III and the current renorming. Other Axis I scales for which users may see an increase in elevated scores are Bipolar: Manic (N) and Drug Dependence (T). On the other hand, users may note a decrease in the number of elevations on the Axis II Masochistic (8B) scale.

Comparisons by Patient Status and Gender

Table 6.2 shows the BR score means and standard deviations for the 24 clinical scales separately for the inpatients and outpatients in the new norm sample. Users may find this information helpful as a reference point for their own patients. Not surprisingly, inpatients scored higher on average on most of the scales, particularly the Axis I scales. The differences attained statistical significance for 12 of the 24 scales, as denoted in Table 6.2.

Means and standard deviations for females and males in the new norm sample are provided in Table 6.3. None of the scale means differed by more than about one-third of a standard deviation. Ten of the comparisons showed statistically significant differences. Consistent with expectations, females scored higher on average on scales reflecting internalizing problems: Depressive (2B), Dependent (3), Somatoform (H), Dysthymia (D), Post-Traumatic Stress Disorder (R), and Major Depression (CC). Conversely, males had higher mean scores on scales tapping externalizing problems: Antisocial (6A), Alcohol Dependence (B), Drug Dependence (T), and, somewhat tangentially, Narcissistic (5). However, though not shown in the table, gender accounted for no more than about 3% of the score variance for any scale.

Table 6.2 Inpatient vs. Outpatient Comparisons of Mean BR Scores

Scale	Inpatients (IP)		Outpatients (OP)		IP Mean - OP Mean
	Mean	SD	Mean	SD	
1 Schizoid	59.2	21.4	52.0	24.9	7.2 ^a
2A Avoidant	56.2	29.7	50.6	30.1	5.6
2B Depressive	61.3	32.3	50.6	32.2	10.7 ^a
3 Dependent	65.3	25.1	58.3	27.2	7.0
4 Histrionic	43.5	23.4	50.9	23.3	-7.4 ^a
5 Narcissistic	53.1	23.7	58.2	22.4	-5.1
6A Antisocial	57.9	22.6	52.4	23.9	5.5
6B Sadistic (Aggressive)	50.9	21.0	49.0	22.8	1.9
7 Compulsive	49.9	22.4	53.8	21.0	-3.9
8A Negativistic (Passive-Aggressive)	56.3	24.5	51.1	26.1	5.2
8B Masochistic (Self-Defeating)	54.0	30.9	48.5	29.5	5.5
S Schizotypal	54.2	26.3	45.0	28.0	9.2 ^a
C Borderline	60.6	27.9	50.0	27.8	10.6 ^a
P Paranoid	49.3	28.7	46.4	27.5	2.9
A Anxiety	70.4	31.8	58.0	35.1	12.4 ^a
H Somatoform	51.9	27.6	44.1	29.5	7.8
N Bipolar: Manic	57.5	26.4	50.5	26.8	7.0
D Dysthymia	61.2	35.4	50.0	33.4	11.2 ^a
B Alcohol Dependence	60.1	26.8	50.4	25.8	9.7 ^a
T Drug Dependence	56.4	27.8	47.3	27.1	9.1 ^a
R Post-Traumatic Stress Disorder	55.8	30.3	44.9	29.3	10.9 ^a
SS Thought Disorder	54.9	27.7	47.1	27.4	7.8 ^a
CC Major Depression	66.3	36.8	51.4	36.3	14.9 ^a
PP Delusional Disorder	41.6	29.2	35.8	27.9	5.8

Note. Inpatient $n = 145$. Outpatient $n = 607$.

^a These mean differences are statistically significant when the Bonferroni correction is used to control the Type I error rate across all comparisons at $p < .05$.

Table 6.3 Female vs. Male Comparisons of Mean BR Scores

Scale		Female (F)		Male (M)		F Mean – M Mean
		Mean	SD	Mean	SD	
1	Schizoid	52.2	25.2	54.8	23.4	–2.6
2A	Avoidant	53.4	30.5	49.8	29.5	3.6
2B	Depressive	57.4	31.8	47.3	32.5	10.1 ^a
3	Dependent	62.9	27.1	56.0	26.4	6.9 ^a
4	Histrionic	48.3	24.7	50.9	22.0	–2.6
5	Narcissistic	53.3	23.3	61.6	21.2	–8.3 ^a
6A	Antisocial	49.8	23.6	57.6	23.4	–7.8 ^a
6B	Sadistic (Aggressive)	49.0	21.8	49.8	23.1	–0.8
7	Compulsive	51.8	21.6	54.4	20.9	–2.6
8A	Negativistic (Passive-Aggressive)	54.2	25.7	49.8	25.9	4.4
8B	Masochistic (Self-Defeating)	52.0	29.6	46.8	29.9	5.2
S	Schizotypal	47.4	27.9	46.0	27.9	1.4
C	Borderline	54.7	28.0	49.1	28.0	5.6
P	Paranoid	47.1	27.9	46.8	27.6	0.3
A	Anxiety	63.6	33.8	56.8	35.6	6.8
H	Somatoform	49.2	28.9	41.6	29.1	7.6 ^a
N	Bipolar: Manic	52.7	27.7	50.9	26.0	1.8
D	Dysthymia	56.7	34.2	47.1	33.1	9.6 ^a
B	Alcohol Dependence	48.8	25.9	56.0	26.2	–7.2 ^a
T	Drug Dependence	44.7	27.3	53.9	26.9	–9.2 ^a
R	Post-Traumatic Stress Disorder	50.8	29.2	42.8	29.9	8.0 ^a
SS	Thought Disorder	51.1	27.1	45.8	28.0	5.3
CC	Major Depression	59.3	36.7	48.6	36.2	10.7 ^a
PP	Delusional Disorder	35.6	29.1	38.4	27.3	–2.8

Note. Female $n = 397$. Male $n = 355$.

^a These mean differences are statistically significant when the Bonferroni correction is used to control the Type I error rate across all comparisons at $p < .05$.

BR Score Transformations for Scales X, Y, and Z

As was true for the original norms, the BR transformations for the Disclosure (X), Desirability (Y), and Debasement (Z) scales were tied to percentiles in the distributions of raw scale scores. For each scale, BR 85 was equated with the raw score that came closest to cutting off the highest-scoring 10% of the distribution in the new norm sample. BR 75 was set at the raw score that came closest to cutting off the next 15% of the distribution. BR 35 was set at the raw score at or above which 85% of the sample scored, leaving 15% scoring below 35. A raw score of 0 (34 for Scale X, the lowest raw score considered valid for that scale) was equated with a BR score of 0, and BR 100 was assigned to the highest raw score attained. Linear interpolation was then used to fill in the BR scores corresponding to

raw scores between these anchor points. The BR transformations for these scales are included with those for the clinical scales in Appendix C.

Inconsistency Scale

Since its initial publication in 1994, the MCMI-III has included a simple index, the V scale, designed to detect random responding. The V scale consists of three items: “I flew across the Atlantic 30 times last year,” “I was on the front cover of several magazines last year,” and “I have not seen a car in the last ten years.” These items were deliberately written so that a True response (the keyed response) is very unlikely to be accurate and therefore indicates lack of responsiveness to the content of the items. (The V scale was formerly named Validity, but has now been renamed Invalidity to reflect the fact that higher scores reflect a greater likelihood of random responding.) The

term *random responding* is used here to refer to responding that is not related to item content; possible reasons for random responding are considered in more detail below.

At the same time as the norms were updated, a new response inconsistency scale was developed for the MCMI-III. This new scale is patterned after measures developed by Tellegen to detect random responding on the Multidimensional Personality Questionnaire (MPQ; Tellegen, 1978, 1982), the MMPI-2, and the MMPI-2-RF. It consists of 44 pairs of item responses, with each inconsistent response pair (True for one item in the pair, False for the other item) contributing 1 raw score point to the scale score. For example, a True response to the item "I'm alone most of the time and I prefer it that way," when combined with a False response to the item "When I have a choice, I prefer to do things alone," contributes 1 point to the scale score. The methods used to select this and the other item response pairs that make up the scale are described in detail below. The rationale behind the scale is that the greater the number of inconsistent pairs of item responses, the greater the likelihood of random responding. Thus, there are now two random response indicators for the MCMI-III, and they work in a complementary fashion, as described in detail later in this chapter.

Scale Development

Selection of the 44 item response pairs that make up the response inconsistency scale was based on both content and empirical considerations. Data analyses were based on a sample of MCMI-III protocols that were sent to Pearson for scoring in 2005, 2006, and 2007. From an initial sample of 5,907 protocols, the 5,119 with no missing item responses were selected. Another 204 protocols were excluded because one or more of the V scale items were answered True, or the raw score on the Disclosure scale indicated protocol invalidity (i.e., was outside the range of 34 to 178). Thus, scale development employed a sample of 4,915 cases that was as "noise-free" as possible according to all available criteria.

The first step in scale development was to inspect correlations between MCMI-III items to identify possible candidates for the scale. Pairs of items that correlated .40 or higher were considered. Most of those pairs had similar content,

and the pairs that did not were excluded from further consideration.

Two-by-two tables of observed and expected frequencies were then examined for all of the remaining (i.e., correlated, content-consistent) item pairs. To illustrate the process, the two-by-two table for the item pair mentioned earlier (Items 27 and 92) is shown in Table 6.4. The table shows the frequency of each of the four possible pairs of responses to the two items in the sample of 4,915 cases (the observed frequency, or *o*). It also shows the frequency that would be expected if there were no relationship whatsoever between responses to the two items (i.e., if the items were uncorrelated). This expected frequency is represented by the letter *e*. Of course, these two items *are* correlated ($r = .45$). Accordingly, there are more True/True and False/False response pairs than expected and fewer True/False and False/True pairs than expected. The ratio *o/e* shows how unusual each of the two different-response pairs is relative to expectations. The combination of Item 27 False and Item 92 True is highly unusual, with $o/e = 0.20$. This is easy to understand from a content standpoint, because responding True to "I'm alone most of the time and I prefer it that way" and False to "When I have a choice, I prefer to do things alone" is semantically inconsistent. This item response pair is therefore included in the scale. However, responding True to "When I have a choice, I prefer to do things

Table 6.4 Example of a Two-by-Two Table Used to Select an Item Response Pair

	Item 92 F	Item 92 T	Total
Item 27 F	o = 2,579 e = 2,131	o = 111 e = 559 o/e = 0.20	2,690
Item 27 T	o = 1,314 e = 1,762 o/e = 0.75	o = 911 e = 463	2,225
Total	3,893	1,022	4,915

Note. o = observed frequency of each item response pair.
e = expected frequency of each item response pair if the items were uncorrelated.

Item 27. When I have a choice, I prefer to do things alone.

Item 92. I'm alone most of the time and I prefer it that way.

alone” and False to “I’m alone most of the time and I prefer it that way” is not necessarily inconsistent, because a person may prefer doing things alone but not be able to be alone most of the time because of his or her job, family, etc. This is reflected in the much higher o/e value of 0.75 for this response pair. The item pairs that make up the response inconsistency scale had o/e values ranging between 0.12 and 0.56 in the scale development sample.

Following this methodology, 44 item response pairs were selected to make up the new scale, Inconsistency (called Scale W to fit with the existing MCMI-III response validity indicators, Scales V, X, Y, and Z). Scoring for the scale is included with that for the other MCMI-III scales in Appendix B. Each item response contributes to only one item response pair that is scored on the scale. For example, the True response to the item “I’m alone most of the time and I prefer it that way” is paired only with the False response to “When I have a choice, I prefer to do things alone.” However, the False response to “I’m alone most of the time and I prefer it that way” is paired with the True response to “A long time ago, I decided it’s best to have little to do with people” in a different pair that is scored on the scale.

Selection of Cutoffs

Cutoffs for the Inconsistency (W) scale were selected to mirror the classifications provided by the V scale. When the raw score on V is 2 or 3 (i.e., two or three of the items are answered True), the test protocol is considered *invalid*. When the raw score is 1, the protocol is considered *questionable*, and the user should consider the possibility of random responding when interpreting the results. These cutoffs are rationally based.

To select Scale W cutoffs for classifying test protocols as invalid or questionable, the frequency distribution of scale scores was examined in a separate sample of 5,444 MCMI-III protocols that were sent to Pearson for scoring in 2007. There is no way to know the actual number of random protocols in this sample (or in any other sample of test data gathered without special instructions or procedures), and, as discussed below, randomness is almost certainly not an all-or-nothing phenomenon. Thus, a conservative approach was taken to selecting cutoffs. Specifi-

cally, the raw score that came closest to cutting off the highest-scoring 0.5% of the protocols in the sample of 5,444 was selected for the *invalid* cutoff, and the raw score that came closest to cutting off the next-highest-scoring 2.0% of the protocols was selected for the *questionable* cutoff. A raw score of 10 or higher (i.e., 10 or more pairs of inconsistent item responses) was obtained by 0.4% of the sample, and a raw score of 8 or 9 was obtained by 1.9% of the sample. These cutoffs are conservative in that they are intended to minimize the number of nonrandom protocols classified as invalid or questionable (i.e., maximizing specificity) at the risk of failing to flag random protocols (i.e., lowering sensitivity). Such conservative cutoffs put more faith in the clinician’s judgment in recognizing test results that “don’t make sense” than in hard-and-fast cut scores that are somewhat arbitrary (though necessarily so).

Scale Validation

The Inconsistency scale (W), the Invalidity scale (V), and an index combining the two scales were evaluated using data sets containing varying degrees of randomization. Both research and common sense indicate that, when test-takers respond in a way that is not related to item content, they usually do so for some of the items in a test rather than all of the items. Although it is true that some individuals refuse to cooperate with the testing process and deliberately answer all items randomly (or amuse themselves by making patterns on the answer sheet), this kind of uncooperativeness is unlikely in most settings in which the MCMI-III is administered. Difficulty understanding the test items due to reading or language limitations is probably the most common cause of noncontent-related responding on the MCMI-III. However, most test-takers who are unable to understand all of the items can understand some of them; hence, their protocols are only partially random. Other reasons for random responding—such as difficulty maintaining attention, cognitive impairment, carelessness, and fatigue—are also likely to affect some test answers rather than all of them. Even marking answers in the wrong spaces on the answer sheet is likely to begin at least partway through the test.

In an article on the effectiveness of the V scale of the MCMI-II under full randomization,

Bagby, Gillis, and Rogers (1991) called for further research involving partially random protocols. Berry et al. (1992) reported that test-takers commonly admitted to giving random responses to some but not all items on the MMPI-2. In recent years, it has become common to evaluate random response indicators using actual test protocols that have been partially randomized (e.g., Handel, Ben-Porath, Tellegen, & Archer, 2007). Thus, the MCMI-III random response indicators were tested using data sets with varying percentages of random responses substituted for actual test responses.

The starting point for the randomization manipulation was the set of 5,444 test protocols that was used to select the cutoffs for Scale W (recall that this sample is different from the one that was used for scale development). First, for each subject, 20% of item responses (i.e., 35 responses) were chosen at random by computer and replaced (again by computer) with a randomly generated True or False response, with a 0.50 probability of each. A different set of 35 responses was randomly chosen for replacement for each of the 5,444 subjects in the sample. This data set is referred to as the “20% randomization” data set. The process was then repeated, replacing 40%, then 60%, then 80% of the item responses for each case with randomly generated True or False responses, thereby creating the “40% randomization,” “60% randomization,” and “80% randomization” data sets. Finally, a “100% randomization” data set was created consisting of 10,000 cases with 175 randomly generated True or False responses.

Table 6.5 shows the classification results for Scale V alone, Scale W alone, and a combined index based on the two scales. Results are shown separately for the original sample of 5,444 protocols, the four partially randomized data sets, and the fully randomized data set. The combined V/W index is straightforward: if *either* V or W meets or exceeds its raw score cutoff for classifying a protocol as invalid (2 or higher for V, 10 or higher for W), the protocol is considered invalid. If the protocol is not invalid but *either* V or W meets its raw score criterion for classifying a protocol as questionable (1 for V, 8 or 9 for W), the protocol is considered questionable. If both V and W are below their respective raw score cutoffs (i.e., 0 for V and less than 8 for

W), the protocol is not flagged for possible random responding. In other words, if the classifications based on the two individual scales alone differ, the classification for the combined index is the more negative of the two.

The results in Table 6.5 can be summarized as follows:

- V alone outperforms W alone at 20% randomization, flagging 28.0% of cases as questionable or invalid, compared to 15.7% for W.
- W alone slightly outperforms V alone at 40% randomization. Both classify about half of the cases as questionable or invalid, but W classifies nearly twice as many as invalid (19.7% vs. 10.4%).
- W alone clearly outperforms V alone at 60%, 80%, and 100% randomization, classifying far more cases as invalid.
- At all levels of randomization, the combined V/W index significantly outperforms either V alone or W alone. This is discussed in more detail below.

One possible ambiguity concerning the results in Table 6.5 is that the sample contained evidence of some degree of randomness *before* actual item responses were replaced with random ones. For example, 2.3% of the sample is classified as questionable or invalid based on the Scale W cutoffs, and 1.0% is classified as questionable or invalid based on Scale V. Thus, it could be argued that the 20% randomized data set includes 20% randomized item responses layered over whatever ambient randomness was already present in the sample, and so on for the other partially randomized data sets.

To address this ambiguity, a subsample of 3,136 cases having no evidence of randomness was chosen from the larger sample. Cases with a Scale W score above the sample mean of 2.38 (i.e., a raw score of 3 or higher) were excluded, as were cases with a Scale V score of 1 or higher, a total of 2,272 cases. An additional 36 cases that met the one other fixed MCMI-III invalidity criterion, a raw Disclosure scale score below 34 or above 178, were excluded from the “purified” subsample. The previous analyses were then repeated in this subsample to provide perhaps a more accurate picture of the ability of the ran-

Table 6.5 Validity of Scale V Alone, Scale W Alone, and the Combined V/W Index for Detecting Various Degrees of Protocol Randomization: All Cases

Indicator		Percent of Item Responses Randomized					
		0%	20%	40%	60%	80%	100%
Scale V Alone	% Below cutoff	99.0	72.0	51.6	33.8	21.0	12.1
	% Questionable	0.9	25.3	38.0	43.5	43.0	38.3
	% Invalid	0.1	2.7	10.4	22.7	36.0	49.5
Scale W Alone	% Below cutoff	97.7	84.3	50.0	23.5	11.8	7.9
	% Questionable	1.9	12.1	30.2	30.8	23.5	20.8
	% Invalid	0.4	3.6	19.7	45.6	64.8	71.4
Combined V/W Index	% Below cutoff	96.9	60.7	26.0	7.7	2.7	0.9
	% Questionable	2.6	33.1	46.3	34.3	20.1	13.6
	% Invalid	0.5	6.1	27.7	57.9	77.1	85.5
Scale V mean		.01	.31	.60	.92	1.21	1.50
Scale W mean		2.38	5.25	7.58	9.29	10.44	10.99

N = 5,444, except for the 100% random sample, which is *N* = 10,000.

dom response indicators to identify protocols with known degrees of randomness.

The results for this subset of cases hypothesized to have negligible randomness prior to item replacement are shown in Table 6.6. The results for Scale V alone are nearly identical to those in the total sample. This is to be expected because very few cases in the full sample (1.0%) had V scores other than 0 in the first place. Scale W, on the other hand, does not perform as well in flagging protocols in the subsample with minimal ambient randomness, although the decrement in its performance is negligible at 60% and 80% randomization. This same pattern applies to the combined V/W index. Overall, the conclusions based on the “purified” subsample are very similar to those based on the total sample:

- V alone clearly outperforms W alone at 20% randomization.
- V alone and W alone perform comparably at 40% randomization. V flags somewhat more cases as questionable or invalid, but W classifies more as invalid.
- W alone clearly outperforms V alone at 60%, 80%, and 100% randomization.
- At all levels of randomization, the combined V/W index outperforms either V alone or W alone.

These results are not surprising. To reach the cutoff of 8 for a *questionable* classification on Scale W requires a much larger than average number of inconsistent pairs of item responses. A mere 20% randomization of item responses does not produce this level of inconsistent responding in most cases. On the other hand, reaching the *questionable* cutoff on Scale V requires only one keyed response. This can be shown mathematically to have a 0.271 probability of occurring at 20% randomization, which indeed exactly matches the result shown in Table 6.6. However, as the degree of randomization increases, the performance of the more complex Inconsistency scale improves more rapidly than does that of the simpler Invalidity scale. In fact, the “ceiling” performance of Scale V (i.e., its performance at 100% randomization) can also be predicted mathematically: 50.0% scoring 2 or 3 (i.e., invalid), 37.5% scoring 1 (questionable), and 12.5% scoring 0 (undetected).

What is perhaps surprising is the extent to which Scales V and W flag different cases as random. As one example, in the full sample with 20% randomization (Table 6.5), 149 cases (2.7%) are classified as invalid by V, 194 (3.6%) are classified as invalid by W, and 334 (6.1%) are classified as invalid by the combined V/W index, meaning that only 9 of the 334 cases

Table 6.6 Validity of V Alone, W Alone, and the Combined V/W Index for Detecting Various Degrees of Protocol Randomization: Subset of Cases Hypothesized to Have Negligible Randomness Before Item Replacement

Indicator		Percent of Item Responses Randomized					
		0%	20%	40%	60%	80%	100%
V Alone	% Below cutoff	100	72.9	51.7	33.8	21.0	12.1
	% Questionable	0	24.9	38.1	44.0	42.5	38.3
	% Invalid	0	2.2	10.2	22.2	36.5	49.5
W Alone	% Below cutoff	100	95.1	58.3	25.4	12.3	7.9
	% Questionable	0	4.4	27.4	31.1	24.2	20.8
	% Invalid	0	0.4	14.3	43.5	63.6	71.4
Combined V/W Index	% Below cutoff	100	69.2	30.4	8.6	2.6	0.9
	% Questionable	0	28.2	47.0	35.7	20.5	13.6
	% Invalid	0	2.6	22.6	55.7	76.9	85.5
Scale V mean		.00	.29	.59	.91	1.22	1.50
Scale W mean		0.88	4.33	7.11	9.15	10.37	10.99

Note. $N = 3,136$, except for the 100% random sample, which is $N = 10,000$.

met the invalidity criterion for both scales. As another example, in the 100% random sample, 12.1% scored below the *questionable* cutoff on V and 7.9% scored below the *questionable* cutoff on W, yet only 0.9% scored below the cutoff on both. Thus, V and W complement each other rather than providing redundant information. As another indicator of this, raw scores on the two scales correlated only $r = .11$ in the full sample of 5,444.

Whether considering the results in Table 6.5 or the more conservative results in Table 6.6, the performance of the combined V/W index can be considered *fair* at the challenging 20% level of randomization, *good* at the 40% level, and *excellent* at the 60% level and above. At each of the three highest levels of randomization, fewer than 10% of cases escape detection. At 100% randomization, fully 86% of cases are classified as invalid by the combined index, and fewer than 1% escape detection entirely. Thus, despite the decision to set conservative cutoffs for the Inconsistency scale, the sensitivity of the combined index appears quite good overall.

Implementation

Based on these results, the combined V/W index is recommended for the detection of random

responding on the MCMI-III and, accordingly, is implemented in computer scoring of the test, as described in the following chapter. Table 6.7 summarizes the classification rules for the index. The net result is that the MCMI-III is now able to identify random responding far more accurately than it could when the V scale was the sole random response indicator.

Table 6.7 Classification Rules for the Combined V/W Index for Detecting Random Responding

Scale W Raw Score	Scale V Raw Score		
	0	1	2-3
0-7	OK	Questionable	Invalid
8-9	Questionable	Questionable	Invalid
10+	Invalid	Invalid	Invalid

As a final note, each of the tests mentioned earlier—the MPQ, MMPI-2, and MMPI-2-RF—contains a scale to detect inconsistent fixed responding to complement its variable response inconsistency scale. Fixed response inconsistency scales measure an indiscriminant tendency to answer True to test items (sometimes called

response *acquiescence*) or False to test items regardless of their content. Developing a scale of this kind requires an item pool with a relatively large number of pairs of items with contradictory content, so that answering True to both items in a pair or False to both items in a pair is semantically inconsistent. The MCMI-III item pool does not contain a sufficient number of semantically contradictory items to allow development of such a scale.

However, other MCMI-III validity indicators should do an adequate job of identifying inconsistent fixed responding. Specifically, a protocol

with indiscriminant True responding is very likely to be detected by Scale V (recall that all three scale items are keyed True). Similarly, a protocol with indiscriminant False responding is likely to produce a raw score on the Disclosure scale that is low enough to meet the criterion for invalidity. For example, an all-False protocol results in a raw Disclosure score of 32, which is below the invalidity cutoff of 34. Thus, the MCMI-III now has a good chance of detecting both random and fixed responding when they reach levels that are likely to invalidate the test protocol.

Chapter 7

Administration and Scoring

Administration

The MCMI-III test can be administered in English or Spanish in several formats. The paper-and-pencil format can be scored by computer or by hand. The test can also be administered and scored by computer. Audio recordings are available for clients with normal comprehension who have visual handicaps or limited reading skills. For more information, see the current Pearson psychological assessments catalog.

Appropriate Examinee Populations

The MCMI-III test was developed for use with adults who are seeking mental health treatment and who have at least eighth-grade reading skills. Use of the test should be limited to populations that are not notably different in background from the samples employed to develop the instrument's base rate norms (e.g., it is not appropriate for adolescent populations). Because the normative sample includes individuals from highly diverse settings, subgroup deviance should be minimal.

Preparation and Setting

Psychologists, counselors, psychiatrists, or their administrative assistants should be fully

acquainted with the format, instructions, and other components of the MCMI-III test before administering it.

Because the MCMI-III test requires no special conditions or instructions beyond those provided with the test, administration can be readily and routinely handled by properly trained assistants in clinic settings. The brevity of the test and the minimal facilities required make it convenient for use in settings where time, space, and privacy are limited. The MCMI-III test should not be mailed to clients or sent home with them for completion.

Optimally, the client should be reasonably comfortable and free of distractions or excessive fatigue while taking the test. The clinician should bear in mind that the MCMI-III test is a self-report instrument and that severe anxiety, confusion, sedation, or drug intoxication will interfere with the individual's ability to respond to the test items. If the instrument is administered under such circumstances, the client should be retested at a later date.

Directions and Explanations

Instructions for completing the MCMI-III test are provided with the test and are largely self-

explanatory. Although the instrument is self-administered, some examinees may require assistance. The MCMI-III test is designed to be nonthreatening to most clients and can be presented as a brief inventory that is used as an aid in helping clinicians understand their patients' situation and problems.

The client should be asked to take a few minutes to read the directions carefully and to fill in the appropriate demographic information. The administrator may help the client complete this information. If there is any doubt about whether the examinee understands the directions, the administrator should read them aloud and attend to any questions that arise. Questions raised by examinees concerning procedures or word clarifications should be answered to ensure accuracy and a proper test-taking attitude. Examinees should be encouraged to answer the test items without assistance. Although some examinees may need to be assured that it is common to have trouble answering certain items, it is crucial that they make their own decisions about which response to choose.

The respondent must be willing to approach the inventory with honesty and seriousness for meaningful results to be achieved. Examinees may need to be reassured that they will be doing themselves the most good by assuming a frank and open attitude in responding to the items. Some individuals may express concern regarding the use of their results. It is consistent with individual rights and proper professional practice to respond to these concerns frankly. For example, it is advisable to inform clients beforehand whether the test data will be employed for clinical assessment, research, or other appropriate uses.

Similarly, clients should be told what measures will be taken to maintain the confidentiality of their results and to ensure the security of test forms and associated reports. An informed and voluntary decision to complete the inventory can best be achieved by providing the client with a basis for weighing the potential gains of participation against the possible risks of self-revelation. Because all psychological testing rests on an implicit contract between a professional diagnostician and an examinee, clinicians have the responsibility to fully apprise examinees of the potential advantages and disadvantages of

testing. Without this information, an informed and voluntary decision regarding participation is not possible.

Checking Answer Sheets

If paper-and-pencil administration is used, the administrator should immediately check the completed answer sheet to see that all required information has been provided and to look for double-marked (both True and False) items and an excessive number of omitted items (12 or more). If errors are found, the examinee should be encouraged to correct them. Moreover, to avoid any processing delays for scoring, it is important to check that all answers were filled in with a No. 2 pencil, to erase any marks outside the circles, and to ensure that the form is free of tears and perforations.

Scoring

The MCMI-III test can be scored by computer or by hand. Both scoring methods are described in detail below.

Computer Scoring and Reports

The MCMI-III test can be computer scored using Pearson software on-site or by mailing the completed answer sheet to Pearson. Computer scoring saves time and effort and ensures accuracy. Computer scoring is required to obtain an interpretive report.

MCMI-III users who choose computer scoring have three report options: an interpretive report, a profile report, and a corrections report. For more details about report options, see the current Pearson psychological assessments catalog.

Interpretive Report

The MCMI-III Interpretive Report includes scale scores and in-depth interpretive text organized in the following format.

1. *Cover Page.* The cover page lists the client's name (optional), ID number, age, gender, assessment setting, race, marital status, and test administration date.
2. *Capsule Summary.* This section notes the appropriate context, limitations, and restrictions of the use of the MCMI-III test. It summarizes the patient's demographics, the

nature of his or her reported complaints, the duration of the present disorder, the probable severity of the patient's difficulties, the probable *DSM-IV* diagnostic assignments for Axes I and II, and some of the patient's major clinical features and their likely course of treatment.

3. *MCMI-III Score Profile*. The profile page shows the client's raw and BR scores for each of the MCMI-III clinical scales and modifying indices. In addition, the patient's ID number, an indication of the validity of the report, and his or her personality code, syndrome code, and demographic code are presented (these codes are described later in this chapter).
4. *Facet Scales (optional)*. The first of these optional pages lists and graphs the three facet scale scores for each of the three highest personality scales with a BR score of 65 or higher (see the section in Chapter 7 about interpreting facet scale scores). The second page lists the raw and BR scores for all 42 facet scales.
5. *Response Tendencies*. This section assesses the probable validity and reliability of the test data on which the report is based and describes any score adjustments that have been made (see the Base Rate Adjustments section later in this chapter).
6. *Axis II: Personality Patterns*. This section consists of several paragraphs describing the basic features that characterize the patient's personality pattern (Axis II) and level of impairment.
7. *Grossman Personality Facet Scales (optional)*. This optional section interprets the highest of the facet scale scores that are included on the facet scales graph described above.
8. *Axis I: Clinical Syndromes*. This section provides a descriptive summary of the patient's current clinical syndromes (Axis I), interpreted in the context of his or her personality pattern and presented in order of relative significance or salience.
9. *Noteworthy Responses*. Item responses that suggest problem areas that require further

evaluation are listed here, grouped into several categories (Health Preoccupation, Interpersonal Alienation, Emotional Dyscontrol, Self-Destructive Potential, Childhood Abuse, and Eating Disorder). See Appendix F for a list of items in each category.

10. *Possible DSM-IV Multiaxial Diagnoses*. Possible diagnoses are organized in a multiaxial diagnostic format, including detailed subsections on Axes I, II, and IV.
11. *Treatment Guide*. The major considerations and recommendations regarding treatment planning for patients who demonstrate clinical features similar to those of the client are presented here.
12. *Item Responses (optional)*. This optional section lists the client's response to each of the 175 MCMI-III items.

A sample interpretive report can be found in Appendix H.

Profile Report

The MCMI-III Profile Report consists of a cover page; an Interpretive Considerations section that briefly summarizes the patient's demographic features, presenting problems, response style, and self-reported Axis I difficulties; the MCMI-III score profile described above; and the optional facet scale results and item responses described above. A sample profile report can be found in Appendix I.

Corrections Report

The MCMI-III Corrections Report includes most of the information that is in the interpretive report. It also includes a special section that summarizes behavioral and characterological features that are relevant to individuals in prison settings. (See the *MCMI-III Corrections Report User's Guide* for detailed information about this report.)

Personality and Syndrome Codes

Although MCMI-III profiles may be summarized informally simply by noting their high-point Axis II and Axis I characteristics, a more explicit system for coding profiles is often desirable. The following paragraphs outline the system that is used to produce the personality and syndrome codes that are printed on the score profile page

of the computerized MCMI-III reports. A sample code is listed below for reference.

Personality Code: 8A 8B 3 ** 2A 2B 6A * - + 6B 7 1 4
" 5 // S ** C * //
Syndrome Code: B A ** T D * // SS ** - * //

Note the structure of the code. The 24 clinical scales are divided into four main sections, each of which is followed by double slashes: the Clinical Personality Patterns scales (1–8B), the Severe Personality Pathology scales (S, C, P), the Clinical Syndromes scales (A, H, N, D, B, T, R), and the Severe Clinical Syndromes scales (SS, CC, PP). The first two sections make up the *personality code*, the second two sections the *syndrome code*.

First, the 11 Clinical Personality Patterns scales are listed in the code in descending order of magnitude. Scales with scores of BR 85 or higher are located to the left of double asterisks (**), those in the score range of BR 75–84 are followed by a single asterisk (*), scales with BR scores from 60 to 74 are followed by a plus sign (+), scales with BR scores from 35 to 59 are followed by quotation marks ("), and scales with scores below 35 are not followed by a symbol but simply close the set to the left of the first double slashes. A hyphen (-) is employed to indicate that no scale obtained a BR score within a particular numerical range. Scale scores that are tied or that differ by only one point are underlined (any intervening symbols are also underlined).

Second, for the three additional sections of the code, only those scores that exceed BR 74 are listed. Again, double asterisks (**) follow scales with scores over BR 84, and a single asterisk (*) follows scales within the range of BR 75–84.

Thus, the sample code shown above translates as follows. Among the Clinical Personality Patterns scales, 8A, 8B, and 3 are highest (in descending order), all with a BR score greater than 84. Scales 2A, 2B, and 6A are tied or within one point of one another and are between BR scores 75 and 84 (inclusive). There are no scales between BR 60 and 74. Scales 6B, 7, 1, and 4 decrease in magnitude and are between BR 35 and 59. Scale 5 is below BR 35. Among the Severe Personality Pathology scales, S is over BR 84, and C is between BR 75 and 84.

In the syndrome code, the Clinical Syndromes scales B and A, in descending order, are above BR 84. Scales T and D are between BR 75 and 84. Among the Severe Clinical Syndromes scales, SS is over BR 84, and there are no scales between BR 75 and BR 84. Complicated as these codes may appear at first, they become more familiar with regular use and help facilitate the communication of clinical profiles.

Demographic Code

The demographic code string, reported on the score profile page of the computer-generated reports, is coded and reported in the following order. Each item in the string is followed by a slash (/). Omitted demographic items are indicated by hyphens (-) in the coded string.

1. Identification number (9 digits maximum)
2. Setting (2 characters):
 - ON Outpatient (never hospitalized)
 - OP Outpatient (previously hospitalized)
 - IM Inpatient (psychiatric hospital)
 - IG Inpatient (general hospital)
 - CI Correctional inmate
 - SC College student counselee
 - OT Other
3. Gender (1 character):
 - M Male
 - F Female
4. Age (2 digits)
5. Race (1 character):
 - W White
 - B Black
 - H Hispanic
 - A Asian
 - I American Indian
 - O Other
6. Marital status (1 character):
 - N Never married
 - F First marriage
 - R Remarried
 - S Separated
 - D Divorced
 - W Widowed
 - C Cohabiting
 - O Other
7. Years of school completed (1–2 digits)

8. Major problems—1st (2 characters), 2nd (2 characters):
 - MA Marital/family
 - JO Job/school/work
 - LO Loneliness
 - MD Moodiness
 - SC Self-confidence
 - IL Ill/tired
 - AL Alcohol
 - DR Drugs
 - SX Sexual problems
 - AN Antisocial behavior
 - OT Other
9. Diagnostic code: *DSM-IV* Axis I (5 digits)
10. Duration of most recent Axis I episode (2 characters):
 - 01 Less than one week
 - 02 One to four weeks
 - 03 One to three months
 - 04 Three to twelve months
 - 05 Periodic; one to three years
 - 06 Continuous; one to three years
 - 07 Periodic; three to seven years
 - 08 Continuous; three to seven years
 - 09 More than seven years
 - 10 Cannot categorize
11. Diagnostic code: *DSM-IV* Axis II (5 digits)

Unscorable Tests

Any of the following three conditions will cause the computerized scoring program to label an MCMI-III test unscorable.

1. The client's identification number, birth date, test date, and gender are not all recorded and valid.
2. The client is younger than 18. The MCMI-III test was validated and normed on adults only.
3. Twelve or more items have missing or double-marked responses. With this many omitted responses, it is not possible to ensure the validity of the scale scores because an adequate number of items may not have been completed for every scale.

Correcting the condition that caused the test to be labeled unscorable (e.g., supplying missing information or correcting invalid information) will allow the test to be scored.

Invalid Tests

An MCMI-III test is considered invalid if any of the following conditions are met.

1. The raw score on the Invalidity scale (Scale V) is greater than 1 or the raw score on the Inconsistency scale (Scale W) is greater than 9. Scales V and W work in conjunction to help detect random responding, as described in detail in Chapter 6.
2. The raw score on Scale X (Disclosure) is less than 34 or greater than 178. Scores this extreme indicate that the respondent may have over- or underreported significant symptoms to such a degree that the results cannot be interpreted.
3. None of the BR scores on the Clinical Personality Patterns scales (1 through 8B) are above 59. In this case, no clear personality pattern emerges from the test data and therefore no interpretation should be attempted.

Hand Scoring

Hand scoring the MCMI-III test is a very complex procedure that commonly results in scoring errors. Each test should be scored twice to increase accuracy. It may be helpful to keep the following in mind during hand scoring: first, the only item that is keyed False and weighted as a prototypal item (i.e., 2 points) is Item 23 on the Alcohol Dependence scale. All other items that are keyed False receive a weight of 1. Second, all items on Scales Y and Z (Desirability and Debasement) have a weight of 1.

The first step in hand scoring is to ensure that the client is at least 18 years old, that fewer than 12 items have missing or double-marked responses, and that the Invalidity scale score is less than 2 and the Inconsistency scale score is less than 10. The other two invalidity conditions described above should be checked at those points in the hand-scoring process when the pertinent scores have been calculated.

Next, raw scores are calculated. Fortunately, shorter scales and a revised item-weighting scheme make raw score calculation easier for the MCMI-III test than for the MCMI-II test. Because the MCMI-III items are weighted depending on their validation evidence and their

centrality to their respective prototypes, raw score calculation is not a simple matter of adding up the number of responses keyed to each scale. Appendix B lists the direction of keying and the scoring weight for each item that is scored on each MCMI-III scale. For example, if the first item on the Schizoid scale, Item 4, is marked True, 1 point is added to the Schizoid raw total. If the next item, Item 10, is marked True, 2 points are added to the Schizoid raw total, and so on for all items keyed True. Item 32 is the first Schizoid item that is keyed False. If this item is marked False, 1 point is added to the Schizoid raw total.

The single exception to this procedure for calculating raw scale scores occurs for Scale X (Disclosure). The raw score for Scale X is the sum of the raw scores for Scales 1 through 8B, except that the score on Scale 5 is weighted by 2/3 because Scale 5 is longer than the other scales. Thus,

$$\text{Scale X} = \text{Scale 1} + \text{Scale 2A} + \text{Scale 2B} + \text{Scale 3} + \text{Scale 4} + \text{Scale 5} \times \frac{2}{3} + \text{Scale 6A} + \text{Scale 6B} + \text{Scale 7} + \text{Scale 8A} + \text{Scale 8B}$$

The value for Scale X obtained from this formula is then rounded to the nearest integer.

Base Rate Transformations

Raw scores are rarely final scores. Because the individual scales of an inventory differ in their number of items and the shape of their distributions, almost all instruments require raw scores to be transformed as a means of putting them on a common metric. Once the MCMI-III raw scores have been calculated, the next step is to convert them to base rate (BR) scores. This involves looking up each raw score in Appendix C. Note that the same BR transformation table is used for males and females. No base rate transformations are provided for the Invalidity and Inconsistency scales because the raw score values are used directly.

Base Rate Adjustments

After the raw scores have been converted to BRs, one or more adjustments may then be required to arrive at final BR scores. These adjustments are summarized below and are described in detail in Appendix G.

Disclosure Adjustment. Scale X (Disclosure) assesses the degree to which a client was frank and self-revealing or reticent and secretive when he or she responded to the MCMI-III items. The Scale X raw score determines whether an adjustment is made to the initial BR scores of Scales 1 through PP. If the Scale X raw score is low (< 61), BRs on Scales 1 through PP are increased. If the Scale X raw score is high (> 123), BRs on Scales 1 through PP are decreased. The specific amount by which the BRs are increased or decreased depends on the value of the Scale X raw score (see Appendix G). For Scale X raw scores between 61 and 123 (inclusive), no adjustments are made to the BRs on Scales 1 through PP.

Anxiety/Depression Adjustment. An individual's scores may be distorted if he or she completed the test while in an acute or intense emotional state (Hirshfeld et al., 1983; Reich, Noyes, Coryell, & O'Gorman, 1986). The Anxiety/Depression adjustment is applied to the five personality scales that are most frequently affected by psychic turmoil: Avoidant (2A), Depressive (2B), Masochistic (8B), Schizotypal (S), and Borderline (C). This adjustment may decrease BR scores on these scales if the BR score on Anxiety (Scale A) or Dysthymia (Scale D) is 75 or higher, the cutting score that indicates the presence of a syndrome. Various combinations of the length of the client's Axis I episode and inpatient versus noninpatient status determine the extent to which the BR scores are reduced.

Inpatient Adjustment. Some recently hospitalized patients tend to deny the obvious severity of their current emotional state. The Inpatient adjustment may increase BR scores on the Thought Disorder (SS), Major Depression (CC), and Delusional Disorder (PP) scales if the client is an inpatient and the Axis I duration is four weeks or less. The Inpatient adjustment is determined using Appendix G.

Denial/Complaint Adjustment. Some personality styles, notably the histrionic (Scale 4), narcissistic (Scale 5), and compulsive (Scale 7), tend to be characterologically defensive and therefore produce flatter profiles in the self-report modal-

ity. To compensate, if one of these scales is the highest among the Clinical Personality Patterns scales (1 through 8B), the BR score for that scale is increased by 8 points. Otherwise, no adjustment is made.

Additional Considerations. No BR score may be adjusted to a value below 0 or above 115. After each adjustment is made, any BR values that are less than 0 are changed to 0, and any BR values that are greater than 115 are changed to 115.



Chapter 8

Interpretation

Three steps are involved in the interpretation of an MCMI-III score profile: first, evaluating the validity of the protocol; second, making single-scale diagnostic decisions; and third, assessing overall clinical functioning based on profile patterns. Additional optional steps include integration of developmental or historical information and examination of test data at the item level.

The interpretive method detailed here is concerned only with information internal to the MCMI-III test. As with all diagnostic or personality tests, it is important to integrate results with information obtained from other sources. Conclusions are more accurate when they are made within the context of an individual's demographic background, history, interview presentation, and other clinical findings. Blind interpretations may be useful during graduate training, but their use in professional clinical settings is unwise. Information concerning gender, age, socioeconomic status, mental status, and other test and interview data provides perspective for interpreting MCMI-III profiles. Moreover, the ability to integrate auxiliary data is an important advantage the clinician has over the automated interpretive report. At best, computer programs take into account the role of potentially relevant group characteristics, but they cannot take into account individually specific moderator variables.

Evaluating Questionable Inventories

Test data are generally appraised as being either valid or invalid. As a result, most test results are either interpreted or discarded. However, the problem of evaluating the validity of an individual protocol is much more complex. Not only may all test profiles be viewed as valid or invalid to some degree, but within a profile some scale scores may be valid and others may be less so. This may occur, for example, when item omissions cluster around themes tapped by a particular construct. In general, clinicians should be aware that the validity of any individual test profile cannot be appraised in an all-or-none fashion regardless of the modality through which the test information was gathered (e.g., self-report, structured interview, etc.).

Clinical assessment literature is replete with studies that suggest factors that may distort the results of self-report inventories. Such distortions may produce test data that either require adjustments to increase their validity or are so invalid that they are totally unusable for measurement or interpretive purposes. Evidence for the importance of these confounding factors is mixed. Our opinion is that deliberate misrepresentation or random responding on personality and clinical

inventories is much less frequent than is commonly thought. Similarly, response sets seem to have a much smaller influence than substantive item content on how individuals respond to self-report items. Efforts to minimize the influence of response sets on substantive scales may significantly dilute the power of those scales to discriminate important personality trait differences. Furthermore, inventories whose scoring is substantially determined by external-criterion research are only minimally affected by problems of stylistic or intentional distortion; potentially confounding items either fail to correlate with external criteria and are thereby parceled out, or they prove to be predictive of those criteria and are retained.

The preceding argument should not be taken as a rationale for dismissing the potential impact of problematic test-taking attitudes. Efforts to control for such biases are part of the tradition of the MCMI test. Four base rate adjustments were discussed in Chapter 6. Each should serve as a sign of potential stylistic, situational, or personologic tendencies and as an index of complications that may affect the reliability or validity of the MCMI-III scores.

Random Response Indicators (Scales V and W)

Two scales, Invalidity (V) and Inconsistency (W), work in conjunction to help detect random responding. Scale V consists of three items for which a True response is highly implausible if the test-taker is responding to the item content. Scale W comprises 44 item response pairs; each keyed pair of responses is semantically contradictory. An index combining Scales V and W is used to identify test protocols that are considered questionable or invalid because of possible random responding. Details concerning the construction, validation, and use of these scales are provided in Chapter 6. Table 6.7 provides a summary of the Scale V and W cutoffs used to classify a protocol as questionable or invalid.

If a protocol is classified as *questionable* based on the random response indicators, the possibility that reading or language limitations, difficulty maintaining attention, cognitive impairment, carelessness, fatigue, or deliberate uncooperativeness affected the patient's test responses should be investigated. Until this is re-

solved, the test results should be interpreted with caution. A protocol classified as *invalid* based on Scales V and W should not be interpreted.

The Disclosure Index (Scale X)

The Disclosure index (Scale X) is the next index that may signify problematic response behavior, notably whether the patient was inclined to be frank and self-revealing or reticent and secretive. The Disclosure index is calculated from the degree of positive or negative deviation from the midrange of an adjusted composite raw score total for Scales 1 to 8B (see Chapter 7). Scale X is the only MCMI-III scale for which a low score is clinically interpretable. When the raw score for this scale is below 34 or above 178, the protocol should be considered invalid. Approximately 10% of respondents obtain scores of BR 85 or above, and approximately 25% of respondents score BR 75 or above.

The Desirability Index (Scale Y)

The Desirability index (Scale Y) assesses the degree to which the patient's results may have been affected by his or her inclination to appear socially attractive, morally virtuous, or emotionally well composed. BR scores over 75 on Scale Y indicate the tendency to place oneself in a favorable and personally appealing light. The higher the score, the more likely it is that the patient concealed some aspect of his or her psychological or interpersonal difficulties.

The Debasement Index (Scale Z)

The Debasement index (Scale Z) generally reflects tendencies that are the opposite of those reflected by Scale Y. However, on occasion both indices are high, especially among patients who are unusually self-disclosing. In general, BR scores above 75 on the Debasement index suggest an inclination to deprecate or devalue oneself by presenting more troublesome emotional and personal difficulties than are likely to be uncovered upon objective review. An especially elevated score deserves close examination not only for the purpose of gaining a more accurate assessment of what might prove to be a distorted level of psychological problems, but also to determine whether it signifies a cry for help from

a patient who is experiencing an especially distressing degree of emotional turmoil. 68-4-80

Response Style Considerations

As with other measures, Scales X, Y, and Z may be interpreted according to their configural pattern. For example, a low BR on Disclosure and Desirability with a high Debasement score suggests a moderate exaggeration of current emotional problems that is likely to have been sufficiently corrected so that it does not affect the test's interpretive validity. In a case in which Scale X is low and Scales Y and Z are high, the patient endorsed antithetical symptoms and characteristics, raising questions about the validity of the profile. This configuration is occasionally obtained by patients with severe depressive disorders, usually of an agitated type. (The computer-generated interpretive report provides an analysis of the meaning of the configuration of Scales X, Y, and Z.)

Other profile patterns are also notable. Examinees who try to portray themselves in the best possible light may have a low score on Scale X and a high score on Scale Y. Research with the MCMI-II test pointed to a tendency for such individuals to have elevated scores on some of the personality scales. Depending on whether they wish to appear cooperative, self-confident, or conscientious, such individuals may peak on the Dependent, Narcissistic, or Compulsive scale, respectively. In contrast, such profiles typically have no elevations on any of the more symptom-oriented scales such as the Severe Personality Pathology scales or the Clinical Syndromes scales. The difficulty of distinguishing between a "fake-good" profile and a valid profile from an individual who actually does have the attributes in question (e.g., dependency or perfectionism) has been noted in the literature (Bagby, Gillis, Toner, & Goldberg, 1991; Retzlaff, Sheehan, & Fiel, 1991).

The other side of the coin is the "cry for help" profile of individuals whose response style makes them look more psychologically impaired than they really are. In those cases, Scales X and Z are elevated as are many of the scales associated with severe psychopathology (Bagby et al., 1991; McNiel & Meyer, 1990; Retzlaff et al., 1991; Van Gorp & Meyer, 1986).

Although the intent of Scales X, Y, and Z

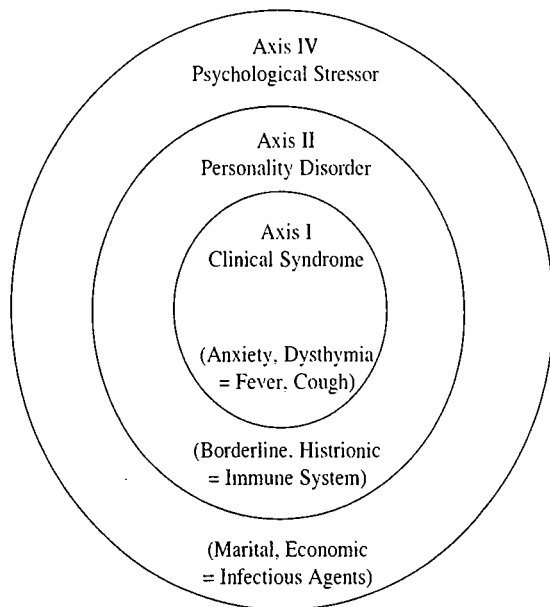
is to compensate for the effect of these three response styles, they cannot make an invalid profile valid. Nor can they truly compensate when the response style is the predominant driving force behind an individual's scores. The discovery of a strong response style may be of clinical relevance in and of itself because it allows the practitioner to form an opinion about how candid or realistic a patient can be about his or her problems. To go beyond the interpretation of a strong response style, the practitioner may need to administer the test again, perhaps after encouraging the individual more emphatically to answer each item as honestly as possible.

Philosophy of the Multiaxial Model

The MCMI-III test is a multiaxial instrument, derived from an integrated model of psychopathology and personality, and coordinated with the multiaxial model of the *DSM-IV*. The interpretive logic of the MCMI-III test follows largely from these two basic facts. Clinicians who use the MCMI-III test should do so with the goal of achieving an understanding of the person as an integrated entity, not as an aggregation of diagnoses.

The movement toward integrationism in the conception of disease is not just an ideal; it is an empirical, historical fact, illustrated by the evolution of the health sciences through two paradigm shifts (neither of which has been completed in psychopathology). The series of concentric circles in Figure 8.1 represents changes that have evolved in medicine over the past century. In the center is Axis I, the so-called clinical syndromes (for example, depression and anxiety). These parallel what characterized the state of medicine a hundred or more years ago. In the early and mid-nineteenth century, physicians defined their patients' ailments in terms of their manifest symptomatology—their sneezes, coughs, boils, and fevers—labeling these "diseases" with terms such as "consumption" and "smallpox." In contrast, the outer ring of Figure 8.1 parallels Axis IV of the *DSM-IV*. The related medical paradigm shift occurred approximately a century ago when illnesses began to be viewed as the result of intrusive microbes that infect and disrupt the body's normal functions. In time, medicine

Figure 8.1 Interactive Nature of the Multiaxial System



began to assign diagnostic labels to reflect this new etiology, replacing the old descriptive terms. “Dementia paralytica,” for example, came to be known as “neurosyphilis.”

Fortunately, medicine has progressed in the past decade or two beyond its turn-of-the-century “intrusion disease” model, an advance that has been most striking in the last 20+ years because of immunological diseases such as AIDS. This progression reflects a growing awareness of the key role of the immune system, the body’s intrinsic capacity to contend with the omnipresent multitude of potentially destructive infectious and carcinogenic agents that pervade the physical environment. Medicine has learned that it is not the symptoms (the sneezes and coughs) or the intruding infections (the viruses and bacteria) that are the key to health or illness. Rather, the ultimate determinant is the competence of the body’s own defensive capacities. So, too, in psychopathology it is not anxiety or depression or the stressors of early childhood or contemporary life that are the key to psychological well-being. Rather, it is the mind’s equivalent of the body’s immune system—that structure and style of psychic processes that represent our overall capacity to perceive and to cope with our psychosocial world—in other words, the psychological construct we call *personality*.

The multiaxial model has been specifically composed to encourage integrative conceptions of the individual’s manifest symptoms in terms of the interaction between longstanding coping styles and psychosocial stressors. Our task as clinicians is to recapitulate the historical progression, so to speak, within the individual person to achieve a conception of each patient’s psychopathology that does not merely diagnose or document his or her boils and sneezes (the Axis I disorders) but instead contextualizes these manifest disorders in terms of the larger context of the individual’s style of perceiving, thinking, feeling, and behaving. The interpretive process may be described in terms of several levels or orders that facilitate such integrated interpretations.

First-Order Interpretive Logic: Making Diagnoses

The first interpretive level is concerned only with providing information for making diagnoses. Diagnostic decisions based on brief self-report inventories are usually justified under three conditions: first, when the cost and time involved in test administration are minimal and the level of accuracy is sufficiently high to serve as a screening device or preliminary assessment procedure; second, when the level of accuracy is at least equal to more time-consuming and costly diagnostic procedures; and third, when the inventory adds a measure of incremental validity to established and efficient ways of assessing patients (Sechrest, 1963). The history of the MCMI test in clinical practice and the validity data presented in Chapter 4 of this manual support the use of the MCMI-III test under each of these three conditions.

The diagnostic level of interpretation is the most basic and potentially the most flawed. At this level, we are interested only in determining which scales are elevated and what diagnoses should be made. For the MCMI-III test, this involves examining which scale elevations exceed the BR 75 and 85 cutting scores. For the Axis II scales, BR 75 may be viewed as indicating the presence of clinically significant personality traits. If an individual obtains BR 75 on the Borderline scale, for example, he or she is likely to have traits of the borderline construct. In con-

trast, elevations at BR 85 or above are likely to indicate pathology pervasive enough to be called a *personality disorder*. For the Axis I scales, BR 75 indicates the *presence* of a *syndrome*, and BR 85 indicates *prominence*. Such elevations form the basis of diagnostic hypotheses that should then be considered in the context of other information outside the inventory, including the patient's verbal report, the therapist's impression of the patient, reports from significant others, background information, and so on. If after considering all these factors the clinician is still uncertain whether a particular diagnosis should be made, the prototypal items of the given scale may be reviewed to determine which *DSM-IV* diagnostic criteria are supported by the subject's responses to MCMI-III items that parallel those criteria.

In view of the end goal—an integrated understanding of the individual and his or her pathologies—a purely diagnostic level of interpretation represents a gross approximation that must be considered simplistic in the context of the total clinical mission. These present-versus-absent judgments present the whole person in a highly distilled form. First, the dichotomous nature of diagnosis neglects the importance of subthreshold disorders that often have legitimate functional implications such as the potential to exacerbate other conditions or to bloom into disorders themselves. Thus, while BR 72 is below the BR 75 threshold, it nevertheless indicates that the client endorsed a number of items that are relevant to the construct for which this score was obtained and the clinician may wish to explore further. Second, a rigid diagnostic focus ignores the configural potential of the inventory. A narcissistic-dependent, for example, is very different from a narcissistic-antisocial, even if the BR of the second personality pattern is less than 85 and therefore likely to be stylistic rather than dysfunctional. And third, the diagnostic model neglects the immunological role that personality plays in the individual's resistance or susceptibility to developing symptoms under conditions of psychosocial stress.

In fact, the very idea that personality can be diagnosed is so potentially damaging to the underlying logic of the multiaxial model that it requires special comment. The terms *diagnosis* and *personality disorder* explicitly invoke the so-

called medical model, implicitly reducing Axis II to Axis I by painting the personality disorders as a variant of disease, a foreign entity or lesion that invades the individual to undermine his or her normal functions. That is, it reduces constructs that are intended to play the immunologically mediating role between psychosocial stress and symptom production to the level of boils and sneezes.

In turn, the medical misconception is reinforced by the idea that personality is a type or substance that fills the vessel of the person in a homogeneous or undifferentiated form (with the further unfortunate implication that the therapist must change the person by emptying and refilling the vessel). In Chapter 2, the personality disorders were operationalized across a number of functional and structural domains. In contrast to the static idea of type, this internally differentiated functional-structural conception of personality—which is unbiased by a psychodynamic, behavioral, cognitive, or even evolutionary perspective—follows directly from the nature of personality as a construct, the idea that personality is concerned with the entire matrix of the person. Accordingly, this perspective is an optimal corrective to the paradigmatic misconceptions perpetuated by the medically evocative term *personality disorder*. This perspective has important interpretive implications to which we will return in a subsequent section. Far better than a purely diagnostic or medical approach to the MCMI-III test is one that interprets scale elevations configurally and contextually.

Configural Interpretation of Personality Scales

The first step in configural interpretation is concerned with achieving an adequate characterization of the subject's personality. Just as a dimensional perspective is necessary to make realistic quantitative appraisals, a configural perspective is necessary to make realistic qualitative appraisals. Rarely do patients present as pure prototypes. Except where the highest personality pattern is elevated above the next by a significant degree, secondary scale elevations are almost always meaningful. Consider the following examples.

Example 1. Two patients have their highest Clinical Personality Patterns score on Dependent (Scale 3). In one case, the second highest is Avoidant (Scale 2A), in the other, Histrionic (Scale 4). Taken alone, a predominantly dependent personality would suggest features such as a need for stronger, nurturing figures, a tendency to be unassertive and placating, a weak and fragile self-concept, and a naive and uncritical attitude toward interpersonal difficulties. However, somewhat different features would be found in a dependent-avoidant, including a defensive withdrawal from social relationships (stemming from a fear of rejection), a tendency to deaden excessive interpersonal sensitivity, a persistent undercurrent of depression and loneliness, and hidden, intense resentment toward those on whom the individual depends. Different features would be found in a dependent-histrionic. In addition to the characteristic compliance, naiveté, and dependency seen in the pure dependent, also evident are attention seeking through self-dramatization, persistent efforts to create interpersonal harmony at the expense of internal values, and occasional periods of hypomanic euphoria and depression. Thus, the addition of a second high-point scale (just one step toward profile analysis) alters the personality picture significantly.

Example 2. Assume that two patients score above the cutting line on Alcohol Dependence (Scale B) but differ in their basic personality style. One is a histrionic-antisocial (4/6A), the second a compulsive-dependent (7/3). These patients are likely to exhibit and deal with their alcoholism in clearly divergent ways. The mixed histrionic-antisocial type is likely to be unconventional and irresponsible, to act out in an expunitive and impulsive manner, and to react to external efforts to control his or her drinking with anger and irritability. In contrast, the mixed compulsive-dependent type is likely to be conventional and controlled (except for the alcoholic episodes), experience deep conflicts over his or her dependency, act in a self-punitive manner, have attempted unsuccessfully to control his or her alcoholism, and react to the concerns of others with contrition and guilt.

As these examples illustrate, different secondary-scale elevations make for quite different interpretations.

Configural Domain Synthesis

In this and the next section, we will sketch the logic of creating rich personality descriptions from MCMI-III profiles. There are, however, certain caveats that modify this ideal procedure. These will be noted in the sections that follow.

As philosophers of science have long noted, the utility of any classification system rests with the capacity of its categories to suggest attributes beyond those that are immediately observed. The eight structural and functional domains listed for each personality construct (see Chapter 2) form the elements that are associated, universally or with high probability, with other clinical attributes within their respective personality patterns. Each personality, then, is a kind of covariance structure of characteristics within these domains. The presence of any one characteristic implies the presence of others with which it is associated within a particular personality.

The clinical attributes discussed in Chapter 2 have been formulated to be both comprehensive, in that they are exhaustive of the domains in which personality is anchored (rather than biased toward any one school), and comparable, in that they exist at approximately equal levels of abstraction. Thus, they become important reference points in the interpretation of the MCMI-III test. We may say with high probability that any individual judged to be narcissistic is likely to be behaviorally haughty, interpersonally exploitive, cognitively expansive, and so on. In general, to the extent that one and only one MCMI-III Axis II scale dominates the profile (i.e., is elevated to the exclusion of other scales), that personality pattern may be said to be prototypal of the individual taking the test, and its associated domain characteristics can be taken from the Chapter 2 tables without significant alteration.

Such prototypal profiles are rare, however. In most cases, the test results will reflect a two-point or three-point code. In this case, it is necessary to integrate or blend the descriptive paragraphs for each domain, weighting them to the extent that each code dominates the profile. Thus, if BR 110 is obtained on the Narcissistic scale with a secondary elevation of BR 76 on the Dependent scale, the clinical report should be written to reflect characteristics that are more consonant with the Narcissistic prototype. The degree to which the clinician chooses to weight

the various features of these disorders depends on a variety of factors, including the availability of auxiliary information.

Deriving Personality Characterizations From Equally Elevated Personality Scales

Configural interpretation becomes more challenging when two or more scales are elevated to approximately the same level (i.e., when there is no real configuration). In this case, the clinical domain attributes of the elevated prototypes all become hypotheses about the client within their respective domains. The choices are to either integrate or blend aspects of the relevant domains or to pull from one more or less to the exclusion of the others.

If, for example, a narcissistic-dependent profile is found with both scales approximately equally elevated, we are faced with two choices in each of eight domains, as illustrated in Table 8.1. The clinician may appeal to two complementary sources of supplemental information when seeking to support specific propositions at the clinical domain level. The MCMI-III item responses are the first source. Recall that the MCMI-III test was constructed in accordance with the prototypal model that is the basis for the *DSM-IV*. An item is assigned a weight of 1 or 2 points depending on its centrality to its respective construct and the strength of its validation evidence. When seeking to sustain or reject the

multitude of domain hypotheses suggested by two or three almost equally elevated personality scales, prototypal items may be reviewed as "critical items" that swing clinical judgment in one direction or the other. Despite objections to the MCMI-III's item weighting system, the prototypal items have an important interpretive and taxonomic or substantive value. However, as a review of the prototypal items for any of the personality scales will show, the MCMI-III test does not contain items that sample each functional and structural domain for every personality. This is in part because some domains are more inferential than others (the defense mechanisms, for example) and are not readily self-reported, and in part because of its coordination to the *DSM-IV*, which is biased toward some domains rather than others (Millon & Davis, 1996). Nevertheless, appeal to item-level information is often decisive when two or three personality scales must be synthesized in the form of a clinical report.

The second source of information that can be used to sustain domain hypotheses lies outside the MCMI-III results in the form of clinical impressions, reports from other individuals who know the client well, the results of other instruments (both objective and projective), and the nature of the referral question. Thus, continuing the example above, if an individual is interviewed and found to be indecisive and lacking in self-confidence, this supports a dependent interpretation over a narcissistic interpretation within the self-image domain. However, if others report that the subject is grandiose and exhibits a sense of specialness about him- or herself, this suggests that evidence for the "inept" self-image of the dependent and the "admirable" self-image of the narcissist should be integrated in some fashion, perhaps through the psychodynamic notion that grandiosity protects against underlying feelings of inferiority. As with any instrument, auxiliary knowledge and psychological theory come together to create rich and plausible interpretations of the individual's character.

Table 8.1 Domains of the Narcissistic and Dependent Prototypes

	Narcissistic	Dependent
Expressive Acts	Haughty	Incompetent
Interpersonal Conduct	Exploitive	Submissive
Cognitive Style	Expansive	Naïve
Self-Image	Admirable	Inept
Object Representations	Contrived	Immature
Regulatory Mechanisms	Rationalization	Introjection
Morphologic Organization	Spurious	Inchoate
Mood/Temperament	Insouciant	Pacific

The Severity and Functionality of Pathological Personalities

Normality and pathology of personality exist on a continuum. The sharp diagnostic dividing line created in the *DSM-IV* and by the BR 75 and 85

thresholds in the MCMI-III test was developed for practical purposes. Such divisions do not exist in reality. How then are multiple elevations above BR 75 to be interpreted, and how do they reflect on the severity and functionality of personality pathology?

Where several personality disorder scales are elevated above BR 75, they do not all deserve equal interpretive weight. Instead, the highest two or three personality scales in the configuration should be given interpretive primacy; these may be interpreted as personality disorders. Other scales that are lower in the configuration but still above BR 75 may be interpreted as personality styles, meaning that in some contexts and at some times the individual is likely to exhibit traits that are characteristic of these disorders in addition to those that form the primary configuration. As a general rule, the greater the number of scales elevated above BR 75, the greater the extent of personality pathology. If still greater precision is desired, three features that characterize pathology may be used to locate a patient on a continuum from the relative normality of personality style to the pathology of personality disorder (Millon, 1969/1983). These features are described next.

The first feature that typically characterizes clinical personality patterns is a tenuous stability under conditions of subjective stress. Like all working systems, normal personalities exhibit functional-structural integration among their various aspects. For example, normal personalities engage in behavior that minimizes the incompatibility between organismic needs and environmental press, a negative feedback process that maintains the integrity of their psychic systems. Pathological personalities, however, practice strategies that inadvertently produce positive feedback, thereby amplifying their adaptive difficulties. Ultimately, given the ease with which already troubled individuals become vulnerable to events that reactivate the past and given their inflexibility and paucity of effective coping mechanisms, they become susceptible to new difficulties and disruptions. Faced with recurrent failures, anxious about old and unresolved conflicts reemerging, and unable to recruit new adaptive strategies, they may revert to pathological ways of coping, to less adequate control over their emotions, and ultimately to increasingly

subjective and distorted perceptions of reality and to the production of clinical symptoms.

The second characteristic that distinguishes pathological patterns is their adaptive inflexibility. Personality systems are open systems in active transaction with their physical, familial, social, and cultural environments. For personalities in the normal range, this means role flexibility, sometimes taking the initiative to modify one's environment and sometimes adapting to what the environment offers, in accordance with the active-passive polarity of the evolutionary theory. Normal individuals exhibit flexibility in these transactions, such that their initiatives or reactions are proportional and appropriate to what the situation requires. Where appropriateness and proportionality no longer exist, however, we might say that the interaction is driven by the person. The alternative strategies that the individual employs for relating to others, for achieving goals, and for coping with stress are usually few in number and rigidly practiced. Often the individual is unable to adapt effectively to the circumstances of his or her life and begins to arrange the environment to avoid objectively neutral events that are perceived as stressful. As a consequence, the individual's opportunities for testing and acquiring new, more adaptive strategies are reduced, and life experiences become even more narrowly circumscribed.

The third characteristic that distinguishes pathological from normal patterns—a tendency to foster vicious circles—is a consequence of the pathological personality's rigidity and inflexibility. The many constraints that personality-disordered individuals bring to their social milieu inevitably result in feedback processes that perpetuate and intensify preexisting difficulties. Protective constriction, cognitive distortion, and overgeneralization are examples of processes by which individuals restrict their opportunities for new learning, misconstrue essentially benign events, and provoke reactions from others that reactivate earlier problems. In effect, then, pathological personality patterns are themselves pathogenic, generating and perpetuating dilemmas, provoking new predicaments, and setting into motion self-defeating sequences with others, thereby causing their already established difficulties not only to persist but to be further aggravated.

The MCMI-III test portrays two different levels of severity, BR 75 and BR 85, that can clearly be used to characterize the degree to which a given personality profile is more or less functional. In general, the higher the personality BR score, the more likely the individual is to exhibit to a greater degree the three characteristics of personality pathology described above. Strict interpretation of these thresholds, however, is complicated by two further characteristics, one of which regards the curvilinear relationship between the normality-pathology continuum and certain personality constructs, and one of which derives from the evolutionary theory. These issues are discussed in the following two sections.

The Histrionic, Narcissistic, and Compulsive Scales

Previous research with the MCMI test has shown that the elevation of three personality scales may at times reflect personality strengths rather than personality pathology. These scales are the Histrionic, Narcissistic, and Compulsive scales. Measurement of these constructs is psychometrically problematic for a number of reasons, the foremost of which is that these personalities excel at minimizing problems, denying difficulties, and presenting themselves in a favorable light. Each is sensitive to admitting foibles and flaws, although for different reasons. Equally troublesome is that these three constructs tend, at modest levels of magnitude, to include traits that are normal or even adaptive, whereas other personalities, such as the negativistic, antisocial, or sadistic, are maladaptive even at moderate levels of expression. Sadistic traits, for example, are always judged to be problematic, but moderate levels of self-esteem and sociability (prototypal components of the narcissistic and histrionic constructs, respectively) are considered beneficial. Thus, the "shape" of these two constructs is curvilinear; that is, high and low levels of each are maladaptive, whereas modest levels are healthful.

Moreover, as an inspection of item assignments shows, these three personality scales include a large number of items that are keyed False. Thus, when an individual without significant personality pathology completes the instrument, the absence of pathology will tend to elevate these three scales. When the elevation of

one or more of these scales is judged to reflect health rather than pathology, the procedure described in the previous section, which advocates the creation of personality characterizations based on the domain descriptions, cannot be followed in a rigorous manner.

Fortunately, a number of factors bear on the hypothesis that an elevation on the Histrionic, Narcissistic, or Compulsive scale reflects healthy personality characteristics. First, the higher the BR, the more likely it is to reflect personality pathology rather than strength. Second, the presence of significant Axis I pathology supports the presence of a personality disorder. Third, the presence of personality pathology can also be judged by the level of the three Severe Personality Pathology scales, considered next.

Interpreting the Severe Personality Pathology Scales

The evolutionary model presented in Chapter 2 holds that the Severe Personality Pathology scales represent extreme dysfunctional variants of the Clinical Personality Patterns scales. Thus, the more dysfunctional variants of the schizoid and avoidant personalities frequently blend into the schizotypal, the more dysfunctional variants of the masochistic and sadistic personalities often blend into either the borderline or the paranoid, the more dysfunctional variants of the dependent and histrionic personalities blend into the borderline, the more dysfunctional variants of the narcissistic and antisocial personalities blend into the paranoid, and the more dysfunctional variants of the compulsive and negativistic personalities blend into either the borderline or the paranoid. Obviously then, when elevations are obtained on the Severe Personality Pathology scales (S, C, and P), the problem of distinguishing health from pathology for the Histrionic, Narcissistic, and Compulsive scales is more easily resolved.

Previously it was noted that interpretations based on the high-point scale in the configuration of basic personality scales should be adapted depending on secondary elevations. Likewise, elevations for basic configurations of personality scales should also be qualified by elevations obtained for the Severe Personality Pathology scales. For instance, a patient with an Avoidant and Negativistic (2A/8A) personality profile who

also scored high on the Schizotypal scale would be characterized by pervasive apprehensiveness, a tendency to be socially remote or isolated, an expectation that others will reject him or her, a strong desire for affection and support that is constrained by fear, frequent numbness of feeling, and a perception of him- or herself as pitiful and alienated. Similarly pathological, but in a different way, is the 2A/8A patient who scored high on the Borderline scale. This individual is likely to behave in an unpredictable, edgy, or sullenly passive manner, to react in ways that are occasionally impulsive and hostile, to anticipate rejection from others, to alternate between profound gloom and irrational negativism, and then to follow these reactions with expressions of genuine remorse and regret.

As with all test data, determining the functional level of a patient's personality structure is greatly assisted by the availability of good historical information, especially from collateral instruments. Individuals in the midst of an acute disturbance often distort historical information. Depressed patients, for example, frequently paint an overly bleak picture of past events and circumstances. The issue is whether the personality traits identified by the MCMI-III test have significantly and chronically impaired the patient's ability to function.

Interpreting the Grossman Facet Scales

As described in Chapter 5, the Grossman facet scales provide a more fine-grained view of the constructs measured by the 11 Clinical Personality Patterns scales and the three Severe Personality Pathology scales. Each of the 14 primary personality scales has three facet scales, each of which in turn reflects one of the most important structural-functional domains of expression (see Chapter 2) for that personality construct. Facet scale interpretation can help generate clinical hypotheses regarding specific clinical domain difficulties in personologic functioning, thus helping point the way toward specific and germane treatment directions. Therapy that flows directly from assessment is a longstanding goal of a mature clinical science (Millon, 1990, 1999; Millon & Davis, 1996; Millon et al., 2004).

In order for the facet scales for a primary

personality scale to be considered interpretable, the personality scale score should be BR 65 or higher. Thus, facet results may be fruitfully interpreted for primary personality scales with scores somewhat below the diagnostic benchmarks of 75 and 85, but not far below those values. Facet scale interpretation should focus on the facets associated with the three highest personality scales at BR 65 and above. (The MCMI-III profile and interpretive reports include a graph of the facets for the three highest personality scales at BR 65 and above; see Chapter 7.)

The first step in facet scale interpretation is to note the client's highest scores among the facets for the three most elevated personality scales at BR 65 or above. Generally, only facet scores that reach BR 75 or higher should be interpreted. Keep in mind that the facet scales are not designed to specify all of the personologic domains for an individual; rather, they are focused on those domains that tend to be most salient and therefore most problematic when he or she is facing psychic difficulties. Usually the most elevated facet scales are the most important in terms of the individual's personality pathology. However, when two or more facets are equally or closely elevated (i.e., when the scores are the same or within 1 or 2 BR points), it will be necessary to consider the referral question, the nature of the patient's presentation, results from other measures, reports from other professionals or from the patient's social context, and clinical impressions.

When examining facet scale elevations, the clinician should look for possible domain overlap. For example, both the Histrionic and Narcissistic personality scales have facet scales that are anchored to the interpersonal functional domain. If these two primary scales are the most elevated of the MCMI-III personality scales (a common combination), it will be necessary to determine whether "interpersonally attention-seeking" (the histrionic interpersonal expression) or "interpersonally exploitive" (the narcissistic interpersonal expression) is the more appropriate description for this person's social interaction style. In doing so, the following three questions should be considered.

1. Which of the two primary personality scales has the higher score elevation, and how outstanding is that elevation? Con-

sider the case where there is a marked difference between the primary scale scores, for example, BR scores of 110 on the Narcissistic and 68 on the Histrionic scale. Here it is likely that the patient manifests a prototypal narcissistic personality disorder, with comparatively subtle histrionic colorization. In this case, many personality facets captured by the second (and third) highest personality scale elevations may be overshadowed by a strong prototypal presentation. (This may not always be true, however; some facets may "colorize" those found on the higher primary elevation.) In profiles with more similar primary scale elevations, greater weight may be given to the facets for the more elevated primary scale, but the interpretation of domain overlap will rest more on the next question.

2. What are the relative elevations of overlapping facets? When the highest primary personality scale elevations are at comparable levels, the relative scores on the facet scales determine which domain description is more applicable. In most instances, a distinction will emerge as a function of these scores. Sometimes, however, overlapping facets may be similarly elevated, and some further determinations are necessary. It is also possible that the individual displays both qualities, either together or at different times. In any case, the third question about clarifying domain overlap remains imperative.
3. What information can be obtained from other measures, including clinical impressions? Keep in mind that the facet scales are designed not as precise diagnostic measures but as hypothesis-building tools for comprehensive personality assessment that has as its aim intervention direction and treatment planning. Thus, some measure of clinical sensitivity is necessary to maximize the scales' utility. Which facet score appears to best reflect the presenting picture? Do multiple descriptors of the domain in question appear to be appropriate, either at different times or in some admixture? Finally, which appears to be the more important, or troubling, facet for

this individual? Answers to these kinds of questions dictate final conclusions.

A final step in facet scale interpretation is to examine the relative facet elevations and begin to make some clinical decisions regarding how these facets may interact, in much the same manner as the primary scale configural interpretation described earlier. Often relationships are easily deducible. For example, it may be fairly obvious why an individual with a 6A/4 elevation (antisocial with histrionic features) displays interpersonal conduct more consonant with the histrionic pattern (i.e., attention-seeking), while maintaining the regulatory mechanism of the antisocial (i.e., acting-out): the person's acting-out defense system fuels a need to be noticed, and lack of recognition tends to trigger acting-out defenses. The interpretation of this reciprocal interaction is fairly straightforward.

Other relationships may not be as obvious. Take, for example, an uncommon 2A/5 elevation (avoidant with narcissistic traits) in which the facet scale results highlight the expansive cognitive style of the narcissistic pattern along with the alienated self-image of an avoidant individual. As these pieces come together, and in conjunction with observations made about the patient, clinical judgment may guide the astute clinician to recognize that the narcissistic (expansive) cognitive style appears to be more of a façade adopted by this individual to combat more pervasive feelings of unhappiness, alienation, fearfulness, etc. Depending on the person's psychic cohesion, it may be prudent to consider whether the cognitive expansiveness, possibly fed by the avoidant pattern's fantasy mechanism, is a precursor to a non-bizarre, mood-disorder-related psychotic episode. This cannot be confirmed, of course, with the information provided by the facet scales alone. However, as this example shows, the facet scales may aid in alerting clinicians to potential hidden problem areas.

Interpreting the Moderate and Severe Axis I Scales

Because personality is an intrinsically integrative construct, profile interpretation of the basic and severe Axis II scales and their facets is a necessity. Profile integration reflects a basic fact of human life: that while individuals may be described

in terms of many different characteristics, these characteristics are organically or substantively synthesized.

In contrast, the Axis I disorders may reflect numerous separate "medical diseases" such that a profile interpretation of the Axis I scales, while certainly possible, is not explicitly required by the nature of these disorders as it is for those of Axis II. In part, this reflects the clinical reality that the comorbidity between several Axis I diagnoses may be mediated through some third variable or may be entirely coincidental. Thus, while each Axis II construct mutually contextualizes the others precisely because personality is thought of as a single organic medium, the "boils and sneezes" of Axis I may reflect genuinely discrete pathological processes, often neurochemically driven.

Accordingly, the interpretation of BR elevations for the Axis I scales is straightforward. BR scores between 60 and 74 are suggestive but not sufficiently indicative of the scale's symptom pathology unless they are the highest scores in this segment of the profile. Scores between 75 and 84 (inclusive) suggest the presence of the scale's clinical syndrome. Scores at 85 and above provide strong support for the presence of the pathological symptom.

Integrating Axis I and Axis II Findings

Recall that Axis II is conceptualized as an integrated immunological context in which the separate Axis I symptoms derive their meaning. Because each person is an individualized composite, clinicians must draw upon clinical theory and their own case experience when weaving together the patient's personality profile, clinical symptoms, and psychosocial history.

Case Study

At this point, we can examine in detail the results obtained by one individual on the MCMI-III test. The case involves Bob S., a 48-year-old divorced white male who was referred for evaluation by his employer. Bob had worked for 10 years as a security guard for a private hospital. He was referred for evaluation because he had been threatening other workers and was seen putting a gun in his locker. Hostile relationships

and outbursts at work had been a problem for several years and had become especially troublesome during the past year. The possibility of termination was being seriously considered.

Bob blamed his problems on differences of opinion that he frequently had with his supervisor. He objected to the low level of training of the other employees and felt that his repeated complaints were responsible for the perception that he was a "troublemaker." He admitted to verbally threatening his coworkers but defended himself with the claim that similar threats were often made by many other members of the crew. He viewed many of his coworkers as "incompetent drunks." He said he had brought the gun to work so that he could take it to a gun shop to be serviced on his way home.

Bob admitted to drinking "a few beers" occasionally and claimed that he did not drink much more than a six pack on weekends. He did not see his drinking as a problem even though he had been convicted twice for driving under the influence and had lost his driver's license for a period of time. He had been receiving outpatient psychotherapy for six months, a condition for continued employment.

An only child, Bob was born and raised in a large metropolitan area. His father left when Bob was quite young. Bob lived with his 65-year-old mother and seemed to be very attached to her. He had a high school education and two years' worth of law enforcement credits at a local junior college. He was married once, but the marriage ended two years later in a "messy" divorce involving allegations of physical abuse by his wife and his claim that his wife had been sexually involved with other men. Bob's MCMI-III results are presented in Figure 8.2.

We first examine the scores on the Random Response Indicators and the Modifying Indices. The Invalidity and Inconsistency scales provide no evidence of random responding. The Disclosure scale (X) score is within the bounds of acceptability. The only Modifying Index that produced significant results is the Desirability scale (Y). This finding is not surprising because the assessment was a fitness-for-duty evaluation. A quick glance at the rest of the profile reveals that Bob's attempt to portray himself in a positive way did not prevent the test from uncovering some of his problem areas. However, the elevation on the

Desirability scale should be kept in mind; it will be used later to raise some clinical concerns in areas in which his scores are not elevated enough to constitute a significant finding.

The personality disorder scales indicate a paranoid (Scale P) outlook with basic narcissistic (Scale 5) and antisocial (Scale 6A) tendencies. A significant amount of hostility is suggested by the cumulative effects of the Paranoid, Negative (8A), and Sadistic (6B) scale scores, which together indicate a person who is antagonistic, angry, and resentful. A touch of inadequate dependency is also hinted at by the profile. The elevation on the Paranoid scale (a Severe Personality Pathology scale) and the high elevations on several other personality disorder scales suggest that his personality structure may be dysfunctional. A diagnosis of paranoid personality disorder is easily confirmed when Bob's history is considered. Historical support includes the allegations of physical abuse, his distrust of his wife, and the chronicity and type of problems he has encountered at work. His interpretive report might read something like the following:

This individual's MCMI-III scores suggest that he suffers from a paranoid personality disorder with hostile, narcissistic, and antisocial elements. These findings are consistent with the presence of significant character pathology and raise concerns about the subject's ability to behave in a collaborative manner, especially when he feels threatened or unsupported.

Individuals obtaining similar MCMI-III profiles tend to fear that other people will cheat or deceive them. In addition, this man is probably often irritable, abrasive, and easily angered, a "touchy" person who presents definite problems in social situations. Similar individuals express fears of losing autonomy and are very resistant to any attempts by others to control their lives in any way. Although their thinking may be internally consistent, it is often based on premises that seem improbable to others.

Behind the more obvious distrusting and aggressive personality elements is an inflated view of the self as a superior or "special" person. A tendency to exaggerate his abilities and positive attributes and to construct argu-

ments to emphasize his own worth and to deprecate others is likely to be present. This tendency is probably externalized through an air of conviction and self-assurance.

Part of his feeling of superiority comes from the tendency that this man shows to look at his environment as a competitive situation. Similar individuals feel that they have to fend for themselves in order to function. They see themselves as assertive, energetic, self-reliant, strong, and realistic. They feel that they have to be tough in order to make it in the "rat race." These individuals usually justify their aggressiveness by pointing to the hostile and exploitive behavior of others. They are contemptuous of compassionate and warm people, feeling that such people are weak and will be taken advantage of. The competitive outlook fits especially well with this man's distrust and feelings of superiority when he is in a situation in which he has a chance of "winning." Insensitivity to the desires and concerns of other people is typically shown by individuals with this personality.

Continuing with the interpretation of the MCMI-III profile, we discover that none of the Clinical Syndromes or Severe Clinical Syndromes scales are elevated. Ordinarily this finding would indicate an absence of additional pathology in the Axis I clinical syndromes of the *DSM-IV*. However, because of the elevation on the Desirability scale, it is possible that some of the subclinical elevations reveal problem areas. The BR score on the Delusional Disorder scale is probably the result of Bob's paranoid stance and may not add any new information; this hypothesis is supported by the absence of delusional thinking in his history. The subclinical elevation on the Alcohol Dependence scale, however, may be more telling. A careful examination of Bob's responses shows that—as expected—he denied having an alcohol problem. However, the fact that he obtained a subclinical elevation on this scale and has a history of two convictions for driving under the influence of alcohol suggests that there are problems in this area. The subclinical elevation on the Post-Traumatic Stress Disorder scale may be an indication of abuse or trauma in his past. This should be investigated

further. The narrative could handle these issues in the following way:

The patient does not see himself as having a drinking problem. However, the subclinical elevation on the Alcohol Dependence scale raises the concern that he may be in denial with regard to an alcohol abuse disorder. The possibility that he has been abused or traumatized in the past should also be investigated.

The test findings are typically used to offer recommendations for treatment. In Bob's case, the treatment recommendations could be phrased as follows:

The patient is reportedly functioning poorly at work. Given his personality, he can be expected to respond maladaptively to further conflicts or attempts to control him. Consideration should be given to removing him from the situation or making an effort to defuse the conflicts.

The subject would also benefit from continued psychotherapy. Given his character pathology, however, he can be expected to have difficulty forming a strong therapeutic alliance or seeing a need for change. The therapist may need to be fairly supportive at first. On the basis of his personality makeup, the patient can be expected to be most comfortable in situations in which he feels looked up to, admired, or at least respected. Similar individuals often need to be the center of attention. If confrontation is used in therapy, great tact must be exercised to avoid injuring his narcissism beyond what he can tolerate. Further difficulties in such confrontations may come from his interpreting them as part of a competitive relationship and fighting them rather than accepting them as useful feedback. On the other hand, the therapist should not be so supportive of the patient's narcissism that no negative feedback is given and growth is not facilitated. It is important to find ways to help the patient accept his fallibilities and work on his problems without feeling disrespected or humiliated.

Once a therapeutic alliance has been established, attempts can be made to gain an understanding of the experiences that have contributed to the patient's personality problems.

As part of this treatment plan, the therapist should investigate the possibility of abuse or trauma in the client's past. The goal is to help the patient become less distrusting and angry and to gain insight into the way his behavior affects others.

Other Clinical Vignettes

Below are several other clinical vignettes that illustrate the interpretive process.

Case 1: Fred W., age 52, married, three children

Fred, a well-dressed engineer working for an aerospace company, was admitted as a voluntary patient to a well-respected alcoholism unit of a metropolitan hospital. He seemed anxious and depressed and expressed dismay about his longstanding inability to control his drinking. He was coherent, of moderately high intelligence, and strongly motivated to overcome his drinking problem. Funding for his inpatient hospital stay was arranged through his employer's Employee Assistance Program. With the exception of his periodic drinking problem, Fred was a responsible and effective worker and a good family man. He was religious and was involved in several community activities. A number of assessment instruments were administered during the first week of his stay in the alcoholism unit. Fred's MCMI-III scores produced the following codes:

Personality Code: 7 ** 3 * 1 + 8A " 4 6A // - ** - * //
Syndrome Code: B ** D * // - ** - * //

The highest clinical syndrome BR score was on the Alcohol Dependence scale, and a secondary high score occurred on Dysthymia; both corresponded to clinical observation. In addition, the Compulsive/Dependent (7/3) pattern on the basic personality scales suggests that Fred's behavior exhibited anxious conformity to the expectations of others, particularly those in authority. Also notable were compulsiveness, fear of losing control, a tendency to blame himself, serious-mindedness, concern about making mistakes, and a preference for repetitive, routine, and familiar activities. Further hypotheses that Fred had a history of persistent tension and functional somatic disorders proved to be correct. High correspondence was found between these MCMI-III

interpretations and the facts that emerged during Fred's three-week stay at the unit.

Case 2: Larry L., age 50, unmarried

Larry had been unemployed since receiving a psychiatric discharge following an extended period of military service. Although his intelligence was above average, Larry was a high school dropout. After he quit school, he lived with his mother and several younger siblings, holding a variety of unskilled jobs. Eventually he joined the army. His discharge papers listed his diagnosis as schizophrenia: catatonic type. After an in-service hospitalization, Larry was admitted to a local VA hospital and continued on an outpatient basis for more than 25 years. Diagnosed as schizophrenic: chronic undifferentiated type, Larry remained at home, watched television most of the day, occasionally visited relatives, and went to the mental health clinic to receive medication and to participate in periodic sessions of supportive group therapy. Larry's MCMI-III codes were as follows:

Personality Code: 3 2A 2B ** - * 8A 1 + 7 " 4 5 // S ** C * //
Syndrome Code: D A ** - * // - ** SS * //

A blind interpretation of this profile suggests the following characteristics.

The patient is docile, inadequate, self-effacing, and noncompetitive. He tends to be dejected and tense, feels helpless, and assumes a passive role in relationships. He appears to lack the means for an autonomous existence and is especially vulnerable to abandonment. He withdraws from personal involvement, turns against himself, and is overly self-critical and self-punitive about his inadequacies and failures. Denying all aspirations and having experienced continuing deprecation and failure, he often succumbs to depressive apathy and exhaustion. Life for him is empty but draining; he reports constant feelings of weariness and worthlessness.

The correspondence between these blind interpretive judgments and the clinician's evaluations appeared to be quite satisfactory.

Case 3: Barbara R., age 42, divorced, two children

Barbara, a divorcee of six years, was referred by her physician to a community mental health

clinic for persistent depression despite pharmacological treatment. A program of various antidepressant medications had been prescribed by her doctor with no appreciable effect. Barbara's husband had left her a few months before the divorce, saying that he could no longer tolerate her moodiness. The divorce was prolonged and stormy. Barbara fell into a clinical depression shortly after the divorce was final. The depression occurred once or twice a year after that. Her physician described her as a tense, agitated woman who reported a history of nervous complaints, unpredictable emotionality, and repeated difficulty in social and family relationships. Barbara's MCMI-III profile is summarized in the following codes:

Personality Code: 8A ** 4 * 6A 2B + 3 2A 7 " 1 5 // - ** - * //
Syndrome Code: D A ** H * // - ** - * //

A blind profile analysis included the following conclusions about this woman.

Her behavior is typified by a high degree of emotional lability, short periods of impulsive acting out, extended depressive complaints, moodiness, and sulking. Also notable are irritable hypersensitivity to criticism and low frustration tolerance. Her moods tend to be brittle and variable, with periods of high excitement alternating with lethargy, fatigue, oversleeping, overeating, and possible overuse of alcohol. If she is cut off from needed external attention, she frantically searches for approval and becomes easily nettled and contentious or dejected and forlorn.

With minor exceptions, this interpretation corresponded closely with clinical observations.

Integrating Developmental Data

Information from the MCMI-III test can be used to derive vivid pictures of personality and psychopathology. However, these pictures are intrinsically cross-sectional in nature. Biographical information and developmental considerations are too rich to be integrated into the structure of the instrument. The clinician must seek this information through the clinical interview and the report of collateral informants.

Fortunately, there are several developmental models that focus on the relationship between the personality disorders and the Axis I clinical

syndromes. In one such model, described earlier, personality is considered the psychological equivalent of the immune system. Hirschfeld and Shea (1992) refer to this as the *predisposition* or *vulnerability* model. In addition, they describe the *complication* model, the *spectrum* model, and the *pathoplasty* model (see Figure 8.3).

According to the vulnerability model, a personality disorder disposes the individual to the development of an Axis I disorder. When coping responses to stressors are limited or impoverished, the probability of developing an Axis I disorder such as anxiety or depression is greatly increased. Moreover, habitual response styles often lead patients with personality disorders to evoke the very stressors under which their Axis I disorders will later develop. Interpersonally, for example, they often become involved in vicious circles that perpetuate stressful conditions and so function to keep their "immune systems" chronically weakened.

In the complication model, the direction of causality is reversed so that an Axis I condition, however initiated, creates a predisposition to personality change. Individuals experiencing their first schizophrenic episode, for example, sometimes have enough insight into the nature of their illness that they become greatly depressed. The resulting pessimism about the future can grow

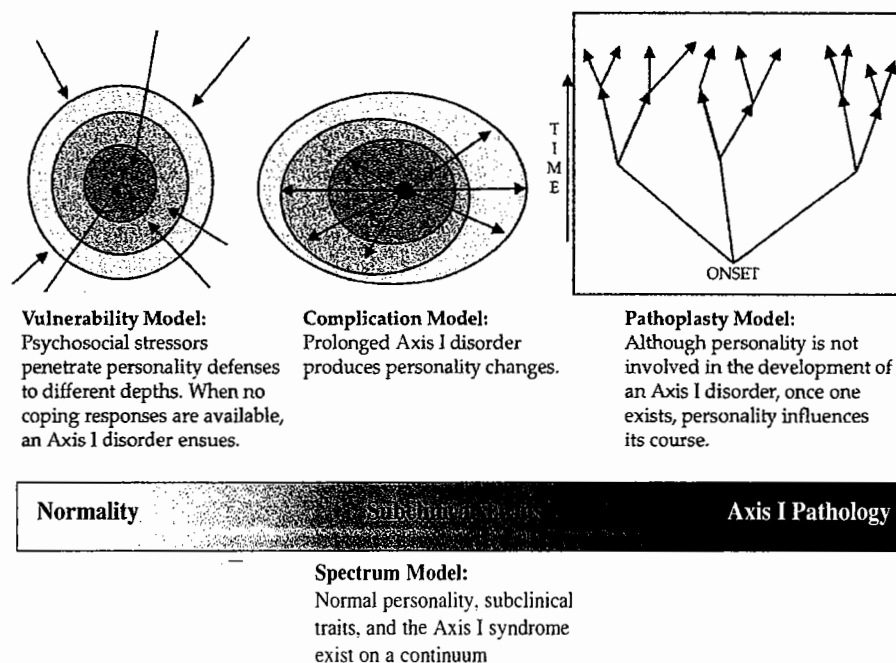
into a personality trait. Patients who experience a severe, long-lasting depression may also internalize pessimism and hopelessness.

In contrast to the vulnerability and complication models, which conceptualize Axis I and Axis II conditions as relatively distinct, the spectrum model sees the disorders on these axes as developing from the same constitutional soil and therefore as existing on a continuum. Biologically based subclinical traits preempt the development of other more adaptive characteristics and effectively become the organizing principle for the entire personality.

Finally, the pathoplasty model holds that personality influences the course of an Axis I disorder but does not in itself dispose the individual toward the development of the disorder. Consequently, if the form of particular Axis I conditions is related to personality, it is only because the course of the disorder tends to be influenced by certain personality features. Within this model, the onset of an Axis I disorder is viewed as independent of personality style.

These models are not mutually exclusive. Different models may be true for different individuals. Alternatively, the partial independence of Axis I and Axis II conditions argues that all may be applicable to a single individual to some degree. For example, an individual with an Axis

Figure 8.3 Possible Relationships Between Axis II and Axis I Disorders



II disorder may be predisposed to develop a particular Axis I condition. Most individuals have some reaction to or feeling about their symptoms, and this awareness influences their personality characteristics in a reciprocal manner. However, neither an Axis I nor an Axis II condition might have been possible at all were it not for constitutional or genetic factors. Once again, clinicians are obligated to bring all available information to bear when attempting to sustain such precise hypotheses.

Using Noteworthy Responses

If an individual responds True to certain MCMI-III items (see Appendix F), the responses are considered "noteworthy." Noteworthy responses are intended to alert clinicians to item-level in-

formation so salient that it deserves interpretive consideration or possibly immediate clinical intervention (any such item that is answered True is noted in the automated interpretive report). Item 132, for example, states "I hate to think about some of the ways I was abused as a child." Item 171 states "I have given serious thought recently to doing away with myself." Depending on the clinical context, a True response to these items may require the clinician to explore the meaning of the response with the patient, perhaps to determine whether there is a significant risk of self-harm. The noteworthy response items are grouped into six content areas: health preoccupation, interpersonal alienation, emotional dyscontrol, self-destructive potential, childhood abuse, and eating disorder.

Chapter 9

Applications of the MCMI Test

Over its nearly 30-year history, the MCMI test has been used in many settings and with many populations. In the 1997 book *The Millon Inventories* (Millon, 1997a), numerous authors associated with the MCMI test or the theory wrote about their experiences and research with the instrument. This chapter provides brief abstracts of some of this material.

Forensic Settings

As expert witnesses, psychologists must demonstrate that their conclusions are valid and founded on evidence beyond their own subjective opinion. Increasingly, psychologists who serve as expert witnesses are viewed as part of the scientific community. Their opinions are subjected to scrutiny according to the dual criteria of the legal requirements for scientific testimony and the profession's own standards for adequate testimony. A complete review of the use of the MCMI test in forensic settings can be found in Dyer (1997) and McCann and Dyer (1996).

A landmark Supreme Court decision (*Daubert v. Merrell Dow Pharmaceuticals*, 1993) changed the way scientific evidence is evaluated for admissibility in federal courts. The court found that the trier of fact (i.e., a judge, jury, or arbitrator) can appropriately determine whether

scientific evidence has sufficient probative value in spite of any lack of general acceptance. This decision paved the way for vigorous cross-examinations by attorneys, who are now free to challenge even generally accepted scientific procedures if it can be demonstrated that they lack sufficient reliability or validity or if they have an unacceptably high error rate. An article published by Heilbrun (1992) for the American Psychology Law Society stressed *relevance* as the ultimate legal issue. Do the constructs measured by an instrument have direct applicability to the ultimate legal issue (i.e., do they help the trier of fact make determinations regarding that issue)?

The issue of relevance is critical as it pertains to the MCMI-III test. This is because the test is recognized for its strength in the assessment of personality disorders, an area that is frequently disputed by attorneys who argue that personality disorders should not be given any weight because they do not appear in the Axis I section of the *DSM-IV*. This argument was challenged in the New Jersey Supreme Court in the case of *State of New Jersey v. Galloway* (1993) where, upon appeal, the court found that a personality disorder does constitute a mental disorder capable of affecting an individual's "cognitive capacities" and therefore the capacity to form a knowing and purposeful mental state.

In addition to relevance, Heilbrun (1992) listed psychometric adequacy as a criterion for an instrument's use in forensic settings. The reliability of the MCMI-III test, especially the scales' internal consistency, compares favorably with that of other objective personality measures. A unique aspect of the MCMI-III test is its use of diagnostic efficiency statistics as evidence of external validity. The advantages of using these operating statistics are numerous. In essence, they make validity results accessible to lay jurors by answering a series of probability questions. Such questions include (a) what are the chances that someone who has X disorder will be identified by the test as having this disorder, (b) what are the chances that someone who does not have X disorder will be correctly classified by the test as not having it, and (c) what are the chances that someone who is found by the test to have X disorder actually does have it?

The MCMI test has been widely employed in two primary forensic applications. First, it has been used extensively in criminal trials, from early phases of criminal confessions, where personality disorders may relate to a subject's vulnerability to coercive interrogation, to the later penalty phase of the trial. In the penalty phase, jurors usually hear testimony concerning aggravating and mitigating factors and then decide, for example, if the defendant should be executed or given a life sentence. Although the psychological evaluation of a defendant may not rise to the level of defense to a crime, it can have significant mitigating value during the penalty phase.

A second forensic use of the MCMI test that has become common is in child custody evaluations. However, this application has recently become controversial. On the one hand, there is an understandable interest in determining each parent's psychological health as one aspect of the ability to adequately discharge parental responsibilities. On the other hand, the MCMI-III test was developed and normed using a group that is far different from child custody litigants in terms of the circumstances surrounding the assessment. The user should become knowledgeable about both sides of this controversy before deciding whether to include the MCMI-III test as part of a child custody evaluation.

Neuropsychology

The MCMI-III test is quite valuable in neuropsychological evaluation. The purpose of a neuropsychological assessment is to determine the cognitive status of an individual. Regardless of how accurate cognitive measures may be, if a deficit is shown to be the result of an emotional problem, this undermines the claim that injury has occurred. Thus, a comprehensive neuropsychological evaluation must include a valid measure of emotional and personality functioning. In addition, it is important to assess the severity of psychopathology because neuropsychological results can be influenced by serious emotional difficulties.

One of the major advantages of the MCMI-III test over other objective tests of psychopathology is that it specifically examines personality characteristics. Therefore, it helps differentiate between long-standing personality styles and emotional dysfunction that would directly affect evaluation. Other important advantages of the MCMI-III test in neuropsychological evaluations include its relative brevity and the availability of an audio version; these are particularly useful given the high incidence of visual impairments and attention difficulties in neuropsychological patients.

Specific applications of the MCMI-III test in neuropsychological evaluations range from assessing the emotional components of brain injury to differentiating between organic and functional conditions. Although extensive research has not been conducted on the MCMI-III test in relation to neuropsychological variables, it is clear that the test can assist in understanding the complex relationship between emotional and cognitive functioning. Further, an understanding of the individual's personality and emotional state is especially important for planning cognitive rehabilitation (Kolitz & Russell, 1997).

Substance Abuse

The MCMI-III test has become an increasingly popular instrument for evaluating substance abuse patients. The value of the test in identifying the presence of substance abuse is well supported (Craig, Verinis, & Wexler, 1985; Flynn & McMahon, 1997; McMahon, Davidson, & Flynn, 1986). In addition to its single-scale diagnostic

potential, the MCMI-III test provides a personality context within which alcohol or drug dependence can be more fully understood. Craig et al. (1985) found that the typical Axis II personality configuration of 106 alcoholics in an inpatient VA unit was quite different from the typical configuration among 100 opiate abusers in a similar VA unit. Alcoholics exhibited significantly higher scores on the Dependent, Avoidant, Schizotypal, and Borderline scales, whereas opiate addicts scored appreciably higher on the Narcissistic, Histrionic, and Antisocial scales. McMahon and Davidson (1985, 1986) used discriminant function analysis with the original MCMI test. Results suggested that depressed alcoholics can be distinguished from their nondepressed counterparts by a detached interpersonal style, distracted cognition, alienated self-image, and mixed depressed-anxious emotionality. Further refinements were made to differentiate alcoholics with transient versus enduring depression.

In recent years, increasing attention has been directed to the specification of personality subtypes among alcoholics (Nerviano, 1976), a movement away from an earlier period when the notion of a unitary alcoholic personality was in vogue. Efforts to derive subtypes based on MCMI profiles have shown a measure of consistency and, for the most part, agree with distinctions developed from the data of other instruments (Bartsch & Hoffman, 1985; Mayer, 1985; Rosario, 1987). A particularly illuminating discussion of cluster analytic findings was presented by Bartsch and Hoffman (1985). Their research led them to propose a series of subtypes. One MCMI variant exhibits high scores on the Antisocial and Compulsive scales, suggesting a central conflict between behaving in a rigid and conscientious manner on the one hand and expressing deep and hidden hostile impulses on the other. Another variant exhibits elevations on the Narcissistic, Histrionic, and Antisocial scales, suggesting a self-centered and manipulative pattern underlying superficial charm and facile sociability, a surface friendliness that is periodically disrupted by outbursts of resentment and anger. A third subtype is associated with elevations on the Dependent, Avoidant, Paranoid, and Borderline personality scales and high scores on the Anxiety and Dysthymia clinical scales. Greater pathology is noted here, especially mood varia-

tion, irritability, ambivalence, and negativism. In a fourth variant, high scores are seen on the Schizoid, Avoidant, Dependent, and Schizotypal measures, indicating the primacy of social apprehension and loneliness. These types correspond rather well to the four variants uncovered by Mayer (1985) and Rosario (1987). In the latter study, the four alcoholic subtypes are conceptualized as anxious-ambivalent, confident-sociable, moody-negativistic, and sensitive-dependent.

The heterogeneous nature of the substance abusing population further demonstrates the importance of understanding personality when considering treatment needs. Moreover, research findings have shown that dual-diagnosis substance abusers have higher relapse rates. Matching substance abusing patients with treatments specifically designed to address their needs may enhance treatment effects.

Post-Traumatic Stress Disorder

Numerous investigators have turned to the use of psychological tests to identify and understand the clinical characteristics of post-traumatic stress disorder (PTSD). The MCMI-III test is different from the previous MCMI versions in several ways, including the addition of a specific PTSD scale. Fairly consistent patterns emerge on MCMI profiles when PTSD patients are contrasted with patients with other psychiatric problems. For example, studies by Hyer and Brandsma (1997) uncovered parallel profile patterns among PTSD patients when compared to a variety of similar psychiatric groups. Several conclusions are warranted. The basic Axis II personality profile that emerged was predominantly passive-aggressive/avoidant, although schizoid and self-defeating features were sometimes included as well. In a separate study using the MCMI-II test (Hyer, Woods, Bruno, & Boudewyns, 1989), the self-defeating personality style predicted treatment problems, early dropout, and compliance problems. Among the Axis I scales, Anxiety and Dysthymia were notably high, as were Alcohol Dependence and Drug Dependence.

Although the investigations just described focused on male war veterans, an equally troubled post-traumatic group is composed primarily of women who were physically or sexually abused as children. Bryer, Nelson, Miller, and Krol (1987) employed the MCMI test to identify

and differentiate psychiatric symptoms among adult survivors of childhood abuse. MCMI scale mean BR scores were obtained for 66 persons admitted to a short-term inpatient psychiatric hospital. Twenty-seven had experienced no abuse, 14 were subjected to sexual abuse only, 10 experienced physical abuse only, and 15 experienced sexual and physical abuse. As with the male PTSD veterans, the Avoidant and Passive-Aggressive scales reached pathological levels (BR 75) and approached or achieved statistical significance, as did the Borderline scale. This overall profile pattern is highly similar to that of Vietnam veterans, suggesting that the long-term psychological effects of profound psychological trauma are much the same for men and women, although their etiology may differ.

Correctional Settings

Psychologists working in prison settings have somewhat different concerns from those working in other clinical settings. The attributes that characterize prison populations differ somewhat from those of general psychiatric patients. Moreover, any inventory employed in prisons must prove useful in settings that call for psychological and behavioral considerations other than those found in most mental health contexts (Millon & Millon, 1997). (There is an MCMI-III Corrections Report designed specifically for use in prison settings. For more information, see the *MCMI-III Corrections Report User's Guide*.)

The MCMI test has proven relevant to the prison setting in several ways. Few characteristics are more relevant to the management of prison behavior than the assessment of violence risk, including tendencies toward domination, impulsive acting out, and rage and brutality. Related to this is the inmate's capacity to restrain urges and momentary desires. Thoughtless and impulsive traits may lead to precipitous and reckless behavior and explosive outbursts. When paired with tendencies toward violence, the potential for frenzied brawls and riotous behavior is increased.

Suicidal potential is another area of focus for the correctional psychologist. Some inmates experience only diffuse and periodic thoughts of suicide and intentional self-injury. Others actually attempt or commit suicide. Although inmates confronted with the bleak prospects of

confinement or with feelings of worthlessness or guilt often think of suicide as an escape, few act or are able to act while incarcerated. Nevertheless, such indications are important for prison psychologists.

Other major considerations in working with inmates include their readiness for and receptivity to psychological and vocational rehabilitation, the likelihood that they will attempt to escape when placed in less restrictive community-based settings, and their preparedness and fitness for parole. The probability of criminal relapse cannot be readily predicted, but it is worth estimating using test results and biographical information.

The MCMI Test and the Rorschach

Few empirical studies have focused on the relationship between the Rorschach and the MCMI test, and those that have demonstrate only a modest relationship between the two instruments (e.g., Dorr, 1997). Nevertheless, the ability to integrate instruments across information modalities (Campbell & Fiske, 1959) may be extremely useful not only for enriching our understanding of patients but also for designing more effective and specific treatment interventions. As noted by Choca et al. (1992), the use of multiple measures can confirm and enhance individual test results. For example, if an elevation on the MCMI-III Major Depression scale is found, the Rorschach might be used to clarify the degree to which the depression influences thinking processes (Dorr, 1997). As another example, a patient with elevated Avoidance and Schizoid scales would be expected to produce Rorschach responses that are lacking in human interaction and not rich in warmth and detail.

Marital Counseling

The MCMI-III test has been used successfully in marital counseling and couples therapy as a means of quickly and efficiently understanding each member's Axis I syndromes and Axis II personality styles or disorders. Profile patterns are often helpful in understanding the intra- and interpersonal dynamics that contribute to a couple's presenting complaints. As noted by Nurse (1997), individual profiles must be considered in terms of the functionality of their personality

patterns for the particular relationship. Paradoxically, the factors that initially attract two individuals may ultimately erode their relationship.

After testing and a joint intake interview, the therapist may discuss with each partner the probable meaning of his or her MCMI-III scores. Next, the implications of these results are considered in the context of the relationship. In a subsequent session, both partners may then share their appraisal of the most relevant findings. The couple and the therapist then establish treatment goals based on this shared information as it pertains to the presenting concerns. The clinician, as in individual therapy, may draw from Millon's theoretical formulation and from his or her own knowledge of personality when designing therapeutic interventions, whether for a specific session or for longer-term therapy.

Treatment Planning and Psychotherapy

Psychological tests are used increasingly by clinicians to tailor interventions to the problems and issues that patients bring to treatment. The MCMI-III automated interpretive report includes a section concerning prognostic and therapeutic implications for the obtained profile pattern. Several authors from varying theoretical perspectives have discussed the role of psychological assessment in treatment planning (Choca et al., 1992; Kiesler, 1986; Maruish, 1994; Neimeyer, 1987; Retzlaff, 1995). Even in the psychodynamic literature (Levy, 1987; Shapiro, 1989), there is a growing recognition that techniques and interventions must be modified on a case-by-case basis. A quick and accurate assessment of symptomatic and personality issues leads to treatment that is also more efficient, focal, and effective. These concerns have become increasingly important in the current mental health climate of managed care and brief treatment intervention (Millon et al., 2000; Millon & Grossman, 2007a, 2007b, 2007c).

Although the very nature of the psychotherapeutic relationship demands respect for uniqueness, flexibility, and individualized integration, several general principles have been formulated to guide the application of MCMI-III results to psychotherapy. First, the Modifying Indices can be examined to determine the client's response style. Although this is not a well-researched

topic, these indices can provide the basis for hypotheses regarding the patient's current experience of his or her problems and how he or she is likely to present in initial psychotherapy sessions. For example, patients with high Disclosure and Debasement scores are often in crisis or need to present themselves as terribly flawed and troubled for some reason. Patients seeking financial compensation for psychiatric disorders or patients who play the role of professional martyrs or victims might also have elevations on these scales. In contrast, other patients might have elevations only on the Desirability scale, with few other symptomatic or personality elevations. A client who has been convicted of child molesting but who protests his innocence might be such a case.

Second, the initial focus should be on helping the client gain some relief from troubling symptoms. The Clinical Syndromes scales provide information about reported symptoms that need to be addressed immediately, such as anxiety, depression, somatic problems, psychosis, alcohol and drug abuse, post-traumatic stress disorder, and activity level. Most patients will experience such problems as unpleasant and ego-dystonic and will welcome mental health treatment. As with any kind of treatment, the initial task is to determine the nature and meaning of the symptoms. If a high score is obtained on the Anxiety scale, the meaning of the score in the patient's life should be determined. The interpersonal aspects of many clinical syndromes are eventually worked out in the context of the therapeutic relationship as well. With an anxious patient, for example, the therapist may find it important to remain calm, to work on cementing the therapeutic alliance, and to avoid overtaxing the patient with demands, confrontations, or deep interpretations of primitive, primary process content.

Third, the Severe Personality Pathology scales should be examined to assess the stability and structure of the client's ego functions and processes. Although there is disagreement in the literature regarding their treatment (Horwitz, 1985; Kernberg, Selzer, Koenigsberg, Carr, & Applebaum, 1989; Westen, 1991), most clinicians agree that serious ego problems such as borderline, paranoid, and schizotypal difficulties require intensive or specially formulated modalities. This might include, in the case of borderline

Chapter 10

Future Directions

As noted at the beginning of this manual, the MCMI test is intended to be an evolving assessment tool, to be refined periodically to reflect substantive progress. In this chapter we review two additional areas in which developments are currently proceeding.

Theoretical Developments: Personality Subtypes

Personality disorders are best thought of as prototypes. Within each prototype lie numerous variations. In other words, there is no one single schizoid or avoidant or depressive or histrionic type, but instead several variations of each, different forms in which the core or prototypal personality expresses itself. In *Disorders of Personality: DSM-IV and Beyond*, Millon and Davis (1996) elaborated a series of personality subtypes for each of the major prototypes deduced from Millon's (1990) evolutionary theory. These subtypes were essentially derived from the modern and historical literature, in convergence with clinical wisdom, cultural myth, and empirical fact. Often they are simply mixtures of major types. The deficient conscience, fraudulent dealings, and arrogant attitude of the *unprincipled narcissist*, for example, are reminiscent

of the con artist or charlatan, a cultural stereotype incarnated as a kind of antisocial narcissist. Likewise, the *nomadic antisocial*, represented as a gypsy-like wanderer, vagrant, or itinerant railroad bum scavenging at the margins of society, essentially reflects a mixture of the schizoid and the antisocial.

In other cases, the subtypes exist at the confluence of Millon's evolutionary theory and other organizing principles. Psychodynamic theorists, for example, have described the *compensating narcissist*, an individual who counteracts deep feelings of inferiority and low self-esteem by creating the illusion of being superior and exceptional. Similarly, the *hypersensitive avoidant* represents a more manifestly cognitive variant of a major type, being intensely wary and suspicious, alternately panicky, terrified, edgy, and timorous, then thin-skinned, high-strung, petulant, and prickly.

An important question when writing the test interpretation sections of this manual concerned whether a detailed description should be included for the more than 60 personality disorder subtypes identified in Millon and Davis (1996). In the end, it was decided to present only a sample, for two reasons. First, these descriptions would easily form a small book on their own and would

dominate the manual. Second, there is no hard empirical support for the existence of these subtypes other than the authors' own reading of the literature and experience with the MCMI test. Table 10.1 presents the sample, which consists of the subtypes of the narcissistic personality disorder and their associated MCMI-III codes.

Besides the obvious relationship of each subtype to its cousins, all the subtypes share an important feature. Each is a statement of a particular truth about the nature of the personologic landscape, a claim that the world contains such-and-such personalities, more by historical accident than ontological necessity. Each is a specific manifestation of a major type in the context of our particular times and culture. If history or culture were different, the subtypes would probably also be different. The major prototypes, on the other hand, are in a sense required by the theory, just as existence, adaptation, and replication are required by the laws of evolution. Although the subtypes lack the deductive inevitability of major prototypes, they nevertheless bring the major types down to a level of unparalleled descriptive specificity and clinical utility.

However, it is precisely because the subtypes are not inevitable products of the evolutionary theory that they not only bear a greater burden of validation evidence than do the major types but also present a greater challenge for reliable and valid assessment. The issue of base rates (i.e., the prevalence of the various subtypes) is of particular importance. Because the base rates for some of the major types (e.g., schizoid and sadistic) are quite low, many of the subtypes are

likely to be quite rare. A particular clinician, in fact, might go years without seeing a particular subtype, making validation research almost impossible in any one clinical setting. All of the problems associated with the prediction of low base rate phenomena such as suicide or murder are also problems here.

Nevertheless, several directions could be pursued in order to advance the assessment utility of the MCMI-III test in relation to the personality subtypes. Millon (1997b), for example, presented MCMI-III code types for each personality disorder subtype. On theoretical grounds, it is expected that the unprincipled narcissist will appear as a 5/6A, whereas the compensating narcissist is more likely to appear as a 5/8A or 5/2A. This code type approach would seem to be the most straightforward, especially when the subtypes exist predominately as a mixture of two or more major types. In the coming years, as clinicians who are already familiar with Millon's theory become well enough acquainted with these subtypes to provide useful external validity ratings, studies that examine the diagnostic efficiency of these MCMI-III profile patterns will become possible. Not all 5/8A code types, for example, are likely to be classified as compensating narcissists. Conversely, some compensating narcissists may produce profiles other than 5/8A. Eventually the MCMI-III items that are most directly related to each subtype profile will be identified and examined against the content of the subtypes, and a feedback process will begin through which the item content of the MCMI-III test and the content of the subtypes themselves

Table 10.1 Subtypes of the Narcissistic Personality

Elitist (5)	Feels privileged and empowered by virtue of special childhood status and pseudo-achievements; entitled facade bears little relation to reality; seeks favored and good life; is upwardly mobile; cultivates special status and advantages by association
Amorous (5,4)	Is sexually seductive, enticing, beguiling, tantalizing, glib, and clever; avoids real intimacy; indulges hedonistic desires; bewitches and inveigles the needy and naive; is a pathological liar and swindler
Unprincipled (5, 6A)	Has a deficient conscience; is unscrupulous, amoral, disloyal, fraudulent, deceptive, arrogant, and exploitive; is a con artist and a charlatan; is dominating, contemptuous, and vindictive
Compensating (5, 8A/5, 2A)	Seeks to counteract or cancel out deep feelings of inferiority and lack of self-esteem; offsets deficits by creating illusions of being superior, exceptional, admirable, and noteworthy; self-worth results from self-enhancement

will be mutually refined. This will allow items for future versions of the MCMI test to be written with accurate classification of the subtypes in mind.

How an emphasis on personality subtypes will affect the length of the MCMI-III test and its exact item composition is, of course, an open question. The answer depends in part on the nature of the empirical problems encountered. Certainly, it seems difficult to provide an accurate assessment of more than 60 subtypes with an MCMI test of only 175 items. Even — if all of the MCMI items were oriented toward Axis II, that would still leave fewer than three unique items per subtype, far too few to ensure discriminant validity. Fortunately, the relatively high base rates of some of the major personalities—the antisocial and borderline, for example—make their subtypes more important than those of lower base rate personalities such as the schizoid. Where compromises in inventory length must be made, at least for paper-and-pencil forms, they will favor the assessment of those subtypes that are most clinically relevant. Future computer-administered adaptive versions of the MCMI test, programmed to optimize the selection of items based on the client's pattern of responses, will not require the entire item bank to be administered and are much more likely to yield highly specific, subtype-relevant information.

Refinement Across Diverse Data Sources

One of the fundamental principles of instrument construction is that validity should be built in from the beginning. Loevinger (1957), Jackson (1970), and Skinner (1986) have discussed test construction as an iterative three-stage process. In the *rational stage*, the content of each construct to be measured is defined as precisely as possible. Items may be written and refined by multiple experts in the theory of the construct, sorted according to their discriminative characteristics, ranked in terms of difficulty level, and so on, all before being administered to actual subjects. In the *internal stage*, the items are administered to actual subjects. Various statistics are then calculated to ensure that the items tap a single dimension. Items with strong relationships to scales for which they were not intended may be deleted or reassigned. In addition, a correla-

tion matrix of the instrument's scales may be examined for anomalies, and items may be reassigned or dropped to bring the intercorrelation pattern in line with theoretical expectations. In the *external stage*, the newly developed scales are assessed against other instruments of established reputation. Items that pass all three stages are said to have been validated theoretically, statistically, and empirically. The Millon inventories have essentially been constructed according to this tripartite logic.

Although inventories should be constructed to mirror our expectations regarding relationships among constructs both internal and external to the instrument, they should also possess a generative value. That is, an inventory should contribute to the accumulation of new knowledge. If a test is intended to assess personality disorders, we might expect, for example, that individuals classified as having a dependent personality will also be classified as experiencing depressive episodes more often than, say, antisocials. This is because dependents are likely to feel helpless and hopeless more often than antisocials, who take the initiative in changing their external world (albeit usually in a destructive way). Together, these internal-internal and internal-external expectations form a set of constraints that an inventory must be constructed to satisfy. Each constraint is an additional point through which the instrument's predictions are triangulated with reality as it is assumed to exist. The larger this set of constraints the better, because if the inventory can satisfy many such constraints at the beginning, we can have greater confidence that it will also meet future challenges.

Sometimes, for example, after an instrument has been constructed, certain variables show an unexpected relationship. Perhaps in the beginning the authors believed that individuals with antisocial personality disorder would report high family distress and a high incidence of alcoholism. Perhaps they also believed that individuals with a dependent personality would report overprotective parents and an inability to end relationships. The authors build expectations into the instrument. However, they may have had no expectations at all about the relationship between dependent personality and the incidence of alcoholism. How do they evaluate this observed but unexpected relationship? Having constructed the

instrument to satisfy multiple constraints in the form of theoretically driven expectations, they are no longer working with just any item pool. The more the test satisfies many different constraints, the greater the authors' expectations of the generalizability of the entire system of instrumentation. Demonstrated validity in diverse areas of the nomological network becomes a promissory note that observed relationships between intervening variables elsewhere are not peculiar to their particular mode of operationalization but are instead representative of nature's structure and therefore worthy of genuine scientific interest. In general, the more constraints an instrument satisfies at the outset, the more confidence we can have in the validity of relationships that are unanticipated.

At the current time, most psychological instruments are developed using a single data source, be it self-report, clinical ratings, observer ratings, or projectives, and then validated against instruments of already established utility based on that same data source. Validation against instruments from other data sources, while important, is usually secondary. Thus, a new self-report scale to measure dependency finds evidence for its validity first against other self-report dependency scales and then against observer ratings and projectives, and so on. Lower correlations between self-report and observer ratings are attributed to method considerations, often rightly so. Unfortunately, this rather haphazard style of instrument development often leads to confusing results when such instruments are paired against each other in a multitrait-multimethod matrix. The narcissistic scale of a self-report instrument may correlate more highly with a histrionic rating scale than with a narcissistic rating scale, and so on. Many such mismatches are likely to be found for any given set of personality instruments, a fact that has undermined confidence in the valid assessment of personality disorders.

A much better method blurs the distinction between internal and external by viewing test construction as a system of relations or constraints to be satisfied across multiple data sources. The goal is to build an entire suite of instruments, each informed by the others across the entire construction process. The more data sources that can be involved, the better. For example, a self-report inventory for personality

disorders, an observer-rated version of the same inventory, and a clinical checklist would be constructed simultaneously, not one at a time. Correlation matrices involving all of the instruments would be inspected at every stage. If during development it was found that the observer-rated narcissistic scale correlated more highly with the self-report histrionic scale than with its parallel self-report narcissistic scale, indicating a lack of convergent-discriminant validity, the items from both pools would be examined not only with regard to their content characteristics and correlations with their assigned scales but also in terms of their cross-method correlations. Violations of multitrait-multimethod considerations would be tracked to particular items, which would be reassigned, dropped, or rewritten. Rather than being examined post hoc, multitrait-multimethod considerations become a fundamental part of the item selection process, considered from the beginning, thus allowing each instrument to strengthen the others. Validity is built into an entire system of instrumentation and not simply into a single instrument. Future versions of the MCMI test are likely to be constructed in this way.

What are the requirements for the construction of such a test suite? The first is essentially practical, reflecting a much larger investment on the part of participants and the need for more iterations in the development process. Patients and their clinicians must be subjected to the inconvenience of providing data for a large item pool. The second is theoretical and is the more difficult of the two, requiring expert clinical judgment concerning the constructs the test seeks to assess. The construction process is begun knowing that correlations between like scales across diverse data sources will be less than perfect and that this reflects informational biases as much as it does measurement error inherent in the respective sources. For example, personalities can be expected to differ in terms of their level of self-knowledge and the extent to which they distort objective information. The egocentricity of narcissists and their tendency to construe reality in ways that are favorable to themselves make them less likely than pessimistic, self-focused depressives to provide valid self-report information. We might hypothesize, then, that the correlation between self-reported and clinician-rated nar-

cissistic personality is likely to be less than the correlation between self-reported and clinician-rated depressive personality. This is the problem the theorist faces. Knowing that the multitrait-multimethod matrix reflects informational biases in addition to measurement error, the theorist must provide a substantive framework within which to evaluate the direction and magnitude of its correlations and to determine the path in which the results are best refined. In turn, this involves knowing (a) those traits or characteristics for which each personality can typically be expected to provide valid self-report information, (b) those traits or characteristics for which each personality cannot be expected to provide valid information, (c) those traits or characteristics for which clinicians can typically be expected to provide valid ratings, and (d) those traits or characteristics which, for whatever reason, clinicians will not be able to rate validly.

Unfortunately, Axis II of the *DSM-IV*, deliberately formulated to be atheoretical and so existing at a descriptive rather than an explanatory level, cannot provide the substantive framework necessary to guide such research. Rather than providing a coherent deductive foundation within which to ask questions about the various personalities, Axis II is best viewed as a body of descriptive material collated across several perspectives but not really coordinated with any of them. While the periodic table is the unique

province of chemistry, the problematic types of behavior that are to be carved up into diagnostic categories are often given to psychopathologists by parties whose standards are extrinsic to psychopathology itself. As scientists, however, we must answer this question: why these particular diagnostic groups rather than others? One cannot merely accept any list of kinds or dimensions as given. Instead, a taxonomic scheme must be justified, and to be justified scientifically, it must be justified theoretically. Taxonomy and theory are intimately linked.

Item writers involved in attempts to construct a coordinated suite of multimethod instruments (e.g., future versions of the MCMI test and the Millon Personality Diagnostic Checklist [Tringone, 1997]) require a strong theoretical basis on which to compare and contrast the personality disorders across different data sources. Relevant questions include, for example, "what traits does the narcissist possess about which he or she is likely to have self-knowledge and will admit to" and "what traits does the narcissist possess about which he or she is unlikely to have self-knowledge, but which can be reliably rated by clinicians?" For the authors, the theoretical basis is Millon's evolutionary theory (Millon, 1990). Guided by continuing developments in this theory, we hope that the future will bring at least as much advancement in assessment technologies as has already occurred.

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Appendix A

Item Content

1. Lately, my strength seems to be draining out of me, even in the morning.
2. I think highly of rules because they are a good guide to follow.
3. I enjoy doing so many different things that I can't make up my mind what to do first.
4. I feel weak and tired much of the time.
5. I know I'm a superior person, so I don't care what people think.
6. People have never given me enough recognition for the things I've done.
7. If my family puts pressure on me, I'm likely to feel angry and resist doing what they want.
8. People make fun of me behind my back, talking about the way I act or look.
9. I often criticize people strongly if they annoy me.
10. What few feelings I seem to have I rarely show to the outside world.
11. I have a hard time keeping my balance when walking.
12. I show my feelings easily and quickly.
13. My drug habits have often gotten me into a good deal of trouble in the past.
14. Sometimes I can be pretty rough and mean in my relations with my family.
15. Things that are going well today won't last very long.
16. I am a very agreeable and submissive person.
17. As a teenager, I got into lots of trouble because of bad school behavior.
18. I'm afraid to get really close to another person because it may end up with my being ridiculed or shamed.
19. I seem to choose friends who end up mistreating me.
20. I've had sad thoughts much of my life since I was a child.
21. I like to flirt with members of the opposite sex.
22. I'm a very erratic person, changing my mind and feelings all the time.
23. Drinking alcohol has never caused me any real problems in my work.
24. I began to feel like a failure some years ago.
25. I feel guilty much of the time for no reason that I know.
26. Other people envy my abilities.
27. When I have a choice, I prefer to do things alone.
28. I think it's necessary to place strict controls on the behavior of members of my family.

29. People usually think of me as a reserved and serious-minded person.
30. Lately, I have begun to feel like smashing things.
31. I think I'm a special person who deserves special attention from others.
32. I am always looking to make new friends and meet new people.
33. If someone criticized me for making a mistake, I would quickly point out some of that person's mistakes.
34. Lately, I have gone all to pieces.
35. I often give up doing things because I'm afraid I won't do them well.
36. I often let my angry feelings out and then feel terribly guilty about it.
37. I very often lose my ability to feel any sensations in parts of my body.
38. I do what I want without worrying about its effect on others.
39. Taking so-called illegal drugs may be unwise, but in the past I found I needed them.
40. I guess I'm a fearful and inhibited person.
41. I've done a number of stupid things on impulse that ended up causing me great trouble.
42. I never forgive an insult or forget an embarrassment that someone caused me.
43. I often feel sad or tense right after something good has happened to me.
44. I feel terribly depressed and sad much of the time now.
45. I always try hard to please others, even when I dislike them.
46. I've always had less interest in sex than most people do.
47. I tend to always blame myself when things go wrong.
48. A long time ago, I decided it's best to have little to do with people.
49. Since I was a child, I have always had to watch out for people who were trying to cheat me.
50. I strongly resent "big shots" who always think they can do things better than I can.
51. When things get boring, I like to stir up some excitement.
52. I have an alcohol problem that has made difficulties for me and my family.
53. Punishment never stopped me from doing what I wanted.
54. There are many times, when for no reason, I feel very cheerful and full of excitement.
55. In recent weeks I feel worn out for no special—
— reason.
56. For some time now I've been feeling very guilty because I can't do things right anymore.
57. I think I am a very sociable and outgoing person.
58. I've become very jumpy in the last few weeks.
59. I keep very close track of my money so I am prepared if a need comes up.
60. I just haven't had the luck in life that others have had.
61. Ideas keep turning over and over in my mind and they won't go away.
62. I've become quite discouraged and sad about life in the past year or two.
63. Many people have been spying into my private life for years.
64. I don't know why, but I sometimes say cruel things just to make others unhappy.
65. I flew across the Atlantic 30 times last year.
66. My habit of abusing drugs has caused me to miss work in the past.
67. I have many ideas that are ahead of the times.
68. Lately, I have to think things over and over again for no good reason.
69. I avoid most social situations because I expect
— people to criticize or reject me.
70. I often think that I don't deserve the good things that happen to me.
71. When I'm alone, I often feel the strong presence of someone nearby who can't be seen.
72. I feel pretty aimless and don't know where I'm going in life.
73. I often allow others to make important decisions for me.

74. I can't seem to sleep, and wake up just as tired as when I went to bed.
75. Lately, I've been sweating a great deal and feel very tense.
76. I keep having strange thoughts that I wish I could get rid of.
77. I have a great deal of trouble trying to control an impulse to drink to excess.
78. Even when I'm awake, I don't seem to notice people who are near me.
79. I am often cross and grouchy.
80. It is very easy for me to make many friends.
81. I'm ashamed of some of the abuses I suffered when I was young.
82. I always make sure that my work is well planned and organized.
83. My moods seem to change a great deal from one day to the next.
84. I'm too unsure of myself to risk trying something new.
85. I don't blame anyone who takes advantage of someone who allows it.
86. For some time now I've been feeling sad and blue and can't seem to snap out of it.
87. I often get angry with people who do things slowly.
88. I never sit on the sidelines when I'm at a party.
89. I watch my family closely so I'll know who can and who can't be trusted.
90. I sometimes get confused and feel upset when people are kind to me.
91. My use of so-called illegal drugs has led to family arguments.
92. I'm alone most of the time and I prefer it that way.
93. There are members of my family who say I'm selfish and think only of myself.
94. People can easily change my ideas, even if I thought my mind was made up.
95. I often make people angry by bossing them.
96. People have said in the past that I became too interested and too excited about too many things.
97. I believe in the saying, "early to bed and early to rise..."
98. My feelings toward important people in my life often swing from loving them to hating them.
99. In social groups I am almost always very self-conscious and tense.
100. I guess I'm no different from my parents in becoming somewhat of an alcoholic.
101. I guess I don't take many of my family responsibilities as seriously as I should.
102. Ever since I was a child, I have been losing touch with the real world.
103. Sneaky people often try to get the credit for things I have done or thought of.
104. I can't experience much pleasure because I don't feel I deserve it.
105. I have little desire for close friendships.
106. I've had many periods in my life when I was so cheerful and used up so much energy that I fell into a low mood.
107. I have completely lost my appetite and have trouble sleeping most nights.
108. I worry a great deal about being left alone and having to take care of myself.
109. The memory of a very upsetting experience in my past keeps coming back to haunt my thoughts.
110. I was on the front cover of several magazines last year.
111. I seem to have lost interest in most things that I used to find pleasurable, such as sex.
112. I have been downhearted and sad much of my life since I was quite young.
113. I've gotten into trouble with the law a couple of times.
114. A good way to avoid mistakes is to have a routine for doing things.
115. Other people often blame me for things I didn't do.
116. I have had to be really rough with some people to keep them in line.
117. People think I sometimes talk about strange or different things than they do.

118. There have been times when I couldn't get through the day without some street drugs.
119. People are trying to make me believe that I'm crazy.
120. I'll do something desperate to prevent a person I love from abandoning me.
121. I go on eating binges a couple of times a week.
122. I seem to make a mess of good opportunities that come my way.
123. I've always had a hard time stopping myself from feeling blue and unhappy.
124. When I'm alone and away from home, I often begin to feel tense and panicky.
125. People sometimes get annoyed with me because they say I talk too much or too fast for them.
126. Most successful people today have been either lucky or dishonest.
127. I won't get involved with people unless I'm sure they'll like me.
128. I feel deeply depressed for no reason I can figure out.
129. Years later I still have nightmares about an event that was a real threat to my life.
130. I don't have the energy to concentrate on my everyday responsibilities anymore.
131. Drinking alcohol helps when I'm feeling down.
132. I hate to think about some of the ways I was abused as a child.
133. Even in good times, I've always been afraid that things would soon go bad.
134. I sometimes feel crazy-like or unreal when things start to go badly in my life.
135. Being alone, without the help of someone close to depend on, really frightens me.
136. I know I've spent more money than I should buying illegal drugs.
137. I always see to it that my work is finished before taking time out for leisure activities.
138. I can tell that people are talking about me when I pass by them.
139. I'm very good at making up excuses when I get into trouble.
140. I believe I'm being plotted against.
141. I feel that most people think poorly of me.
142. I frequently feel there's nothing inside me, like I'm empty and hollow.
143. I sometimes force myself to vomit after eating.
144. I guess I go out of my way to encourage people to admire the things I say or do.
145. I spend my life worrying over one thing or another.
146. I always wonder what the real reason is when someone is acting especially nice to me.
147. There are certain thoughts that keep coming back again and again in my mind.
148. Few things in life give me pleasure.
149. I feel shaky and have difficulty falling asleep because painful memories of a past event keep running through my mind.
150. Looking ahead as each day begins makes me feel terribly depressed.
151. I've never been able to shake the feeling that I'm worthless to others.
152. I have a drinking problem that I've tried unsuccessfully to end.
153. Someone has been trying to control my mind.
154. I have tried to commit suicide.
155. I'm willing to starve myself to be even thinner than I am.
156. I don't understand why some people smile at me.
157. I have not seen a car in the last ten years.
158. I get very tense with people I don't know well because they may want to harm me.
159. Someone would have to be pretty exceptional to understand my special abilities.
160. My current life is still upset by flashbacks of something terrible that happened to me.
161. I seem to create situations with others in which I get hurt or feel rejected.
162. I often get lost in my thoughts and forget what's going on around me.

163. People say I'm a thin person, but I feel that my thighs and backside are much too big.
164. There are terrible events from my past that come back repeatedly to haunt my thoughts and dreams.
165. Other than my family, I have no close friends.
166. I act quickly much of the time and don't think things through as I should.
167. I take great care to keep my life a private matter so no one can take advantage of me.
168. I very often hear things so well that it bothers me.
169. I'm always willing to give in to others in a disagreement because I fear their anger or rejection.
170. I repeat certain behaviors again and again, sometimes to reduce my anxiety and sometimes to stop something bad from happening.
171. I have given serious thought recently to doing away with myself.
172. People tell me that I'm a very proper and moral person.
173. I still feel terrified when I ~~think~~ of a traumatic experience I had years ago.
174. Although I'm afraid to make friendships, I wish I had more than I do.
175. There are people who are supposed to be my friends who would like to do me harm.

RS BS

Scale 1 - 4 - 48
2A - 2 - 24 - 1
2B - 0 - 0 - 1
3 - 2 - 20

60 - 15 - 45

Histrionic 4 - 20 - 72 + 8
Narcissistic 5 - 18 - 81 + 8 + 8 = 93
Antisocial 6A - 10 - 71

Report with sequence

Validity, etc

X42 -

1 to 8B

Severe

Clinical

Some clinical

6B - 4 - 48

7 - 10 - 38

8A - 4 - 40

8B - 0 - 0 - 1

9 - 0 - 0 - 1

10 - 1 - 12 - 1

11 - 0 - 0 - 1

12 - 0 - 0 - 1

13 - 0 - 0 - 1

14 - 0 - 0 - 1

15 - 3 - 36

16 - 0 - 0 - 1

17 - 4 - 63 - Marochotic

18 - 5 - 67 - Drug dependence

19 - 0 - 0 - 1

20 - 0 - 0 - 1

21 - 0 - 0 - 1

22 - 0 - 0 - 1

23 - 0 - 0 - 1

24 - 18 - 80 - A. scamblerby

25 - 0 - 0 - 1

26 - 0 - 0 - 1

27 - 0 - 0 - 1

28 - 0 - 0 - 1

29 - 0 - 0 - 1

30 - 0 - 0 - 1

31 - 0 - 0 - 1

32 - 0 - 0 - 1

33 - 0 - 0 - 1

34 - 0 - 0 - 1

35 - 0 - 0 - 1

36 - 0 - 0 - 1

37 - 0 - 0 - 1

38 - 0 - 0 - 1

39 - 0 - 0 - 1

Proposed

clinical work of

Pl. personality and

Psychopathology

MMPI by MEMS
Psychopathology Traits

1/2
1/2
1/2

Scale X: $1 + 2A + 2B + 3 + 4 + 5 \times \frac{3}{2} + 6A + 6B + 7 + 8A + 8B$

$= 4 + 2 + 0 + 2 + 20 + 12 + 10 + 4 + 10 + 11 + 0$

$X = [68] = BR. 40$

Whom to?

Appendix B

Scale Composition and Item Weighting

(See Appendix D for scoring information for the Grossman facet scales.)

Scale 1 Schizoid (16 items)

True prototypal items (weight = 2)

10, 27, 46, 92, 105, 148, 165

True nonprototypal items (weight = 1)

38, 48, 101, 142, 156, 167

False (weight = 1)

32, 57

12
4
16

Scale 2A Avoidant (16 items)

True prototypal items (weight = 2)

18, 40, 69, 84, 99, 127, 141, 174

True nonprototypal items (weight = 1)

47, 48, 146, 148, 151, 158

False (weight = 1)

57, 80

2
2 / 4

Scale 2B Depressive (15 items)

True prototypal items (weight = 2)

20, 25, 47, 112, 123, 133, 143, 151

True nonprototypal items (weight = 1)

24, 43, 83, 86, 142, 148, 154

6
4 10

Scale 3 Dependent (16 items)

True prototypal items (weight = 2)

16, 35, 45, 73, 94, 108, 135, 169

True nonprototypal items (weight = 1)

47, 66, 84, 120, 133, 141, 151

False (weight = 1)

82

10
1 11

Scale 4 Histrionic (17 items)

True prototypal items (weight = 2)

12, 21, 32, 51, 57, 80, 88

False (weight = 1)

10, 24, 27, 48, 69, 92, 99, 123, 127, 174

6
3 9

Scale 5 Narcissistic (24 items)

True prototypal items (weight = 2)

5, 26, 31, 67, 85, 93, 144, 159

True nonprototypal items (weight = 1)

21, 38, 57, 80, 88, 116

False (weight = 1)

35, 40, 47, 69, 84, 86, 94, 99, 141, 169

6
3
16**Scale 6A Antisocial (17 items)**

True prototypal items (weight = 2)

17, 38, 53, 101, 113, 139, 166

True nonprototypal items (weight = 1)

7, 13, 14, 21, 41, 52, 93, 122, 136

False (weight = 1)

172

(2)

Scale 6B Sadistic (Aggressive) (20 items)

True prototypal items (weight = 2)

9, 14, 28, 64, 87, 95, 116

True nonprototypal items (weight = 1)

7, 13, 17, 33, 36, 39, 41, 49, 53, 79, 93, 96, 166

6
4
10**Scale 7 Compulsive (17 items)**

True prototypal items (weight = 2)

2, 29, 59, 82, 97, 114, 137, 172

False (weight = 1)

7, 14, 22, 41, 53, 72, 101, 139, 166

12
7
14**Scale 8A Negativistic (Passive-Aggressive) (16 items)**

True prototypal items (weight = 2)

7, 15, 22, 36, 50, 60, 79, 115, 126

True nonprototypal items (weight = 1)

6, 42, 83, 98, 122, 133, 166

12
1
13**Scale 8B Masochistic (Self-Defeating) (15 items)**

True prototypal items (weight = 2)

19, 43, 70, 90, 104, 122, 161

True nonprototypal items (weight = 1)

18, 24, 25, 35, 40, 98, 148, 169

(4)

Scale S Schizotypal (16 items)

True prototypal items (weight = 2)

8, 48, 71, 76, 117, 138, 156, 158, 162

True nonprototypal items (weight = 1)

69, 99, 102, 134, 141, 148, 151

4
2
6**Scale C Borderline (16 items)**

True prototypal items (weight = 2)

30, 41, 72, 83, 98, 120, 134, 142, 154

True nonprototypal items (weight = 1)

7, 22, 122, 135, 161, 166, 171

6
1
7**Scale P Paranoid (17 items)**

True prototypal items (weight = 2)

6, 33, 42, 49, 89, 103, 146, 167, 175

True nonprototypal items (weight = 1)

8, 48, 60, 63, 115, 138, 158, 159

9
1
12**Scale A Anxiety (14 items)**

True prototypal items (weight = 2)

58, 75, 124, 147, 164, 170

True nonprototypal items (weight = 1)

40, 61, 76, 108, 109, 135, 145, 149

(12)

Scale H Somatoform (12 items)

True prototypal items (weight = 2)

4, 11, 37, 55, 74

True nonprototypal items (weight = 1)

1, 75, 107, 111, 130, 145, 148

(12)

Scale N Bipolar: Manic (13 items)

True prototypal items (weight = 2) 3, 54, 96, 106, 125
 True nonprototypal items (weight = 1) 22, 41, 51, 83, 117, 134, 166, 170

Scale D Dysthymia (14 items)

True prototypal items (weight = 2) 24, 56, 62, 86, 111, 130
 True nonprototypal items (weight = 1) 15, 25, 55, 83, 104, 141, 142, 148

Scale B Alcohol Dependence (15 items)

True prototypal items (weight = 2) 52, 77, 100, 131, 152
 True nonprototypal items (weight = 1) 14, 41, 64, 93, 101, 113, 122, 139, 166
 False prototypal item (weight = 2) 23

Scale T Drug Dependence (14 items)

True prototypal items (weight = 2) 13, 39, 66, 91, 118, 136
 True nonprototypal items (weight = 1) 7, 21, 38, 41, 53, 101, 113, 139

Scale R Post-Traumatic Stress Disorder (16 items)

True prototypal items (weight = 2) 109, 129, 149, 160, 173
 True nonprototypal items (weight = 1) 62, 76, 83, 123, 133, 142, 147, 148, 151, 154, 164

Scale SS Thought Disorder (17 items)

True prototypal items (weight = 2) 34, 61, 68, 78, 102, 168
 True nonprototypal items (weight = 1) 22, 56, 72, 76, 83, 117, 134, 142, 148, 151, 162

Scale CC Major Depression (17 items)

True prototypal items (weight = 2) 1, 44, 107, 128, 150, 171
 True nonprototypal items (weight = 1) 4, 34, 55, 74, 104, 111, 130, 142, 148, 149, 154

Scale PP Delusional Disorder (13 items)

True prototypal items (weight = 2) 63, 119, 140, 153
 True nonprototypal items (weight = 1) 5, 38, 49, 67, 89, 103, 138, 159, 175

Scale Y Desirability (21 items)

True (weight = 1) 32, 51, 57, 59, 80, 82, 88, 97, 137, 172
 False (weight = 1) 20, 35, 40, 69, 104, 112, 123, 141, 142, 148, 151

Scale Z Debasement (33 items)

True (weight = 1) 1, 4, 8, 15, 22, 24, 30, 34, 36, 37, 44, 55, 56, 58, 62, 63, 70, 74, 75, 76, 83, 84, 86, 99, 111, 123, 128, 133, 134, 142, 145, 150, 171

Scale V Invalidity (3 items)

True (weight = 1)

65, 110, 157

**Scale W Inconsistency
(44 item response pairs)**

Add 1 raw score point for each

1T-4F, 1F-4T, 8F-141T, 13T-66F, 13F-66T, 15T-133F,
20T-112F, 20F-112T, 22T-83F, 24F-151T, 25T-56F,
25F-56T, 27F-92T, 32T-80F, 35T-84F, 35F-84T,
39T-118F, 39F-118T, 41F-166T, 44T-86F, 44F-150T,
48T-92F, 49T-146F, 52F-152T, 55F-130T, 57F-80T,
61F-76T, 62F-86T, 68F-162T, 69T-99F, 70F-104T,
72F-142T, 74F-107T, 77T-131F, 91T-136F, 91F-136T,
108T-135F, 109F-164T, 123F-128T, 129T-173F,
133T-145F, 147F-149T, 160T-164F, 160F-173T

Appendix C

Base Rate

Transformation Tables

(The base rate transformation tables for the Grossman facet scales can be found in Appendix E.)

Appendix C.1 Base Rate Transformations for All Scales Except X (Males and Females)

Raw Score	BR Scale Scores														P
	1	2A	2B	3	4	5	6A	6B	7	8A	8B	S	C		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	12	12	10	10	4	5	12	12	4	10	20	20	12	15	15
2	24	24	20	20	8	10	24	24	8	20	40	40	24	30	30
3	36	36	30	30	12	15	36	36	12	30	60	60	36	45	45
4	48	48	40	40	16	20	48	48	16	40	63	62	48	60	60
5	60	60	50	50	20	25	60	60	20	50	66	64	60	61	61
6	62	63	60	60	24	30	62	62	24	60	69	66	63	63	63
7	64	67	65	65	28	35	64	64	28	62	72	67	66	64	64
8	66	71	70	70	32	40	66	66	32	64	75	68	69	66	66
9	68	75	75	75	36	44	69	68	36	66	78	69	72	67	67
10	70	77	77	78	40	48	71	69	39	68	80	70	75	69	69
11	72	79	79	81	44	52	73	70	42	70	82	71	77	70	70
12	75	81	81	83	48	56	75	71	45	72	85	72	79	72	72
13	78	83	83	85	51	60	79	72	48	75	88	73	81	73	73
14	81	85	85	87	54	63	82	73	51	77	91	74	83	75	75
15	85	88	89	89	57	67	85	74	54	79	94	75	85	77	77
16	89	91	92	91	60	71	89	75	57	81	97	78	88	79	79
17	93	94	96	94	63	75	92	78	60	83	100	81	91	81	81
18	97	97	99	97	66	85	96	80	63	85	103	85	94	83	83
19	101	100	103	100	69	89	99	83	66	89	106	90	97	85	85
20	105	103	106	103	72	93	103	85	69	93	109	95	100	90	90
21	109	106	109	106	75	97	106	90	72	97	112	99	103	95	95
22	112	109	112	109	79	101	109	95	75	101	115	103	106	100	100
23	115	112	115	112	83	105	112	100	79	105	107	107	109	105	105
24		115		115	88	110	115	105	83	110	111	111	112	110	110
25				115		115		110	87	115	115	115	115	115	115
26								115							

Appendix C.1 (continued)

Raw Score	BR Scale Scores											
	A	H	N	D	B	T	R	SS	CC	PP	Y	Z
0	0	0	0	0	0	0	0	0	0	0	0	0
1	20	15	12	12	20	20	15	15	15	30	5	18
2	40	30	24	24	40	40	30	30	30	60	10	35
3	60	45	36	36	60	60	45	45	45	62	15	38
4	75	60	48	48	63	63	60	60	60	65	20	40
5	77	62	60	60	67	67	62	62	65	68	25	42
6	79	64	63	62	71	71	63	64	70	70	30	45
7	81	66	66	64	75	75	65	66	75	72	35	47
8	83	68	69	66	77	76	66	67	78	75	39	49
9	85	70	72	69	80	77	68	68	81	80	43	52
10	87	72	75	72	83	78	69	69	85	85	47	54
11	89	73	80	75	85	79	71	70	87	90	51	56
12	91	74	85	78	88	81	73	71	89	95	55	59
13	94	75	90	80	91	83	75	72	91	100	59	61
14	97	80	95	82	95	85	77	73	93	105	63	63
15	100	85	100	85	99	90	79	74	95	110	67	66
16	103	100	105	91	103	95	81	75	97	115	71	68
17	106	115	110	97	107	100	83	79	99		75	70
18	109		115	103	111	105	85	82	101		80	73
19	112			109	115	110	95	85	-103		85	75
20	115			115		115	105	93	106		93	76
21							115	100	109		100	78
22								108	112			79
23								115	115			81
24												82
25												84
26												85
27												88
28												90
29												92
30												94
31												96
32												98
33												100

Appincha C-2. Base Rate Transformation

ations for Scale X (Males and Females)

$\frac{98}{95}$
 $\frac{98}{95}$

1034-38

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63-64

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66-67

68

69-70

71

72-73

74

75-76

77

78-79

80

81-82

83

84-85

86

87-88

89

90-91

92

BR Score

Raw Score

BR Score

0

93-94

57

2

95

58

3

96-97

59

5

98

60

6

99-100

61

8

101

62

9

102-103

63

11

104

64

12

105-106

65

14

107

66

15

108-109

67

17

110

68

18

111-112

69

20

113

70

21

114-115

71

23

116

72

24

117-118

73

26

119-120

74

27

121-123

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124-126

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127-129

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130-132

78

33

133-135

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136-138

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139-141

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37

142-144

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145-146

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147-148

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149-150

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151-152

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154-155

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156-157

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158-159

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161-162

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163-164

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165-166

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167-168

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169-170

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171-172

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174

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175-178

100

56

Appendix D

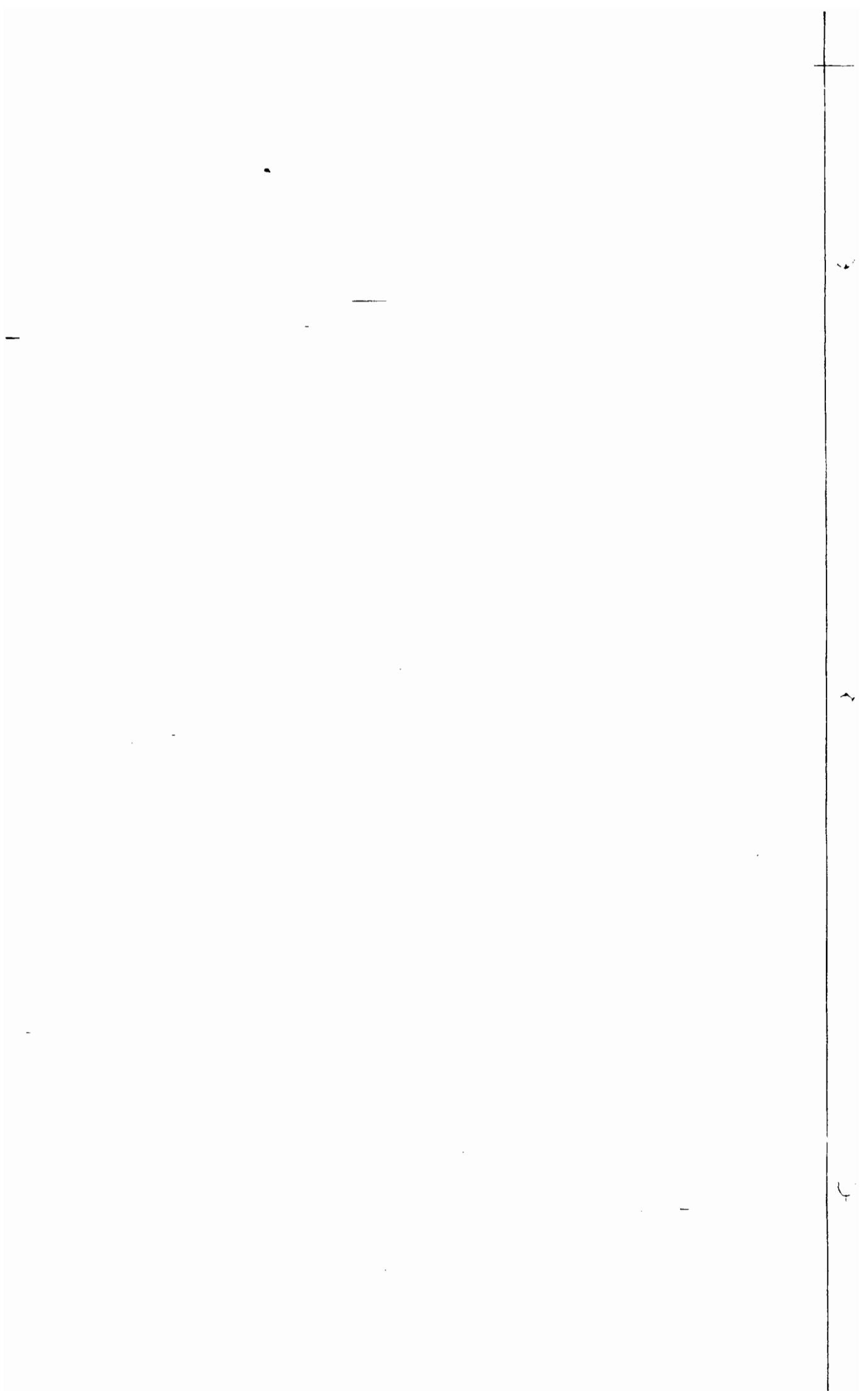
Grossman Facet Scale Scoring

For the 14 primary personality scales, each scale's three facet scales are listed along with the True-keyed and False-keyed items that make up each facet. Each keyed response counts 1 point toward the facet scale raw score. Appendix E provides tables for converting the facet scale raw scores to base rate scores.

	Items Keyed True	Items Keyed False
Scale 1 Schizoid		
1.1 Temperamentally Apathetic	1, 4, 46, 142, 148, 156, 162	51
1.2 Interpersonally Unengaged	27, 48, 92, 105, 141, 165	32, 57, 80, 88
1.3 Expressively Impassive	10, 29, 35, 38, 84, 101, 167	12
Scale 2A Avoidant		
2A.1 Interpersonally Aversive	19, 27, 48, 69, 92, 127, 146, 158, 161	
2A.2 Alienated Self-Image	20, 35, 40, 47, 70, 84, 104, 141, 148, 151	
2A.3 Vexatious Representations	10, 18, 46, 99, 105, 174	57, 80
Scale 2B Depressive		
2B.1 Temperamentally Woeful	1, 86, 112, 123, 130, 142, 148	
2B.2 Worthless Self-Image	19, 24, 25, 43, 47, 70, 104, 122, 151	
2B.3 Cognitively Fatalistic	15, 20, 62, 133, 145, 150, 154, 171	

	Items Keyed True	Items Keyed False
	35, 40, 47, 56, 84, 122, 133, 151 2, 16, 45, 73, 94, 120, 169, 172 108, 127, 135, 141, 142, 165	
	51, 88 31, 32, 57, 80, 144 12, 21, 54	69, 99, 105, 127, 174 46, 48, 92 10, 40, 123, 148
5.1	5, 31, 57, 80	35, 40, 47, 69, 99, 141
5.2 Cognitive	21, 26, 42, 67, 87, 139, 144	
5.3 Interpersonally sensitive	9, 33, 38, 64, 85, 93, 95, 101, 116, 159	
Scale 6A Antisocial		
6A.1 Expressively Impulsive	9, 12, 33, 38, 41, 87, 122, 166	172
6A.2 Acting-Out Mechanism	7, 13, 17, 21, 39, 51, 52, 53, 113, 136	
6A.3 Interpersonally Irresponsible	14, 85, 91, 93, 101, 116, 126, 139	
Scale 6B Sadistic (Aggressive)		
6B.1 Temperamentally Hostile	9, 14, 33, 36, 53, 64, 79, 113	
6B.2 Eruptive Organization	13, 17, 39, 41, 51, 87, 96, 166	
6B.3 Pernicious Representations	6, 7, 28, 38, 49, 89, 93, 95, 103, 116, 175	
Scale 7 Compulsive		
7.1 Cognitively Constricted	2, 114	22, 51, 84, 139, 147, 166
7.2 Interpersonally Respectful	59, 82, 97	7, 38, 53
7.3 Reliable Self-Image	16, 137, 172	17, 41, 101, 113
Scale 8A Negativistic (Passive-Aggressive)		
8A.1 Temperamentally Irritable	7, 9, 22, 36, 41, 79, 83, 87, 98, 166	
8A.2 Expressively Resentful	6, 15, 33, 42, 50, 115, 126	
8A.3 Discontented Self-Image	25, 47, 60, 70, 104, 122, 133	
Scale 8B Masochistic (Self-Defeating)		
8B.1 Discredited Representations	18, 19, 48, 98, 120, 127, 135, 161	
8B.2 Cognitively Diffident	16, 24, 35, 40, 45, 94, 133, 169	
8B.3 Undeserving Self-Image	25, 43, 47, 70, 90, 104, 122, 141, 148, 151	

	Items Keyed True	Items Keyed False
Scale S Schizotypal		
S.1 Estranged Self-Image	8, 18, 40, 48, 49, 69, 99, 141, 158, 165, 175	
S.2 Cognitively Autistic	71, 76, 78, 102, 109, 117, 138, 149, 153, 162	
S.3 Chaotic Representations	10, 22, 34, 134, 148, 151, 156, 173	
Scale C Borderline		
C.1 Temperamentally Labile	20, 22, 25, 30, 36, 83, 98, 133, 154, 171	
C.2 Interpersonally Paradoxical	7, 19, 41, 43, 115, 120, 122, 161, 166	
C.3 Uncertain Self-Image	24, 70, 72, 90, 104, 134, 135, 142, 151	
Scale P Paranoid		
P.1 Cognitively Mistrustful	8, 28, 49, 63, 89, 138, 140, 175	
P.2 Expressively Defensive	42, 48, 53, 146, 153, 158, 167	
P.3 Projection Mechanism	6, 26, 31, 33, 60, 85, 103, 115, 159	



Appendix E

Base Rate Transformation Tables for the Grossman Facet Scales

Appendix E.1 Base Rate Transformations for Grossman Facet Scales (Males and Females)

Facet Scale	Raw Scores											
	0	1	2	3	4	5	6	7	8	9	10	11
1.1	0	32	48	63	73	84	93	98	100			
1.2	0	27	42	55	65	75	82	90	95	98	100	
1.3	0	26	47	63	77	90	97	99	100			
2A.1	0	42	59	69	78	83	91	95	98	100		
2A.2	0	37	51	61	68	75	82	87	93	96	100	
2A.3	0	37	52	64	76	85	93	98	100			
2B.1	0	46	56	66	74	82	91	100				
2B.2	0	42	54	64	73	79	85	91	96	100		
2B.3	0	37	52	63	74	83	92	97	100			
3.1	0	38	51	61	69	78	86	94	100			
3.2	0	15	32	53	72	84	93	98	100			
3.3	0	56	73	83	92	98	100					
4.1	0	11	21	33	49	73	92	100				
4.2	0	9	17	29	43	65	89	98	100			
4.3	0	9	21	36	60	81	95	100				
5.1	0	9	15	22	31	43	58	76	93	98	100	
5.2	0	35	59	77	89	97	99	100				
5.3	0	51	69	79	88	94	96	98	99	99	100	
6A.1	0	23	42	57	70	82	91	97	99	100		
6A.2	0	34	55	70	79	86	92	95	98	99	100	
6A.3	0	49	68	82	90	96	98	99	100			
6B.1	0	43	61	75	85	92	97	99	100			
6B.2	0	34	54	69	83	91	96	99	100			
6B.3	0	44	60	72	81	88	94	97	99	99	99	100
7.1	0	3	9	18	31	50	69	89	100			
7.2	0	7	20	43	66	85	100					
7.3	0	6	16	34	54	77	92	100				
8A.1	0	28	42	53	62	71	80	87	93	98	100	
8A.2	0	46	64	77	87	94	98	100				
8A.3	0	41	55	66	77	85	93	100				
8B.1	0	52	66	76	83	91	96	98	100			
8B.2	0	26	43	57	71	81	89	96	100			
8B.3	0	43	55	66	73	78	83	88	93	98	100	
S.1	0	46	58	68	75	81	88	92	95	97	99	100
S.2	0	50	65	76	84	89	94	97	98	99	100	
S.3	0	43	57	68	78	86	93	98	100			
C.1	0	38	51	61	71	78	85	92	96	98	100	
C.2	0	36	49	64	77	84	91	96	99	100		
C.3	0	46	57	66	73	79	86	91	96	100		
P.1	0	63	77	87	92	96	98	99	100			
P.2	0	55	70	84	93	97	99	100				
P.3	0	42	60	76	87	93	96	99	99	100		

Appendix F

Noteworthy Responses

Any item below that is marked True is reported as a Noteworthy Response.

Health Preoccupation

- 1. Lately, my strength seems to be draining out of me, even in the morning.
- 4. I feel weak and tired much of the time.
- 11. I have a hard time keeping my balance when walking.
- 37. I very often lose my ability to feel any sensations in parts of my body.
- 55. In recent weeks I feel worn out for no special reason.
- 74. I can't seem to sleep, and wake up just as tired as when I went to bed.
- 75. Lately, I've been sweating a great deal and feel very tense.
- 107. I have completely lost my appetite and have trouble sleeping most nights.
- 130. I don't have the energy to concentrate on my everyday responsibilities anymore.
- 149. I feel shaky and have difficulty falling asleep because painful memories of a past event keep running through my mind.

Interpersonal Alienation

- 10. What few feelings I seem to have I rarely show to the outside world.
- 18. I'm afraid to get really close to another person because it may end up with my being ridiculed or shamed.
- 27. When I have a choice, I prefer to do things alone.
- 48. A long time ago, I decided it's best to have little to do with people.
- 63. Many people have been spying into my private life for years.
- 69. I avoid most social situations because I expect people to criticize or reject me.
- 92. I'm alone most of the time and I prefer it that way.
- 99. In social groups I am almost always very self-conscious and tense.
- 105. I have little desire for close friendships.
- 161. I seem to create situations with others in which I get hurt or feel rejected.

Interpersonal Alienation (continued)

- 165. Other than my family, I have no close friends.
- 167. I take great care to keep my life a private matter so no one can take advantage of me.
- 174. Although I'm afraid to make friendships, I wish I had more than I do.

Emotional Dyscontrol

- 9. I often criticize people strongly if they annoy me.
- 14. Sometimes I can be pretty rough and mean in my relations with my family.
- 22. I'm a very erratic person, changing my mind and feelings all the time.
- 30. Lately, I have begun to feel like smashing things.
- 34. Lately, I have gone all to pieces. —
- 77. I have a great deal of trouble trying to control an impulse to drink to excess.
- 83. My moods seem to change a great deal from one day to the next.
- 87. I often get angry with people who do things slowly.
- 96. People have said in the past that I became too interested and too excited about too many things.
- 116. I have had to be really rough with some people to keep them in line.
- 124. When I'm alone and away from home, I often begin to feel tense and panicky.
- 134. I sometimes feel crazy-like or unreal when things start to go badly in my life.

Self-Destructive Potential

- 24. I began to feel like a failure some years ago.
- 44. I feel terribly depressed and sad much of the time now.
- 112. I have been downhearted and sad much of my life since I was quite young.
- 128. I feel deeply depressed for no reason I can figure out.
- 142. I frequently feel there's nothing inside me, like I'm empty and hollow.
- 150. Looking ahead as each day begins makes me feel terribly depressed.
- 151. I've never been able to shake the feeling that I'm worthless to others.
- 154. I have tried to commit suicide.
- 171. I have given serious thought recently to doing away with myself.

Childhood Abuse

- 81. I'm ashamed of some of the abuses I suffered when I was young.
- 132. I hate to think about some of the ways I was abused as a child.

Eating Disorder

- 121. I go on eating binges a couple of times a week.
- 143. I sometimes force myself to vomit after eating.
- 155. I'm willing to starve myself to be even thinner than I am.
- 163. People say I'm a thin person, but I feel that my thighs and backside are much too big.

Appendix G

Score Adjustments

1. Disclosure Adjustment

This section computes the Disclosure adjustment, which is used to adjust the scores for the Clinical Personality Patterns scales (1, 2A, 2B, 3, 4, 5, 6A, 6B, 7, 8A, 8B), the Severe Personality Pathology scales (S, C, P), the Clinical Syndromes scales (A, H, N, D, B, T, R), and the Severe Clinical Syndromes scales (SS, CC, PP).

NOTE: This adjustment is based on the Scale X **RAW** score (**not** the BR score).

- 1-1:** In Table 1, locate the raw score range that contains the respondent's Scale X **raw** score.
- 1-2:** Find the 1-8B and the S-PP adjustments that correspond to the respondent's **raw** score on Scale X.
- 1-3:** Apply the 1-8B adjustment to the BR scores for Scales 1-8B. Apply the S-PP adjustment to the BR scores for Scales S-PP. After the adjustments are applied, any BR values that are less than 0 should be changed to 0, and any BR values that are greater than 115 should be changed to 115.

2. Anxiety/Depression (A/D) Adjustment

The Anxiety/Depression (A/D) adjustment is based on the respondent's Scale A (Anxiety) and Scale D (Dysthymia) scores. The A/D adjustment is applied to Clinical Personality Patterns Scales 2A, 2B, and 8B and to Severe Personality Pathology Scales S and C.

- 2-1:** Compute the Anxiety/Depression value, which is used to select the appropriate A/D adjustment based on the BR scores for Scales A and D.
 - **if A and D are both less than 75:**
No A/D adjustment is necessary. Proceed to section 3, "Inpatient Adjustment."
 - **if A is greater than or equal to 75 and D is less than 75:**
Calculate $(A - 75) = \text{A/D value}$. Proceed to step 2-2.
 - **if D is greater than or equal to 75 and A is less than 75:**
Calculate $(D - 75) = \text{A/D value}$. Proceed to step 2-2.
 - **if A and D are both greater than or equal to 75:**
Calculate $[(A - 75) + (D - 75)] = \text{A/D value}$.

Table 1 Disclosure Adjustment Values

Scale X Raw Score	1-8B Adjustment	S-PP Adjustment	Scale X Raw Score	1-8B Adjustment	S-PP Adjustment
0-33 ^a	+20	+10	138	-6	-4
34	+20	+10	139	-7	-4
35	+20	+10	140	-7	-4
36	+20	+10	141	-7	-4
37	+19	+10	142	-8	-5
38	+18	+10	143	-8	-5
39	+17	+9	144	-9	-5
40	+17	+9	145	-9	-5
41	+16	+9	146	-9	-5
42	+15	+8	147	-10	-6
43	+14	+8	148	-10	-6
44	+13	+7	149	-11	-6
45	+13	+7	150	-11	-6
46	+12	+7	151	-11	-6
47	+11	+6	152	-12	-7
48	+10	+6	153	-12	-7
49	+9	+5	154	-13	-7
50	+9	+5	155	-13	-7
51	+8	+5	156	-13	-7
52	+7	+4	157	-14	-8
53	+6	+4	158	-14	-8
54	+5	+3	159	-15	-8
55	+5	+3	160	-15	-8
56	+4	+3	161	-15	-8
57	+3	+2	162	-16	-9
58	+2	+2	163	-16	-9
59	+1	+1	164	-17	-9
60	+1	+1	165	-17	-9
61-123	None	None	166	-17	-9
124	-1	-1	167	-18	-10
125	-1	-1	168	-18	-10
126	-1	-1	169	-19	-10
127	-2	-2	170	-19	-10
128	-2	-2	171	-19	-10
129	-3	-2	172	-20	-11
130	-3	-2	173	-20	-11
131	-3	-2	174	-20	-11
132	-4	-3	175	-20	-11
133	-4	-3	176	-20	-11
134	-5	-3	177	-20	-11
135	-5	-3	178	-20	-11
136	-5	-3	179+ ^a	-20	-11
137	-6	-4			

^aAn individual's MCMI-III results are considered invalid if the raw score on Scale X is less than 34 or greater than 178. Scoring and/or interpretation of invalid results is not recommended.

Proceed to step 2-2.

- 2-2:** Using the result of the preceding calculation (the A/D value), find the A/D adjustments in the appropriate table. Table 2 should be used when the client is not an inpatient, or when the client is an inpatient and the duration of his or her Axis I disorder is more than four weeks. Table 3 should be used when the client is an inpatient and his or her Axis I disorder has been present for less than one week. Table 4 should be used when the client is an inpatient and his or her Axis I disorder has been present for one to four weeks.
- 2-3:** Apply the 2B, 8B, C adjustment factor to the BR scores for Scales 2B, 8B, and C. Apply the 2A, S adjustment factor to the BR scores for Scales 2A and S. After the adjustments are applied, any BR values that are less than 0 should be changed to 0, and any BR values that are greater than 115 should be changed to 115.

3. Inpatient Adjustment

The Inpatient adjustment is computed using the client's setting and Axis I duration. It is used to adjust the BR scores for the Severe Clinical Syndromes scales (SS, CC, PP).

- 3-1:** Determine the setting and Axis I duration for the client.
- 3-2:** Determine whether an adjustment should be applied.

If the client:

- is being evaluated using the correctional norms in the *MCMI-III Corrections Report User's Guide*, or
- is not an inpatient, or
- is an inpatient and his or her Axis I symptoms have been present for more than four weeks: No Inpatient adjustment is necessary. Proceed to section 4, "Denial/Complaint Adjustment."

If the client:

- is an inpatient and his or her Axis I symptoms have been present less than one week or one to four weeks: Proceed to step 3-3.

Table 2 Non-Inpatient, or Inpatient and Axis I Duration of More Than Four Weeks

2B, 8B, C		2A, S	
A/D Value	Adjustment	A/D Value	Adjustment
0	-1	0	-1
1	-1	1	-1
2	-1	2	-1
3	-1	3	-1
4	-1	4	-1
5	-1	5	-1
6	-1	6	-1
7	-1	7	-1
8	-1	8-15	-1
9	-1	16-23	-2
10	-2	24-31	-3
11	-2	32-39	-4
12	-2	40-47	-5
13	-2	48-55	-6
14	-2	56-63	-7
15-19	-3	64-71	-8
20-24	-4	72-79	-9
25-29	-5	80	-10
30-34	-6		
35-39	-7		
40-44	-8		
45-49	-9		
50-54	-10		
55-59	-11		
60-64	-12		
65-69	-13		
70-74	-14		
75-80	-15		

- 3-3:** In Table 5, select the appropriate row to find the Inpatient adjustment.

- 3-4:** Apply the Inpatient adjustment to the BR scores for Scales SS, CC, and PP. After the adjustment is applied, any BR values that are less than 0 should be changed to 0, and any BR values that are greater than 115 should be changed to 115.

Table 3 Inpatient and Axis I Duration of Less Than One Week

2B, 8B, C		2A, S	
A/D Value	Adjustment	A/D Value	Adjustment
0	-1	0	-1
1	-1	1	-1
2	-1	2	-1
3	-1	3	-1
4	-1	4	-1
5	-2	5	-1
6	-2	6	-1
7	-2	7	-1
8	-3	8-11	-2
9	-3	12-15	-3
10-12	-4	16-19	-4
13-14	-5	20-23	-5
15-17	-6	24-27	-6
18-19	-7	28-31	-7
20-22	-8	32-35	-8
23-24	-9	36-39	-9
25-27	-10	40-43	-10
28-29	-11	44-47	-11
30-32	-12	48-51	-12
33-34	-13	52-55	-13
35-37	-14	56-59	-14
38-39	-15	60-63	-15
40-80	-15	64-80	-15

Table 4 Inpatient and Axis I Duration of One to Four Weeks

2B, 8B, C		2A, S	
A/D Value	Adjustment	A/D Value	Adjustment
0-3	-1	0-5	-1
4-6	-1	6-10	-1
7-9	-2	11-15	-2
10-13	-3	16-21	-3
14-16	-4	22-26	-4
17-19	-5	27-31	-5
20-23	-6	32-37	-6
24-26	-7	38-42	-7
27-29	-8	43-47	-8
30-33	-9	48-53	-9
34-36	-10	54-58	-10
37-39	-11	59-63	-11
40-43	-12	64-69	-12
44-46	-13	70-74	-13
47-49	-14	75-79	-14
50-53	-15	80	-15
54-80	-15		

4. Denial/Complaint Adjustment

The Denial/Complaint adjustment is computed using a rank ordering of the current BR scores for the Clinical Personality Patterns scales. It is used to adjust the latest BR scores for Scales 4, 5, and 7.

- 4-1:** Determine which of the Clinical Personality Patterns scales (1, 2A, 2B, 3, 4, 5, 6A, 6B, 7, 8A, 8B) is highest based on their current **adjusted BR** scores. Any ties between scales should be broken by giving a higher ranking to the scale that appears first in the following

list: 2B, 6A, 8B, 6B, 2A, 8A, 7, 4, 5, 3, 1.

- 4-2:** The Denial/Complaint adjustment is applied as follows: if the highest ranked Clinical Personality Patterns scale is 4, 5, or 7, the adjustment value is 8. If the highest ranked Clinical Personality Patterns scale is anything else, no adjustment is made.
- 4-3:** Add the 8-point adjustment factor from step 4-2 **only** to the highest ranking scale of Scales 4, 5, and 7. After the adjustment is applied, if the BR value is less than 0, it should be changed to 0. If the BR value is greater than 115, it should be changed to 115.

The resulting BR values are the final BR scores. They are used for all profiling and interpretation.

Table 5 Inpatient Axis I Duration Status

Inpatient/Axis I Duration Status	SS Adjustment	CC Adjustment	PP Adjustment
If setting = Inpatient and Axis I duration = Less than one week	+6	+10	+4
If setting = Inpatient and Axis I duration = One to four weeks	+4	+8	+2

Appendix H

Sample Interpretive Report



MCMI-III™
MILLON™ CLINICAL
MULTIAXIAL INVENTORY-III

MCMI-III™

Millon™ Clinical Multiaxial Inventory-III

Interpretive Report with Grossman Facet Scales

Theodore Millon, PhD, DSc

Name:	Sample Interpretive Report
ID Number:	98765
Age:	21
Gender:	Female
Setting:	Outpatient Never Hospitalized
Race:	White
Marital Status:	Never Married
Date Assessed:	03/03/2009

TRADE SECRET INFORMATION

Not for release under HIPAA or other data disclosure laws that exempt trade secrets from disclosure.

[7.0/83/2.3.10]

CAPSULE SUMMARY

MCMI-III reports are normed on patients who were in the early phases of assessment or psychotherapy for emotional discomfort or social difficulties. Respondents who do not fit this normative population or who have inappropriately taken the MCMI-III for nonclinical purposes may have inaccurate reports. The MCMI-III report cannot be considered definitive. It should be evaluated in conjunction with additional clinical data. The report should be evaluated by a mental health clinician trained in the use of psychological tests. The report should not be shown to patients or their relatives.

Interpretive Considerations

The client is a 21-year-old single white female with 11 years of education. She is currently being seen as an outpatient, and she reports that she has recently experienced problems that involve her job or school and use of alcohol. These self-reported difficulties, which have occurred for an undetermined period of time, may take the form of an Axis I disorder.

Unless this patient is a well-functioning adult with only minor life stressors, her responses suggest a need for social approval or naivete about psychological matters. This interpretive report should be read with these characteristics in mind.

Profile Severity

On the basis of the test data, it may be reasonable to assume that the patient is experiencing a moderately severe mental disorder; further professional study may be advisable to assess the need for ongoing clinical care. The text of the following interpretive report may need to be modulated only slightly upward or downward given this probable level of severity.

Possible Diagnoses

She appears to fit the following Axis II classifications best: Antisocial Personality Disorder, with Histrionic Personality Features, and Paranoid Personality Features.

Axis I clinical syndromes are suggested by the client's MCMI-III profile in the areas of Alcohol Abuse and Psychoactive Substance Abuse NOS.

Therapeutic Considerations

Superficially gregarious and friendly, this patient can readily become ill-humored and touchy if subjected to persistent social discomfort and external demands. She is disinclined to persevere in routine tasks such as long-term therapeutic compliance, but there may be considerable gain by using short-term treatment regimens that focus on specific goals and time-limited techniques.

MILLON CLINICAL MULTIAXIAL INVENTORY - III
CONFIDENTIAL INFORMATION FOR PROFESSIONAL USE ONLY

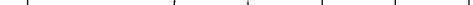
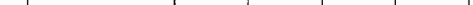
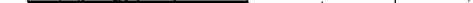
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PERSONALITY CODE: 6A ** - * 4 5 6B 8A 8B 3 + 7 2A " 1 2B ' ' // - ** - * //
SYNDROME CODE: B ** T * // - ** - * //
DEMOGRAPHIC CODE: 98765/ON/F/21/W/N/11/JO/AL/-----/10/-----/

CATEGORY		SCORE		PROFILE OF BR SCORES					DIAGNOSTIC SCALES
		RAW	BR	0	60	75	85	115	
MODIFYING INDICES	X	92	56						DISCLOSURE
	Y	17	75						DESIRABILITY
	Z	4	40						DEBASEMENT
CLINICAL PERSONALITY PATTERNS	1	2	24						SCHIZOID
	2A	3	36						AVOIDANT
	2B	1	10						DEPRESSIVE
	3	6	60						DEPENDENT
	4	20	72						HISTRIONIC
	5	15	67						NARCISSISTIC
	6A	17	92						ANTISOCIAL
	6B	8	66						SADISTIC
	7	12	45						COMPULSIVE
	8A	9	66						NEGATIVISTIC
SEVERE PERSONALITY PATHOLOGY	8B	4	63						MASOCHISTIC
	S	4	62						SCHIZOTYPAL
	C	5	60						BORDERLINE
CLINICAL SYNDROMES	P	9	67						PARANOID
	A	1	20						ANXIETY
	H	0	0						SOMATOFORM
	N	7	66						BIPOLAR: MANIC
	D	4	48						DYSTHYMIA
	B	12	88						ALCOHOL DEPENDENCE
	T	11	79						DRUG DEPENDENCE
SEVERE CLINICAL SYNDROMES	R	0	0						POST-TRAUMATIC STRESS
	SS	1	15						THOUGHT DISORDER
	CC	0	0						MAJOR DEPRESSION
SYNDROMES	PP	4	65						DELUSIONAL DISORDER

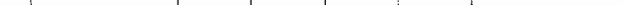
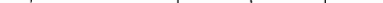
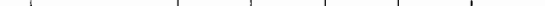
MILLON CLINICAL MULTIAXIAL INVENTORY - III
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FACET SCORES FOR HIGHEST PERSONALITY SCALES BR 65 OR HIGHER

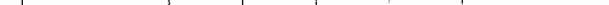
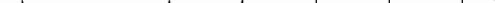
HIGHEST PERSONALITY SCALE BR 65 OR HIGHER: SCALE 6A Antisocial

SCALE	SCORE		PROFILE OF BR SCORES							FACET SCALES
	RAW	BR	0	60	70	80	90	100		
6A.1	4	70								Expressively Impulsive
6A.2	8	98								Acting-Out Mechanism
6A.3	2	68								Interpersonally Irresponsible

SECOND-HIGHEST PERSONALITY SCALE BR 65 OR HIGHER: SCALE 4 Histrionic

SCALE	SCORE		PROFILE OF BR SCORES							FACET SCALES
	RAW	BR	0	60	70	80	90	100		
4.1	7	100								Gregarious Self-Image
4.2	5	65								Interpersonally Attention-Seeking
4.3	5	81								Expressively Dramatic

THIRD-HIGHEST PERSONALITY SCALE BR 65 OR HIGHER: SCALE P Paranoid

SCALE	SCORE		PROFILE OF BR SCORES							FACET SCALES
	RAW	BR	0	60	70	80	90	100		
P.1	3	87								Cognitively Mistrustful
P.2	1	55								Expressively Defensive
P.3	3	76								Projection Mechanism

MILLON CLINICAL MULTIAXIAL INVENTORY - III
CONFIDENTIAL INFORMATION FOR PROFESSIONAL USE ONLY

COMPLETE LISTING OF MCMII-III GROSSMAN FACET SCALE SCORES

	RAW	BR		RAW	BR
1 Schizoid			6B Sadistic		
1.1 Temperamentally Apathetic	0	0	6B.1 Temperamentally Hostile	2	61
1.2 Interpersonally Unengaged	1	27	6B.2 Eruptive Organization	5	91
1.3 Expressively Impassive	2	47	6B.3 Pernicious Representations	3	72
2A Avoidant			7 Compulsive		
2A.1 Interpersonally Aversive	1	42	7.1 Cognitively Constricted	6	69
2A.2 Alienated Self-Image	1	37	7.2 Interpersonally Respectful	4	66
2A.3 Vexatious Representations	0	0	7.3 Reliable Self-Image	1	6
2B Depressive			8A Negativistic		
2B.1 Temperamentally Woeful	0	0	8A.1 Temperamentally Irritable	3	53
2B.2 Worthless Self-Image	2	54	8A.2 Expressively Resentful	2	64
2B.3 Cognitively Fatalistic	0	0	8A.3 Discontented Self-Image	2	55
3 Dependent			8B Masochistic		
3.1 Inept Self-Image	3	61	8B.1 Discredited Representations	1	52
3.2 Interpersonally Submissive	3	53	8B.2 Cognitively Diffident	4	71
3.3 Immature Representations	0	0	8B.3 Undeserving Self-Image	1	43
4 Histrionic			S Schizotypal		
4.1 Gregarious Self-Image	7	100	S.1 Estranged Self-Image	4	75
4.2 Interpersonally Attention-Seeking	5	65	S.2 Cognitively Autistic	0	0
4.3 Expressively Dramatic	5	81	S.3 Chaotic Representations	0	0
5 Narcissistic			C Borderline		
5.1 Admirable Self-Image	7	76	C.1 Temperamentally Labile	0	0
5.2 Cognitively Expansive	2	59	C.2 Interpersonally Paradoxical	5	84
5.3 Interpersonally Exploitive	1	51	C.3 Uncertain Self-Image	1	46
6A Antisocial			P Paranoid		
6A.1 Expressively Impulsive	4	70	P.1 Cognitively Mistrustful	3	87
6A.2 Acting-Out Mechanism	8	98	P.2 Expressively Defensive	1	55
6A.3 Interpersonally Irresponsible	2	68	P.3 Projection Mechanism	3	76

For each of the Clinical Personality Patterns and Severe Personality Pathology scales (the scale names shown in **bold**), scores on the three facet scales are shown beneath the scale name.

RESPONSE TENDENCIES

Unless this patient is a demonstrably well-functioning adult who is currently facing minor life stressors, her responses suggest (1) a well-established need for social approval and commendation, evident in tendencies to present herself in a favorable light, or (2) a general naivete about psychological matters, including a possible deficit in self-knowledge. The interpretation of this profile should be made with these characteristics in mind.

No adjustments were made to the BR scores of this individual to account for any undesirable response tendencies.

AXIS II: PERSONALITY PATTERNS

The following paragraphs refer to those enduring and pervasive personality traits that underlie this woman's emotional, cognitive, and interpersonal difficulties. Rather than focus on the largely transitory symptoms that make up Axis I clinical syndromes, this section concentrates on her more habitual and maladaptive methods of relating, behaving, thinking, and feeling.

The MCMI-III profile of this woman suggests a veneer of friendliness and sociability overlying a deeper contempt for conventional morals. Although she is able to make a good impression on casual acquaintances, there is a characteristic unreliability, impulsiveness, restlessness, and moodiness that may be seen frequently by family members and close associates. There is the possibility that she is untrustworthy and unreliable, persistently seeking attention and excitement and often engaging in seductive and self-dramatizing behavior. Her relationships may be shallow and fleeting, and she may fail to meet routine responsibilities. Interactions may be disrupted by caustic comments and hostile outbursts. Not infrequently, she may act impetuously with insufficient deliberation and poor judgment. She also tends to exhibit short-lived enthusiasm followed by disillusionment and resentment at having been misled. The referring clinician may wish to corroborate these hypotheses as well as those in subsequent paragraphs.

This woman is unlikely to admit responsibility for personal or family difficulties, possessing what may be an easily circumvented conscience. Moreover, she may be quite facile in denying the presence of psychological tension or conflicts. Interpersonal problems are likely to be rationalized, especially those that she engenders, and blame may readily be projected onto others. Although she is prone to be self-indulgent and insistent on gaining the center of attention, she may reciprocate this attention with only minimal loyalty and affection.

When her actions are criticized or she is subjected to minor pressures or faced with potential embarrassment, she may be inclined to abandon her responsibilities, possibly with minimal guilt or remorse. Unfettered by the restrictions of social conventions or the restraints of personal loyalties, she may be quick to free herself from unwanted obligations. Her superficial affability may easily collapse, and she may be readily inclined to jettison anyone who might undermine her autonomy. Although infrequent, her temper outbursts may turn into uncontrollable rages. More typically, she is impetuous and imprudent, driven by a need for excitement and an inability to delay gratification, with minimal regard for consequences. Stimulus-seeking, she may restlessly chase one capricious whim after another, and she may have traveled an erratic course of irresponsibility, perhaps even delighting in defying social

conventions. She appears to have a poor prognosis for staying out of trouble.

GROSSMAN PERSONALITY FACET SCALES

The Grossman facet scales are designed to aid in the interpretation of elevations on the Clinical Personality Patterns and Severe Personality Pathology scales by helping to pinpoint the specific personality processes (e.g., self-image, interpersonal relations) that underlie overall scale elevations. A careful analysis of this patient's facet scale scores suggests that the following characteristics are among her most prominent personality features.

Most notable is her view of herself as a socially stimulating and charming person, one who seeks to pursue a busy and pleasure-oriented lifestyle and is invariably perceived by others as appealing and attractive. She probably lacks insight, however, failing to recognize, or to admit recognizing, her deeper insecurities and her desperate need to garner attention and to be well liked. Signs of inner turmoil, weakness, depression, or hostility are almost invariably denied, suppressed so as not to be part of her sense of self.

Also salient are her failure to constrain or postpone the expression of offensive thoughts or malevolent actions, a deficit in guilt feelings, and a consequent disinclination to refashion repugnant impulses in sublimated form. Given her perception of the environment, she does not feel the need to rationalize her outbursts, which she believes are fully justified as a response to the supposed malevolence of others. She experiences herself as the victim, an indignant bystander subjected to persecution and hostility. Through this intrapsychic maneuver, she not only disowns her malicious impulses but attributes the evil to others. As a persecuted victim, she feels free to counterattack and gain restitution and vindication.

Also worthy of attention are her suspiciousness regarding the motives of others and her tendency to misconstrue innocuous events as signifying proof of duplicity or conspiratorial intent. Her learned feelings and attitudes have produced deep mistrust and pervasive suspiciousness of others. She is notoriously oversensitive and disposed to detect signs of trickery and deception everywhere. She is preoccupied with these thoughts, actively picking up minute cues, then magnifying and distorting them so as to confirm her worst expectations. To further complicate matters, events that fail to confirm her suspicions are evidence in her mind of just how deceitful and clever others can be.

Early treatment efforts are likely to produce optimal results if they are oriented toward modifying the personality features just described.

AXIS I: CLINICAL SYNDROMES

The features and dynamics of the following Axis I clinical syndromes appear worthy of description and analysis. They may arise in response to external precipitants but are likely to reflect and accentuate several of the more enduring and pervasive aspects of this woman's basic personality makeup.

The evidence is strong that this woman exhibits an alcoholic disorder, probably contained within the context of a pervasive substance-abuse syndrome. Her excessive drinking may also be seen, as well as understood, within the context of a broad-based, somewhat self-indulgent, and excitement-seeking

lifestyle. Hedonistic and manipulative, she may use alcohol to maintain the camaraderie of youthful socializing. In addition to accommodating her immature, pleasure-oriented, and exploitive traits, alcoholism permits her to express a number of narcissistic attitudes, antiauthority resentments, an unwillingness to tolerate the limits of conventional society, and rejection of traditional family constraints.

An addictive disposition, probably involving active use of illicit or street agents, seems highly probable in this woman, who is hedonistic and exploitive. That drug use fits her recreational pattern of adolescent-like stimulus seeking and narcissistic indulgence is likely. Also consonant with her personality is the use of drugs as a symbol of disdain for conventional social values as well as an image of flouting authority that includes the posturing of independence from her family.

NOTEWORTHY RESPONSES

The client answered the following statements in the direction noted in parentheses. These items suggest specific problem areas that the clinician may wish to investigate.

Health Preoccupation

No items endorsed.

Interpersonal Alienation

- 48. A long time ago, I decided it's best to have little to do with people. (True)
- 63. Many people have been spying into my private life for years. (True)

Emotional Dyscontrol

- 14. Sometimes I can be pretty rough and mean in my relations with my family. (True)

Self-Destructive Potential

- 24. I began to feel like a failure some years ago. (True)

Childhood Abuse

No items endorsed.

Eating Disorder

No items endorsed.

POSSIBLE *DSM-IV*® MULTIAXIAL DIAGNOSES

The following diagnostic assignments should be considered judgments of personality and clinical prototypes that correspond conceptually to formal diagnostic categories. The diagnostic criteria and items used in the MCMI-III differ somewhat from those in the *DSM-IV*, but there are sufficient parallels in the MCMI-III items to recommend consideration of the following assignments. It should be noted that several *DSM-IV* Axis I syndromes are not assessed in the MCMI-III. Definitive diagnoses must draw on biographical, observational, and interview data in addition to self-report inventories such as the MCMI-III.

Axis I: Clinical Syndrome

The major complaints and behaviors of the patient parallel the following Axis I diagnoses, listed in order of their clinical significance and salience.

- 305.00 Alcohol Abuse
- 305.90 Psychoactive Substance Abuse NOS

Axis II: Personality Disorders

Deeply ingrained and pervasive patterns of maladaptive functioning underlie Axis I clinical syndromal pictures. The following personality prototypes correspond to the most probable *DSM-IV* diagnoses (Disorders, Traits, Features) that characterize this patient.

Personality configuration composed of the following:

- 301.70 Antisocial Personality Disorder
with Histrionic Personality Features
and Paranoid Personality Features

Course: The major personality features described previously reflect long-term or chronic traits that are likely to have persisted for several years prior to the present assessment.

The clinical syndromes described previously tend to be relatively transient, waxing and waning in their prominence and intensity depending on the presence of environmental stress.

Axis IV: Psychosocial and Environmental Problems

In completing the MCMI-III, this individual identified the following problems that may be complicating or exacerbating her present emotional state. They are listed in order of importance as indicated by the client. This information should be viewed as a guide for further investigation by the clinician.

Job or School Problems; Use of Alcohol

TREATMENT GUIDE

If additional clinical data are supportive of the MCMI-III's hypotheses, it is likely that this patient's difficulties can be managed with either brief or extended therapeutic methods. The following guide to treatment planning is oriented toward issues and techniques of a short-term character, focusing on matters that might call for immediate attention, followed by time-limited procedures designed to reduce the likelihood of repeated relapses.

Worthy of note is the possibility of a troublesome alcohol and/or substance-abuse disorder. If verified, appropriate short-term behavioral management or group therapy programs should be rapidly implemented.

Once this patient's more pressing or acute difficulties are adequately stabilized, attention should be directed toward goals that would aid in preventing a recurrence of problems, focusing on circumscribed issues and employing delimited methods such as those discussed in the following paragraphs.

Essential to the success of a short-term approach with this woman is the therapist's readiness to see things from the patient's point of view and to convey a sense of trust and to create a feeling of alliance. To achieve reasonable short-term goals, this building of rapport must not be interpreted as a sign of the therapist's capitulation to the patient's bluff and arrogance. Brief treatment with her will require a balance of professional firmness and authority, mixed with tolerance for the patient's less attractive traits. By building an image of a fair-minded and strong authority figure, the therapist may successfully employ cognitive methods that will encourage the patient to change her expectations. Through reasoned and convincing comments, the therapist may provide a model for the patient to learn the mix of power, logic and fairness.

Less confrontive cognitive approaches may provide the patient with opportunities to vent her anger, even in short-term therapy. Once drained of these hostile feelings, she may be led to examine her habitual behavior and cognitive attitudes and be guided into less destructive perceptions and outlets than before. Interpersonal methods, such as those of Benjamin and Kiesler, may provide a means to explore more socially acceptable behaviors. As far as group methods are concerned, until the patient has incorporated changed cognitions and actions, she may intrude and disrupt therapeutic functions. On the other hand, she may become a useful catalyst for short-term group interaction and gain some useful insights and a few constructive skills.

A useful short-term goal for this woman is to enable her to tolerate the experience of guilt or to accept blame for the turmoil she may cause. Cognitive methods using a measure of confrontation may help undermine her tendency to always trace problems to another person's stupidity, laziness, or hostility. When she does accept responsibility for some of her difficulties it is important that the therapist be prepared to deal with the patient's inclination to resent the therapist for supposedly tricking her into admitting it. Similarly, the therapist should be ready to be challenged and avoid efforts to outwit her. The patient may try to set up situations to test the therapist's skills, to catch inconsistencies, to arouse ire and, if possible, to belittle and humiliate the therapist. Restraining impulses to express condemning attitudes can be a major task for the therapist, but one that can be used for positive gains, especially if tied into the application of combined cognitive (e.g., Beck, Ellis) and interpersonal interventions.

It should be noted that the precipitant for this woman's treatment is probably situational rather than internal. Hence, she is unlikely to have sought therapy voluntarily, and she may be convinced that if she were just left alone, she could work matters out on her own. Such beliefs will have to be confronted, albeit carefully. Similarly, if treatment is self-motivated, it probably was inspired by a series of legal entanglements, family problems, social humiliations, or achievement failures. Whatever its source, a firm cognitive and behavior-change approach would seem required. For this domineering and often intimidating woman, complaints are likely to be expressed in the form of irritability and restlessness. To succeed in her initial disinclination to be frank with authority figures, she may wander from one superficial topic to another. This inclination should be monitored and prevented. Moreover, contact with family members may be advisable because they may report matters quite differently than the patient. To ensure that she takes discussions seriously, she may have to be confronted directly with evidence of her contribution to her troubles. Treatment is best geared to short-term goals, reestablishing her psychic balance, and strengthening her previously adequate coping behavior with cognitive methods, unless her actions are frankly antisocial. In general, short-term approaches with this patient are best directed toward building controls rather than insights, toward the here and now rather than the past, and toward teaching her ways to sustain relationships cooperatively rather than with dominance and intimidation.

End of Report

NOTE: This and previous pages of this report contain trade secrets and are not to be released in response to requests under HIPAA (or any other data disclosure law that exempts trade secret information from release). Further, release in response to litigation discovery demands should be made only in accordance with your profession's ethical guidelines and under an appropriate protective order.

ITEM RESPONSES

1: 2	2: 1	3: 1	4: 2	5: 2	6: 1	7: 1	8: 1	9: 2	10: 2
11: 2	12: 2	13: 1	14: 1	15: 2	16: 1	17: 1	18: 2	19: 2	20: 2
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Appendix I

Sample Profile Report



MCMI-III™
MILLON™ CLINICAL
MULTIAXIAL INVENTORY-III

MCMI-III™

Millon™ Clinical Multiaxial Inventory-III

Profile Report with Grossman Facet Scales

Theodore Millon, PhD, DSc

Name:	Sample Profile Report
ID Number:	12345
Age:	34
Gender:	Female
Setting:	Outpatient Never Hospitalized
Race:	Hispanic
Marital Status:	First Marriage
Date Assessed:	03/11/2009

TRADE SECRET INFORMATION

Not for release under HIPAA or other data disclosure laws that exempt trade secrets from disclosure.

[7.0 / 83 / 2.3.10]

INTERPRETIVE CONSIDERATIONS

MCMI-III reports are normed on patients who were in the early phases of assessment or psychotherapy for emotional discomfort or social difficulties. Respondents who do not fit this normative population or who have inappropriately taken the MCMI-III for nonclinical purposes may have inaccurate reports. The MCMI-III report cannot be considered definitive. It should be evaluated in conjunction with additional clinical data. The report should be evaluated by a mental health clinician trained in the use of psychological tests. The report should not be shown to patients or their relatives.

The client is a 34-year-old married Hispanic female with 13 years of education. She is currently being seen as an outpatient, and she reports that she has recently experienced problems that involve illness or fatigue and low self-confidence. These self-reported difficulties, which have occurred in the last one to four weeks, may take the form of an Axis I disorder.

The BR scores reported for this individual have been modified to account for the psychic tension indicated by the elevation on Scale A (Anxiety).

MILLON CLINICAL MULTIAXIAL INVENTORY - III

CONFIDENTIAL INFORMATION FOR PROFESSIONAL USE ONLY

INVALIDITY (SCALE V) = 0 INCONSISTENCY (SCALE W) = 5

PERSONALITY CODE: 3 ** - * 2A 7 8B 2B 1 6A + 4 " 8A 5 6B ' ' // - ** - * //

SYNDROME CODE: A ** - * // - ** CC * //

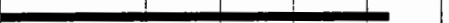
DEMOGRAPHIC CODE: 12345/ON/F/34/H/F/13/IL/SC/30030/2/-----/

CATEGORY		SCORE		PROFILE OF BR SCORES					DIAGNOSTIC SCALES
		RAW	BR	0	60	75	85	115	
MODIFYING INDICES	X	87	53						DISCLOSURE
	Y	13	59						DESIRABILITY
	Z	13	61						DEBASEMENT
CLINICAL PERSONALITY PATTERNS	1	5	60						SCHIZOID
	2A	8	69						AVOIDANT
	2B	8	67						DEPRESSIVE
	3	15	89						DEPENDENT
	4	11	44						HISTRIONIC
	5	5	25						NARCISSISTIC
	6A	5	60						ANTISOCIAL
	6B	2	24						SADISTIC
	7	20	69						COMPULSIVE
	8A	3	30						NEGATIVISTIC
	8B	7	69						MASOCHISTIC
SEVERE PERSONALITY PATHOLOGY	S	3	58						SCHIZOTYPAL
	C	3	33						BORDERLINE
	P	0	0						PARANOID
CLINICAL SYNDROMES	A	12	91						ANXIETY
	H	7	66						SOMATOFORM
	N	2	24						BIPOLAR: MANIC
	D	6	62						DYSTHYMIA
	B	3	60						ALCOHOL DEPENDENCE
	T	4	63						DRUG DEPENDENCE
	R	3	45						POST-TRAUMATIC STRESS
SEVERE CLINICAL SYNDROMES	SS	5	62						THOUGHT DISORDER
	CC	7	75						MAJOR DEPRESSION
	PP	0	0						DELUSIONAL DISORDER

MILLON CLINICAL MULTIAXIAL INVENTORY - III
CONFIDENTIAL INFORMATION FOR PROFESSIONAL USE ONLY

FACET SCORES FOR HIGHEST PERSONALITY SCALES BR 65 OR HIGHER

HIGHEST PERSONALITY SCALE BR 65 OR HIGHER: SCALE 3 Dependent

SCALE	SCORE		PROFILE OF BR SCORES						FACET SCALES
	RAW	BR	0	60	70	80	90	100	
3.1	5	78							Inept Self-Image
3.2	6	93							Interpersonally Submissive
3.3	1	56							Immature Representations

SECOND-HIGHEST PERSONALITY SCALE BR 65 OR HIGHER: SCALE 2A Avoidant

SCALE	SCORE		PROFILE OF BR SCORES						FACET SCALES
	RAW	BR	0	60	70	80	90	100	
2A.1	1	42							Interpersonally Aversive
2A.2	5	75							Alienated Self-Image
2A.3	2	52							Vexatious Representations

THIRD-HIGHEST PERSONALITY SCALE BR 65 OR HIGHER: SCALE 7 Compulsive

SCALE	SCORE		PROFILE OF BR SCORES						FACET SCALES
	RAW	BR	0	60	70	80	90	100	
7.1	6	69							Cognitively Constricted
7.2	5	85							Interpersonally Respectful
7.3	4	54							Reliable Self-Image

MILLON CLINICAL MULTIAXIAL INVENTORY - III
CONFIDENTIAL INFORMATION FOR PROFESSIONAL USE ONLY

COMPLETE LISTING OF MCMII-III GROSSMAN FACET SCALE SCORES

	RAW	BR		RAW	BR
1 Schizoid			6B Sadistic		
1.1 Temperamentally Apathetic	3	63	6B.1 Temperamentally Hostile	0	0
1.2 Interpersonally Unengaged	4	65	6B.2 Eruptive Organization	2	54
1.3 Expressively Impassive	4	77	6B.3 Pernicious Representations	0	0
2A Avoidant			7 Compulsive		
2A.1 Interpersonally Aversive	1	42	7.1 Cognitively Constricted	6	69
2A.2 Alienated Self-Image	5	75	7.2 Interpersonally Respectful	5	85
2A.3 Vexatious Representations	2	52	7.3 Reliable Self-Image	4	54
2B Depressive			8A Negativistic		
2B.1 Temperamentally Woeful	2	56	8A.1 Temperamentally Irritable	1	28
2B.2 Worthless Self-Image	4	73	8A.2 Expressively Resentful	1	46
2B.3 Cognitively Fatalistic	2	52	8A.3 Discontented Self-Image	4	77
3 Dependent			8B Masochistic		
3.1 Inept Self-Image	5	78	8B.1 Discredited Representations	1	52
3.2 Interpersonally Submissive	6	93	8B.2 Cognitively Diffident	7	96
3.3 Immature Representations	1	56	8B.3 Undeserving Self-Image	3	66
4 Histrionic			S Schizotypal		
4.1 Gregarious Self-Image	4	49	S.1 Estranged Self-Image	2	58
4.2 Interpersonally Attention-Seeking	4	43	S.2 Cognitively Autistic	1	50
4.3 Expressively Dramatic	4	60	S.3 Chaotic Representations	1	43
5 Narcissistic			C Borderline		
5.1 Admirable Self-Image	3	22	C.1 Temperamentally Labile	2	51
5.2 Cognitively Expansive	1	35	C.2 Interpersonally Paradoxical	1	36
5.3 Interpersonally Exploitive	0	0	C.3 Uncertain Self-Image	3	66
6A Antisocial			P Paranoid		
6A.1 Expressively Impulsive	1	23	P.1 Cognitively Mistrustful	0	0
6A.2 Acting-Out Mechanism	3	70	P.2 Expressively Defensive	0	0
6A.3 Interpersonally Irresponsible	0	0	P.3 Projection Mechanism	0	0

For each of the Clinical Personality Patterns and Severe Personality Pathology scales (the scale names shown in **bold**), scores on the three facet scales are shown beneath the scale name.

NOTEWORTHY RESPONSES

The client answered the following statements in the direction noted in parentheses. These items suggest specific problem areas that the clinician may wish to investigate.

Health Preoccupation

- 1. Lately, my strength seems to be draining out of me, even in the morning. (True)
- 4. I feel weak and tired much of the time. (True)
- 74. I can't seem to sleep, and wake up just as tired as when I went to bed. (True)
- 75. Lately, I've been sweating a great deal and feel very tense. (True)
- 107. I have completely lost my appetite and have trouble sleeping most nights. (True)

Interpersonal Alienation

- 27. When I have a choice, I prefer to do things alone. (True)
- 99. In social groups I am almost always very self-conscious and tense. (True)

Emotional Dyscontrol

- 34. Lately, I have gone all to pieces. (True)
- 124. When I'm alone and away from home, I often begin to feel tense and panicky. (True)

Self-Destructive Potential

- 24. I began to feel like a failure some years ago. (True)

Childhood Abuse

No items endorsed.

Eating Disorder

No items endorsed.

End of Report

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161: 2	162: 2	163: 2	164: 2	165: 2	166: 2	167: 2	168: 2	169: 1	170: 1
171: 2	172: 1	173: 2	174: 2	175: 2					

Appendix J

Descriptions of the Grossman Facet Scales

Schizoid (Scale 1) Facets

- 1.1 *Temperamentally Apathetic:* Is emotionally unexcitable, exhibiting an intrinsic unfeeling, cold, stark quality. Reports weak affectionate or erotic needs, rarely displaying warm or intense feelings. Is apparently unable to experience most affects—pleasure, sadness, or anger—in any depth.
- 1.2 *Interpersonally Unengaged:* Seems indifferent and remote, rarely responsive to the actions or feelings of others. Chooses solitary activities and has minimal “human” interests. Fades into the background. Is aloof or unobtrusive. Neither desires nor enjoys close relationships, preferring a peripheral role in social, school, and family settings.
- 1.3 *Expressively Impassive:* Appears to be in an inert emotional state, lifeless, undemonstrative, and lacking in energy and vitality. Is unmoved, boring, unanimated, robotic, phlegmatic. Displays deficits in activation, motoric expressiveness, and spontaneity.

Avoidant (Scale 2A) Facets

- 2A.1 *Interpersonally Aversive:* Distances him- or herself from activities that involve intimate personal relationships and reports an extensive history of social anxiety and distrust. Seeks acceptance but is unwilling to get involved unless certain to be liked, maintaining distance and privacy to avoid being shamed and humiliated.
- 2A.2 *Alienated Self-Image:* Sees him- or herself as socially inept, inadequate, and inferior, thereby justifying his or her isolation from and rejection by others. Feels personally unappealing, devalues his or her achievements, and reports a persistent sense of aloneness and emptiness.
- 2A.3 *Vexatious Representations:* Has internalized representations that are composed of readily reactivated, intense, and conflict-ridden memories of problematic early relations, limited avenues for experiencing or recalling gratification, and few mechanisms to channel needs, bind impulses, resolve conflicts, or deflect external stressors.

Depressive (Scale 2B) Facets

- 2B.1 *Temperamentally Woeful:* Is typically melancholic, gloomy, tearful, joyless, and morose. Worries and broods continually. Low spirits and dysphoric state rarely remit.

- 2B.2 *Worthless Self-Image*: Judges him- or herself to be of no account, valueless to self or others, inadequate, and unsuccessful in all aspirations. Feels barren, sterile, and impotent. Sees self as inconsequential and reproachable if not contemptible, a person who should be criticized and derogated and who should feel guilty for having no praiseworthy traits or achievements.
- 2B.3 *Cognitively Fatalistic*: Has defeatist and fatalistic attitudes about almost all matters. Sees things in their blackest form and invariably expects the worst. Feels weighed down, discouraged, and bleak. Gives the gloomiest interpretation of current events, feeling that things will never improve.

Dependent (Scale 3) Facets

- 3.1 *Inept Self-Image*: Views self as weak, fragile, and inadequate. Exhibits lack of self-confidence by belittling own attitudes and competencies and hence is not capable of doing things on his or her own.
- 3.2 *Interpersonally Submissive*: Needs excessive advice and reassurance. Subordinates him- or herself to a stronger, nurturing figure, without whom he or she may feel anxiously alone and helpless. Is compliant, conciliatory, and placating, afraid of being left to care for him- or herself.
- 3.3 *Immature Representations*: Has internalized representations that are composed of infantile impressions of others, unsophisticated ideas, incomplete recollections, rudimentary drives, childlike impulses, and minimal competencies to manage and resolve stressors.

Histrionic (Scale 4) Facets

- 4.1 *Gregarious Self-Image*: Views him- or herself as sociable, stimulating, and charming. Enjoys the image of attracting acquaintances by physical appearance and by pursuing a busy and pleasure-oriented life.
- 4.2 *Interpersonally Attention-Seeking*: Actively solicits praise and manipulates others to gain needed reassurance, attention, and approval. Is demanding, flirtatious, vain, and seductively exhibitionistic, especially when wishing to be the center of attention.
- 4.3 *Expressively Dramatic*: Is overreactive, volatile, provocative, and engaging. Is intolerant of inactivity, resulting in impulsive, highly emotional, and theatrical responsiveness. Describes penchant for momentary excitement, fleeting adventures, and shortsighted hedonism.

Narcissistic (Scale 5) Facets

- 5.1 *Admirable Self-Image*: Believes him- or herself to be meritorious, special, unique, and deserving of great admiration. Acts in a grandiose or self-assured manner, often without parallel achievements. Has a very strong sense of self-worth despite being seen by others as egotistic, inconsiderate, and arrogant.
- 5.2 *Cognitively Expansive*: Has an undisciplined imagination and exhibits a preoccupation with immature and self-glorifying fantasies of success, beauty, or love. Is minimally constrained by objective reality, takes liberties with facts, and often lies to redeem illusions about him- or herself.
- 5.3 *Interpersonally Exploitive*: Feels entitled, is unempathic, and expects special favors without assuming reciprocal responsibilities. Shamelessly takes others for granted and uses them to enhance self and indulge desires.

Antisocial (Scale 6A) Facets

- 6A.1 *Expressively Impulsive*: Is impetuous and irrepressible, acting hastily and spontaneously in a restless, spur-of-the-moment manner. Is short-sighted, careless, and imprudent, failing to plan ahead, consider alternatives, or heed consequences.

- 6A.2 *Acting-Out Mechanism*: Avoids inner tensions through the unconstrained expression of offensive thoughts and malevolent actions. Does not refashion socially repugnant impulses in sublimated forms but discharges them directly in precipitous ways, usually without guilt or remorse.
- 6A.3 *Interpersonally Irresponsible*: Is untrustworthy and unreliable, failing to meet or intentionally negating personal obligations. Actively intrudes on and violates the rights of others. Transgresses established social codes through deceitful or illegal behavior.

Sadistic (Scale 6B) Facets

- 6B.1 *Temperamentally Hostile*: Has an excitable and irritable temper that flares readily into contentious arguments and physical belligerence. Is cruel, mean-spirited, and fractious, willing to do harm or even persecute others to get his or her way.
- 6B.2 *Eruptive Organization*: Although he or she has a generally cohesive morphologic structure composed of routinely adequate modulating controls, defenses, and expressive channels, surging powerful and explosive energies of an aggressive and sexual nature threaten to produce precipitous outbursts that periodically overwhelm and overrun otherwise competent restraints.
- 6B.3 *Pernicious Representations*: Has internalized representations of the past that are distinguished by early relationships that have generated strongly driven aggressive energies and malicious attitudes as well as by a contrasting paucity of sentimental memories, tender affects, internal conflicts, shame, or guilt feelings.

Compulsive (Scale 7) Facets

- 7.1 *Cognitively Constricted*: Constructs the world in terms of rules, regulations, schedules, and hierarchies. Is rigid, stubborn, and indecisive and is notably upset by unfamiliar or novel ideas and customs.
- 7.2 *Interpersonally Respectful*: Exhibits unusual adherence to social conventions and propriety. Is scrupulous and overly conscientious about matters of morality and ethics. Prefers polite, formal, and proper personal relationships, usually insisting that subordinates adhere to personally established rules and methods.
- 7.3 *Reliable Self-Image*: Sees him- or herself as devoted to work, industrious, reliable, meticulous, and efficient, largely to the exclusion of leisure activities. Is fearful of error or misjudgment and hence overvalues aspects of him- or herself that exhibit discipline, perfection, prudence, and loyalty.

Negativistic (Scale 8A) Facets

- 8A.1 *Temperamentally Irritable*: Is frequently touchy, temperamental, and peevish, followed by sullen and moody withdrawal. Is often petulant and impatient. Unreasonably scorns those in authority and reports being easily annoyed or frustrated by others.
- 8A.2 *Expressively Resentful*: Resists fulfilling others' expectations, frequently exhibiting procrastination, inefficiency, and obstinate, contrary, and irksome behavior. Reveals gratification in demoralizing and undermining the pleasures and aspirations of others.
- 8A.3 *Discontented Self-Image*: Sees him- or herself as misunderstood, luckless, unappreciated, jinxed, and demeaned by others. Recognizes being characteristically embittered, disgruntled, and disillusioned with life.

Masochistic (Scale 8B) Facets

- 8B.1 *Discredited Representations*: Has object representations that are composed of failed past relationships and disparaged personal achievements, of positive feelings and erotic drives transposed into their least attractive opposites, of internal conflicts intentionally aggravated, and of mechanisms for reducing dysphoria being subverted by processes that intensify discomfort.

- 8B.2 *Cognitively Diffident*: Is hesitant to interpret observations positively for fear that, in doing so, they may not take problematic forms or achieve troublesome and self-denigrating outcomes. As a result, he or she has a habit of repeatedly expressing attitudes and anticipation that are contrary to favorable beliefs and feelings.
- 8B.3 *Undeserving Self-Image*: Is self-abasing, focusing on his or her very worst personal features and thereby asserting that he or she is worthy of being shamed, humbled, and debased. Feels that he or she has failed to live up to others' expectations and hence deserves to suffer painful consequences.

Schizotypal (Scale S) Facets

- S.1 *Estranged Self-Image*: Experiences recurrent social perplexities and illusions and experiences of depersonalization, derealization, and dissociation. Sees him- or herself as forlorn, with repetitive thoughts of life's emptiness and meaninglessness.
- S.2 *Cognitively Autistic*: Has a capacity to "read" the thoughts and feelings of others that is markedly dysfunctional. Mixes social communication with personal irrelevancies, circumstantial speech, ideas of reference, and metaphorical asides. Is often ruminative, appearing self-absorbed and lost in daydreams with occasional magical thinking, bodily illusions, obscure suspicions, odd beliefs, and a blurring of reality and fantasy.
- S.3 *Chaotic Representations*: Has internalized representations that consist of a piecemeal jumble of early relationships and affects, random drives and impulses, and uncoordinated channels of regulation that are only fitfully competent for binding tensions, accommodating needs, and mediating conflicts.

Borderline (Scale C) Facets

- C.1 *Temperamentally Labile*: Fails to accord unstable mood level with external reality. Has marked shifts from normality to depression to excitement or has periods of dejection and apathy interspersed with episodes of inappropriate and intense anger and brief spells of anxiety or euphoria.
- C.2 *Interpersonally Paradoxical*: Although needing attention and affection, is unpredictably contrary, manipulative, and volatile, frequently eliciting rejection rather than support. Frantically reacts to fears of abandonment and isolation but often in angry, mercurial, and self-destructive ways.
- C.3 *Uncertain Self-Image*: Experiences the confusion of an immature, nebulous, or wavering sense of identity, often with underlying feelings of emptiness. Seeks to redeem precipitate actions and changing self-presentations with expressions of contrition and self-punitive behavior.

Paranoid (Scale P) Facets

- P.1 *Cognitively Mistrustful*: Is unwarrantedly skeptical, cynical, and mistrustful of the motives of others, including relatives, friends, and associates, construing innocuous events as signifying hidden or conspiratorial intent. Tends to read hidden meaning into benign matters and to magnify tangential or minor difficulties into proof of duplicity and treachery, especially regarding the fidelity and trustworthiness of a spouse or intimate friend.
- P.2 *Expressively Defensive*: Is vigilantly guarded, alert to anticipate and ward off expected derogation, malice, and deception. Is tenacious and firmly resistant to sources of external influence and control.
- P.3 *Projection Mechanism*: Actively disowns undesirable personal traits and motives and attributes them to others. Remains blind to his or her own unattractive behavior and characteristics, yet is overalert to and hypercritical of similar features in others.

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